



ICT-215282 STP ROCKET

Reconfigurable OFDMA-based Cooperative Networks Enabled
by Agile Spectrum Use

D7

3D2 – Multi-User Cooperative Transmission techniques for OFDMA Multi-hop Cellular Networks

Contractual Date of Delivery to the CEC: August 2009	
Actual Date of Delivery to the CEC:	November 2009
Author(s):	Reza Hoshyar, Fabien Hélot, Muhammad Imran (UniS), Adrian Agustin, Josep Vidal, Olga Muñoz (UPC), Sébastien Simoens, Patrick Labbe (MOT), Enrico de Marinis, Giovanni Imponente (DUNE), Isabelle Icart, Pascale Fouillot (TCF)
Participant(s):	UniS, UPC, MOT, DUNE, TCF
Workpackage:	3
Est. person months:	46.93
Security:	Restricted
Dissemination Level:	Public (PU)
Version:	1
Total number of pages:	220

Abstract:

This deliverable presents the outcomes of the activity 3A2. The main objective is the development and design of novel Multi-User (MU) single-cell non-cooperative and cooperative transmission techniques in the context of Orthogonal Frequency Division Multiple Access (OFDMA) multi-hop cellular networks. This deliverable extends the work that was undertaken in [ROCKET-3D1] for the single user case to the MU case by considering none, one or two relays, half-duplex relays, relays working either in Decode-and-Forward (DF) or Amplify and Forward (AF) mode, transmission schemes that are based on one-way, two-way and two-path relaying, UpLink (UL) or DownLink (DL) setting, and full or imperfect Channel State Information (CSI). The first group of techniques that are presented in this deliverable is the MU non-relaying ones, where low-complexity UL & DL MU pairing and best band selection schemes are designed by either considering a fully coherent or a more practical approach. The second part of the deliverable introduces novel MU cooperative resource allocation techniques for the one-way, two-way and two-path relay channel in both UL and DL settings. Moreover, practical issues concerning the Media Access Control (MAC) parameter definition and modification for implementing our newly proposed techniques and the level of complexity of our various novel MU schemes are also discussed.

Keyword list: Multi-user, single-cell systems, cooperation, one-way relay channel, two-way relay channel, two-path relay channel, resource allocation

DISCLAIMER

The work associated with this report has been carried out in accordance with the highest technical standards and the ROCKET partners have endeavoured to achieve the degree of accuracy and reliability appropriate to the work in question. However since the partners have no control over the use to which the information contained within the report is to be put by any other party, any other such party shall be deemed to have satisfied itself as to the suitability and reliability of the information in relation to any particular use, purpose or application.

Under no circumstances will any of the partners, their servants, employees or agents accept any liability whatsoever arising out of any error or inaccuracy contained in this report (or any further consolidation, summary, publication or dissemination of the information contained within this report) and/or the connected work and disclaim all liability for any loss, damage, expenses, claims or infringement of third party rights.

Executive Summary

This deliverable presents advanced Multi-User (MU) non-cooperative and cooperative techniques for single cell communication systems. It therefore extends the concepts that were developed in the activity 3A1 of ROCKET for the single-user scenario to MU communication scenario, where Base Station (BS) and Relay Station (RS) serve jointly multiple terminals. Three examples are:

- The Multiple Access Relay Channel (MARC), to be used in the UpLink (UL).
- The Broadcast Relay Channel (BRC), to be used in the Downlink (DL).
- The MU two-way and two-path relay channel.

In addition, the simultaneous transmission from multiple relays is studied. In the MU context, this means that the cooperative relays are now shared among several users. One obvious solution is to allocate orthogonal resources for different relays (an approach to be considered as a benchmark). However, depending on the degree of knowledge at the relays, more efficient solutions can be envisioned.

Resource allocation is first studied in a non-cooperative context for both UL and DL settings. Novel techniques explore the reuse of resources when superposed transmission is utilised, i.e., multiple users share the same resource, in coherent and more realistic scenarios. In this context, using pairing is a key aspect of the resource allocation and a low complexity UL MU pairing and best band selection scheme for transmit BeamForming (BF) is first studied. This study provides insights on the optimisation of user distribution over Orthogonal Frequency Division Multiple Access (OFDMA) chunks, in order to maximise the overall throughput and the band allocation for Single-Input Multiple-Output (SIMO) and Multiple-Input Multiple-Output (MIMO) systems. This study, which shows great benefits in a coherent setting, is then extended to inhomogeneous setting where the user mobility is taken into account. The presence of mobility steered toward a re-design of the frame format and a sub-optimal aggregate methodology for the MU channel estimation at the BS. The user pairing concept is also analysed in the DL MU setting in terms of reuse benefits, where a pair of users with diverse mean channel strengths competes in order to be served by a single Base Station (BS) over two allocated subcarriers. Each subcarrier is either exclusively used by only one user at a time or reused with one additional user sharing the same subcarrier by simultaneously using it along with the other user. Furthermore, several criteria are investigated in order to optimise the user pairing, such as the Weighted Sum-Rate (WSR) and the Proportional Fairness (PF), and the optimal resource allocation is derived for these criteria.

Resource allocation is then studied in a cooperative context for UL and DL settings, different numbers of relay and various transmission strategies. Optimal resource allocation schemes are designed under various transmit power constraints for different transmissions schemes involving the One-Way Relay Channel (OWRC), Two-Way Relay Channel (TWRC) and superposed transmission. System level evaluation of the cellular throughput will be used to measure the performance of one of the scheduling algorithm that groups the users sharing the same resources. The benefit of using superposed transmission is also investigated. Moreover, novel power allocation algorithms are also designed under various levels of CSI at the BS and Relay Station (RS) for a UL MU MIMO AF system with multiple carriers by assuming different optimisation criteria. Finally, a comprehensive study of the performance of the One-Way Two-Path-Cooperative Relaying (OWTP-CR) protocol in a single-cell MU OFDMA context is undertaken. The study aims at evaluate the performance gains that can be obtained by using a two-way relay cooperative protocol in comparison with the Direct Transmission (DT) and other conventional single-relay assisted protocols in a cellular context. The applicability of this protocol is also investigated in future WiMAX cellular networks.

Practical issues concerning the Media Access Control (MAC) parameter definition and the level of complexity and adaptability of our various novel MU schemes will also be discussed.

Table of Contents

LIST OF TABLES	12
LIST OF FIGURES	13
1 INTRODUCTION.....	18
2 CELLULAR SCENARIOS AND MAC ASSUMPTIONS.....	20
2.1 SCENARIOS	20
2.2 MAC SUPPORT OF PHY	22
2.2.1 General assumptions	22
2.2.2 OFDMA radio access and definitions	23
3 ADVANCED NON-RELAYING MULTI-USER TECHNIQUES	26
3.1 UPLINK MULTI-USER PAIRING	28
3.1.1 Best band selection for transmit beamforming	28
3.1.2 Impact of the mobility and inhomogeneous scenarios	39
3.1.3 Complexity	61
3.2 DOWNLINK MULTI-USER PAIRING	64
3.2.1 User pairing in OFDM systems	64
3.2.2 Scheduling criteria for downlink superposed transmissions	71
4 ADVANCED RELAYING MULTI-USER TECHNIQUES.....	97
4.1 MULTI-USER ONE-WAY & TWO-WAY RELAY CHANNEL WITH A COMMON RS	97
4.1.1 Optimal resource allocation for relay-assisted transmission	97
4.1.2 Cellular throughput evaluation of multi-user relay-assisted transmissions	124
4.2 UPLINK MULTI-USER POWER ALLOCATION FOR AF COOPERATIVE MIMO SYSTEMS... 132	
4.2.1 Introduction	132
4.2.2 Single-user power allocation methods for nonregenerative cooperative MIMO communication under various levels of CSI	133
4.2.3 Uplink multi-user power allocation methods for nonregenerative cooperative MIMO communication	151
4.2.4 Conclusion	156
4.3 MULTI-USER ONE-WAY TWO-PATH RELAY CHANNEL WITH AF	157
4.3.1 Introduction	157
4.3.2 Network	157
4.3.3 Waveform design	159
4.3.4 OMNeT++ simulation framework for ROCKET	175
4.3.5 Performance evaluation	185
4.3.6 Complexity	197
4.3.7 Conclusions and outlook	198
5 MAC PROTOCOL SPECIFICATION.....	200
5.1 MAC FRAME CONTROL	200
5.2 RECOMMENDATIONS FOR MAC PROTOCOL SPECIFICATION	202
5.2.1 Multi-user pairing in the UL	202
5.2.2 RRM for superposed transmission for DL non-relay techniques	205
5.2.3 RRM for superposed transmission for relay-assisted techniques	205
5.2.4 Uplink multi-user power allocation for non-regenerative cooperative MIMO	206
5.2.5 Multi-user one-way two-path relay channel with AF	206
6 COMPLEXITY AND ADAPTABILITY OF THE PROPOSED ALGORITHMS	208
7 CONCLUSIONS	214
8 APPENDIX: CHANNEL MODEL AND SYSTEM CONFIGURATION	216

References & Standards

- [Agus-08] A.Agustin, "Relay-Assisted Transmission and Radio Resource Management for Wireless Networks", PhD Thesis, Technical University of Catalonia (UPC), Barcelona, Spain, May 2008.
- [Agus-09] A.Agustin, J.Vidal, O.Muñoz, "*Protocols and resource allocation for the two-way relay channel with half-duplex terminals*", in Proc. of IEEE International Conference on Communications, Dresden, Germany, June 2009.
- [Alam-98] S. M. Alamouti, "A simple Transmit Diversity Technique for Wireless Communications," *IEEE Journal on Selected Areas in Communications*, vol. 16, n°8, pp. 1451-1458, Oct. 1998.
- [Anders-00] Jorgen Bach Anderson. Array Gain and Capacity for Known Random Channels with Multiple Element Arrays at Both Ends. *IEEE Journal of Selected Areas in Communications*, 18(11):2172–2178, November 2000.
- [Andrews-05] Andrews, J.G. Interference cancellation for cellular systems: a contemporary overview. *Wireless Communications, IEEE*, 12(2):19-29, 2005.
- [Anghel-03] P.A. Anghel, G. Leus, M. Kaveh, "Multi-user space-time coding in cooperative networks, " Proc. of ICASSP 2003, Vol. 4, pp. IV-73-IV-76, April 2003.
- [Bergmans-74] P. P. Bergmans and T. M. Cover. Cooperative Broadcasting. *IEEE Transactions on Information Theory*, IT-20:317--324, 1974.
- [Bhas-08] S.R.Bhaskaran, "Gaussian Degraded Relay Broadcast Channel", *IEEE Trans. on Information Theory*, vol. 54, no.8, August 2008.
- [Boyd-04] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge Univ. Press, Cambridge, UK, 2004.
- [Calvo-09] E.Calvo, J.Vidal and J.R.Fonollosa, "Optimal Resource Allocation in Relay-assisted Cellular Networks with Partial CSI", *IEEE Trans. on Signal Processing*, vol.57, no.7, July 2009.
- [Chen-08] X.Chen, H.Hu, H.Wang, M.Guizani, "Double proportional fair user pairing algorithm for uplink virtual MIMO systems", *IEEE Trans. on Wireless Comm*, vol.7, no.7, July 2008.
- [Cheong-99] Cheong Yui Wong and Cheng, R.S. and Lataief, K.B. and Murch, R.D. Multiuser OFDM with adaptive subcarrier, bit, and power allocation. *Selected Areas in Communications, IEEE Journal on*, 17(10):1747-1758, 1999.
- [Choi-07] J.G.Choi, S.Bahk, "Cell-Throughput Analysis of the proportional fair scheduler in the single-cell environment", *IEEE Trans. on Vehicular Technology*, vol.56, no.2, March 2007.

- [Cost-81] M.H.M.Costa, "Writing on dirty paper", *IEEE Trans. on Information Theory*, vol.27, no.3, pp.508-511, Jul. 1981.
- [Cove-91] T.M.Cover, J.A.Thomas, *Elements of Information Theory*, 2nd edition, Wiley, 1991.
- [Esli-08a] C.Esli, A.Wittneben, "Multiuser MIMO Two-way Relaying for cellular communications", in *Proc. IEEE Personal Indoor Mobile Radio Communications (PIMRC)*, Sep. 2008.
- [Esli-08b] C.Esli, A.Wittneben, "One- and Two-Way Decode-and-forward relaying for Wireless Multi-user MIMO Networks", in *Proc. IEEE Global Communications Conf. (GLOBECOM)*, Dec.2008.
- [Fan-07] Y. Fan and J. Thompson, "MIMO Configurations for Relay Channels: Theory and Practice," *IEEE Trans. Wireless Commun.*, vol. 6, no. 5, pp. 1774–1786, May 2007.
- [FWRKS-2D2] Flexible Relay Wireless OFDM-based networks, (IST-27675 STP FIREWORKS) consortium, "2D2- Advanced radio resource management algorithms for relay-based networks", July 2006. Available online: www.ist-fireworks.eu
- [Goldsmith-97] A. J. Goldsmith. The capacity of downlink fading channels with variable rate and power. *IEEE Transactions on Vehicular Technology*, 46:569--580, 1997.
- [Goldsmith-05] A. J. Goldsmith. *Wireless Communications*. Cambridge University Press, 2005.
- [Han-03] L. Hanzo, M. Munster, B. J. Choi, and T. Keller, OFDM and MCCDMA for broadband multi-user communications, WLANs and broadcasting, IEEE Press, John Wiley&Sons, Ltd., 2003.
- [Hara-06] Yoshitaka Hara, Loic Brunel, and Kazuyoshi Oshima. Uplink Spatial Scheduling with Adaptive Transmit Beamforming in Multiuser MIMO system. In *Proceedings of the IEEE Personal Indoor Mobile Radio Communications Conference*, pages 1–5, Helsinki, September 2006.
- [Héliot-09a] F. Héliot, R. Hoshyar, R. Tafazolli, "An Asymptotical Approximation of Outage Probability for Distributed MIMO Systems," in *proc. ICT Mobile Summit 09*, Santander, Spain.
- [Héliot-09b] F. Héliot, R. Hoshyar, R. Tafazolli, "A Closed-Form Approximation of the Outage Probability for Distributed MIMO Systems," in *proc. IEEE SPAWC 09*, Perugia, Italy.
- [Héliot-09c] F. Héliot, R. Hoshyar, R. Tafazolli, "Receive Knowledge Only Power Allocation for Nonregenerative Cooperative MIMO Communication," in *proc. IEEE SPAWC 09*, Perugia, Italy.
- [HKim-05] H.Kim, Y.Han, "A proportional fair scheduling for Multicarrier Transmission Systems", *IEEE Communications Letters*, vol.9, no.3, March 2005.

- [Hott-08] A.Hottinen, E.Viterbo, "Optimal user pairing in downlink MU-MIMO with transmit precoding", in *4th Workshop on Resource Allocation in Wireless Networks (RAWNET)*, Berlin, Germany, March 2008.
- [HVis-03] H.Viswanathan, S. Venkatesan, H. Huang, "Downlink capacity evaluation of cellular networks with known-interference cancellation", *IEEE Journal on Selec. Areas in Communications*, vol.21, no.5, September 2003.
- [IEEE802.16e-06] IEEE Computer Society and IEEE Microwave Theory and Techniques Society. Part 16: Air interface for fixed and mobile broadband wireless access systems, amendment 2: Physical and medium access control layers for combined fixed and mobile operation in licensed bands and corrigendum 1. Technical report, IEEE Standards Department, February 2006.
- [IEEE802.16m-09] IEEE 802.16m System Description Document (SDD), IEEE 802.16m-09/0034r1, September 2009.
- [EEE802.16m-EMD-09] IEEE 802.16m Evaluation Methodology Document (EMD), IEEE 802.16m-08/004r5, January 2009.
- [Jaga-07] K.Jagannathan, S.Borst, P.Whiting, E.Modiano, "Scheduling of multi-antenna broadcast systems with heterogeneous users", *IEEE Journal on Selec. Areas in Communications*, vol.25, no.7, September 2007.
- [Jind-05] N.Jindal, A.Goldsmith, "Dirty-Paper Coding versus TDMA for MIMO Broadcast channels", *IEEE Trans. on Information Theory*, vol.51, no.5, May 2005.
- [Kell-98] F.P.Kelly, A.K.Maulloo, D.K.H.Tan, "Rate control for communication networks: shadow prices, proportional fairness and stability", *Journal of the Operational Research Society*, vol.49, pp.237-252, 1998.
- [Kim-08] S.J.Kim, P.Mitran, V.Tarokh, "Performance Bounds for Bidirectional Coded Cooperation Protocols", *IEEE Trans. on Information Theory*, vol.54, no.11, Nov. 2008.
- [Kram-05] G.Kramer, M.Gastpar, P.Gupta, "Cooperative strategies and capacity theorems for relay networks", *IEEE Trans. on Information Theory*, vol. 51, no. 9, Sept. 2005.
- [Lane-03] J. Laneman and G. Wornell, "Distributed Space-Time Coded Protocols for Exploiting Cooperative Diversity in Wireless Networks," *IEEE Trans. on Information Theory*, Oct 2003.
- [Le-05] A. T. Le, X. N. Tran, and T. Fujino, "Combined MMSE and ML multiuser detection for multiuser STBC-OFDM Systems," Technical Report of IEICE, WBS2004-137, vol. 104, no. 674, 91-96, March 2005.
- [Li-01] L.Li,A.Goldsmith, "Capacity and optimal resource allocation for fading broadcast channels–part I: Ergodic Capacity", *IEEE Trans. on Information Theory*, vol.47, no.3, March 2001.

- [Liang-07] Y.Liang, G.Kramer, "Rate Regions for Relay Broadcast Channels", *IEEE Trans. on Information Theory*, vol.53, no.10, October 2007.
- [Liu-08] E.Liu, K.K.Leung, "Fair Resource Allocation under Rayleigh and/or Rician Fading Environments", in *Proc. of IEEE Personal Indoor Mobile Radio Communications (PIMRC)*, September 2008, Cannes, France.
- [Mazet-07] Laurent Mazet, Stéphanie Rouquette-Léveil, Mohamed Kamoun, Patrick Labbé, and Marc de Courville. Achievable phy rates in the IEEE.16m/IMT Advanced framework. WiNext deliverable Win012, Motorola, 2007.
- [Mo-00] J.Mo, J.Waland, "Fair End-to-end window-based congestion control", *IEEE/ACM Trans. Networking*, vol.8, no.5, pp.556-557, Oct. 2000.
- [Muñoz-07] O. Muñoz-Medina, J. Vidal, and A. Agustin, "Linear Transceiver Design in Nonregenerative Relays with Channel State Information," *IEEE Trans. Signal Processing*, vol. 55, no. 6, pp. 2593–2604, June 2007.
- [Nabar-04] R. U. Nabar, H. Bölcskei, and F. W. Kneubühler. "Fading relay channels: Performance limits and space-time signal design". *IEEE Journal on Selected Areas in Communications*, June 2004.
- [PVis-02] P.Viswanath, D.N.C.Tse, R.Laroia, "Opportunistic beamforming using dumb antennas", *IEEE Trans. on Information Theory*, vol. 48, no.6, pp.1277-1294, June 2002.
- [ROCKET-1D1] ICT-215282 STP ROCKET (Reconfigurable OFDMA-based Cooperative Networks Enabled by Agile Spectrum Use) - 1D1: "Report on Operational Scenarios and Simulation Methodology", June 2008.
- [ROCKET-3D1] Reconfigurable OFDMA-based Cooperative Networks Enabled by Agile Spectrum Use (ICT-215282 STP ROCKET) consortium, "3D1- Single User Cooperative Transmission Techniques for OFDMA Multi-hop Cellular Networks", Jan 2009. Available online: www.ict-rocket.eu
- [ROCKET-2D1] Reconfigurable OFDMA-based Cooperative Networks Enabled by Agile Spectrum Use (ICT-215282 STP ROCKET) consortium, "2D1- Performance evaluation and final specification of the enhanced MAC protocol efficiently supporting increased assigned spectrum", Feb 2009. Available online: www.ict-rocket.eu
- [ROCKET-2D2] Reconfigurable OFDMA-based Cooperative Networks Enabled by Agile Spectrum Use (ICT-215282 STP ROCKET) consortium, "2D2- Performance evaluation and final specification of the enhanced MAC protocol for dynamic spectrum usage", Jan 2010. Available online: www.ict-rocket.eu
- [Smith-03] P. J. Smith, S. Roy, and M. Shafi, "Capacity of MIMO Systems With Semicorrelated Flat Fading," *IEEE Transactions on Information Theory* 49 (10): 2781-2788, October 2003.
- [Starr-99] T.Starr, J.M.Cioffi, P.J.Silverman, Understanding digital subscriber line technology, Prentice Hall PTR, Upper Saddle River, NJ, USA, 1999.

- [Tarokh-98] V. Tarokh, N. Seshadri, and A. R. Calderbank, "Space-time codes for high data rate wireless communication: Performance analysis and code construction," *IEEE Transactions on Information Theory* 44 (2): 744–765, March 1998.
- [Telatar-99] Emre Telatar. Capacity of Multi-Antenna Gaussian Channels. *European Transactions on Telecommunications*, 10(6):585–595, November 1999.
- [Tran-04] X. N. Tran, T. Fujino and Y. Karasawa, "On performance of multiuser OFDM Systems with transmit diversity," Proc. The IEEE Symposium on Signal Processing and Information Technology (ISSPIT'04), Rome, Italy, 2004.
- [Tran-05] X. N. Tran, T. Fujino and Y. Karasawa, "An MMSE detector for multiuser space-time block coded OFDM", *IEICE Trans. on Commun.*, vol.E88-B, no.1, pp. 141–149, January 2005.
- [Tse-05] D.Tse, P. Viswanath, *Fundamentals of Wireless Communication*, Cambridge University Press, 2005.
- [Vish-04] S. Vishwanath, N. Jindal, A. Goldsmith, "On the duality of Gaussian Multiple-Access and Broadcast channels", *IEEE Trans. on Information Theory*, vol. 50, no. 5, May 2004.
- [Vite-08] E.Viterbo, A.Hottinen, "Optimal user pairing for multiuser MIMO", in *Proc. Intl. Symposium on Spread Spectrum Techniques and Applications (ISSSTA)*, August 2008, Bologna, Italy.
- [Wein-06] H.Weingarten, Y.Steinberg, S.Shamai, "The capacity region of the Gaussian multiple-input multiple-output broadcast channel", *IEEE Trans. on Information Theory*, vol.52, no.9, Sep.2006.
- [Weng-07] L.Weng, R.D.Murch, "Multi-user MIMO relay system with self-interference cancellation", in *Proc. IEEE Wireless Communications and Networking Conf. (WCNC)*, March 2007.
- [WIMAX] Wimax Forum. <http://www.wimaxforum.org>.
- [WIMAX-07] WiMAX Forum members. WiMAX system evaluation methodology. Technical report, WiMAXForum, 2007.
- [Wolf] "Wolfram Mathworld Website," [Online]. Available:<http://mathworld.wolfram.com/NewtonsMethod.html>.
- [Xi-06] Songnan Xi and Michael D. Zoltowski. Uplink Multiuser MIMO Transceiver Design with Transmitting Beamforming under Power Constraint. In *Proceedings of the IEEE Military Communications Conference*, pages 1–7, October 2006.
- [Yang-07] S. Yang, J-C. Belfiore, "Optimal Space-Time Codes for the MIMO Amplify-and-Forward Cooperative Channel," *Transactions on Information Theory*, Vol. 53, Issue 2, Page 647-663, Feb. 2007.

[Ying-04]

Ying Jun Zhang and Letaief, K.B. Multiuser adaptive subcarrier-and-bit allocation with adaptive cell selection for OFDM systems. *Wireless Communications, IEEE Transactions on*, 3(5):1566-1575, 2004.

List of abbreviations & symbols

AF	Amplify-and-Forward
AMC	Adaptive Modulation and Coding
BC	Broadcast Channel
BER	Bit-Error Rate
BF	BeamForming
BRC	Broadcast Relay Channel
BS	Base Station
CBR	Constant Bit Rate
CDF	Cumulative Density Function
CIR	Channel Impulse Response
CSI	Channel State Information
DF	Decode-and-Forward
DL	DownLink
DPC	Dirty Paper Coding
DSTC	Distributed Space-Time Code
DT	Direct Transmission
ECC	Error Correction Code
EEM	Equal EigenMode
EG	Equal Gain
FCSI	Full CSI
FCH	Frame Control Header
FDMA	Frequency Division Multiple Access
FEC	Forward Error Correction
FS	Frequency Selective
FOST	Framework for OmneT++ Simulations in Thales
GB	Guaranteed Burst
MAC	Media Access Control
MACH	Multiple Access Channel
MARC	Multiple Access Relay Channel
MFR	Matched Filter based Relaying
MIMO	Multiple-Input Multiple-Output
MISO	Multiple-Input Single-Output
MMSE	Minimum Mean Square Error
MMSEF	MMSE Filtering
MRC	Maximal Ratio Combiner
MS	Mobile Station
MU	Multi-User
NCR	Network Coding Relaying
OFDM	Orthogonal Frequency Division Multiplexing
OFDMA	Orthogonal Frequency Division Multiple Access
OWOP	One-Way One-Path
OWOP-CR	OWOP-Cooperative Relaying
OWOP-SR	OWOP-Simple Relaying
OWRC	One-Way Relay Channel
OWTP	One-Way Two-Path
OWTP-CR	OWTP-Cooperative Relaying
OWTP-SR	OWTP-Simple Relaying
PDU	Protocol Data Unit
PDU-ER	PDU-Error Rate
PF	Proportional Fairness
PHY	Physical layer
PL	Propagation Loss

PUM	Packing Unpacking Manager
PUSC	Partial Usage of Sub-Carriers
RCPC	Rate Compatible Punctured Convolutional Code
RS	Relay Station
RU	Resource Unit
Rx	Receive antenna
SAR	Segmentation And Reassembly
SCME	Spatial Channel Model Extended
SDMA	Spatial Division Multiple Access
SDU	Service Data Unit
SE	Spectral Efficiency
SIC	Successive Interference Cancellation
SIMO	Single-Input Multiple-Output
SINR	Signal-to Interference plus Noise Ratio
SNR	Signal-to-Noise Ratio
STBC	Space-Time Block Code
STC	Space-Time Code
SVD	Singular Value Decomposition
TCP	Transmission Control Protocol
TCSI	Transmit CSI
TDD	Time-Division Duplex
TDMA	Time Division Multiple Access
TSM	Transmission Scheme Manager
TWRC	Two-Way Relay Channel
Tx	Transmit antenna
TxBF	Transmit BF
UDP	User Datagram Protocol
UL	UpLink
UMaHL	Urban Macrocell High to Low
UMaHM	Urban Macrocell High to Medium
UMiML	Urban Microcell Medium to Low
WSR	Weighted Sum-Rate
XLI	cross-Layer Interface
ZF	Zero Forcing

List of Tables

Table 1: System Parameters	32
Table 2: Simulation environment summary	34
Table 3: Average number of served users for the different schedulers (Qualcomm, WSR and 2-user PF)	89
Table 4: Variables employed under the different power constraints.....	102
Table 5: Power variables and power as a function of the power constraint	103
Table 6: Variables employed under the different power constraints.....	107
Table 7: Power variables and power as a function of the power constraint	109
Table 8: Variables employed under the different power constraints.....	112
Table 9: Power variables and power as a function of the power constraint	113
Table 10: Variables employed under the different power constraints.....	116
Table 11: Power variables and power as a function of the power constraint	118
Table 12: DT protocol.....	165
Table 13: OWOP-SR protocol	165
Table 14: Alamouti DSTC cooperative relaying protocol	167
Table 15: OWOP-CR protocol.....	168
Table 16: OWTP-CR protocol	169
Table 17: Noise term v	171
Table 18: Complexity of the proposed techniques.....	209
Table 19: System parameters	216

List of Figures

Figure 1: DL Scenario that is used to compare the performance of exclusive use and re-use schemes for subcarrier allocation.....	20
Figure 2: MU scenario with one relay.....	21
Figure 3: MU UL scenario with one relay.....	21
Figure 4: From left to right and top to bottom: (1) DT, (2) OWOP-SR, (3) OWOP-CR, (4) OWTP non-orthogonal cooperative relaying scenarios.....	22
Figure 5: Time decomposition of the WiMAX OFDMA frame.....	23
Figure 6: ROCKET Distributed channelization algorithm.....	25
Figure 7: Simplified view of the WiMAX DL sub-frame (left) vs. ROCKET DL sub-frame (right).....	25
Figure 8: Superposed vs. orthogonal transmission in the DL, bitrate of MS_1 as a function of the bitrate of MS_2 : Left) $SNR_{MS1}=12$ dB, $SNR_{MS2}=0$ dB, Right) $SNR_{MS1}=3$ dB, $SNR_{MS2}=0$ dB.....	27
Figure 9: Superposed vs. orthogonal transmission in the UL, bitrate transmitted from MS_2 as a function of the bitrate transmitted from MS_1 : Left) $SNR_{MS1}=12$ dB, $SNR_{MS2}=0$ dB, Right) $SNR_{MS1}=3$ dB, $SNR_{MS2}=0$ dB.....	27
Figure 10: Distribution of capacity with (a) ZF receiver, (b) MMSE detector, (c) Joint MMSE receiver, (d) SIC receiver.....	33
Figure 11: Time scheduling, frequency scheduling and user pairing.....	36
Figure 12: MUs versus single user for 1Tx 2Rx configurations.....	37
Figure 13: MUs versus single user for 2Tx 2Rx configurations.....	38
Figure 14: Channel coherence time vs. speed for a transmission with a 2GHz carrier. The horizontal line corresponds to the time needed to transmit 200 preambles (i.e. 100 users equipped with 2 TX antennas).....	40
Figure 15: UEs spatial distribution and MCS assigned to be compliant with the single link SNR; pathloss model HATA suburban (BS height 30 m, UE height 2 m).....	42
Figure 16: SNR (sorted) with and without mobility: carrier 1 GHz, 200 UEs in the scenario depicted in Figure 15, 30% of UEs with random mobility up to the value indicated in the legend, Hata suburban pathloss model, 1 GHz.....	43
Figure 17: Nearly-orthogonal pairing metric $\left \mathbf{v}_j^H \mathbf{v}_k \right $ for 3 users and a channel coherence time equal to 200 time slots.....	43
Figure 18: Vehicular type A channel for the 768 data subcarriers of a 802.16e compliant 1024 subcarriers OFDM symbol.....	44
Figure 19: Difference between the true channel transfer function and the one estimated when a) all the subcarriers are transmitted b) one out of five subcarriers are transmitted and the not-transmitted subcarriers are interpolated at the RX side. IEE 902.16e 1024 subcarriers OFDM symbol with SNR=20 dB and channel Vehicular Type A.....	45
Figure 20: Throughput achievable by the MU MIMO introduced in Section 3.1.1 vs. distance, assuming all the users experiencing the same mean SNR.....	47
Figure 21: Scenario with 300 UEs; Shadowing Hata suburban, 2 GHz. Leftmost plot: supported MCS for single link (no coherent BF) and (rightmost plot) corresponding SE in b/s/Hz.....	47
Figure 22: Left plot: number of users experiencing a given SNR, assuming 300 users uniformly distributed in space. Right plot mean SNR vs. the distance of the UEs. Scenario from Figure 21 PL model given by (252). ..	48
Figure 23: SNR matrix from (32) for a scenario with uniform SNR distribution, 300 single antenna UEs and 2 antennas BS.....	50
Figure 24: Joint SE matrix from (39) for a scenario with uniform SNR distribution, 300 single antenna UEs and 2 antennas BS.....	50
Figure 25: leftmost plot: orthogonal condition $\mathbf{v}_k^H \mathbf{v}_j$ matrix and (rightmost) best values achieved for each UE. Scenario with uniform SNR distribution, 300 single antenna UEs and 2 antennas BS.....	51
Figure 26: Left plot: single channel realisation, SE (ranked) achievable by each user for coupling procedures based on the most orthogonal, the best joint SE, the best single coupled SNR and the best single coupled SE. Right plot: mean value of the SE over 200 channel realisations for most orthogonal and best joint SE coupling.....	51
Figure 27: selection frequency of each UE over a period equal to the number of UEs: each UE is selected at least 1 time and the maximal fairness is 2 times. Left plot is for most orthogonal coupling, right plot is for best joint SE coupling.....	52
Figure 28: Mean value of the selection frequency over 60 random channel realisations: both coupling methods converge toward the best fairness value (2 selections every 300 UL phases).....	52
Figure 29: SNR matrix from (32) for the scenario with not-uniform SNR distribution of Figure 21; single antenna UEs and 2 antennas BS.....	53

Figure 30: Joint SE matrix from (39) for the scenario with not-uniform SNR distribution of Figure 21; single antenna UEs and 2 antennas BS.....	53
Figure 31: Left plot: single channel realisation SE (ranked) achievable by each user for coupling procedures based on the most orthogonal, the best joint SE, the best single coupled SNR and the best single coupled SE. Right plot: the mean value over 200 channel realisations for the best joint SE and the most orthogonal coupling. Scenario from Figure 21; single antenna UEs and 2 antennas BS.	54
Figure 32: Mean value of the selection frequency over 200 random channel realisations (left plot: unsorted user index, right plot: sorted by increasing selection frequencies): both coupling methods have a mean value close to the best fairness value (2 selections every 300 UL phases), but the inhomogeneity introduces a lower bound to the variance of the scheduling frequency for the best joint SE coupling method.	55
Figure 33: Mean value of the selection frequency over 200 random channel realisations plotted against the SNR experienced of the corresponding UE. For the best joint SE coupling, the lower limit of the scheduling frequency variance is related with the SNR distribution, whereas the most orthogonal coupling is independent of it.....	55
Figure 34: Mean SE over 200 channel realisations plotted against the SNR that is experienced by the corresponding UE. Red plot: best joint SE; blue plot: most orthogonal. Scenario from Figure 21; single antenna UEs and 2 antennas BS.	56
Figure 35: SNR matrix from (32) for the scenario with not-uniform SNR distribution of Figure 21. BS and UEs antennas (3, 2).	56
Figure 36: Joint SE matrix from (39) for the scenario with not-uniform SNR distribution of Figure 21; BS and UEs antennas (3, 2).	57
Figure 37: Mean value of the selection frequency over 200 random channel realisations plotted against the SNR experienced of the corresponding UE. For the best joint SE coupling, the lower limit of the scheduling frequency variance is related with the SNR distribution, whereas the most orthogonal coupling is independent of it	58
Figure 38: Mean SE over 200 channel realisations. Left plot: against the SNR that is experienced by the corresponding UE; right plot: sorted by ascending SE. Scenario from Figure 21. BS and UEs antennas (3, 2)...	58
Figure 39: Difference between the ideal SNR matrix (i.e. no mobility) given by (32) and the actual matrix SNR (80%@25 km/h max) given by (31).....	59
Figure 40: Mean SE over 200 channel realisations. Right plot: against the SNR experienced by the corresponding UE; left plot: sorted by ascending SE. Scenario from Figure 21. BS and UEs antennas (3, 2).....	59
Figure 41: Mean SE over 200 channel realisations. Right plot: against the SNR experienced by the corresponding UE; left plot: sorted by ascending SE. Scenario from Figure 21. BS and UEs antennas (3, 2).....	60
Figure 42: Representation of the processing building blocks for the best joint SE protocol for the UL pairing with the indication of the parallelisation order (Pord) of the processing block, where M_U and N_P are the number of UEs and the number of subcarriers, respectively.	61
Figure 43: Depiction of orthogonal subcarrier allocation scheme and power optimisation over the pair of subcarriers.	67
Figure 44: Depiction of reuse subcarrier allocation with power optimisation over the two users and two subcarriers.	68
Figure 45: Sum-rate comparison for exclusive allocation (Orthogonal Use: dashed line) and subcarrier reuse allocation (Reuse: solid line) for different weights given to the strong-channel user	69
Figure 46: The rate shares (user 1 and user 2) for the two schemes. For exclusive allocation (Orthogonal Use: dashed line) and subcarrier reuse allocation (Reuse: solid line). The rate for strong-channel user (user 1) is indicated by an up-pointing triangle marker. The rate for weak-channel user (user 2) is indicated by a down-pointing triangle marker.....	70
Figure 47: The performance comparison of the two schemes if the target rate for one user is fixed and both schemes can use any weight vector to obtain the targeted rate	70
Figure 48: WSR (left) and sum-rate (right) as a function of ratio between SNRs of the two served users (ξ , in linear scale). $\rho_2=10$ dB, $(\mu_1, \mu_2)=\{(0.3,0.7), (0.1,0.9)\}$	74
Figure 49: Additive sum-rate gain of BC over TDMA as a function of ratio between SNRs of the served users (ξ)(linear scale). $\rho_2=10$ dB, $(\mu_1, \mu_2)=\{(0.28,0.72),(0.38,0.62)\}$	76
Figure 50: Additive sum-rate gain of BC over TDMA as a function of ratio between SNRs of the users served (ξ)(linear scale) (x-axis) and the ratio between priorities (μ_2/μ_1) (y-axis). $\rho_2=10$ dB,	76
Figure 51: Left) Additive sum-rate and Right) percentage-wise gain of the sum-rate of BC over TDMA (see(67)) as a function of μ_1 when the weak user presents $\rho_2=\{10,0\}$ dB and $\mu_2=0.8$. $\nu=100$ (20 dB).	77
Figure 52: Sum-rate of BC, FDMA and TDMA as a function of ratio between SNRs of the served users (ξ)(linear) $\rho_2=10$ dB, $(\mu_1, \mu_2)=(0.3,0.7)$. ξ_{TDMA} introduced in (64) and S_{weak} , S_{strong} in (74).	79
Figure 53: Achievable Rate regions of the TDMA, FDMA and BC for weak user equal to $\rho_2=10$ dB and a strong user with $\xi=25$ (left) and $\xi=40$ (right). The operational points attained due to the priorities $(\mu_1, \mu_2)=(0.3,0.7)$ are shown for a comparison with Figure 52.	80

Figure 54: WSR, J_{WSR} in (50), for the TDMA, FDMA, and BC as a function of the priorities of the users. Weak user equal to $\rho_2=10$ dB and a strong user $\rho_1=\rho_2+10\log_{10}(\xi)$ with $\xi=25$.	80
Figure 55: Left) Additive sum-rate and Right) Percentage-wise gain of the sum-rate of BC over FDMA (see (79)) as a function of μ_1 when the weak user presents $\rho_2=\{10,0\}$ dB and $\mu_2=0.8$. $\nu=100$ (20 dB).	81
Figure 56: Sum-rate attained by the BC using the PF criterion, (89) as a function of ratio of SNRs between the strong and weak users (ξ), for $(\mu_1, \mu_2)=\{(0.3,0.7), (0.1,0.9)\}$, $\tau=\{0.5,0.1\}$, (85). $\rho_2=10$ dB.	84
Figure 57: CDF: User throughput (left) and per-user delay (right) for the Qualcomm scheduler (single-user PF) and 2-user PF scheduler with $L=\{2,3,9\}$ candidates. $\tau=0.05$.	87
Figure 58: CDF: User throughput (left) and per-user delay (right) for the Qualcomm scheduler (single-user PF) and 2-user PF scheduler with $L=\{2,3,9\}$ candidates. $\tau=0.7$.	88
Figure 59: CDF of the Cell Throughput for the Qualcomm scheduler (single-user PF) and 2-user PF scheduler with $L=\{2,3,9\}$ candidates. $\tau=\{0.05, 0.7\}$.	88
Figure 60: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=30$.	89
Figure 61: Left) 10%-outage throughput ($p_{out}=0.1$, (100)) and Right) 90%-outage throughput ($p_{out}=0.9$) as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=30$.	90
Figure 62: Left) 10%-outage delay ($p_{out}=0.1$, (100)) and Right) 90%-outage delay ($p_{out}=0.9$) as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=30$.	91
Figure 63: Left) 10%-outage cell throughput ($p_{out}=0.1$, (100)) and Right) 90%-outage cell throughput ($p_{out}=0.9$) as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=30$.	91
Figure 64: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=90$.	92
Figure 65: Percentage-wise cell throughput gain of the weighed sum-rate and 2-user PF over the Qualcomm scheduler as a function of the parameter τ , (85). $N_u=\{30,90\}$.	92
Figure 66: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, 3-user WSR and 3-user PF. $N_u=30$. $L=5$.	93
Figure 67: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, 3-user WSR and 3-user PF. $N_u=90$. $L=5$.	94
Figure 68: Percentage-wise cell throughput gain of the 2-user PF and 3-user PF over the Qualcomm scheduler as a function of the parameter τ , (85). $N_u=\{30,90\}$. $L=5$.	94
Figure 69: Percentage-wise cell throughput gain of the 2-user and 3-user WSR schedulers over the Qualcomm algorithm as a function of the parameter τ , (85). $N_u=\{30,90\}$. $L=5$.	95
Figure 70: Frame format for the OWRC forwarding and protocol I	99
Figure 71: BRC during the first 2 phases of the frame (Figure 70)	100
Figure 72: MARC during the last 2 phases of the frame (Figure 70)	101
Figure 73: BRC during the first 2 phases of the frame (Figure 70) under protocol I	104
Figure 74: MARC during the last 2 phases of the frame (Figure 70) under protocol I	106
Figure 75: Transmissions in the TWRC under forwarding protocol	109
Figure 76: Transmissions in the TWRC under protocol I	114
Figure 77: Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Channel configuration A, Top-right) Channel configuration B, Bottom-left) Channel configuration C and Bottom-right) Channel Configuration D, (163). Traffic Asymmetry = 3, Sum-average power constraint.	119
Figure 78: Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Channel configuration A, Top-right) Channel configuration B, Bottom-left) Channel configuration C and Bottom-right) Channel Configuration D, (163). Traffic Asymmetry = 3, Max-power constraint.	120
Figure 79: Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Channel configuration A, Top-right) Channel configuration B, Bottom-left) Channel configuration C and Bottom-right) Channel Configuration D, (163). Traffic Asymmetry = 3, Indv-average power constraint.	121
Figure 80: Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Traffic Asymmetry=1, Top-right) Traffic Asymmetry=3, Bottom-left) Traffic Asymmetry=5 and Bottom-right) Traffic Asymmetry=10. Channel configuration A, (163), Sum-average power constraint.	122
Figure 81: Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Traffic Asymmetry=1, Top-right) Traffic Asymmetry=3, Bottom-left) Traffic Asymmetry=5 and Bottom-right) Traffic Asymmetry=10. Channel configuration A, (163), Indv-average power constraint.	123

Figure 82. Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Traffic Asymmetry=1, Top-right) Traffic Asymmetry=3, Bottom-left) Traffic Asymmetry=5 and Bottom-right) Traffic Asymmetry=10. Channel configuration A, (163), Max-power constraint.....	124
Figure 83. Scenario assumed for the cellular throughput evaluation.....	126
Figure 84. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: $2 \times 2 \times 1$	129
Figure 85. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: $2 \times 2 \times 2$	129
Figure 86. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration $4 \times 4 \times 2$	130
Figure 87. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: $2 \times 2 \times 1$	131
Figure 88. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: $4 \times 4 \times 2$	131
Figure 89: Single-user cooperative MIMO system model.....	133
Figure 90: Mutual information performance of various power allocation algorithms for $n = q = r = 2$, $SNR_0 = 0$ dB and $SNR_2 = 10$ dB.....	145
Figure 91: Non-cooperative (a) and Cooperative (b) Mutual information performance of various power allocation algorithms for $n = q = r = 4$, $SNR_0 = 0$ dB and $SNR_2 = 10$ dB.....	146
Figure 92: Cooperative mutual information performance of various power allocation schemes for $SNR_0 = 0$ dB and $SNR_2 = 10$ dB, (a) MISO $n=q=2$, $r=1$, (b) MISO $n=q=4$, $r=1$, (c) MISO $n=q=64$, $r=1$, (d) MIMO $n=q=r=4$, (e) MIMO $n=q=4$, $r=8$, (f) MIMO $n=q=r=8$	147
Figure 93: Cooperative mutual information performance of various power allocation schemes for $SNR_0 = 0$ dB and $SNR_1 = 10$ dB, (a) MISO $n=q=4$, $r=1$, (b) MIMO $n=q=4$, $r=4$, (c) MIMO $n=q=4$, $r=8$	148
Figure 94: Cooperative mutual information performance of various power allocation schemes for $SNR_1 = 10$ dB and $SNR_2 = 10$ dB, (a) MISO $n=q=4$, $r=1$, (b) MIMO $n=q=4$, $r=4$, (c) low SNR magnified version of (b), (d) MIMO $n=q=4$, $r=8$, (e) low SNR magnified version of (d).....	149
Figure 95: Cooperative mutual information performance of various power allocation methods for the SISO case with a) 8, b) 64 and c) 256 subcarriers, $SNR_0 = 0$ dB, and $SNR_2 = 10$ dB, car scenario, 50km/h.....	150
Figure 96: Cooperative mutual information performance of various power allocation methods for the MIMO case with a) $n=q=r=2$, 4 subcarriers, b) $n=q=r=2$, 32 subcarriers and c) $n=q=r=4$, 32 subcarriers, $SNR_0 = 0$ dB, and $SNR_2 = 10$ dB, car scenario, 50km/h.....	151
Figure 97: UL MU scenario.....	151
Figure 98: Cooperative sum-rate performance of various power allocation schemes for $SNR_0 = 0$ dB and $SNR_2 = 10$ dB, (a) 4-users MIMO $n=4$, $q=r=8$, (b) 8-users MIMO $n=1$, $q=r=8$	154
Figure 99: Cooperative sum-rate performance of various power allocation methods for the SISO case with 4 users and a) 8, b) 64 and c) 128 subcarriers, $SNR_0 = 0$ dB, and $SNR_2 = 10$ dB, car scenario, 50km/h.....	155
Figure 100: Cooperative mutual information performance of various power allocation methods for the MIMO case with 4 users and a) $n=q=r=2$, 4 subcarriers, b) $n=q=r=2$, 32 subcarriers and c) $n=q=r=4$, 32 subcarriers, $SNR_0 = 0$ dB, and $SNR_2 = 10$ dB, car scenario, 50km/h.....	155
Figure 101: Cooperative mutual information performance of various power allocation methods for the MIMO case with 4 users and a) 4 users, $n=q=r=4$, 1subcarriers, Rayleigh fading channel b) $n=q=r=2$, 32 subcarriers, car scenario, 50km/h, $SNR_0 = 0$ dB, and $SNR_2 = 10$ dB.....	156
Figure 102: Channel modelling.....	159
Figure 103: Burst allocation algorithm.....	160
Figure 104: burst allocation: examples.....	161
Figure 105: Transmission scheme representation.....	162
Figure 106: OWTP-CR cooperative relaying protocol.....	169
Figure 107: left: DT beacon, right: DT transmission.....	173
Figure 108: left: OWOP-SR beacon, right: OWOP-SR transmission.....	174
Figure 109: left: OWOP-CR beacon, right: OWOP-CR transmission.....	174
Figure 110: OWTP-CR transmission.....	175
Figure 111: OWTP-CR beacon information.....	175
Figure 112: OMNeT++ simulation architecture.....	176
Figure 113: Data message encapsulation.....	177
Figure 114: Routing table updating process.....	178
Figure 115: SAR layer operations.....	179
Figure 116: SAR-PDU header.....	179

Figure 117: Structure of the FIFO queues stage.....	180
Figure 118: data retrieving from FIFO queues.....	181
Figure 119: DL-PDU header.....	182
Figure 120: PHY architecture	183
Figure 121: PHY – sequence diagram in transmission	184
Figure 122: PHY – sequence diagram in reception.....	184
Figure 123: Relay positioning.....	186
Figure 124: OWTP transmission scenarios.....	187
Figure 125: OWTP BER analysis for $d(sr)=1000m$ and $d(sd)=2000m$	188
Figure 126: Relay positioning.....	189
Figure 127: OWTP-CR BER versus the inter-relay distance for different source-relay distances.....	189
Figure 128: Optimal $d(sr)$ for a given $d(r1r2)$	190
Figure 129: Simulation scenarios.....	190
Figure 130: Comparison of different transmission schemes BER versus relay position.....	191
Figure 131: Cell configuration.....	192
Figure 132: Relay positions in the cell.....	193
Figure 133: Maximum relative gain in percentage for different scenarios	194
Figure 134: Average BER in the cell for different scenarios	195
Figure 135: BER standard deviation in the cell for different scenarios	195
Figure 136: BER per user in the cell.....	196
Figure 137: Relay frame format considered in [IEEE802.16m-09]	201
Figure 138: Operations in the currently adopted frame format.	203
Figure 139: Pictorial view of a possible frame format for reducing the time between the CSI estimation by the BS and its use by the MSs.....	204
Figure 140: mean SNR as a function of the distance from the BS for the shadowing models Hata urban suburban and open and for the simplified two-slope model. BS height 30 m, UE height 2 m, 2 GHz, Tx power 30 dBm, no antenna gain.	218

1 INTRODUCTION

This deliverable reports the consortium findings that are related to activity 3A2 of the ROCKET project. The aim of this activity is to design advanced Multi-User (MU) non-cooperative and cooperative transmission techniques in the context of Orthogonal Frequency Division Multiple Access (OFDMA) Multi-hop Cellular Networks. This deliverable follows up [ROCKET-3D1], where only single user techniques have been proposed and it investigates transmission techniques to deal with a MU configuration. To this end, this work develops and design novel transmission techniques considering:

- None, one or two half duplex relays.
- The relays work under decode-and-forward (DF) or amplify and forward (AF).
- The transmission scheme is based on One-Way, Two-Way and Two-Path Relaying
- The transmission techniques encompass the UpLink (UL), DownLink (DL) or both settings.
- Full or imperfect Channel State Information (CSI).

Some of the scenarios, which have been defined in [ROCKET-1D1] and used in this deliverable to support our novel techniques, will be presented in section 2. They will be utilised to evaluate the performance of the different advanced MU non-cooperative and cooperative transmission techniques that will be studied in sections 3 and 4, respectively.

Section 3 is devoted to the MU non-relaying techniques in UL and DL. The novel techniques explore the reuse of resources when superposed transmission is utilised. In such a case, multiple users transmit simultaneously using the same resources. The benefits that are provided by those techniques depend on the user selection or *user pairing*. Section 3.1 presents UL techniques, while section 3.2 is devoted to DL techniques. First, a low complexity UL MU pairing and best band selection scheme for transmit BeamForming (BF) is studied in section 3.1.1. In the context of cellular broadband MU Multiple-Input Multiple-Output (MIMO) UL modes for band Adaptive Modulation and Coding (AMC) like IEEE802.16e/m systems, the study provides answers on how to distribute the users on the OFDMA chunks in order to maximise the overall throughput and to optimise the band allocation for Single-Input Multiple-Output (SIMO) and MIMO systems. This scheme is then extended in section 3.1.2 for “nearly” orthogonal and best user selection in coherent BF taking into account the user mobility and inhomogeneous scenarios, since possible drawbacks might limit the applicability of a fully-coherent approach for the user pairing and best band selection. On the other hand, section 3.2 tackles the user pairing in an Orthogonal Frequency Division Multiplexing (OFDM) system in the DL, where a pair of users with diverse mean channel strengths competes in order to be served by a single Base Station (BS) over two allocated subcarriers. This scheme exhibits reuse benefits. Delving into the user grouping in the DL, section 3.2.2 investigates different criteria for scheduling the best pair of users. Those criteria are the Weighted Sum-Rate (WSR) and the Proportional Fairness (PF). The optimal resource allocation is provided for those criteria. Results show that in spite of the channel strength of the users, the priority of each one, commonly dependent on the average served throughput, is another important parameter for selecting a pair of users in order to share the same resources.

On the other hand, section 4 investigates the MU cooperative techniques with different numbers of relay and transmission strategies. First, section 4.1 develops the optimal resource allocation schemes for different transmissions schemes involving the One-Way Relay Channel (OWRC) and Two-Way Relay Channel (TWRC) and superposed transmission. In all cases, the DL and UL traffic is allocated under Time Division Duplexing (TDD). The benefits that are provided by the MU techniques will be measured at system level by a cellular throughput evaluation, where the scheduling algorithm is based on the MU PF criterion for grouping the users that will share the same resources. Then, section 4.2 presents novel power allocation methods for non-regenerative cooperative scheme. First, various schemes operating with different levels of CSI for single user, single and multi antennas, and single and multi carriers are investigated. Building on this work, new MU power allocation methods relying

on receive and transmit knowledge for the UL setting of single or multi-carriers AF systems are proposed under two optimisation criteria, namely, the sum-rate and the WSR. Next in section 4.3, a comprehensive study of the performance of the One-Way Two-Path-Cooperative Relaying (OWTP-CR) protocol in a single-cell MU OFDMA context is undertaken. The aim of this study is to evaluate the gain in performance that can be expected from the proposed two-relays cooperative protocol, with respect to Direct Transmission (DT) and other conventional single-relay assisted protocols (cooperative or non-cooperative), in a cellular context, and to determine its applicability in future WiMAX cellular networks. An OmneT++ simulation is built to measure the BER performance of this protocol and its concurrent protocols in both a single-user and MU context. An extensive comparison is made, based on these results.

Finally, MAC layer parameters and recommendations will be discussed in section 5, the complexity and adaptability of the proposed algorithms are explained in section 6 and conclusions are drawn in section 7.

2 CELLULAR SCENARIOS AND MAC ASSUMPTIONS

2.1 SCENARIOS

The definition of ROCKET operational scenarios can be found in [ROCKET-1D1]. These scenarios serve as a basis for the performance evaluation of algorithms that have been developed in the different work packages. In this section, we list the various scenarios of [ROCKET-1D1] that have been considered in each section of this deliverable and we justify their selections.

Section 3 presents MU transmission for non-relaying systems in the UL and DL. Under that configuration, the resources are shared simultaneously by multiple users that employ the superposed transmission. The general case is assumed to be the two-user case. The benefits that are provided by superposed transmission are dependent on the user selection. To this end, section 3 investigates different algorithms to select the best users, i.e. user pairing. In section 3.1.1, we consider an UL transmission where two single streams of a two-Transmit antenna (Tx) Mobile Station (MS) transmit simultaneously to a two-Receive antennas (Rx) BS. Each of the MS employ per carrier Singular Value Decomposition (SVD) - MIMO Transmit BF (TxBF) to the BS. In order to schedule two active users at the same time, which are picked among a larger pool of N users, we propose to select the two users that have their channel eigenvectors – corresponding to the maximum eigenvalue of their channels – the most orthogonal to each other. The activity planned in section 3.1.2 focuses on the reformulation of the approaches proposed in section 3.1.1 in the context of inhomogeneous scenarios with mobility. The spatial distribution of the subscribers is assumed statistically uniform in space (i.e. uniform users density) and the speed distribution is assumed to be basically from pedestrian to urban vehicular speeds (e.g. from 3 to 50 km/h). The assumed uniform spatial density of the users entails a not-uniform Signal-to-Noise Ratio (SNR) distribution, as the subscribers experiencing lower SNR exponentially grow with the distance from the BS.

On the other hand, Section 3.2.1, and 3.2.2 consider the multiple users in DL. We first consider that the users are paired such that the users with a very strong channel (mean value of magnitude squared channels over all subcarriers) pair with the users of a very weak channel allowing for a diverse channel ratio in each pair. Each pair is assumed to be allocated with two subcarriers such that it can either use exclusively for each user in the pair or that it can use for both users in the pair and then perform interference cancellation at the receiver end, where it is possible. CSI is assumed to be available at the transmitter end and each subcarrier channel is a Rayleigh fading channel. The power at the BS transmitter is assumed to be constrained and this power is allocated to the users and subcarriers in both schemes with the objective of maximising the WSR for the given user weights. The scenario is depicted in Figure 1. The performance of two schemes is compared. Additionally, section 3.2.2 investigates the optimal power allocation when using the WSR and proportional fair (PF) criteria.

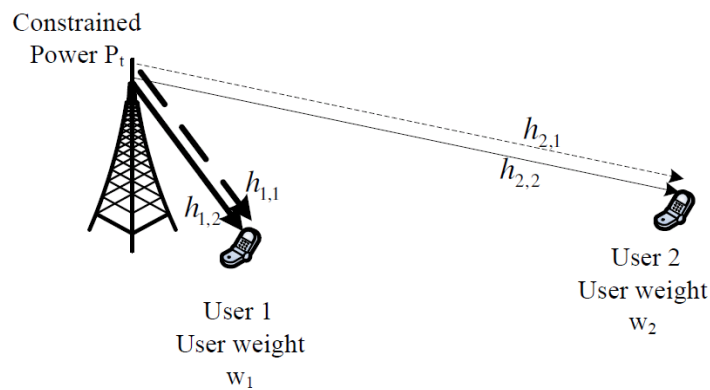


Figure 1: DL Scenario that is used to compare the performance of exclusive use and re-use schemes for subcarrier allocation

In section 4.1, the MU transmission techniques of the TWRC are analysed in a scenario consisting of a BS, a single Relay Station (RS) and multiple MSs. This scenario is depicted in Figure 2 and it has been introduced in [ROCKET-1D1]. The investigated techniques have been evaluated at system level taking into account the Winner Models for a sub-urban environment presented in [ROCKET-1D1] (C1 type for BS-MS, RS-MS links and B5 for BS-RS link) and baseline test models considered in [IEEE802.16m-EMD-09] (Baseline Test Scenario at 2.5 GHz for BS-MS, RS-MS links and 802.16j EVM type D for the BS-RS link)

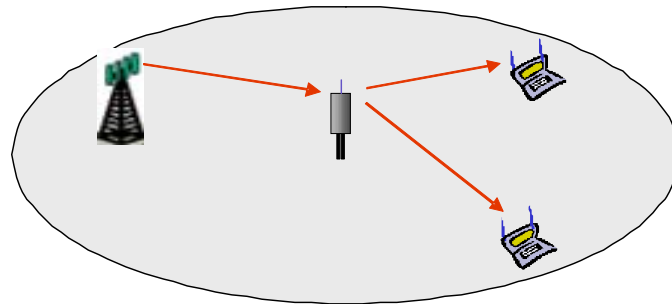


Figure 2: MU scenario with one relay

In the work about power allocation methods for non-regenerative cooperative scheme that is presented in section 4.2, we consider two scenarios:

- First, we consider a classic single BS, RS, MS scenario.
- Then, we consider a MU UL scenario for the main analysis, where several MSs communicate with a single BS and RS, as it is illustrated in Figure 3.

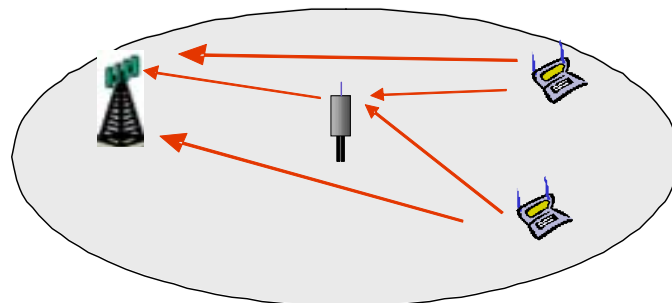


Figure 3: MU UL scenario with one relay

In single-carrier simulation, we assume that the channels between the MSs, RS, and BS, are single path Rayleigh fading channel. In multi-carrier setting, we consider the car scenario, where each MS-RS channel is modelled by the UMiML channel, each MS-BS channel is modelled by the UMaHL channel, and the RS-BS channel is modelled by the Urban Macrocell High to Medium (UMaHM). All these channels have been defined in [ROCKET-1D1].

The contributions in section 4.3 will focus on the single-cell MU extension of the OWTP cooperative relaying scenario that is depicted in Figure 13, p. 84 of the Technical annex and is recalled on Figure 4 below (drawing number (4)). Only DL transmissions will be considered in the simulations, meaning that the BS is the source of all communications taking place in the network, with the destination being any one of the multiple terminals. Relays are supposed to be half-duplex TDD, so that there have dedicated relay-transmit and relay-receive time slots in the transmission frame.

For the sake of performance comparison, the following transmission protocols will also be considered (will be further detailed in paragraph 4.3.3.2.2):

- DT (Figure 4, (1)),

- One-Way One-Path-Simple Relaying (OWOP-SR) protocol (Figure 4, (2))
- OWOP-Cooperative Relaying (OWOP-CR) scenario (Figure 4, (3)).

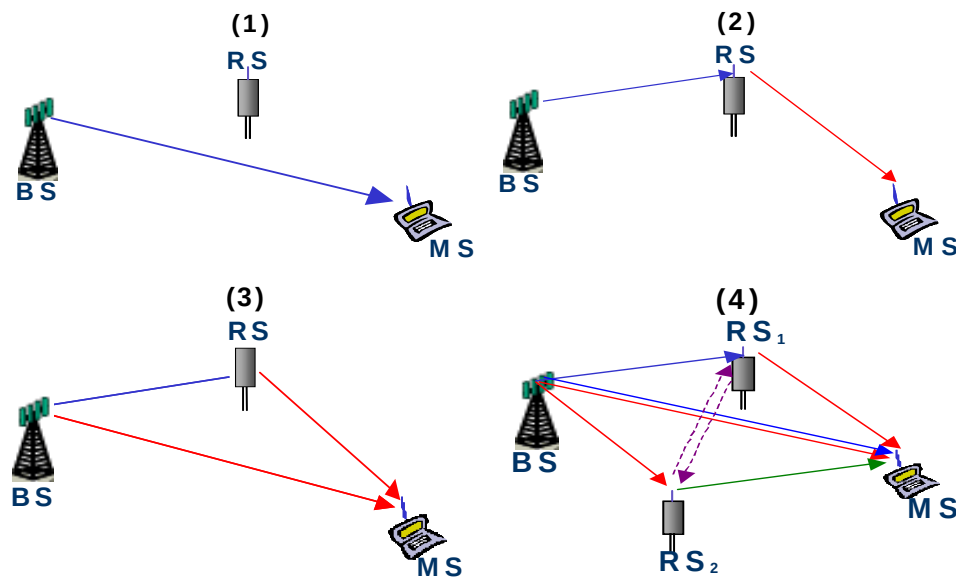


Figure 4: From left to right and top to bottom: (1) DT, (2) OWOP-SR, (3) OWOP-CR, (4) OWTP non-orthogonal cooperative relaying scenarios

The Physical layer (PHY)/ Media Access Control (MAC) layer is based on OFDMA radio access and the propagation environment for the BS/RS and RS/MS links is supposed to be macrocell urban. Path loss models will be derived from the Winner C2 scenario and small-scale fading will be simulated using a multipath Rayleigh fading model.

2.2 MAC SUPPORT OF PHY

2.2.1 General assumptions

The techniques investigated in this work assume the following features in order to be supported by the MAC layer:

- The relay terminals can be time-division duplex devices (acting as half-duplex AF or DF devices).
- The MAC frame structure assumes time-division duplexing for the UL and DL. In both cases, access is TDMA or OFDMA-TDMA.
- Each UL or DL subframes are subdivided in sub-subframes so that relay transmission are accommodated.
- The duration of these sub-subframes can be adjusted on a per-frame basis. The duration of the subframes can only be adjusted to multiple values of the slot duration, which is a multiple of OFDM symbol duration in 802.16e.
- The modulation scheme and the coding rates can be adjusted independently in the transmissions from the source and the relay terminals
- A preamble is used in transmissions from the source, as well as for the transmissions from the relay. This allows fast synchronization at the receiver.
- Pilot tones are required to independently evaluate the channel in the path source-destination and relay-destination (and vice versa). A precise knowledge of the channel in reception is required for coherent detection and best information rate performance.

- Proper guard times are allocated between reception at the relay and transmission, so that the relay terminal can fully decode the message on the received signal before re-encoding and transmitting.
- MIMO technologies are available at the source, relay and destination. This implies that DSTCs are available at the source and the relay terminals. For those protocols in which both transmit simultaneously, STCs are split and per-antenna codewords distributed.
- Depending on the type of transmission scheme considered and the coherence time associated to each cooperative link, the level of precision of CSIT is different. This implies that the rate at which the channel state is reported might exceed the feedback rate of the system if the coherence time is too short. For those cases, coding across carriers is the preferred option, (related to PUSC/FUSC carrier allocation modes in 802.16e). In this case, the metric that describes completely the capacity of the channel is the average SNR across the carriers.

2.2.2 OFDMA radio access and definitions

The previous specifications are supported by the IEEE 802.16e OFDMA radio access. The WiMAX OFDMA frame of 5 ms duration contains a total of 48 OFDM symbols of 102.8 μ s. As represented in Figure 5, it consists of an UL subframe and a DL subframe, with adjustable DL/UL ratio. For the simulations, the DL/UL ratio will be set to 2:1, leading to a DL sub-frame of 3.33 ms duration containing a total of 32 OFDM symbols in the time dimension, as it is sketched in Figure 7. As this work focuses on DL transmissions and for simplicity reasons, the word “frame” will be used instead of “DL subframe” in the following.

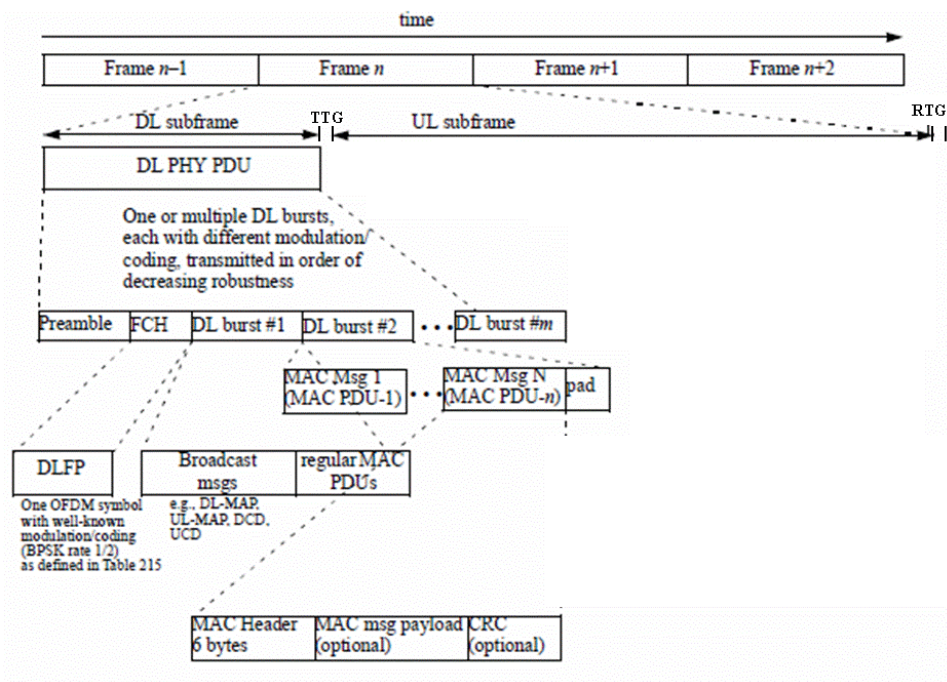


Figure 5: Time decomposition of the WiMAX OFDMA frame

OFDMA sub-channelization. The transmission bandwidth is 10 MHz, and thus, according to the Mobile WiMAX 802.16e settings, the total number of subcarriers is 1024, 720 of which are dedicated to data transmission (data subcarriers), while 120 are used for synchronization and channel estimation purposes (pilot subcarriers) and 184 as nulls/guard and DC subcarriers. Subcarrier spacing is $\Delta f = 10.94 \text{ kHz}$.

In DL PUSC, the pilot tones are allocated first. The remaining subcarriers are reserved for user data transmission and are divided into 30 elementary sub-channels of 24 data subcarriers. Each of the OFDMA subcarrier permutation modes (FUSC, PUSC,...) has its own definition for a sub-channel. Basically, the sub-channel corresponds to the minimum transmission unit in an OFDMA symbol.

There are two different subcarrier allocation (or sub-channelization) modes:

- The first type, called distributed allocations, pseudo-randomly distributes the subcarriers over the available bandwidth. This type of allocation provides frequency diversity in frequency-selective fading channels and inter-cell interference averaging and is well suited for mobile applications. The 802.16e OFDMA PHY specifies different types of distributed allocation modes: FUSC (Full Use of Sub-Channels), OFUSC (Optional Full Use of Sub-Channels), PUSC (Partial Use of Sub-Channels), OPUSC (Optional Partial Use of Sub-Channels) and TUSC (Tile Use of Sub-Channels).
- The second type of subcarrier allocation, called adjacent allocation, groups subcarriers that are adjacent to each other in the frequency domain. Adjacent allocation is useful for frequency nonselective and slowly fading channels, and for implementing AMC. Adjacent allocation is typically used in fixed and low mobility networks.

The most encountered sub-channelization modes in practical systems are: PUSC, FUSC, and AMC. A single OFDMA frame may use one or more sub-channelization modes. The DL sub-frame requires at least one zone, always starting with PUSC but other allocation schemes may be used in addition, depending on the network conditions.

In ROCKET simulations, we will use a simplified PUSC-like distributed allocation technique, which is sketched in Figure 6. In the left drawing, logical sub-channels correspond to blocks of 24 consecutive sub-carriers. These subcarriers are mapped to a physical channel consisting of N_p elementary sub-channels of p sub-carriers (in Figure 6, we took $p = 3$ and $N_p = 240$ for readability reasons), where p is chosen to match the coherence bandwidth CBW of the radio channel $p \Delta f = CBW$. Considering a typical mean coherence bandwidth of about 500 kHz in urban macrocell environment, we find $p \approx 45$ and $N_p \approx 16$. These are the values that will be used in the simulations.

The subcarrier permutation algorithm performs random permutations between OFDMA subcarriers. For each index n of the original OFDMA subcarriers' table (Figure 6, left), the algorithm draws a random number p between 1 and 720 and swaps elements at indexes n and p of the table to provide the new reordered subcarriers' table presented on the right drawing of Figure 6 (left table).

Slot. A slot in the OFDMA PHY has both a time and frequency dimension. It is the minimum possible data allocation unit in the 802.16 standard. The definition of an OFDMA slot depends on the OFDMA symbol structure, which varies for UL and DL and also according to the selected sub-channelization mode. In the DL PUSC case, slot dimensions are 1 sub-channel x 2 OFDMA symbols.

Burst. In Mobile WiMAX OFDMA, a burst is a two-dimensional allocation of a group of slots, i.e., a group of contiguous subchannels in a group of contiguous OFDMA slots. 802.16e supports bursts of various sizes, with no time relationship, in both the UL and DL sub-frames, as it is sketched in Figure 7, left-hand side, which provides a generic time/frequency view of the mobile WiMAX frame.

For the sake of simplicity and due to actual limitations of our FOST simulation framework, which would require a huge amount of work to be overcome, we will only consider bursts of equal size that are aligned in time (see Figure 7, right); this will not have any significant impact on the simulation results.

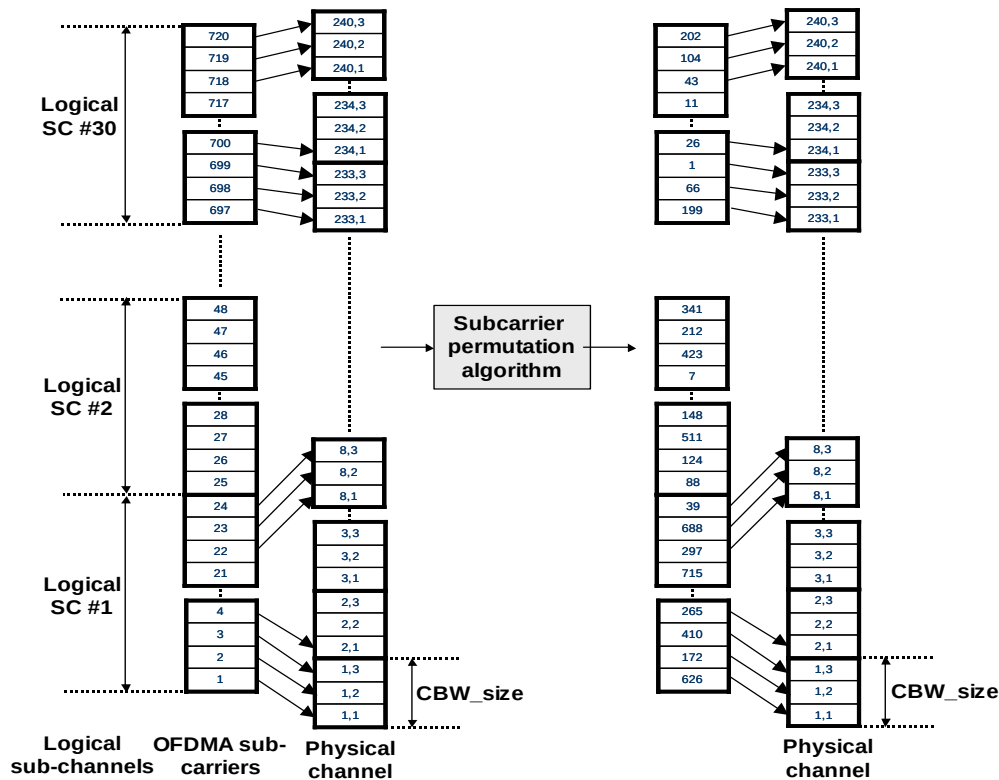


Figure 6: ROCKET Distributed channelization algorithm

Burst size. The time (N slots) and frequency (M sub-channels) dimensions of a burst are chosen, taking into account the following considerations:

- M determines the maximum number of simultaneous users in the time duration of a burst.
- N is chosen such that the number of bursts in a frame is an integer. M shall not be taken too small; otherwise, the overhead that is incurred by the burst headers would considerably lower the transmission SE.

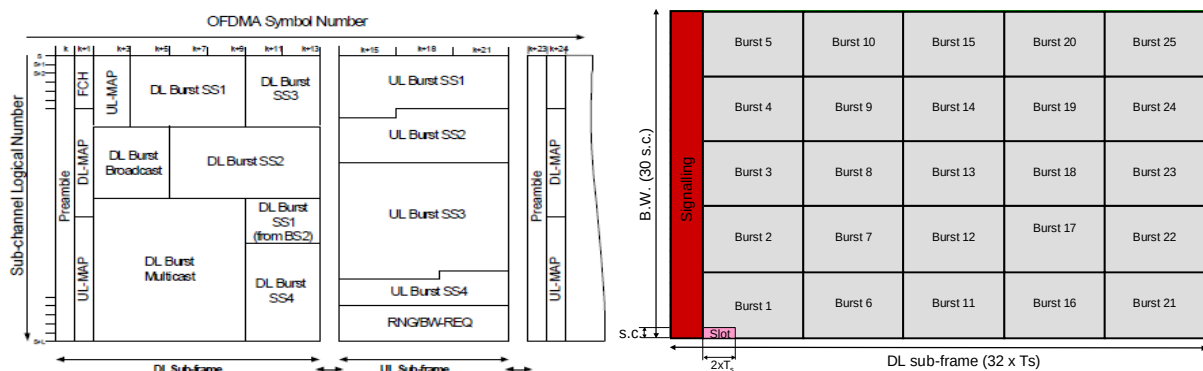


Figure 7: Simplified view of the WiMAX DL sub-frame (left) vs. ROCKET DL sub-frame (right)

3 ADVANCED NON-RELAYING MULTI-USER TECHNIQUES

Currently, the wireless transmission technologies utilise orthogonal transmissions in the DL and UL to accommodate MU communication. This solution is known to be inefficient, but it can be easily implemented by using simple transmitters and receivers at the terminals.

When there are multiple users to be served or multiple transmitting users, the DL and UL are described by the Broadcast Channel (BC) and the Multiple Access Channel (MACH), respectively. The optimal transmission scheme for both types of channel is based on the superposed transmission, as it is explained in [Cove-91]. Hence, the signals from the BS that are intended to the MSs (DL) or those ones that are originated from MSs and are intended to the BS (UL) are transmitted simultaneously. To this end, we review the achievable rate regions that can be attained by the superposed transmission and we compare them with the achievable rates under the orthogonal transmission strategy.

➤ Downlink with Two User

In the BC, there is a single transmitter, i.e., the BS, simultaneously transmits two independent messages to two receivers, the MSs, [Cove-91]. The BS distributes its available power among the two transmitted signals. When the scenario is *physically degraded*, for instance if all terminals are equipped with single antennas, we can classify the MSs as *strong* or *weak* users in function of each receiver SNR. In those circumstances, the *weak* user receives its intended signal interfered by the signal devoted to the *strong* user. Likewise, the strong user is able to decode the signal devoted to the weak user, subtract its contribution from the received signal and decode its devoted message without interference, since it has a better SNR than the weak user. If MS₁ is the *strong* user and MS₂ stands for the *weak* one, the achievable rate region is given by

$$B_{BC} \left\{ \begin{array}{l} R_{bs1} \leq \log_2 (1 + \gamma_1 \rho_{bs-ms1}) \\ R_{bs2} \leq \log_2 \left(1 + \frac{\gamma_2 \rho_{bs-ms2}}{1 + \gamma_1 \rho_{bs-ms2}} \right) \end{array} \right. \quad (1)$$

where R_{bs1} , R_{bs2} are the data rates of the messages intended to MS₁, MS₂, respectively, ρ_{bs-ms1} , ρ_{bs-ms2} denote the SNR of the BS-MS₁ and BS-MS₂ links and γ_1, γ_2 stand for the fraction of power allocated to each transmitted signal. The optimal design for the bitrate and the power allocation is tackled in [Li-01]. The achievable rate region introduced in (1) is also valid for a single-carrier Multiple-Input Single-Output (MISO) configuration with uninformed transmitter (no, CSI), where the BS is equipped with multiple antennas, while destinations only have one antenna. The achievable rate region is obtained by solving the following optimisation problem

$$\begin{aligned} \max \quad & \mu_1 R_{bs1} + \mu_2 R_{bs2} \\ \text{s.t.} \quad & \gamma_1 + \gamma_2 = 1 \\ & R_{bs1}, R_{bs2} \in B_{BC} \end{aligned} \quad (2)$$

where B_{BC} is defined in (1) and μ_1, μ_2 are a given weight factors. By varying those weight factors (or user priorities) we get the achievable rate regions depicted in Figure 8 for two different cases; in Figure 8-left, the users to be served present a large SNR difference, and in Figure 8-right, both users have similar SNRs. We can observe that superposed transmission is beneficial when both users have a different SNR. Otherwise, the orthogonal transmission is almost optimal. On the other hand, if we are interested in maximising the sum-rate ($\mu_1 = \mu_2$), the optimal solution leads to dedicate all the resources (in terms of power) to the strongest user [Tse-05]. Notice that in the orthogonal transmission setup, the total transmission time can be divided among the two users, albeit the optimal solution devotes all the resources to one of the users as a function of the weight factors. Nevertheless, in the superposed transmission setup, both users employ the total transmission time and the source distributes its power.

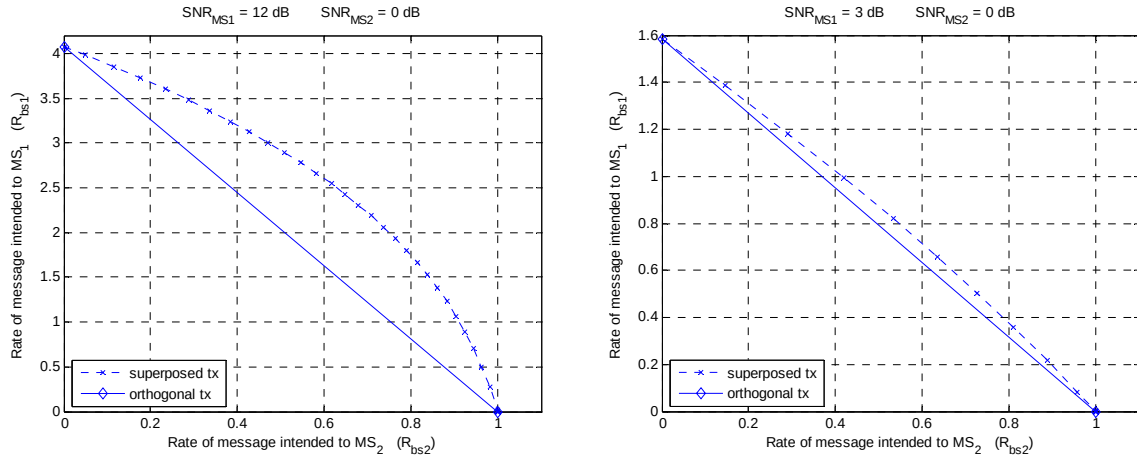


Figure 8: Superposed vs. orthogonal transmission in the DL, bitrate of MS_1 as a function of the bitrate of MS_2 : Left) $SNR_{MS1}=12$ dB, $SNR_{MS2}=0$ dB, Right) $SNR_{MS1}=3$ dB, $SNR_{MS2}=0$ dB

➤ Uplink with Two User

In the MACH, two MSs simultaneously transmit data to a single destination, BS. The bit-rate of the messages transmitted by the MS are constrained by [Cove-91],

$$B_{MAC} \begin{cases} R_{ms1} & \leq \log_2(1 + \rho_{ms1}) \\ R_{ms2} & \leq \log_2(1 + \rho_{ms2}) \\ R_{ms1} + R_{ms2} & \leq \log_2(1 + \rho_{ms1} + \rho_{ms2}) \end{cases}, \quad (3)$$

where R_{ms1} , R_{ms2} stand for the bitrate of the messages transmitted by MS_1 and MS_2 , respectively, and ρ_{ms1} , ρ_{ms2} denote the SNR of the BS- MS_1 and BS- MS_2 links. The rate region that is presented in (3) is also valid for a single-carrier SIMO configuration, where the MSs have one antenna and the BS has multiple antenna elements. In this case the MSs transmit independent messages.

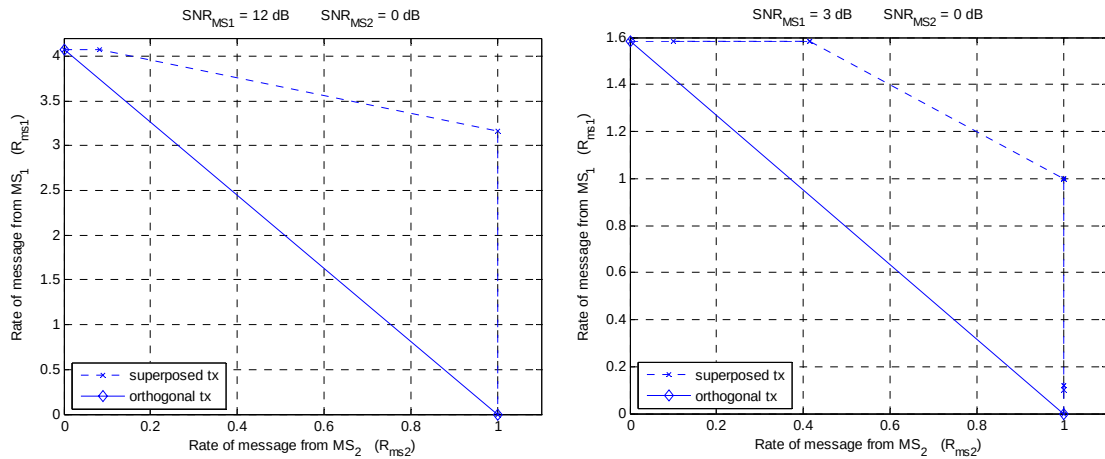


Figure 9: Superposed vs. orthogonal transmission in the UL, bitrate transmitted from MS_2 as a function of the bitrate transmitted from MS_1 : Left) $SNR_{MS1}=12$ dB, $SNR_{MS2}=0$ dB, Right) $SNR_{MS1}=3$ dB, $SNR_{MS2}=0$ dB

The achievable rate region is depicted by the trapezoidal shape shown in Figure 9. One case assumes that both MSs have a large SNR difference (Figure 9-left), and the other case considers that MSs have similar SNR (Figure 9-right). In contrast to the BC, there is always a gain in terms of bitrate in the MACH, irrespective of the SNR of the users. Likewise, under a sum-rate constraint ($\mu_1=\mu_2$) we allow

both users to simultaneously transmit, because the achievable rate is defined by the third rate constraint in (3).

➤ *Principal remarks*

For the UL channel, a significant gain, which is due to the superposed transmission and depends on the user selection, is always observed. A scheduling strategy based on a combinatorial optimisation problem has been derived in [Vite-08]. It selects the best users in a MIMO MU scenario. Additionally, a new criterion named double proportional fair user pairing that tackles the fairness among the users was introduced in [Chen-08]. We have shown the performance of the orthogonal transmission and the superposed transmission strategies in the DL (BC) and UL (MACH). The superposed transmission is the optimal transmission scheme over the BC and MACH channel, but it imposes a more complex receiver, because it requires Successive Interference Cancellation (SIC) at the receiver

Therefore, the scheduler at the DL/UL must select the best users that will use the same resources simultaneously under the superposed transmission criterion, i.e. *user pairing*. To this end section 3.1 is devoted to investigate the user pairing in the UL, while section 3.2 looks into the user pairing in the DL.

In order to delve into the gains in the UL, section 3.1.1 investigates the case of multiple antennas at the BS and MSs. It derives a low complexity UL MU pairing, which selects the pair of users according to the eigenvectors of the current channel state. The algorithm selects those users having the most orthogonal eigenvectors; as a consequence the BS is able to receive the messages that are transmitted by both MSs with a simple receiver. Additionally, section 3.1.2 analyses the algorithm proposed in section 3.1.1 in an inhomogeneous scenario with user mobility. The user mobility is an important aspect to be considered because it limits the applicability of the algorithm proposed in section 3.1.1. The channel state values that are used for deciding which users are paired and the channel state values that are employed when these users are actually served change as function of the user mobility.

The gain of the superposed transmission over the orthogonal transmission in the DL channel arises when the served users (or the paired users) present a large SNR difference and the optimisation is not based on the sum-rate (different weight factors or user priorities). To this end, section 3.2.1 investigates user pairing in the DL of OFDM systems by considering the WSR scheduling criterion, elucidating how to distribute the power among the data carriers as a function of the paired users. Finally, section 3.2.2 looks into the optimal resource allocation for the sum-rate and PF scheduling criteria, which will depend on the SNR of the users and their priorities.

3.1 UPLINK MULTI-USER PAIRING

3.1.1 Best band selection for transmit beamforming

3.1.1.1 Introduction

The work in this section considers the general framework of cellular broadband MU MIMO UL modes for band AMC like IEEE802.16e/m systems and addresses the following key questions:

- How to distribute the users on the OFDMA chunks in order to maximise the overall throughput?
- What is the best band allocation scheme?
- How does this modification evolve from SIMO to MIMO?
- Is mono-user multi-stream better than mono-stream MU in MIMO OFDMA?

Performance is illustrated for one specific UL MU-MIMO schemes: user pairing strategies for Spatial Division Multiple Access (SDMA) at BS, 2 TxBF MSs. The goal is to evaluate the impact of user

pairing and best band selection on performance by using multi-link PHY simulator. This simulator provides technical insights in order to derive efficient MAC scheduling algorithms. This study is in line with NEH business needs, since they have included SDMA in their product roadmap.

Last generation of Wi-Fi IEEE 802.11n and WiMax IEEE 802.16e wireless systems rely on multiple transmit and receive antenna schemes (*a.k.a.* MIMO) to improve their link robustness or throughput by implementing closed loop method taking advantage of the channel knowledge at the transmitter side. Though initially applied for single link designs, MU schemes such SDMA are now considered but few contributions deal with the impact of UL scheduling on the overall system performance [Xi-06], [Hara-06].

Our work focuses on a low arithmetical complexity Zero Forcing (ZF) based UL-scheduling technique for a two-Rx BS, which simultaneously receives two single streams from a 2Tx antennas MS selected in a larger pool of users. Performance gain is evaluated against a reference Time Division Multiple Access (TDMA) case where each users are transmitting sequentially based on information theory capacity metric. This metric is designed according to IEEE 802.16m parameters for a band AMC carrier allocation.

User pairing is achieved by optimising the orthogonality of the ZF maximum ratio combining single user decoding vectors, which tremendously reduces the search complexity compared to a full matrix dual user Minimum Mean Square Error (MMSE) approach (approximate complexity gain is n^2 versus $n!$). Moreover, the proposed method avoids the computation of the sum capacity for each pair of users, which results in an additional arithmetical complexity saving.

Simulation results are provided for various receiver structures, i.e., ZF, MMSE, joint MMSE and SIC. They highlight a scheduling gain from 5% to 13% on the achievable capacity/throughput.

3.1.1.2 Multiple user MIMO scheme

3.1.1.2.1 Notations

We denote by T and H respectively the transpose and the complex conjugate transpose operators. Capital bold face is used to indicate matrices, small boldface letters are used to indicate vectors and small letters are used to indicate complex or real scalars.

We consider the UL of a MU system, with K simultaneous transmitting MSs to a single BS. Each MS is equipped with N_T antennas to perform BF, and the receiver has N_R antennas. Let us first consider the single carrier case with a MIMO channel composed of independent Rayleigh fading sub-channels. The received signal vector at the receiver, called $\mathbf{r}$ can be written as

$$\mathbf{r} = \sum_{i=1}^K \sqrt{P_i^{(S)}} \mathbf{H}_i \mathbf{u}_i s_i + \mathbf{n}, \quad (4)$$

where s_i is the transmitted symbol from the i -th transmitter with $E\{|s_i|^2\} = 1$ and $P_i^{(S)}$ represents the total transmit power of each user, $\mathbf{H}_i$ is the $N_R \times N_T$ channel matrix from the i -th transmitter to the receiver and $\mathbf{n}$ is the noise vector with $E\{\mathbf{n}\mathbf{n}^H\} = P^{(n)}\mathbf{I}$. We define $\mathbf{u}_i$, a $N_T \times 1$ vector, as the TxBF vector used by MS i . It is well-known that the maximum channel capacity is obtained when $\mathbf{u}_i$ is the vector corresponding to the maximum eigenvalue of $\mathbf{H}_i^H \mathbf{H}_i$ (see [Anders-00]). Additionally, $\mathbf{H}_i$ can be written as

$$\begin{aligned} \mathbf{H}_i &= \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{U}_i^H \\ &= \lambda_{i,k} \mathbf{v}_{i,k} \mathbf{u}_{i,k}^H + \sum_{k \neq i} \lambda_{i,k} \mathbf{v}_{i,k} \mathbf{u}_{i,k}^H, \end{aligned} \quad (5)$$

where $\mathbf{u}_{i,k}$ and $\mathbf{v}_{i,k}$ are respectively the column vector of $\mathbf{U}_i$ and $\mathbf{V}_i$, and $\lambda_{i,k}$, $\mathbf{u}_{i,k}$ and $\mathbf{v}_{i,k}$ are the strongest singular value and its associated vectors. As a consequence, equation (4) yields

$$\mathbf{r} = \sum_{i=1}^K \sqrt{P_i^{(s)}} \lambda_i \mathbf{v}_i s_i + \mathbf{n}, \quad (6)$$

3.1.1.2.2 Scheduling algorithm description

As we mentioned before, scheduling is done to authorise two users that have *orthogonal* or *near to orthogonal* channel eigenvectors – associated to the maximum channel eigenvalue – to send data at the same time. The procedure can be summarised as follows:

1. Over all active users, compute eigenvectors associated to maximum eigenvalue of MU channels at the BS.
2. For each eigenvector, compute the inner product with all available eigenvectors.
3. The chosen couple is the one with the least value of inner product, *i.e.* the *most orthogonal*.
4. Eliminate the chosen couple from the list and go to step 2. and repeat till all the couples are computed.
5. Authorize each pre-computed couple to send consecutively data at the same time.

3.1.1.3 Detection scheme and capacity evaluations

Several equalizers are considered in this section and the corresponding capacity expressions are detailed. These capacity formulas are used to evaluate the various scheduling methods.

3.1.1.3.1 Projection: Zero-Forcing

When using the receive weights $\mathbf{v}_k^H$ for ZF detection, the output of the k -th receive BF is such that

$$\begin{aligned} \hat{s}_k &= \mathbf{v}_k^H \mathbf{r} \\ &= \sqrt{P_k^{(s)}} \lambda_k s_k + \mathbf{v}_k^H \sum_{i \neq k} \sqrt{P_i^{(s)}} \lambda_i \mathbf{v}_i s_i + \mathbf{v}_k^H \mathbf{n}, \end{aligned} \quad (7)$$

Since our scheduling algorithm aims at pairing users such that $\mathbf{v}_k$ and $\mathbf{v}_i$ are orthogonal or near to orthogonal, the second term in the above equation is close to zero. The throughput of the system can therefore be evaluated as ([Telatar-99])

$$\begin{aligned} C &= \log_2 (1 + SNIR) \\ &= \log_2 \left(1 + \frac{P_k^{(s)} \lambda_k^2}{\sum_{i \neq k} P_i^{(s)} \lambda_i^2 \|\mathbf{v}_k^H \mathbf{v}_i\|^2 + P^{(n)}} \right), \end{aligned} \quad (8)$$

with $\|\mathbf{v}_k^H \mathbf{v}_i\|$ being close to zero. This detection method has the advantage of being simple and almost interference free.

3.1.1.3.2 MMSE

Here the MMSE equalizer aims at performing a scalar MMSE detection per user considering the interference as Gaussian noise. Let us denote $\mathbf{g}_k$ the MMSE equalizer corresponding to the system model in (4). It can be written as

$$\mathbf{g}_k^H = \left(\lambda_k^2 \mathbf{v}_k^H \mathbf{v}_k \frac{P^{(n)}}{P_k^{(s)}} \mathbf{I} \right)^{-1} \lambda_k \mathbf{v}_k, \quad (9)$$

and consequently,

$$\begin{aligned} \hat{s}_k &= \mathbf{g}_k^H \mathbf{r} \\ &= \sqrt{P_k^{(s)}} \lambda_k \mathbf{g}_k^H \mathbf{v}_k s_k + \mathbf{g}_k^H \sum_{i \neq k} \sqrt{P_i^{(s)}} \lambda_i \mathbf{v}_i s_i + \mathbf{g}_k^H \mathbf{n}. \end{aligned} \quad (10)$$

Derivation of the capacity is straightforward and it is not mentioned here.

3.1.1.3.3 Joint MMSE

We define the Joint MMSE equalizer as the full matrix MMSE equalizer. Equation (4) can be written in a matrix form as

$$\mathbf{r} = \mathbf{V}\mathbf{s} + \mathbf{n}, \quad (11)$$

with $\mathbf{V} = [\sqrt{P_1^{(s)}}\lambda_1\mathbf{v}_1, \sqrt{P_2^{(s)}}\lambda_2\mathbf{v}_2, \dots, \sqrt{P_K^{(s)}}\lambda_K\mathbf{v}_K]$ of size $N_R \times N_T$ and $\mathbf{s} = [s_1, s_2, \dots, s_K]^T$. Joint MMSE detection can be computed as

$$\begin{aligned} \hat{\mathbf{s}} &= \mathbf{G}^H \mathbf{r} \\ &= \mathbf{M}\mathbf{s} + \mathbf{G}^H \mathbf{n} \end{aligned} \quad (12)$$

Here $\mathbf{G}$ is the linear MMSE equalizer: $\mathbf{M} = \mathbf{G}^H \mathbf{V}$. In this case, channel throughput is computed as follows ([Telatar-99])

$$\begin{aligned} C &= \mathcal{H}(\mathbf{r}) - \mathcal{H}(\mathbf{r}|\mathbf{s}) \\ &= \log_2 \left(\det(\mathbf{M}^H \mathbf{M} + \mathbf{G}^H \mathbf{G}) \right) - \log_2 \left(\det(\mathbf{G}^H \mathbf{G}) \right), \end{aligned} \quad (13)$$

where $\mathcal{H}$ is the entropy.

3.1.1.3.4 Successive Interference cancellation

The capacity of a SIC is usually derived considering perfect interference removal. Since interferences are successively cancelled, one term only remains in the capacity formula, which can be expressed as follows

$$C = \log_2 \left(\det(\mathbf{I} + \tilde{\mathbf{H}}^H \tilde{\mathbf{H}}) \right). \quad (14)$$

Furthermore, it can be simplified as

$$C = \log_2 \left(1 + \frac{P_1^{(s)}}{P^{(n)}} \lambda_1^2 + \frac{P_2^{(s)}}{P^{(n)}} \lambda_2^2 + \frac{P_1^{(s)}}{P^{(n)}} \lambda_1^2 \frac{P_2^{(s)}}{P^{(n)}} \lambda_2^2 \left(1 - |\langle \mathbf{v}_1, \mathbf{v}_2 \rangle|^2 \right) \right), \quad (15)$$

for two user.

3.1.1.4 Efficient implementation to find orthogonal eigenvectors

3.1.1.4.1 Basic selection approach: scalar product based metric

In this section, we present an efficient implementation method that can considerably reduce the hardware implementation complexity for the case of two Tx antennas ($N_T = 2$).

Let us consider two arbitrary eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ for the two Tx antennas' case. They can be written as follows

$$\mathbf{v}_1 = \begin{bmatrix} x_1 \\ y_1 + iz_1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} x_2 \\ y_2 + iz_2 \end{bmatrix}. \quad (16)$$

If no complexity reduction is used, in an exhaustive search approach, then the inner product of the eigenvectors is computed and then two orthogonal or almost orthogonal eigenvectors are obtained that minimise the inner product value. The inner product of $\mathbf{v}_1$ with $\mathbf{v}_2$ can be written as

$$\begin{aligned} \langle \mathbf{v}_1, \mathbf{v}_2 \rangle &= \mathbf{v}_1^H \mathbf{v}_2 \\ &= x_1 x_2 + y_1 y_2 - z_1 z_2 + i(z_1 y_2 - y_1 z_2) \end{aligned} \quad (17)$$

Final metric computation is given by

$$\text{metric} = |\langle \mathbf{v}_1, \mathbf{v}_2 \rangle|^2. \quad (18)$$

Therefore, seven real multiplications and four real additions are required.

3.1.1.4.2 A lower complexity metric

One can further reduce the complexity of the metric used for the eigenvectors selection, since in the two Tx antennas' case, the expression of orthogonal vectors can be directly derived. For example, the orthogonal vector of $\mathbf{v}_1$ is given by¹

$$\mathbf{v}_1^\perp = \begin{bmatrix} -y_1 + iz_1 \\ x_1 \end{bmatrix}. \quad (19)$$

Once the orthogonal vector $\mathbf{v}_1^\perp$ is obtained, we just need to compute the distance between this vector and other vectors in order to directly obtain metrics

$$\begin{aligned} d(\mathbf{v}_2, \mathbf{v}_1^\perp) &= \left\| \begin{bmatrix} x_2 + y_1 - iz_1 \\ x_1 + iz_2 - y_2 \end{bmatrix} \right\|^2 \\ &= \underbrace{x_1^2 + y_1^2 + z_1^2}_{=1} + \underbrace{x_2^2 + y_2^2 + z_2^2}_{=1} + 2(y_1 x_2 - x_1 y_2). \\ &= 2 + 2(y_1 x_2 - x_1 y_2) \end{aligned} \quad (20)$$

For the metric computation, we will have

$$\text{metric} = y_1 x_2 - x_1 y_2, \quad (21)$$

which corresponds only to two multiplications and one addition. Consequently, a large amount of complexity reduction is achieved.

3.1.1.5 System model and simulation results

Here a single cell deployment is considered that serves a given number of users. The antennas at the BS are supposed isotropic; from two up to eight antennas configurations are captured. A two slopes classical path loss model is adopted, which is provided in (253) in the Appendix, section 8. The different parameters of the system are recalled in Table 1.

Shadowing is taken into account in the simulations through a ray-tracing approach, which is more suited for a single cell deployment. A set of obstacles are dropped over the cell with a random attenuation and aperture (the distribution is defined in the channel parameters). Assuming a dominant line of sight component, the losses due to shadowing effect are directly computed by summing the loss of each obstacle on the path.

Table 1: System Parameters

<i>Syst. assumptions</i>	<i>802.11n</i>	<i>802.16m</i>
Center frequency	5.2GHz	2.4GHz
Emitting power at the BS	0.2W	10W
Signal bandwidth	20MHz	20MHz
Break Point distance (BPd)	10m	10m
Path loss exponent after BPd	3.5	2.5
Shadowing Std Dev before BPd	3	3
Shadowing Std Dev after BPd	5	5
Antenna Gain (Rx Tx)	0dB	0dB
Noise Figure at the receiver	10dB	10dB

¹ We have considered also the fact that the eigenvector $\mathbf{v}_1$ is unitary.

3.1.1.5.1 Capacity results

For the simulations that are presented below, a set of 20 users is considered. Several detection strategies are benchmarked; the simplest receiver is based on projection (ZF); the second receiver is a simple MMSE receiver; the third one is a joint MMSE receiver; the last receiver is a complex SIC receiver in a 2×2 MIMO context. Note that the influence of the number of obstacles is not discriminating on the final capacity gain (the impact is less than 0.3%). The results are averaged using 100 sets of users and 100 channel realisations per user.

In order to highlight the interest of such pairing/scheduling techniques, we also implemented an arbitrary method that performs a random user coupling/sequencing. Both methods were first compared according to the gain provided over a classical TDMA scheme; a null gain corresponds to no improvement over TDMA, whereas a 100% gain is equivalent to perfect spatial multiplexing with no interference (or perfect interference cancellation).

Figure 10 presents the comparison of the capacity gain distribution observed for the various detection schemes for both an arbitrary and the proposed pairing/scheduling method. The vertical lines appearing on the plots mark the location of the average gain associated to each distribution. For all receiver structures studied, the proposed pairing/sequencing method outperforms the random scheduling. The average gain varies depending on the receiver complexity: 13% for the projection, 12% for the MMSE and 5% for the joint MMSE or SIC structures. Though the gain is modest for full MMSE receivers, note that it comes at a cost of a very small additional complexity paid in the pairing process.

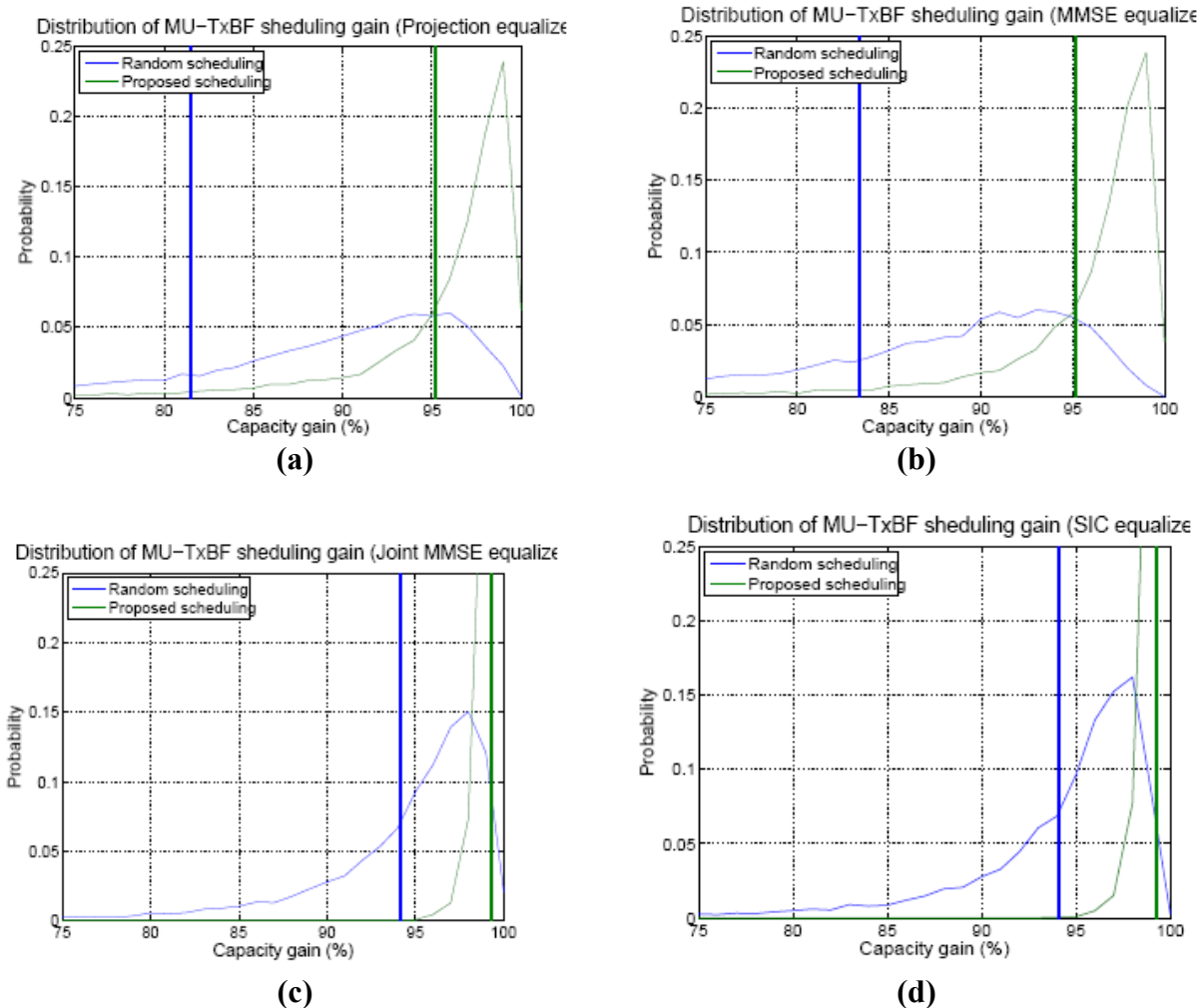


Figure 10: Distribution of capacity with (a) ZF receiver, (b) MMSE detector, (c) Joint MMSE receiver, (d) SIC receiver

3.1.1.5.2 Throughput results

We compare the sum throughput of single-user SVD-MIMO OFDMA versus MU TxBF OFDMA for different antenna configurations and user pairing strategies, based on the evaluation methodology described in [Mazet-07].

In the simulations conducted, perfect channel knowledge is assumed at the transmitter. The PHY parameters are derived from the WiMAX Release-1 reference system specification [WIMAX-07], [WIMAX]. Bit interleaved convolutionally encoded modulation for BPSK, QPSK, 16QAM, 64QAM constellations are used. We consider 1536 carriers for a bandwidth of 20MHz. One or two spatial streams (subscribe stations is assumed to support up to two antennas) are allocated to one to two users using BF on their main signal value vectors. Spatial Channel Model Extended (SCME) urban-micro channel models are used applying COST-231 Hata path loss model.

Note that, for the sake of simplicity, we do not take into account the interference between cells, and all users are assumed to be localised at equal distance from the base-station in the range of assessment.

System deployment model

The system simulation summary appears in Table 2. This model applies to any duplexing mode proposed for 802.16e amendment [IEEE802.16e-06].

Table 2: Simulation environment summary

<i>Parameters</i>	<i>Value</i>
Operating frequency	2500MHz
BS-to-BS distance	1.5km
Minimum mobile-to-BS distance	35m
BS height	32m
Mobile terminal height	1.5m
Propagation model	COST-231 modified Hata (suburban) model
BS thermal noise figure	5dB
Mobile terminal noise figure	7dB
Thermal noise density	-174dBm/Hz
BS antenna gain incl. cable Loss	15dB
Mobile terminal Maximum PA power	200mW (23dBm)
Signal bandwidth	20MHz
# of BS Tx/Rx antennas	1 or 2
# of MT Tx/Rx antennas	2
Modulation constellation	BSPK, QPSK, 16QAM and 64QAM
FEC	Convolutionnal code (o171, o133)
Puncturing rate	1/2, 2/3, 3/4 and 5/6
Performance target	PER of 5% over 1kB packet

Pathloss model

The proposed path loss model is based on the well-known COST-231 modified Hata model for systems supporting carrier frequencies less than or equal to 2.5GHz. The COST-231 modified Hata model at 2.0GHz is used to simulate the IEEE 802.16m reference system in suburban mode ($L = 126.2 + 36 \log_{10}(R)$) for a 2.5Ghz carrier frequency.

Evaluation methodology

A band AMC like OFDMA system with sub-channels composed of 14 adjacent carriers is used for the simulations; in 20MHz, 100 chunks are considered for a total of 1536 useful carriers transmitted. In

order to evaluate the throughput versus distance behaviour of the proposed user pairing and best band selection scheme, we assume that all users have the same path loss (are at the same distance from the BS) but experience a different channel. We also assume that the CSI is perfectly known at the BS and at the MS side. Statistical BF transmission averaged over the chunk is used to limit the overhead for explicit vectors feedback. For user i , when considering frequency slot k , we defined statistical BF as the weight vectors extracted from the following $\mathbf{V}_{k,i}$ matrix

$$\begin{aligned}\mathbf{R}_{k,i} &= \frac{1}{|\mathcal{S}_k|} \sum_{l \in \mathcal{S}_k} \mathbf{H}_i^H(l) \mathbf{H}_i(l), \\ &= \mathbf{V}_{k,i} \mathbf{\Lambda}_{k,i} \mathbf{V}_{k,i}^H\end{aligned}\quad (22)$$

where $\mathbf{R}_{k,i}$ is an estimate of the correlation matrix of the channel $\left| \mathbf{H}_i(l) \right|_{l \in \mathcal{S}_k}$ seen by user i over the set of subcarrier $\mathcal{S}_k$ that defines the frequency chunk k . Matrix $\mathbf{V}_{k,i}$ is composed of the eigenvectors of $\mathbf{R}_{k,i}$.

All simulations are performed for the UL and the base-station is equipped with a linear equalisation scheme (joint MMSE per carrier). This means that the equalizer $\mathbf{G}_{k,(i,j)}(l)$ on subcarrier $\bullet$ is chosen as

$$\begin{aligned}\forall l \in \mathcal{S}_k, \quad \tilde{\mathbf{H}}_{i,j}(l) &= \begin{bmatrix} \mathbf{H}_i(l) \tilde{\mathbf{V}}_{k,i} & \mathbf{H}_j(l) \tilde{\mathbf{V}}_{k,j} \end{bmatrix} \\ \mathbf{G}_{k,(i,j)}(l) &= \left(\tilde{\mathbf{H}}_{i,j}(l) * \tilde{\mathbf{H}}_{i,j}^H(l) + \sigma^2 \mathbf{I} \right)^{-1} * \tilde{\mathbf{H}}_{i,j}(l)\end{aligned}\quad (23)$$

for two paired-users, where $\tilde{\mathbf{H}}_{i,j}$ is the equivalent channel for user i and j BF respectively by $\tilde{\mathbf{V}}_{k,i}$ and $\tilde{\mathbf{V}}_{k,j}$. Note that $\tilde{\mathbf{V}}_{k,i}$ is a matrix of one or two columns depending on the number of used stream extracted from $\mathbf{V}_{k,i}$ for single-user configuration (*i.e.* no user j). In the case of MU configuration, $\tilde{\mathbf{V}}_{k,i}$ and $\tilde{\mathbf{V}}_{k,j}$ are respectively the main eigenvectors extracted from $\mathbf{V}_{k,i}$ and $\mathbf{V}_{k,j}$.

Best band selection and pairing algorithm evaluation

We consider a larger pool of users (*i.e.* 1000 users) than the number of users served per OFDM symbol; the total number of users is allocated over five different time slots when using SDMA user pairing and 10 different time slots when no spatial multiplexing is used. The bandwidth is subdivided into 100 frequency chunks of 14 adjacent carriers each. Note that single-user multi-stream transmissions are considered when no UL SDMA is applied. Channel Impulse Response (CIR) is almost constant on each chunk since the occupied bandwidth is much smaller than the frequency coherent bandwidth of the SCME channel. The evaluation carried out is performed on a chunk basis, but in reality, one would allocate several sub-channels per user. The expected impact in terms of performance shift is anticipated to be marginal.

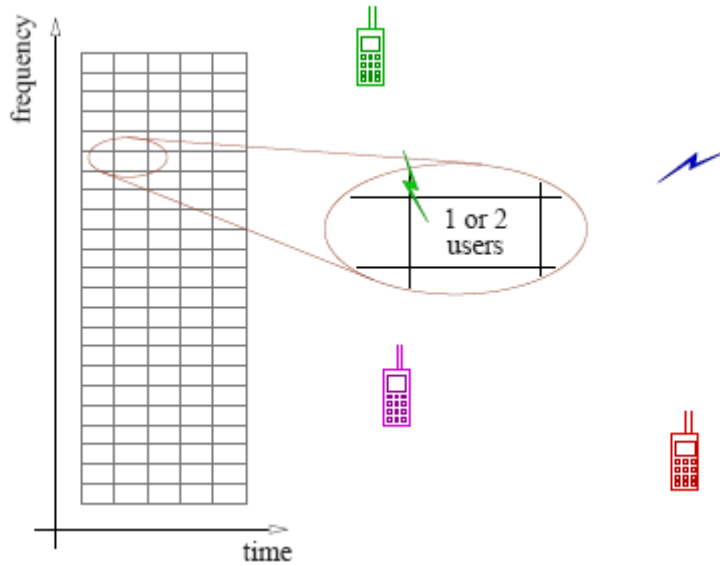


Figure 11: Time scheduling, frequency scheduling and user pairing

Four user allocation strategies are evaluated in the single user context and MU context:

1. The first scheduler arbitrary allocates one or two users per time and frequency slot. It corresponds to a lower bound on the achievable performance. It is referred as *Arbitrary*, which are the following curve legends.
2. The second scheduler starts by allocating 10 users per frequency slot arbitrary; it then applies our user pairing method by packing 10 users two by two over the 5 consecutive time slots. It is referred as *User Pairing*.
3. The third scheduler performs only a best band selection by optimising the user allocation per frequency slot without taking into account that two users transmitting simultaneously may interfere in some cases. It is referred as *Frequency Selective (FS)*. Frequency slot allocation is made by sequentially maximising the average capacity; if $\tilde{c}_i(k)$ is defined as

$$\tilde{c}_i(k) = \frac{1}{|\mathcal{S}_k|} \sum_{l \in \mathcal{S}_k} \log_2 \left(\det \left(\mathbf{I} + \sigma^{-2} \mathbf{H}_i(l) \mathbf{H}_i^H(l) \right) \right) \quad (24)$$

is maximum, user i gets slot k . Once a user is served, it cannot claim for another slot. Only one other user can potentially get it, however, the second user is selected using the same criteria.

4. The fourth scheduler starts by performing a frequency scheduling and then applies our user pairing method, packing 10 users two by two over five consecutive time slots. It is referred as *FS+UP*.
5. The last scheduler operates a best band selection taking into account user interference. This means that, for slot k , the scheduler chooses user i that maximises $\tilde{c}_i(k)$ and it then looks for user j that maximises the average capacity of the MIMO channel created by the two users $\tilde{c}_{i,j}(k)$.

$$\begin{aligned} \forall l \in \mathcal{S}_k, \quad \mathbf{H}_{(i,j)}(l) &= \begin{bmatrix} \mathbf{H}_i(l) & \mathbf{H}_j(l) \end{bmatrix} \\ \tilde{c}_{(i,j)}(l) &= \frac{1}{|\mathcal{S}_k|} \sum_{l \in \mathcal{S}_k} \log_2 \left(\det \left(\mathbf{I} + \sigma^{-2} \mathbf{H}_{(i,j)}(l) \mathbf{H}_{(i,j)}^H(l) \right) \right) \end{aligned} \quad (25)$$

It is referred as *MU FS*. This scheme is extremely time consuming but yields the best performance.

We assume that all MSs are at equal distance of the base-station (i.e. experience the same path loss), in order to present curves of aggregated PHY over the air throughput versus distance. The assumption that a certain number of users will be seen by the BS to have the same SNR is realistic; this is one of the goals of the ranging procedure for performing transmit power control in order to avoid AGC recalibration when switching users during the UL phase at the BS.

Throughput for 1x2 antenna configurations

Figure 12 presents aggregated PHY over the air throughput versus distance for configurations using 1 Tx antenna and 2 Rx antennas.

Frequency Selective and User Pairing (1x2, SCME UrbMic)

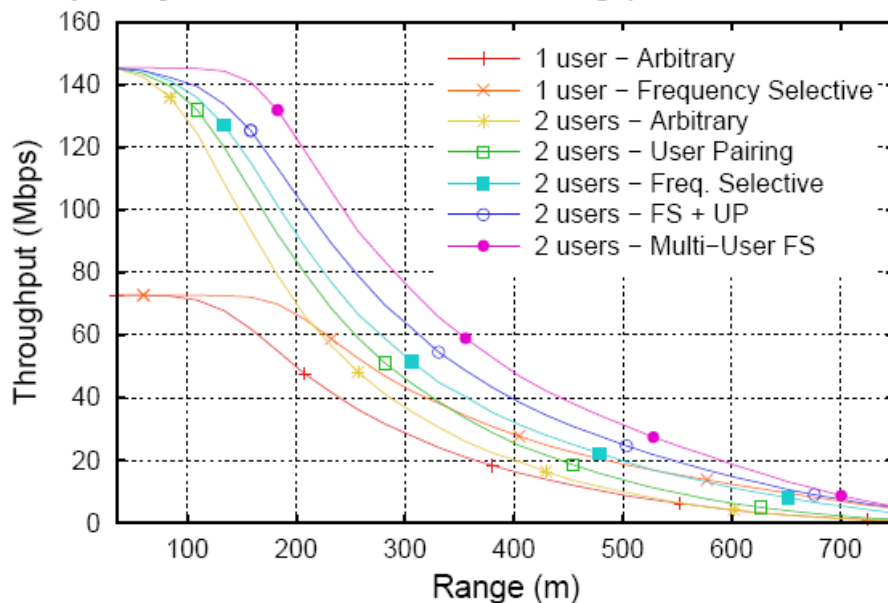


Figure 12: MUs versus single user for 1Tx 2Rx configurations

Many comparisons can be drawn from this figure:

1. The yellow curve, denoted by '2 users - Arbitrary', compared to the red curve, denoted by '1 user - Arbitrary', shows that the gain of user pairing is large when users are close to the BS (around twice the aggregated throughput) but is significantly reduced at cell border where the inter-user interference level becomes negligible compared to the noise level.
2. The orange curve, denoted by '1 user - FS', compared to the red curve, denoted by '1 user - Arbitrary', and also the cyan curve, denoted by '2 users - FS', compared to the yellow curve, denoted '2 users - Arbitrary', illustrate the gains of best band selection. At 200m, aggregated throughputs are respectively increased by a factor of 1.5 and 1.3. Over most of the cell, this yields a throughput gain of 10Mbps to 20Mbps.
3. The green curve, denotes by '2 users - User Pairing' compared to the yellow curve, denoted by '2 users - Arbitrary', and also the blue curve, denoted by '2 users - FS + US', compared to the cyan curve, denoted by '2 users - FS', highlight the gain of our low complexity user pairing technique. At 200m, aggregated throughputs are increased by 15% and, over most of the cell, there is a throughput gain of 12Mbps.
4. The magenta curve, denoted by '2 users - MU FS' is the best considered technique when one can afford its complexity; it performs jointly user pairing and best band selection providing an idea on an upper bound on achievable throughput.

Throughput for 1x2 antenna configurations

Figure 13 presents aggregated PHY over the air throughput versus distance for configurations using 2 Tx antenna and 2 Rx antennas.

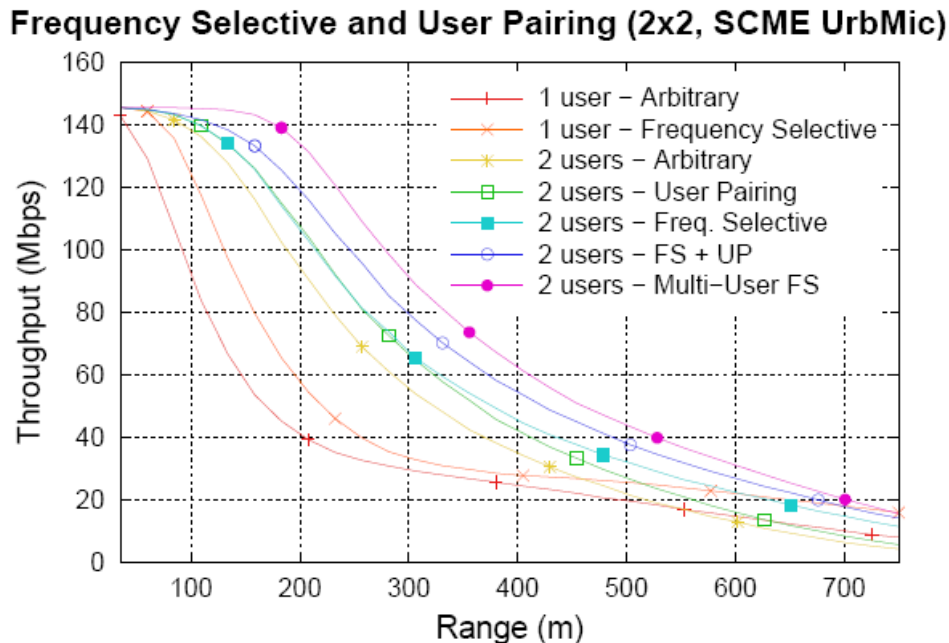


Figure 13: MUs versus single user for 2Tx 2Rx configurations

This figure compares single user multi-stream UL transmissions versus MU single stream UL transmission. The yellow curve, denoted by '2 users - Arbitrary', crosses the red curve, denoted by '1 user - Arbitrary', at low SNR. It appears that, for ranges exceeding half the cell radius, it is better to use single user multi-stream transmission over mono-stream MU (SDMA). Moreover, by comparing the cyan curve, denoted by '2 users - FS' and the green curve, denoted by '2 users - User Pairing', it tells that, for high SNR, user pairing or best band selection slot selection individually behaves the same, but at cell border (low SNR) best band selection supersedes user pairing.

From these performance curves, we conclude that

1. It is always better to serve two users rather than one,
2. Best band selection gives us an huge improvement in SIMO configurations,
3. User pairing improves performances in MIMO systems
4. Joint user pairing and best band selection is the golden nugget on which we will focus in our future studies.

3.1.1.6 Conclusions

In this Section, we detailed a scheduling scheme for the UL of the MIMO – channel aware systems – that supports *simultaneous transmitting* users with a very little amount of interference; it helps us to benefit from a very simple receiver scheme.

Performance assessments demonstrate the potential interest of MU MIMO in broadband wireless access systems, and it has to be proposed for standardization. In this regards, IEEE 802.16m is the perfect landing zone for such a technology. Moreover, this mechanism will create some hooks for future and more complex MU strategies by inserting it into IEEE 802.16m.

Performance assessments also provide an interesting insight: MUs diversity gain is restored in MIMO when considering user pairing on top of best band allocation. MAC schedulers will need to capture these considerations to fully exploit MU-MIMO SDMA in OFDMA systems (16e/m and beyond) and

that leads to potential differentiators for Motorola WiMAX base-stations. Some new investigations have been started on low-complexity user pairing and best band selection for other MIMO schemes.

3.1.2 Impact of the mobility and inhomogeneous scenarios

The major goal of this activity targets at giving a follow-up and at extending the activity of Section 3.1.1. The pursued objectives are to investigate low complexity UL MU user pairing and best band selection for TxBF mobiles, based on “nearly” orthogonal and best user selection and coherent BF. The rationale for this activity relies in the foreseen possible drawbacks that might limit the applicability of a fully coherent approach for the user pairing and best band selection. The major drawbacks highlighted so far are related with the following aspects:

- *Mobility*: in the presence of mobility, the time lag between the UL of the training sequences and data the UL causes a mismatch between the complex weight employed on the UE and BS antennas and those that would have been employed to insure the matching to the channel realisation;
- *Inhomogeneity*: the users in a realistic scenario are likely to have SNR distribution driven by an exponential model and this might introduce the need of additional care for the coupling process, as different performance are to be expected when low SNR UEs are coupled with high SNR UEs with respect to the case of coupling among similar (homogeneous) SNR distributions.

It will be showed that the presence of mobility poses severe limitations to the effectiveness of the methodology and indicates both a re-design of the frame format (specified in section 5) and a sub-optimal aggregate methodology for the MU channel estimation at the BS. The strong inhomogeneity in realistic scenarios has been faced by introducing a different pairing and scheduling algorithm with respect to the one proposed in section 3.1.1.2.2, capable of (nearly) insuring the maximal Spectral Efficiency (SE) that can be offered by the system, meanwhile maintaining an acceptable level of fairness among the active users. In the following sections a more detailed analysis of the above factors will be presented.

3.1.2.1 Effect of mobility

A first constraint of the scheme comes from the training time/resources needed for the estimation of the $M_{BS} \times N_U$ channel matrix of U UEs by the BS, M_{BS} and N_U being the number of antennas at the BS and at the UE, respectively: this requires in principle $U \times M_{BS} \times N_U$ not-overlapping preambles (but could be reduced to $U \times M_U$ by proper preamble design). Another constraint of a fully coherent technique comes from the channel coherence time: indeed, no matter the adopted scheduling approach, the scheduling information has to be communicated and the whole UL phase for the selected users has to fall well within the channel coherence time. The failure of this constraint implies that the users selected by the scheduler are associated with joint MIMO complex weights computed by the BS assuming an estimated channel in a time interval that has little or no statistical relation with the actual channel realization at the scheduled time for the transmission. This constraint does not jeopardise the techniques proposed in Section 3.1.1, but gives indications about the upper mobility limits beyond which the MU MIMO UL is not properly performing. Another consideration comes from the trade-off between the achievement of a throughput increase and the fairness of the system: in inhomogeneous scenarios (assuming a uniform spatial distribution of users) the users experiencing low SNR are expected to be significantly more than the users with high SNR (see also section 2). As far as the maximisation of SE is considered, the inhomogeneity of the SNR distribution could entail unfair user selection, as the (less) users with high SNR would be paired and scheduled more frequently than the (more) users with low SNR, causing a significant increase of the latency for most of the users belonging to the BS. In the following, some additional detail is given about the constraints coming from the training for the channel estimation and from the inhomogeneity.

In Figure 14, the channel coherence time T_{CO} vs. mobility is reported assuming a carrier frequency of 2 GHz, assuming the simplified relation

$$T_{CO} = \frac{c}{2f_c v}, \quad (26)$$

where c and v are the speed of the light and of the mobile, respectively and f_c is the carrier frequency. On the other hand, assuming $U=100$ users in the scenario, each equipped with $N_U=2$ antennas, the time needed to transmit 200 UL preambles, $\sim 100 \mu s$ each, in compliance with the IEEE 802.16e-2005 preamble duration, is approximately 2 ms. This value is reported in Figure 14 as a solid green line. In this perspective, the coherence relation between the largest eigenvalues and corresponding eigenvectors to be employed for the MU MIMO in the UL starts to loose its matching and effectiveness from speeds above 10 km/h (or less) and the selection and pairing operated by the BS on the basis of the “nearly orthogonal” eigenvectors might be misleading for speeds above the pedestrian mobility.

Quantitatively, when the time difference between the channel estimation phase and the UL phase (in which the antenna weights computed from the estimated channel are employed) is indicated by t_d , the following relation holds for the generic k^{th} UE:

$$\mathbf{H}_k(t_d) = \rho_k(t_d)\mathbf{H}_k + (1 - \rho_k(t_d))\mathbf{H}'_k, \quad (27)$$

where $\mathbf{H}_k(t_d)$ is the $(M_{BS} \times N_U)$ channel matrix realization after t_d seconds from its estimation $\mathbf{H}_k$ and $\mathbf{H}'_k$ is channel matrix realisation uncorrelated with $\mathbf{H}_k$, $\rho_k(t_d)$ is a coherence factor taking into account the ratio between t_d and T_{CO} that are given by, although approximate as a 1st expansion order, a functional form suitable for simulation

$$\rho(t_d) \approx J_0\left(\frac{2t_d}{T_{CO}}\right), \quad (28)$$

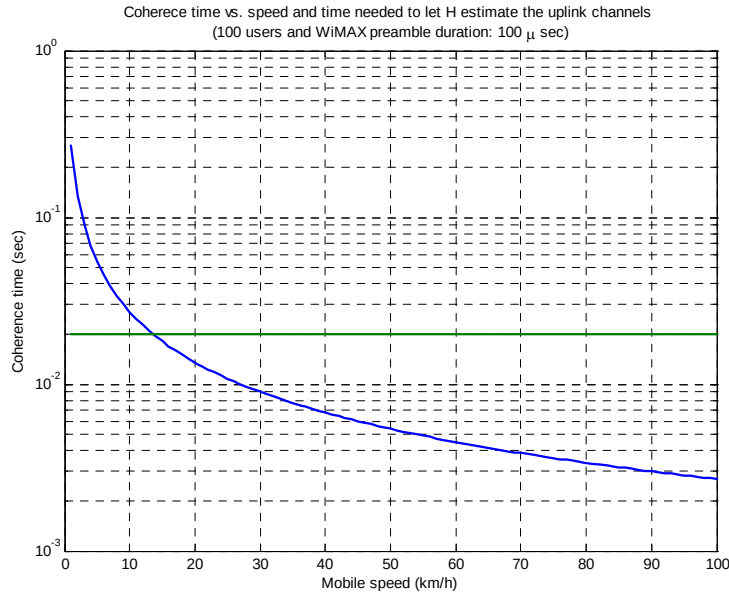


Figure 14: Channel coherence time vs. speed for a transmission with a 2GHz carrier. The horizontal line corresponds to the time needed to transmit 200 preambles (i.e. 100 users equipped with 2 TX antennas)

where J_0 is the Bessel function of 1st type. From (4), (5) and (6), the ZF projection² estimation of the k^{th} UE can be rewritten in the case of a simple coupling between as

$$\hat{s}_k = \mathbf{v}_k^H \left(\sqrt{P_k} \mathbf{H}_k(t_d) \mathbf{u}_k s_k + \sqrt{P_j} \mathbf{H}_j(t_d) \mathbf{u}_j s_j + \mathbf{n} \right), \quad (29)$$

as $\mathbf{H}_k(t_d)$ can be rewritten as $\mathbf{H}_k(t_d) = \mathbf{H}_k + \Delta$, from (29) we have $\Delta = (1 - \rho_k(t_d))(\mathbf{H}_k - \mathbf{H}_k')$ and (27) can be reformulated as

$$\begin{aligned} \hat{s}_k = & s_k \sqrt{P_k} \left(\rho_k \lambda_k + (1 - \rho_k) \mathbf{v}_k^H \mathbf{H}_k' \mathbf{u}_k \right) + \\ & s_j \sqrt{P_j} \left(\rho_j \lambda_j \mathbf{v}_k^H \mathbf{v}_j + (1 - \rho_j) \mathbf{v}_k^H \mathbf{H}_j' \mathbf{u}_j \right) + \mathbf{v}_k^H \mathbf{n} \end{aligned} \quad (30)$$

where the dependence from t_d has been omitted to simplify the notation. As far as $\mathbf{H}_k'$ is unknown, the self-interference term has to be managed as interference and not as a component of useful signal to be exploited in raising the SNR. By disregarding the terms driven by $\rho(1 - \rho)$; the SNR can be thus approximated as

$$SNR_{k,j} = \frac{\rho_k^2 P_k |\lambda_k|^2}{\delta_k^2 P_k |g_k|^2 + \delta_j^2 P_j |q_{k,j}|^2 + \rho_j^2 P_j |\lambda_j \mathbf{v}_k^H \mathbf{v}_j|^2 + P_n}, \quad (31)$$

where $\delta_k = (1 - \rho_k)$, $g_k = \mathbf{v}_k^H \mathbf{H}_k' \mathbf{u}_k$ e $q_{k,j} = \mathbf{v}_k^H \mathbf{H}_j' \mathbf{u}_j$. From (31) it is straightforward to notice that, in absence of mobility, $\rho_k(t_d) = 1$ and the relation reduces to

$$SNR_{k,j} = \frac{P_k |\lambda_k|^2}{P_j |\lambda_j \mathbf{v}_k^H \mathbf{v}_j|^2 + P_n} \quad (32)$$

that is the known relation for a simple pairing. It has to be remarked that, even if the k - j coupling were ideal (i.e. $\mathbf{v}_k^H \mathbf{v}_j = 0$), both a cross-interference and a self-interference term still survive in the general formulation of (31): both are physically due to the mismatch between the actual channel and the employed principal (left, right) eigenvectors (that are computed from $\mathbf{H}_k$, but employed for the channel matrix realisation $\mathbf{H}_k(t_d)$). As $\mathbf{H}_k'$ is uncorrelated with $\mathbf{H}_k$, so are its relations with $\mathbf{u}_k$ and $\mathbf{v}_k$. As far as the channel matrix has normalised IID Gaussian entries, the scalars g_k and $q_{k,j}$ in (31) are random numbers with the same variance of the entries of $\mathbf{H}_k$, and the solely driving parameter from the interference perspective is the coherence factor $\rho_k(t_d)$.

From (31) and (28) it is possible to point out that, when $t_d \gg T_{CO}$, then $\rho_k(t_d) \rightarrow 0$ and consequently $SNR_{k,j} \rightarrow 0$, not because of the increase of the interference component, but because of the unpredictability of the estimated symbol, that turns to be random and uncorrelated with the transmitted information.

The effect of (31) on the adopted coherent BF in UL can be seen by adopting the UEs spatial distribution reported in Figure 15 (carrier 1 GHz, HATA suburban pathloss model eq. (252), channel 3GPP Vehicular type A). The resulting SNR for the UL coherent BF are reported, ranked for increasing values of SNR, in Figure 16 for 30% of the UEs with random speeds up to 0, 25 and 50 km/h and assuming a time lag t_d between the channel estimation and the UL phase equal to 2 WiMAX frames (i.e. 10 ms), in compliance with the modified MAC protocol and frame format specification introduced in section 5.

Stated in a different way, the solely “best pairing” approach would not be able to exploit the MU diversity if the scheduling was not suited to have the UL phase within a time interval below the

² In this context, other linear and not-linear estimation filtering (MMSE, Subsequent cancellation, etc.) will be not considered, as the focus is not the best performing estimator, but the relationship between mobility, inhomogeneity, structured nature of the interference and the coupling strategy.

coherence time from the channel estimation (i.e. $t_d \ll T_{CO}$). To have an idea of what this constraints involves, the subsequent Figure 17 reports the metric employed for the user pairing ($|\mathbf{v}_j^H \mathbf{v}_k|$, $\mathbf{v}_n$ being the eigenvector associated with the largest eigenvalue of the UL channel matrix for the n^{th} user) in the (just illustrative) case of three users. The abscissa represents the time slot index and the employed coherence time is 200 time slots for all the users (so to have a representation independent of the carrier frequency and mobile speed). From the figure, it is apparent that the nearly-orthogonal condition maintains its validity over a time period well below the channel coherence time and thus the channel estimation; scheduling and UL phases must be accomplished in a time interval below this limit. Figure 17 illustrates, for instance, that the pair (1,2) would have been selected at the time index 900 or the pair (2,3) would be selected at the time index 1500, but the nearly-orthogonal condition is lost after few tens of time slots. As far as the time interval between the channel estimation and the UL phase exceeds the time interval in which the nearly-orthogonal condition is approximately valid, the pursued MU diversity gain would be actually lost.

From the considerations outlined above, it is apparent that the mobility is a key factor that negatively impacts on the performance. To partially face it, the solely major countermeasure seems to be the implementation of methodologies to decrease as possible the value of t_d w.r.t. T_{CO} . This can be accomplished by introducing ad-hoc modifications to both the PHY and frame structure. The latter are introduced and described in section 5; for the former, a strategy it is planned to decrease the time needed at the BS to receive the training sequences of all the UEs active in the scenario.

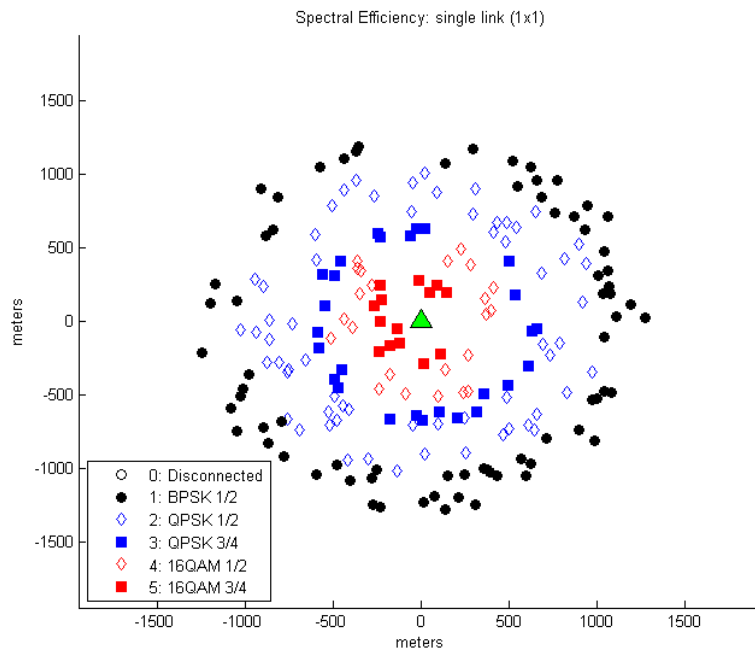


Figure 15: UEs spatial distribution and MCS assigned to be compliant with the single link SNR; pathloss model HATA suburban (BS height 30 m, UE height 2 m)

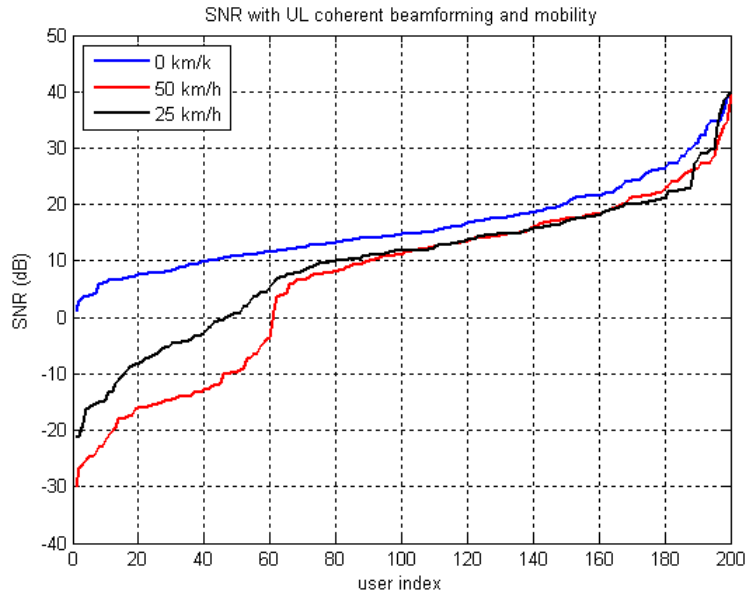


Figure 16: SNR (sorted) with and without mobility: carrier 1 GHz, 200 UEs in the scenario depicted in Figure 15, 30% of UEs with random mobility up to the value indicated in the legend, Hata suburban pathloss model, 1 GHz.

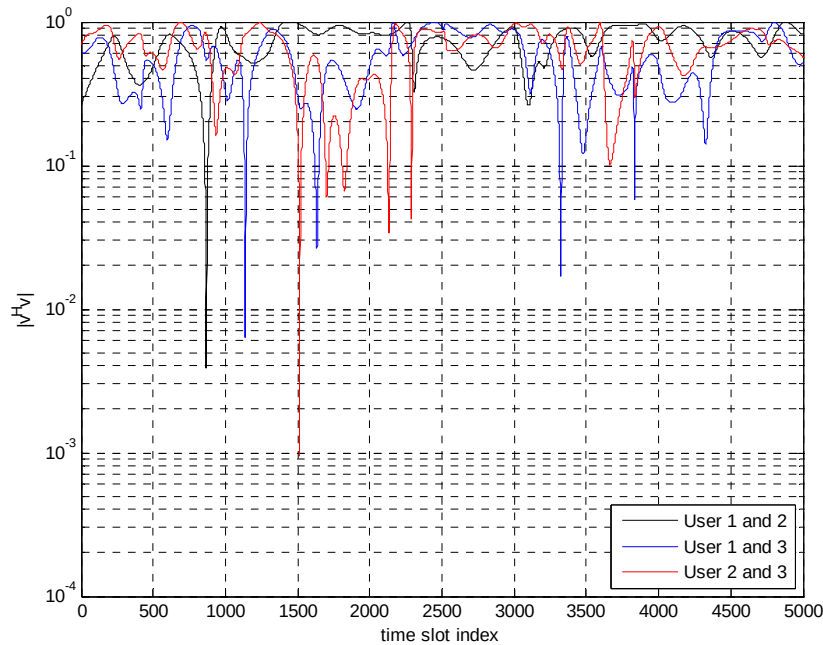


Figure 17: Nearly-orthogonal pairing metric $|\mathbf{v}_j^H \mathbf{v}_k|$ for 3 users and a channel coherence time equal to 200 time slots

Training sequences and channel estimation

For instance, a speed of 25 km/h with a 2 GHz carrier frequency would imply a half-coherence time (in which we can assume that an acceptable amount of coherence still is maintained) of approximately 6 ms, that corresponds to 60 OFDM symbols, $\sim 100 \mu\text{s}$ each (approx. compliant with the WiMAX 802.16e-2005). This suggests that a method capable of accommodating 100 users in 60 OFDM symbols cannot dedicate a full preamble for each user, as the training phase would need 100 OFDM

symbols, well above the coherence time, involving that the next UL phase has no coherence at all with the estimated eigenvectors. A possible strategy to decrease the load associated with the time needed by the training phase could rely on the fact that a realistic multipath channel involves smooth frequency variations of the channel transfer function, so that a sparse set of training subcarriers (e.g. one out of five) followed by interpolation at the receiver side could suffice for each subscriber.

From a physical point of view, this approach drives toward a training scheme in which users (antennas) can be grouped: for instance, five antennas (e.g. belonging to 5 single-antenna UEs) are scheduled to send at the same time their training symbol; each user of the group transmits a preamble made up of unitary power symbols (e.g. taken from a QPSK constellation) on a comb of subcarrier not overlapping with the other UEs of the group; for instance, the subcarrier indices vector $\mathbf{k}_m$ of the m^{th} user can be defined as $\mathbf{k}_m = \text{mod}(\mathbf{n}, 5 + m)$, where $\mathbf{n}$ is the vector listing all the data subcarrier indices. In this way all training subcarriers of the 5-dimensional group of users do not overlap at the BS, although their training preambles do. All the channels are then estimated by repeating the process for all the UEs sequentially.

In this way, each OFDM symbol employed as resource for channel estimation would accommodate 5 subscribers and 20 preambles would suffice to let the BS estimate the (interpolated) channels of all the active users. The feasibility and effectiveness of the adopted approach can be seen from the next Figure 18 and Figure 19. In Figure 18 the transfer function of a single realisation of a Vehicular Type A channel is reported for the 768 data carriers of a 802.16e compliant OFDM symbol with 1024 subcarriers. In the subsequent Figure 19, for an SNR=20 dB, the difference between the true and estimated channel transfer function is reported for the two cases of no interpolation (full set of pilots is transmitted) and for a 1 out of 5 transmitted pilot subcarriers; it can be seen that no significant increase in the estimated error is present by comparing the two cases.

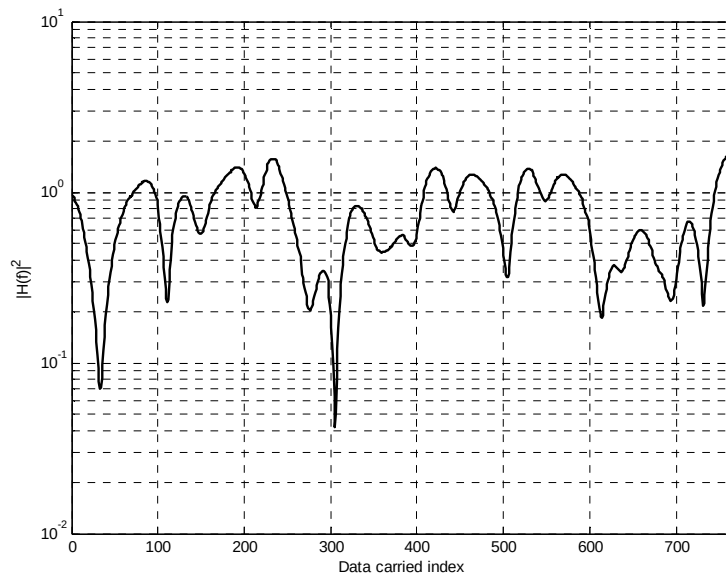


Figure 18: Vehicular type A channel for the 768 data subcarriers of a 802.16e compliant 1024 subcarriers OFDM symbol

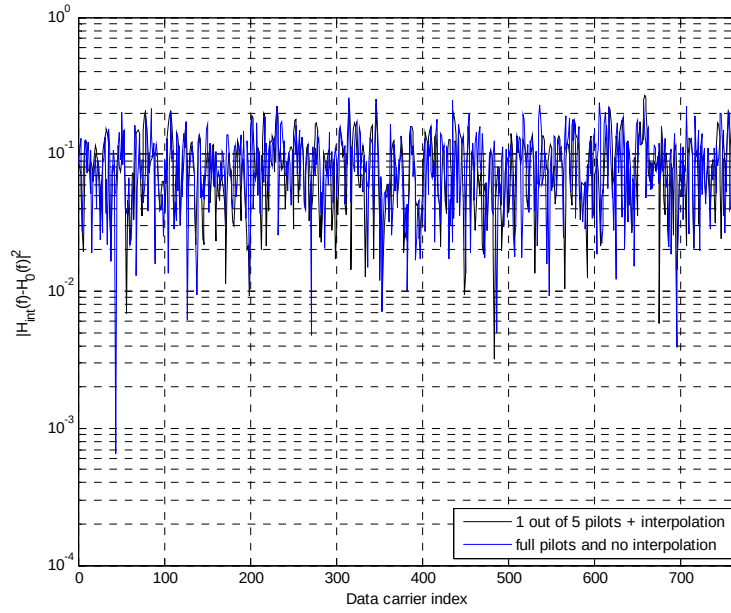


Figure 19: Difference between the true channel transfer function and the one estimated when a) all the subcarriers are transmitted b) one out of five subcarriers are transmitted and the not-transmitted subcarriers are interpolated at the RX side. IEE 902.16e 1024 subcarriers OFDM symbol with SNR=20 dB and channel Vehicular Type A

More specifically, we assume a system with U UEs, each equipped with N_U antennas and a BS equipped with N_B antennas, with $(N_B > N_U)$ and $(N_B < UN_U)$. For sake of simplicity, we drop the notation on the frequency subcarrier, assuming that the following considerations are to be meant “per subcarrier”; this is allowed because an OFDM system is ISI-free and the contributions on different subcarriers are separable. In principle, the system as a whole can be viewed as a MIMO system with UN_U TX antennas and N_B RX antennas. The UN_U TX antennas transmit a set of N_P preambles represented by the (UN_U, N_P) preamble matrix $\mathbf{X}$. The (N_B, N_P) received matrix $\mathbf{Y}$ is thus related with the (N_B, UN_U) MIMO channel as follows:

$$\mathbf{Y} = \mathbf{H}\mathbf{X} + \mathbf{N} \quad (33)$$

where $\mathbf{N}$ is the (N_B, UN_U) received noise matrix. The LS estimation of the global MIMO channel is given by

$$\hat{\mathbf{H}} = \mathbf{Y}\mathbf{X}^P = (\mathbf{H}\mathbf{X} + \mathbf{N})\mathbf{X}^P \quad (34)$$

where $\mathbf{X}^P$ is the More-Penrose left pseudoinverse given by

$$\mathbf{X}^P = (\mathbf{X}^H \mathbf{X})^{-1} \mathbf{X}^H \quad (35)$$

A necessary condition for (34) not to be ill-conditioned (rank deficient) is that $N_P \geq UN_U$. and (34) becomes

$$\hat{\mathbf{H}} = \mathbf{Y}\mathbf{X}^P = \mathbf{H} + \mathbf{N}\mathbf{X}^P \quad (36)$$

For a generic preamble sequence, no parallelism can be exploited, and the computation of $\mathbf{X}^P$ is a batch processing involving $O(N_P^2 UN_U + N_P (UN_U)^2 + \min(N_P, UN_U)^3)$ operations. However, in principle there is no a priori restriction on the design of the training sequences matrix $\mathbf{X}$. And it is straightforward to verify that a proper design of $\mathbf{X}$ can dramatically scale down the complexity and enable the parallelism in computation. We select the number of preambles equal to the overall number of TX antennas (i.e. $N_P = UN_U$); among all the possible square matrices, we can always select training sequences such that are Unitary (i.e. $\mathbf{X}\mathbf{X}^H = \mathbf{X}^H \mathbf{X} = \mathbf{I}$). In this way the computation of the pseudoinverse simply reduces to $\mathbf{X}^P = \mathbf{X}^H$, that needs no computational complexity and the channel matrix estimation reduces to.

$$\hat{\mathbf{H}} = \mathbf{Y}\mathbf{X}^H = \mathbf{H} + \mathbf{N}\mathbf{X}^H \quad (37)$$

The computation of each element of $\hat{\mathbf{H}}$ still requires UN_U complex multiplications and sums but, if we choose, among all the possible unitary matrices $\mathbf{X}=\mathbf{X}^H=\mathbf{I}$, then we have, in a element-wise notation

$$\hat{h}_{j,k}(n;u) = r_{j,k}(n;u) = h_{j,k}(n;u) + n_{j,k}(n;u) \quad (38)$$

Where $r_{j,k}(n;u)$ and $n_{j,k}(n;u)$ are the signal and the noise, respectively, received by the j^{th} antenna of the BS when the k^{th} antenna of the u^{th} user is transmitting its unitary value symbol on the n^{th} subcarrier; $\hat{h}_{j,k}(n;u)$ is the channel estimated on the n^{th} subcarrier for the link between the k^{th} antenna of the u^{th} user and the j^{th} antenna of the BS. The same holds also for other estimation schemes in OFDM such as MMS, that are frequency-wise separable. The proposed training sequence is not Standard compliant, but enables a degree of saving of the computational effort in dense scenarios.. In this way on the receiver side the complexity reduces virtually to no operation but memory storage.

Redesign of the frame format

The next necessary amendment to the MAC management has to be the communication to the users of the eigenvector that each user has to employ in the next UL phase that has to be the eigenvector corresponding to the largest eigenvalue, along with the scheduling of the resources that the BS has computed for the best band assignment and pairing. Indicating with U the number of UEs, the DL (broadcast) communication will require few resources, as the overall amount of bytes to be sent should be limited to $4U$ bytes ($3U$ bytes for the communication of the 2-elements complex eigenvector (1 real entry and 1 complex entry) and U bytes (i.e. 1 byte per user) for the Partial Usage of Sub-Carriers (PUSC) allocation and time slot to be employed for the UL phase). This communication can be accomplished (with some spare space) by three OFDM symbols (e.g. QPSK rate 3/4 and 1024 carriers with 768 data carriers). The remaining 37 OFDM symbols can be thus employed for the UL phase of the users in OFDMA. Assuming a fair sharing of the available resource, all the users will have assigned about 570 data carriers (i.e. $768 \cdot 37 / 50$), shared at the same time by two users at time. In this way, as the channel estimations phase and the UL phase fall within the coherence time and the estimated eigenvectors will be timely employed and the supposed nearly-orthogonal condition is close to be actually fulfilled in the UL phase. The proposed approach strongly leverages on a substantial modification of the MAC management with respect to the current version adopted for the IEEE 802.16e-2005 and foreseen for the IEEE 802.16m. These modifications will be detailed in the relevant Section 5. It is worth remarking also that the above example is tailored on a typical urban speed of 25 km/h, corresponding to a half-coherence time of 6 ms; however just doubling the speed, rising it to 50 km/h would imply a half-coherence time of about 3 ms, indicating that the above timing and re-design of the frame format (with the proportional reductions) given in Section 5 can cope with such a speed, but accommodating about 66 users.

3.1.2.2 Effect of inhomogeneity

The situation outlined from the analysis of the previous section might be even less encouraging, as it does take into account the additional MAC overhead needed for the control plane data in the frame. The inhomogeneous SNR distribution among the users also poses additional constraints/limits to the coherent MU UL MIMO approach. In the subsequent Figure 20, the mean throughput achievable with the MU UL coherent MIMO techniques presented in section 3.1.1 is reported vs. the distance of the users. The underlying hypothesis in the plots is that all the mobile subscribers are at equal distance from the BS (i.e. experience the same path-loss) in order to present curves of the aggregated PHY throughput vs. distance.

Frequency Selective and User Pairing (1x2, SCME UrbMic)

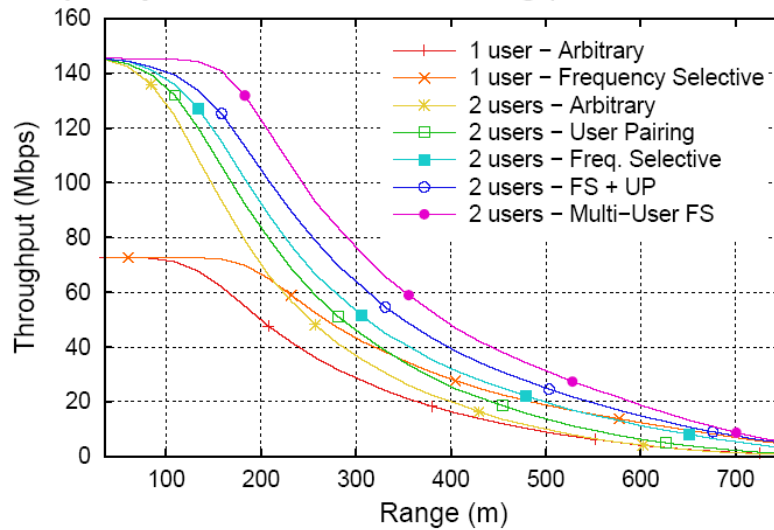


Figure 20: Throughput achievable by the MU MIMO introduced in Section 3.1.1 vs. distance, assuming all the users experiencing the same mean SNR

The results from Figure 20 indicate a significant increase of the achievable throughput, but these are obtained under the assumption that the scheduling operates, for each range, within a pool of the same number of users experiencing the same mean SNR. The situation can be expected to be different when (as in realistic cases) the mobile subscribers have co-existing SNR distributions over orders of magnitude from the centre to the edge of the cell. The following plots might further clarify the situation to be faced. The adopted mean PL model is from (252). In this case, 300 users uniformly distributed in space as depicted in Figure 21 will have an SNR distribution corresponding to the feasible MCS assigned to each user and spectral efficiencies as also reported in the same Figure 21.

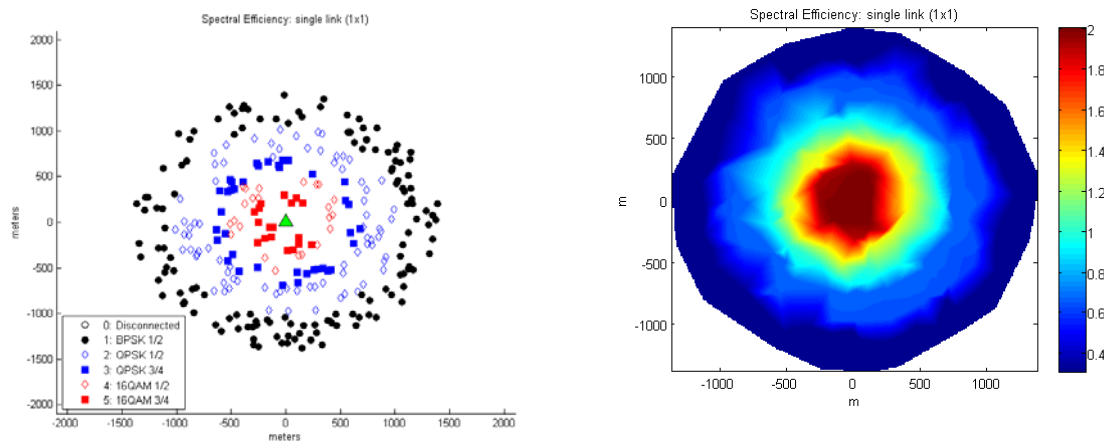


Figure 21: Scenario with 300 UEs; Shadowing Hata suburban, 2 GHz. Leftmost plot: supported MCS for single link (no coherent BF) and (rightmost plot) corresponding SE in b/s/Hz.

As the subscribers are spatially uniform in the cell, their density is constant, but the closer is the distance, the fewer are the users experiencing a given pathloss. This is better depicted in the next Figure 22, where the mean SNR distribution in space and among UEs (excluding further shadowing effects) is represented for the scenario of Figure 21.

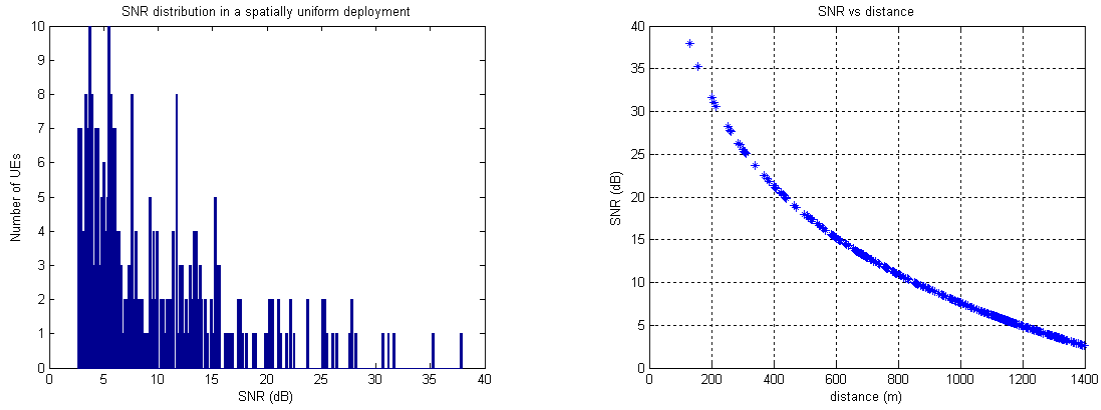


Figure 22: Left plot: number of users experiencing a given SNR, assuming 300 users uniformly distributed in space. Right plot mean SNR vs. the distance of the UEs. Scenario from Figure 21 PL model given by (252).

The effect of the spatially uniform deployment of the UEs (i.e. inhomogeneous SNR distribution) has an impact on the performance of the coherent BF and coupling strategy. This can be better seen when the (32) (that disregards the effect of mobility) is analysed. As far as the PL is similar for all the UEs, the interference is basically driven by the solely cross-user orthogonal factor $\mathbf{v}_k^H \mathbf{v}_j$; in this case the scheduling and pairing strategy outlined in section 3.1.1.2.2 makes perfect sense. This metrics however seems not suitable in (realistic) scenarios in which the SNR varies among the UEs over orders of magnitude and where very few users have SNR significantly higher than the others (see Figure 22). In such a case, the eigenvalues of the channel matrix λ_k (that incorporate the PL effect) can overwhelm the $\mathbf{v}_k^H \mathbf{v}_j$ factor measuring how orthogonal the couples are. Indeed, for a given value of $\mathbf{v}_k^H \mathbf{v}_j$, from (32) the resulting SNR can be higher for a less orthogonal couple, but with a better λ_k/λ_j ratio. Another consideration arises from the consideration that now the statistical symmetry between couple (k, j) w.r.t. the couple (j, k) is broken. Indeed, as far as the solely factor driving the pair selection is how orthogonal they are, the symmetry is maintained, as $\mathbf{v}_k^H \mathbf{v}_j$ has the same value exchanging k with j . From the achievable SE point of view this doesn't hold true any longer in the presence of inhomogeneous SNR distributions, as (always from (32)), it is likely that, once fixed the user k , it is found that the coupling with user, say j , gives the best $\text{SNR}_{k,j}$. This would suggest first selecting the couple (k, j) and then going ahead with the scheduling procedure, as it is presented in section 3.1.1.2.2. However it has to be considered that the overall goal is NOT the selection of the best $\text{SNR}_{k,j}$ – *per se* – but the achievement of the highest SE given by the selected couple. To face this task we have to consider that, in a real system as well as from an information theoretic point of view, the SE of a couple depends on the MCS assigned to each user of the couple and the Protocol Data Unit-Error Rate (PDU-ER) of each user. The aggregated SE $S_{k,j}$ of the coupling (k, j) can be written as

$$S_{k,j} = S_k(\gamma_{k,j})(1 - P(\gamma_{k,j})) + S_j(\gamma_{j,k})(1 - P(\gamma_{j,k})) \quad (39)$$

where $\gamma_{k,j}$ is the reception SNR of the user k when couples with user j . $S_k(\gamma)$ is the maximum SE for the user k with the MCS assigned as a function of the SNR value γ . $P_k(\gamma)$ is the PDU error rate achieved as a consequence of the SNR value γ . From (39) it is apparent that, once fixed user k and selecting the user j as the one insuring the best $\text{SNR}_{k,j}$ doesn't automatically insures the maximal joint SE, as there could be another user, say n , for which $\text{SNR}_{k,n} < \text{SNR}_{k,j}$, but $S_{k,n} > S_{k,j}$.

Beside these considerations, the strong SNR inhomogeneity in the scenario poses another problem about the fairness of the system. Indeed, from a statistical point of view, the few closest UEs with very high SNR would tend to be frequently selected, as they would insure an additional benefit due to the higher SE associated with them, no matter if the orthogonal term (e.g. coming from farthestmost users) is poor, as the injected interference would be still low enough to insure a dominating SNR.

To face the above issue posed by the inhomogeneous distribution of the SNR in the scenario, two additional scheduling procedures are introduced. Both disregard the selection of the most orthogonal users, but focus on either the maximisation of the resulting SNR from the coupling (suboptimal) or on the maximisation of the joint SE:

- **Scheduling procedure based on best SNR selection**

Scheduling is done to authorise two users that guarantee the best SNR value to send data at the same time. The procedure can be summarised as follows:

1. Over all active users, compute eigenvectors associated to maximum eigenvalue of MU channels at the BS.
2. Select a generic user k and compute the expected $\text{SNR}_{k,j}$ for all the couples (k, j) by (32).
3. Choose the couple (k, j) as the one insuring the best $\text{SNR}_{k,j}$.
4. Eliminate k from the pool and go to step 2. Repeat till the pool is empty.
5. Authorize the computed couples to send at the same time consecutively.

- **Scheduling procedure based on best SE selection**

Scheduling is done to authorise two users that guarantee the best joint SE value to send data at the same time. The procedure can be summarised as follows:

1. Over all active users, compute eigenvectors associated to maximum eigenvalue of MU channels at the BS.
2. Select a generic user k and compute the expected $S_{k,j}$ for all the couples (k, j) by (39).
3. Choose the couple (k, j) as the one insuring the best $S_{k,j}$.
4. Eliminate k from the pool and go to step 2. and repeat till the pool is empty.
5. Authorize each pre-computed couple to send at the same time consecutively.

Both procedures are sub-optimal, and involve a slightly increase of complexity w.r.t. the scheduling based on the solely “most orthogonal” condition, but have the attractive advantage of insuring either a partial optimality on SNR (with lower complexity) or an optimality on the joint SE (with a slightly higher complexity). Another point to be highlighted is about the fairness: both scheduling procedures eliminate from the pool one UE at time, not the couple. If we indicate with N the number of UEs in the scenario, thus insures that each UE will be selected for the UL at least with a selection frequency equal to $1/N$ (the best fairness with a coupling scheme is $2/N$) although specific UEs can be selected more frequently.

Case 1: homogeneous SNR (2x1)

To have a representation of the results achievable by the proposed methodologies, we can start considering a simplified scenario with 300 UEs and no mobility, uniformly distributed with a mean SNR value of 20 dB, BS equipped with 2 antennas and the UEs with a single antenna. The resulting (sorted) values of the $\text{SNR}_{k,j}$ matrix, computed by (32), is represented in Figure 23 and the joint SE matrix computed by (39) is represented in Figure 24.

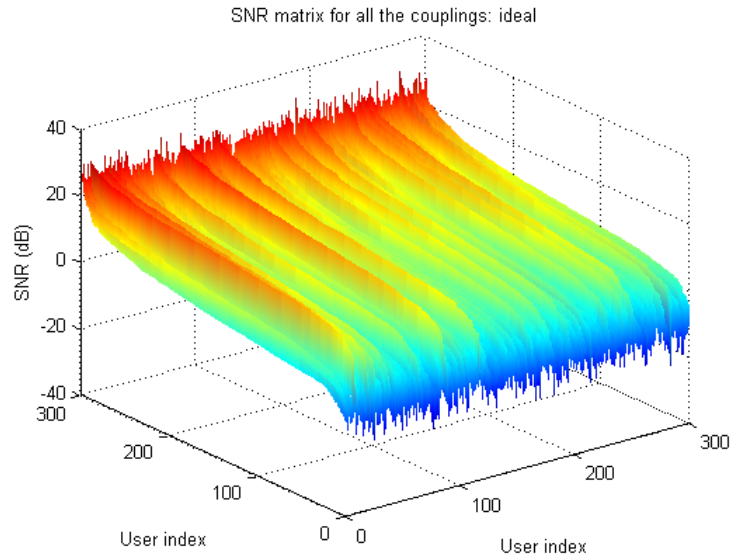


Figure 23: SNR matrix from (32) for a scenario with uniform SNR distribution, 300 single antenna UEs and 2 antennas BS.

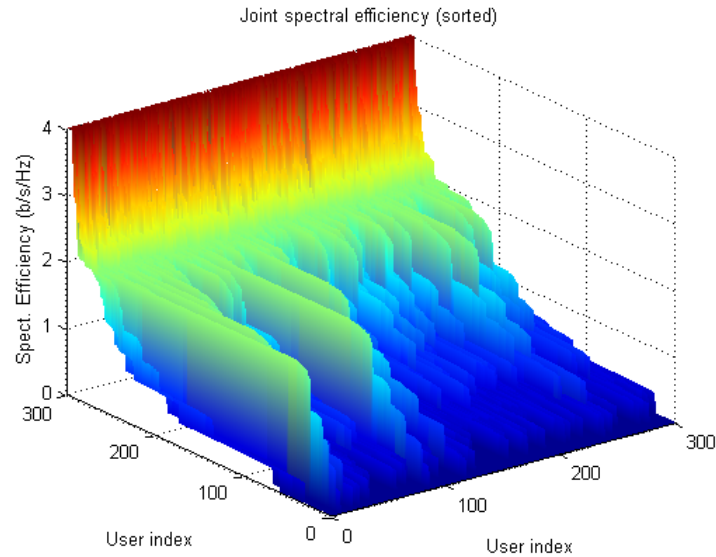


Figure 24: Joint SE matrix from (39) for a scenario with uniform SNR distribution, 300 single antenna UEs and 2 antennas BS.

It can be noticed that in this (simplified) scenario, nearly all the UEs can be assigned the best MCS employed in the simulations (i.e. 16QAM rate 3/4) with a SE equal to 2 b/s/Hz. The joint SE after coupling is 4 b/s/Hz for almost all the selected couples (Figure 24). The analysis of both Figure 23 and Figure 24 indicate that, for each user, there is at least one coupling insuring an acceptable high level of $SNR_{k,j}$ and a maximal value of the joint SE. On the other hand, also the orthogonal condition is acceptably met (Figure 25:), with a mean value of the minimum of -25 dB, that is approximately compliant with the expected mean value, equal to $1/300$ (number of UEs), that is 24.7 dB.

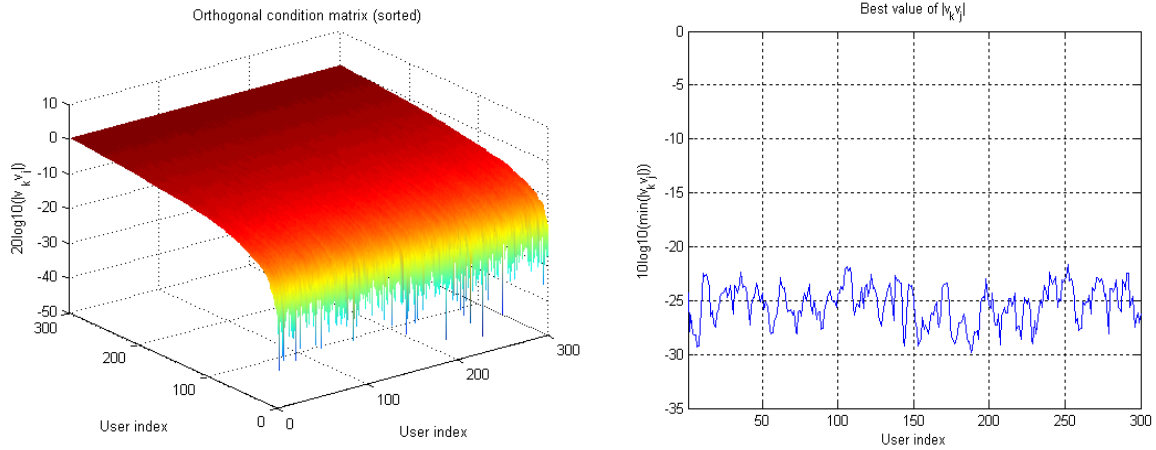


Figure 25: leftmost plot: orthogonal condition $v_k^H v_j$ matrix and (rightmost) best values achieved for each UE. Scenario with uniform SNR distribution, 300 single antenna UEs and 2 antennas BS.

What matters most is now to verify the overall throughput or the overall SE achieved as a consequence of the adopted scheduling strategy. In this specific case, where no inhomogeneity is introduced (apart the randomness of the channel) we do not expect any significant difference among the scheduling based on the most orthogonal condition, the best joint SE, the best single coupled SNR or the best single coupled SE. Indeed this is what can be observed in Figure 26 (both left and right plots, single channel and mean value over 200 realisations, respectively), where all the coupling strategies exhibit a similar behaviour and there is no methodology with a superior performance against the other; the differences are basically due to the actual realization of the random channels.

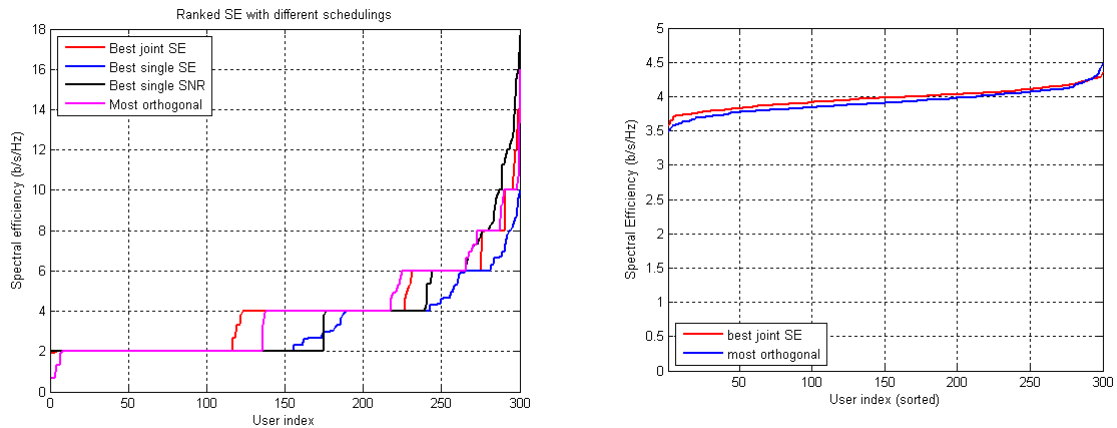


Figure 26: Left plot: single channel realisation, SE (ranked) achievable by each user for coupling procedures based on the most orthogonal, the best joint SE, the best single coupled SNR and the best single coupled SE. Right plot: mean value of the SE over 200 channel realisations for most orthogonal and best joint SE coupling.

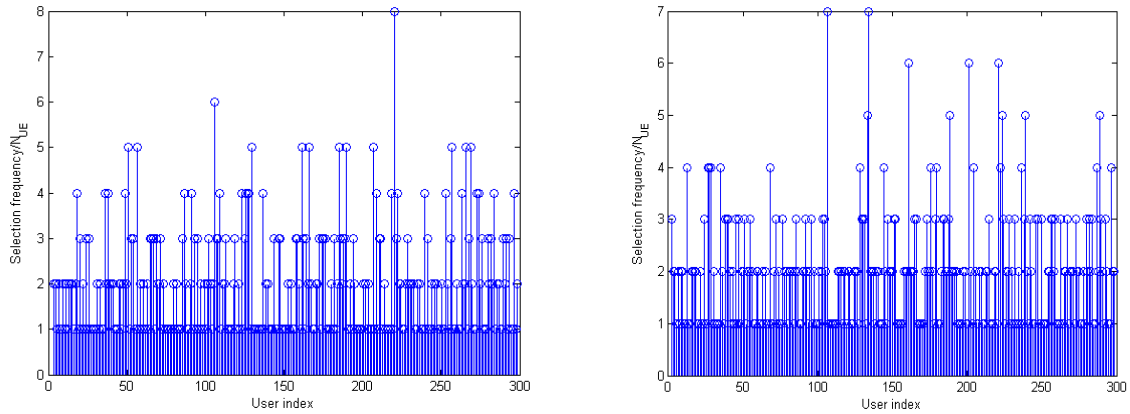


Figure 27: selection frequency of each UE over a period equal to the number of UEs: each UE is selected at least 1 time and the maximal fairness is 2 times. Left plot is for most orthogonal coupling, right plot is for best joint SE coupling.

The last point to analyse is the fairness of the scheduling and coupling schemes; the maximal fairness corresponds to selecting each UE 2 times over a period transmission slots equal to the number of UEs. In this simplified case, we expect no specific difference in fairness among the tested coupling schemes and this is testified by the plots in Figure 27: (relevant to a single channel realisation), where the most orthogonal and the best joint SE exhibit the same behaviour and both are fair in the sense that no UE is different from any other UE and the system is fair in mean, over an ensemble of different channel realisation. This is shown in Figure 28, where it can be noticed that both coupling methods exhibit the same convergence toward the best fairness (2 selections over 300 UL phases), with the same variance of the scheduling frequency among the UEs.

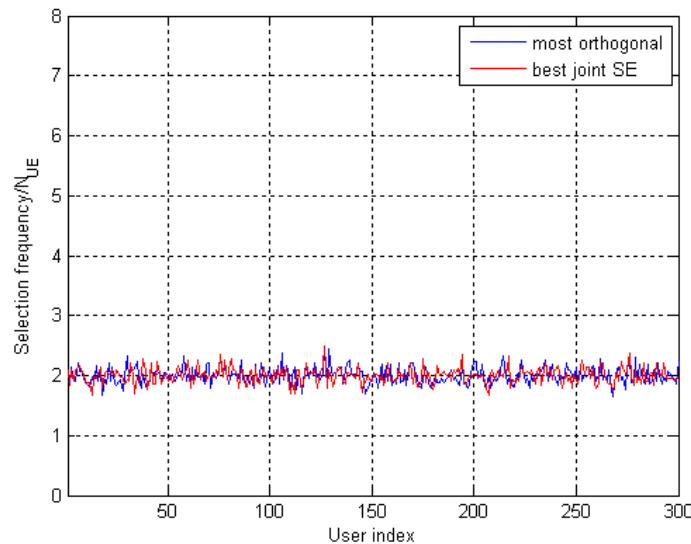


Figure 28: Mean value of the selection frequency over 60 random channel realisations: both coupling methods converge toward the best fairness value (2 selections every 300 UL phases).

Case 2: inhomogeneous SNR, (2x1)

This situation is expected to be substantially different when a realistic inhomogeneous SNR distribution (coming from uniform UEs density) is adopted. In this case we can refer to the scenario depicted in Figure 21, with the SNR distribution given in Figure 22. As in the previous case, we have 300 UEs and no mobility, BS equipped with 2 antennas and the UEs with a single antenna. The

resulting (sorted) values of the $\text{SNR}_{k,j}$ matrix, computed by (32), is represented in Figure 29 and the joint SE matrix computed by (39) is represented in Figure 30.

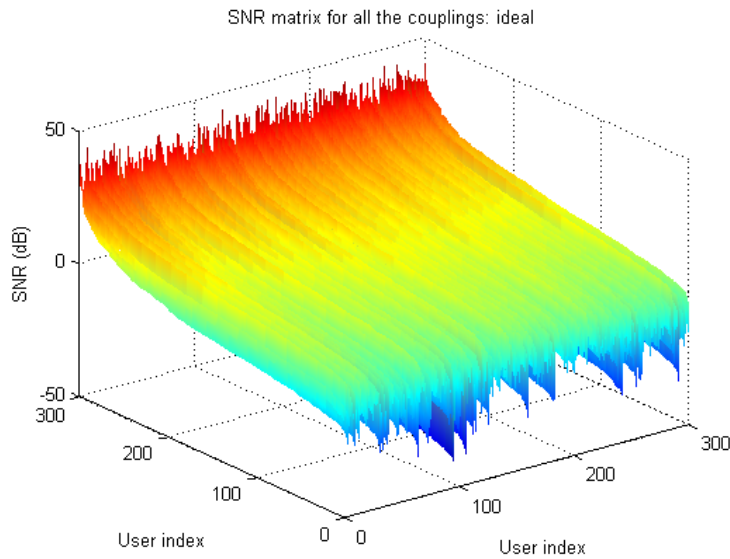


Figure 29: SNR matrix from (32) for the scenario with not-uniform SNR distribution of Figure 21; single antenna UEs and 2 antennas BS.

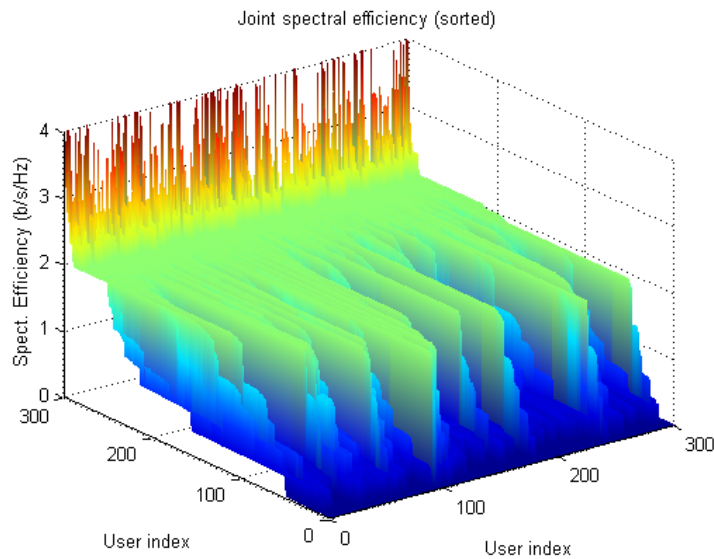


Figure 30: Joint SE matrix from (39) for the scenario with not-uniform SNR distribution of Figure 21; single antenna UEs and 2 antennas BS.

Comparing Figure 24 with Figure 30, it can be noticed that, differently from the homogeneous case, the best score of the joint SE matrix is significantly lower, but this happens not because of a lower probability of orthogonality among UEs (solely driven by the number of UEs, for a given MU channel), but because of the significant rise in the interference term.

In such a scenario, it is thus expected a significant difference of the overall throughput or the overall SE achieved with respect to the Case 1 (homogeneous) as a consequence of the adopted scheduling strategy. In this specific case, where significant inhomogeneity is introduced because of the shadowing model we expect significant differences among the scheduling based on the most orthogonal

condition, the best joint SE, the best single coupled SNR or the best single coupled SE. Indeed this is what can be observed in Figure 31, where all the coupling strategies exhibit different behaviours; in particular it can be noticed that the coupling based on the best joint SE performs better than all the other techniques. This is not surprising, as this coupling strategy is designed to insure the best SE achievable by over all the possible couples, but constrained to schedule at least one time each user over a number of UL phases equal to the number of active UEs. This constraint makes the method sub-optimal from the sum-rate point of view, but let the system gain fairness.

It is now interesting to compare the different behaviours of the fairness of the scheduling and coupling schemes between the homogeneous and inhomogeneous case. We recall (from Figure 28) that in the homogeneous case the frequency of the selection of each UE approached in mean (computed over different channel realisation) to the optimal fairness value (2 selections over 300 UL phases), for both the most orthogonal and the best joint SE (and the other methods, although not reported). In this case, Figure 32 reports the selection frequency of the UEs (sorted and unsorted) and it can be seen that, differently from the most orthogonal coupling, the best joint SE exhibit a lower limit in the variance of the selection frequency (the variance does not vanishes by increasing the dimension of the statistical ensemble). As the solely difference between Figure 28 and Figure 32 is the inhomogeneity of the scenario, it is easy to infer that this effect is introduced by the SNR distribution. This is confirmed in Figure 33, where the scheduling frequency is reported against the SNR experienced by each UE and can be seen that the selection frequency of the most orthogonal coupling is independent of the SNR distribution (as expected), whereas in the case of the best joint SE coupling, the selection frequency depends on the SNR distribution. Qualitatively, this can be explained by observing that UEs with too low SNR will be seldom selected, as the associated gain in SE is expected to be smaller than the one guaranteed by UEs with higher SNR; on the opposite, the (few) UEs with the highest SNR are not so frequently selected, because the interference injected by them in the coupling process is higher than the one associated with UEs with lower SNR. This qualitatively leads to an expected frequency selection distribution with lower values at both lower and higher SNR and a grater selection frequency for intermediate values of the SNR; indeed this is what is observed in Figure 33.

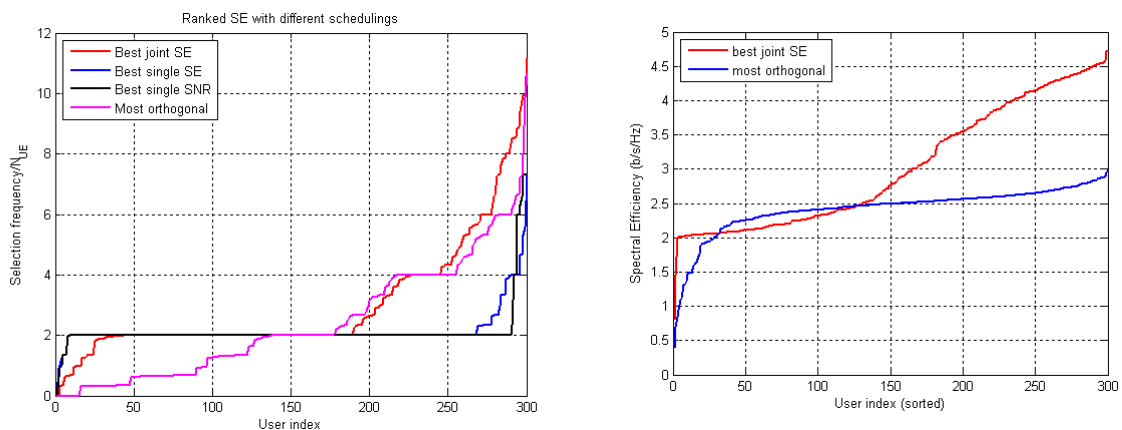


Figure 31: Left plot: single channel realisation SE (ranked) achievable by each user for coupling procedures based on the most orthogonal, the best joint SE, the best single coupled SNR and the best single coupled SE. Right plot: the mean value over 200 channel realisations for the best joint SE and the most orthogonal coupling. Scenario from Figure 21; single antenna UEs and 2 antennas BS.

The scheduling frequency has a direct counterpart in the achievable SE by each UE. As told, Figure 31 reports the SE sorted by increasing values, but a better insight can be given in Figure 34, where the SE (mean over 200 channel realisations) is plotted against the mean SNR experienced by the corresponding UE without the pairing: this parameter has been considered as more meaningful, because it doesn't mix contributions with different relative weights (i.e. joint orthogonality and power)

and can be also more directly interpreted as a function of the distance of the UE from the BS. Also in this case the behaviour is qualitatively explained by considering that the most orthogonal coupling insures a nearly optimal fairness, but the higher is the SNR of the UEs and the higher is the interference term injected in coupling, resulting in a decrease of the assigned MCS and increase of the PDU error rate. On the opposite, the best joint SE, although suffering of the same impairment, schedules less frequently both the low and the high SNR UEs, thus decreasing the loss in SE associated to them and increasing the SE associated with the intermediate values of SNR. This is paid by the best joint SE scheduler by a departure from the optimal fairness, but keeping it within a restricted range around the optimal value (see Figure 32).

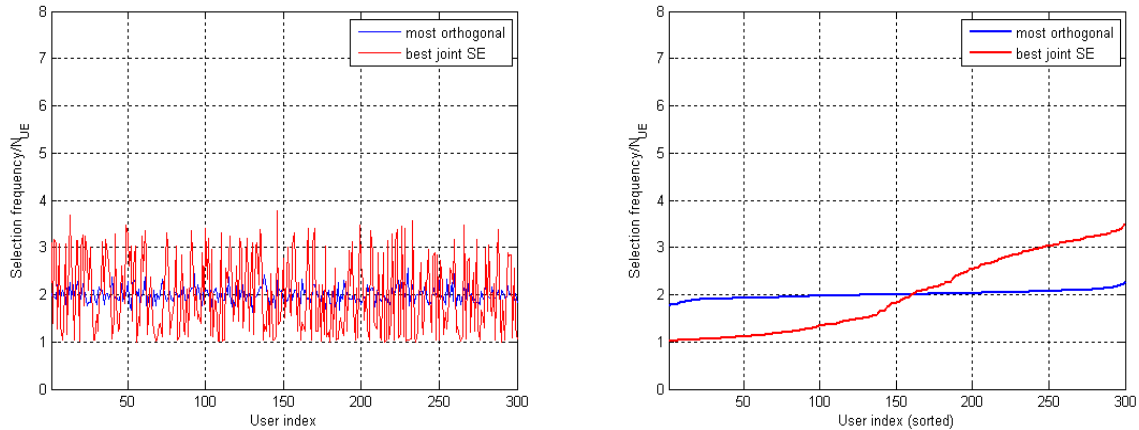


Figure 32: Mean value of the selection frequency over 200 random channel realisations (left plot: unsorted user index, right plot: sorted by increasing selection frequencies): both coupling methods have a mean value close to the best fairness value (2 selections every 300 UL phases), but the inhomogeneity introduces a lower bound to the variance of the scheduling frequency for the best joint SE coupling method.

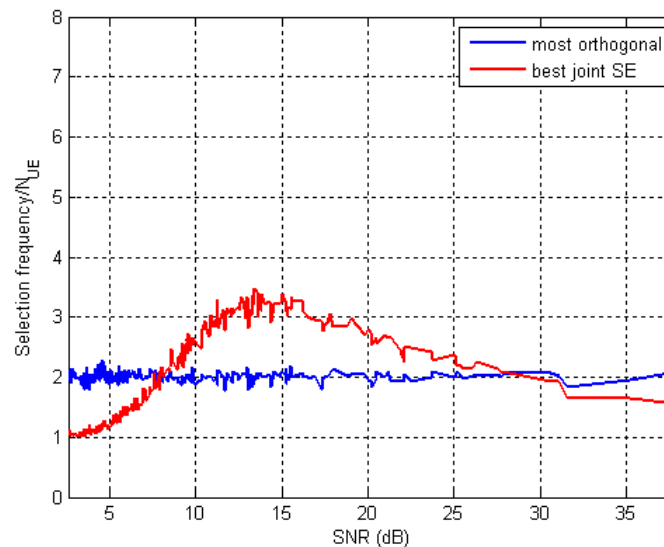


Figure 33: Mean value of the selection frequency over 200 random channel realisations plotted against the SNR experienced of the corresponding UE. For the best joint SE coupling, the lower limit of the scheduling frequency variance is related with the SNR distribution, whereas the most orthogonal coupling is independent of it.

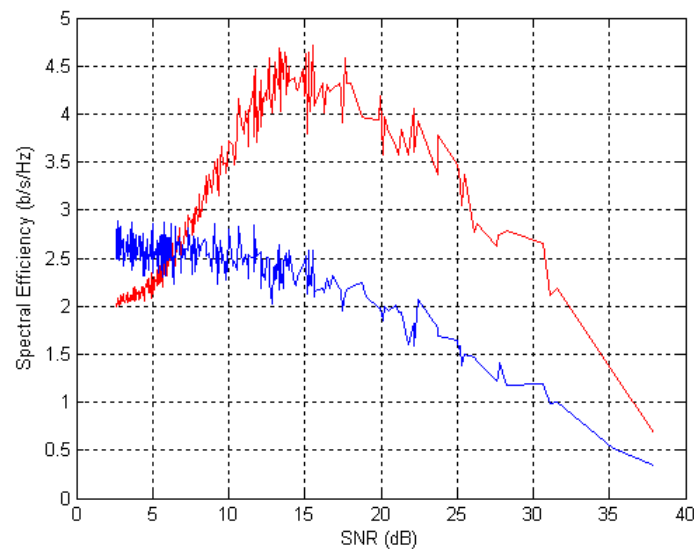


Figure 34: Mean SE over 200 channel realisations plotted against the SNR that is experienced by the corresponding UE. Red plot: best joint SE; blue plot: most orthogonal. Scenario from Figure 21; single antenna UEs and 2 antennas BS.

Case 3: inhomogeneous SNR (3x2)

The comparison between Case1 and Case 2 spotted the main differences arising between homogeneous and inhomogeneous scenarios for the same antenna configuration at the BS and MSs (2x1). The major remarks indicate that in realistic scenarios the coupling based on the most orthogonal coupling is not to be preferred, as it still insures nearly optimal fairness, but at the cost of a significant decrease of the overall sum-rate of the system. On the opposite, the best joint SE coupling scheme (although suboptimal from both the aggregated sum-rate and the fairness point of view) has shown to be able to insure a significant increase in the overall SE, meanwhile maintaining the fairness level around the optimal value.

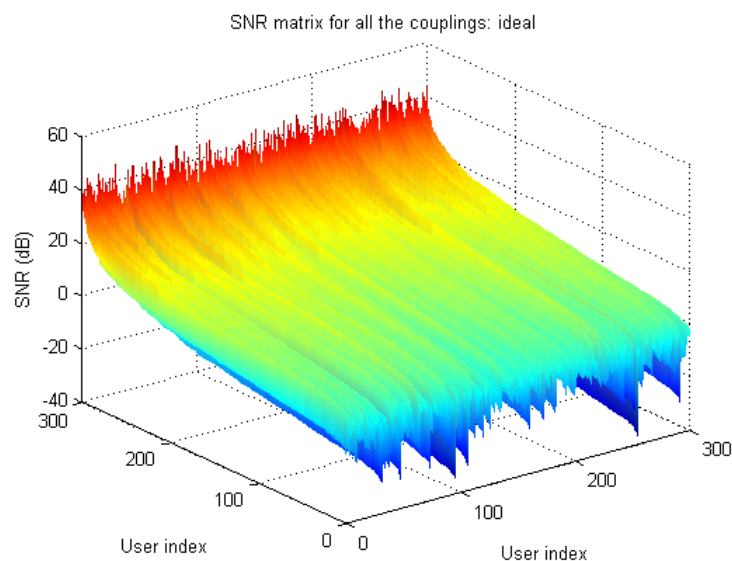


Figure 35: SNR matrix from (32) for the scenario with not-uniform SNR distribution of Figure 21. BS and UEs antennas (3, 2).

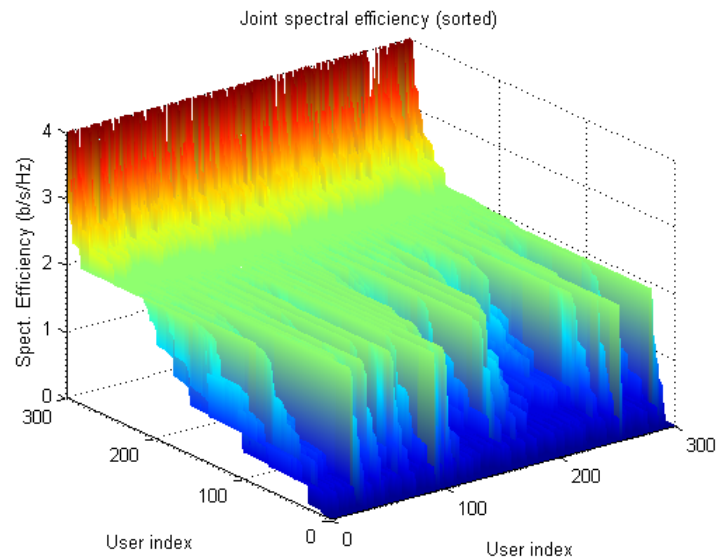


Figure 36: Joint SE matrix from (39) for the scenario with not-uniform SNR distribution of Figure 21; BS and UEs antennas (3, 2).

It is worth considering now a comparison of the case 2 with the same scenario, but with 2 antennas at the UEs and 3 antennas at the BS. From the comparison between Figure 35 and Figure 29 it can be observed an overall increase of the SNR matrix, compliant with the higher dominant eigenvalues associated with the (3,2) antenna configuration against the (2,1) of Figure 29. However the major difference is reflected in the joint SE matrix computed by (39) and here reported in Figure 36, to be compared with Figure 30. It can be observed that, as expected, the maximum achievable SE value is higher and more stable in the (3,2) antenna configuration than in the (2,1) and in this case most of the couplings will insure the maximum achievable SE (i.e. corresponding to the adoption of the 16QAM rate 3/4) for each element of the couple, leading to a mean SE close to 4 b/s/Hz, whereas in the (2,1) configuration is around 3 b/s/Hz (see Figure 31).

The change in the antenna configuration also impacts on the fairness of the system: in Figure 37 the mean value of the UEs selection frequency is reported against the SNR experienced by each UE. A comparison between this figure and the Figure 33 (relevant to the (2,1) configuration) indicates that the variance around the optimal value of the scheduling frequency of the best joint SE coupler is reduced in the (3,2) antenna configuration and the system has a better fairness.

Similar considerations can be done by comparing the mean value of the achieved spectral by each UE in Figure 38 plotted against the SNR that is experienced by the UEs (left plot, to be compared with Figure 34) and sorted by ascending SE (right plot, to be compared with Figure 31): the increase in the number of antennas leads the best joint SE coupler to attain a mean value close to the maximum achievable by the system (i.e. 4 b/s/Hz, corresponding to 2 users coupled with 16QAM rate 3/4 allowed to transmit at the same time).

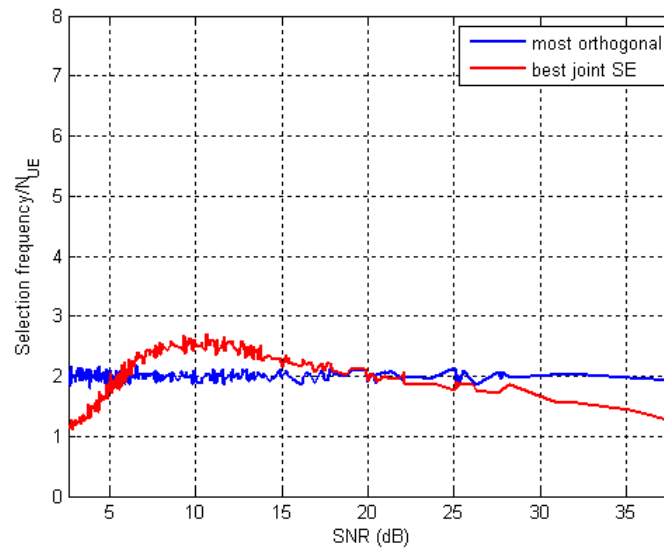


Figure 37: Mean value of the selection frequency over 200 random channel realisations plotted against the SNR experienced of the corresponding UE. For the best joint SE coupling, the lower limit of the scheduling frequency variance is related with the SNR distribution, whereas the most orthogonal coupling is independent of it

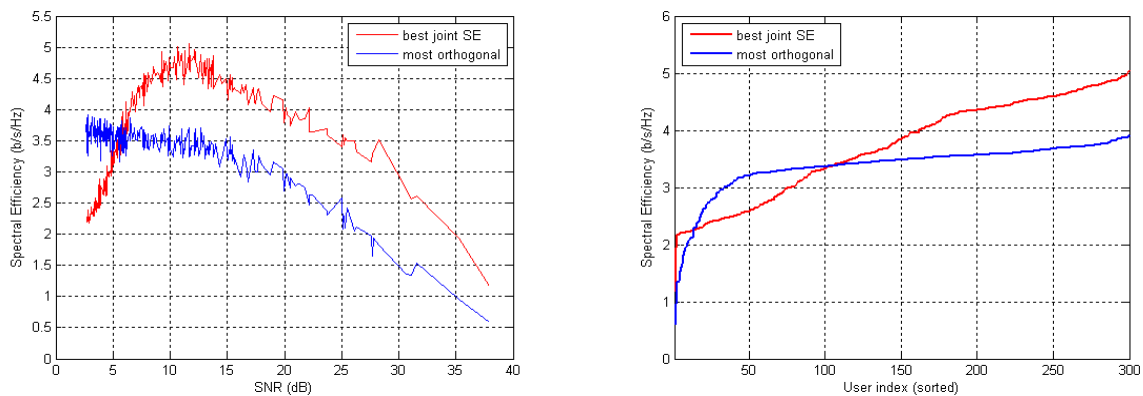


Figure 38: Mean SE over 200 channel realisations. Left plot: against the SNR that is experienced by the corresponding UE; right plot: sorted by ascending SE. Scenario from Figure 21. BS and UEs antennas (3, 2).

3.1.2.3 Joint effect of mobility and inhomogeneity

In the previous sections the effect of mobility and inhomogeneity have been faced as separate, not-overlapping problems, giving guidelines to minimise the effect of the former and designing a different pairing paradigm for the latter. In this subsection some results of the effect of the joint presence of mobility and inhomogeneity will be presented.

In the simulations a given percentage of UEs are not affected by mobility, and the others have mobility values with uniform random distribution up to a specified value of speed. The caption “80%@25 km/h” means that 80% of the UEs have speeds uniformly distributed up to 25 km/h (0 km/h excluded), whereas 20% of the UEs are not affected by mobility.

The first remarkable effect (anticipated by the formulation of the equations (31) and (32)) is the general loss of the SNR matrix due to the presence of mobility (80% of the users with uniformly

distributed speeds up to 25 km/h for 1 GHz carrier), with antenna configuration (3,2) and the same scenario as in Figure 21. In these simulations it has been assumed that the time lag between the channel estimation and the UL phases is 2 frames, that is 10 ms.

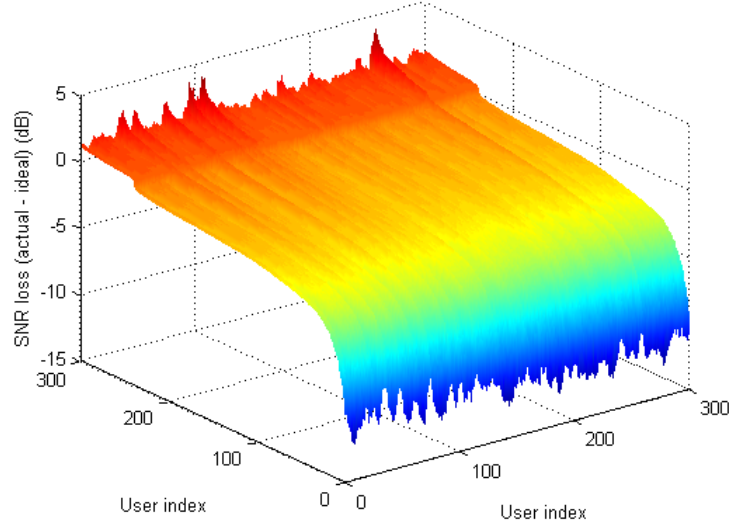


Figure 39: Difference between the ideal SNR matrix (i.e. no mobility) given by (32) and the actual matrix SNR (80%@25 km/h max) given by (31).

Although not straightforward, the decrease effect on the SNR matrix impacts on the performance achievable by the pairing and scheduling algorithms; in the next Figure 40 the mean SE is plotted for both the ideal case (no mobility, solid line) and the case with the aforesaid mobility (dotted line): the right plot reports the sorted values of the mean SE, the right plot as a function of the mean SNR experienced by each UE. It is apparent that the mobility deeply impacts on the performance of both pairing methodologies; in particular, the best joint SE still performs better than the most orthogonal scheduler, but mobility reduces the difference of the performance between the two methodologies.

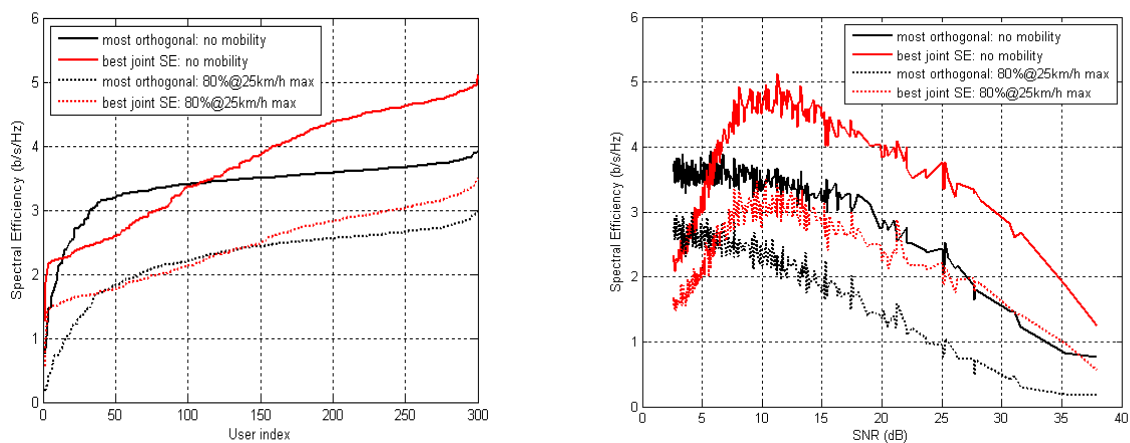


Figure 40: Mean SE over 200 channel realisations. Right plot: against the SNR experienced by the corresponding UE; left plot: sorted by ascending SE. Scenario from Figure 21. BS and UEs antennas (3, 2).

The criticality of the joint effect of the mobility and the inhomogeneity is basically driven by the mobility: indeed the inhomogeneity of the SNR distribution has been shown to be managed by the

introduction of the best joint SE scheduler for pairing. To have a further proof of the critical dependence of the performance on mobility, the next Figure 41 reports the same performance of Figure 40, but increasing the maximal value of the speed to 50 km/h (i.e. a typical urban vehicular speed distribution). It can be noticed that the overall SE dramatically drops to about 1 b/s/Hz (as aggregated mean value), that is below the value that would have been attained by scheduling a one-by-one UE in UL with no pairing.

In conclusion, it has been shown that a real enhancement in the aggregate SE can be actually achieved with an acceptable fairness of the system by adopting the proposed best joint SE pairing algorithm, but only for scenarios in which the speeds distribution slightly above the pedestrian speeds and with at least 100 users in the scheduling pool. This can be translated into the fact that, also in scenarios with higher speeds, the method can be effectively applied by selecting for the pairing pool only the UEs with speeds below 10 km/h (for a 2 GHz carrier).

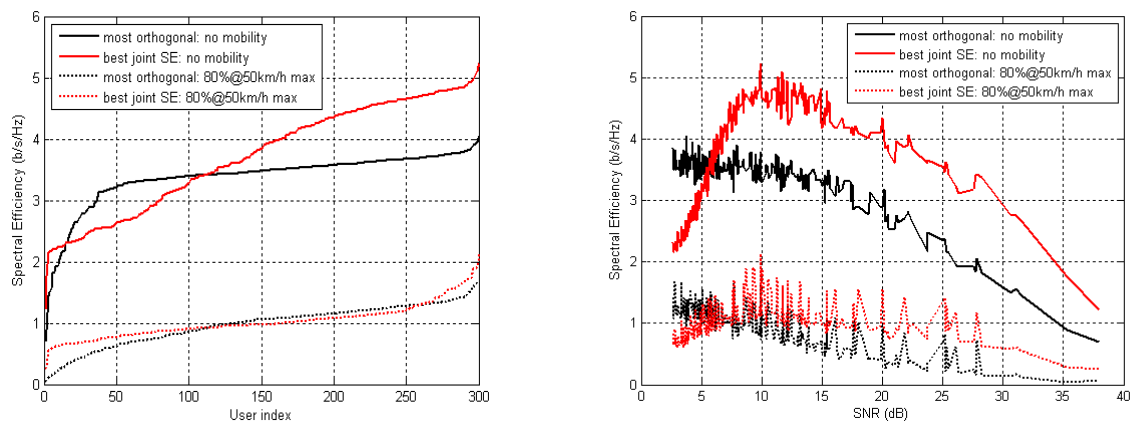


Figure 41: Mean SE over 200 channel realisations. Right plot: against the SNR experienced by the corresponding UE; left plot: sorted by ascending SE. Scenario from Figure 21. BS and UEs antennas (3, 2).

3.1.2.4 Conclusions

It has been showed that the presence of mobility poses severe limitations to the effectiveness of the user pairing in UL based on coherent joint BF at the UEs and at the BS. In addition, also the inhomogeneous distribution of the SNR in a realistic scenario poses additional challenges to the proper working of the scheduling/pairing protocol outlined in section 3.1.1.2.2.

The presence of mobility steered toward a re-design of the frame format (specified in section 5) and a sub-optimal aggregate methodology for the MU channel estimation at the BS.

The strong inhomogeneity in realistic scenarios has been faced by introducing a different pairing and scheduling algorithm with respect to the one proposed in 3.1.1.2.2.

It has been demonstrated that, in the absence of mobility, the proposed scheduling/pairing protocol is capable of (nearly) insuring the maximal SE offered by the system, meanwhile maintaining an acceptable level of fairness among the active users. However the presence of mobility indicates that the achievable aggregate SE gains are jeopardised for speeds distribution above 10 km/h (for 2 GHz carrier), thus limiting the effectiveness of the technique to a pool of UEs (selected among all the active UEs) fulfilling this requirement.

3.1.3 Complexity

The ensemble of methodologies for the UL pairing strategy presented in sections 3.1.1 and 3.1.2, along with the achieved results indicate that the best suited and performing pairing and scheduling approach is the best joint SE selection in conjunction with a re-design of the frame structure: the coupling metric is given by (39) and the proposed comb for UL and DL phases is presented in section 5. As the mere difference in the frame structure does not involve a specific difference in system complexity, the complexity evaluation will focus on the structure, nature of the computations and requirements of the proposed UL pairing protocol. The scheduling procedure is schematically represented in Figure 42, along with the indication of the parallelisation order (Pord) of each building block, where for Pord represents the number of elementary processing units that can be run in parallel, where M_U and N_P are the number of UEs and the number of subcarriers, respectively.

As it can be noticed from the Figure 42, the processing scheme is inherently not-iterative, as the closed-loop part of the scheme is a sequential scheme processing unit with a fixed and known number of operations and it represents a computational tradeoff, adopted to avoid the direct implementation of an exhaustive global-optimisation process (NP-hard problem). In the following, the complexity associated with each processing building block will be discussed.

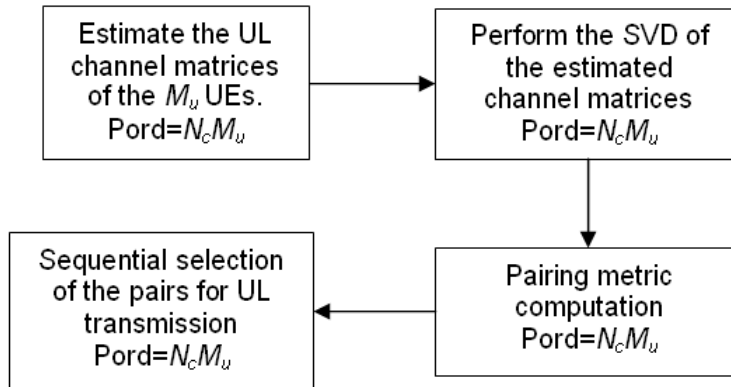


Figure 42: Representation of the processing building blocks for the best joint SE protocol for the UL pairing with the indication of the parallelisation order (Pord) of the processing block, where M_U and N_P are the number of UEs and the number of subcarriers, respectively.

3.1.3.1 Estimation of the UL channel matrices

In principle, this processing unit should not be included in evaluation of the complexity of the scheduling protocol, as it does not involve additional complexity or impose additional requirements to those that the BS already has from the normal receiving procedures, where each UE sends its preamble(s), needed for the equalisation and decoding of the PDUs. However, to better cope with mobility, a different scheme for the “grouped” transmission of the training symbols is proposed in section 3.1.2.1, also impacting on the new frame structure proposed in 5.2.1. This addresses to a re-examination of the complexity associated with the MU channel estimation, as it depends not only on the specific estimation algorithm employed at the BS, but also on the adopted training sequence strategy. The proposed training strategy, as specified in section 3.1.2.1, involves a complexity $O(N_P^2 UN_U + N_P (UN_U)^2 + \min(N_P, UN_U)^3)$, for a system with N_F subcarriers, U UEs, each equipped with N_U antennas and a BS equipped with N_B antennas, with $(N_B > N_U)$ and $(N_B < UN_U)$.

The proposed training sequence is not Standard compliant, but enables a degree of saving of the computational effort in dense scenarios. In this way on the receiver side the complexity reduces virtually to no operation but memory storage. Assuming that each channel value is represented (because the subsequent step is the SVD, that performs better with a 32 bit representation of the complex numbers) by 4 bytes (2 for the real and 2 for the imaginary component), the overall storage

requirement is approximately $4 N_B U N_U$ bytes; for instance, a 300 UEs equipped with two antennas and one BS equipped with 4 antennas lead to a modest storage requirement of about 9.3 kbyte. It has to be remarked that the focus on the simplification, minimisation of training length and parallelisation comes at the cost of not taking into account more costly, but more accurate estimation schemes involving either the introduction of redundancy in the preamble sequence length and/or other estimation schemes (e.g., maximum a posteriori estimators) not separable on N_B and N_U .

3.1.3.2 SVD of the UL channel matrices

In general, we cannot expect any square Hermitian symmetry in the structure of the individual UE UL channel matrix (involving $O(N_B^2)$) complexity, so the SVD computation (or at least the extraction of the left and right principal eigenvectors) cannot exploit the reduced-computation schemes applicable in such cases. In addition, as both principal eigenvectors are needed (the left to be transmitted at the UE for its beamforming and the right to be employed at the BS), the decreased-complexity formulation (when only eigenvalues and left OR right eigenvectors are needed), with complexity order $O(N_U N_B^2 + N_B^3)$, cannot be exploited.

In general, the full rank SVD problem requires $O(N_U N_B^2 + N_B N_U^2 + N_B^3)$ operations. However it has to be remarked that the driving order is due to the double QR decomposition performed before (iterative) eigenvalue solution finding problem, each involving cubic order of operations, whereas the eigenvalue problem involves $O(n)$ iterations, $O(n)$ operations each, where $n = \max(N_U, N_B)$. At this point, it is important to notice that the main feature of the considered scheduling protocol is the exploitation of only one singular (left and right) eigenvector associated to the highest eigenvalue: the full SVD adoption would compute all of them, but only one is actually employed in the algorithm for the coupling metric computation. This suggests the adoption of the TSVD (Truncated SVD) algorithm (e.g. implemented in LINPACK, LAPACK libraries), in such a case (depending on the actual implementation of the TSVD) the computational complexity can be reduced down to $O(4N_U N_B + 4k)$ (for a truncation order of k) but the saving in complexity is not really significant when non-Hermitian symmetry is employed; for instance, a typical configuration involving $N_U=1$ and $N_B=2$, would lead to similar complexity for the computation of full SVD and TSVD truncated at the 1st order. Therefore, due to the inherently reduced-size of the problem (both are limited to few units), we'll consider the asymmetric full SVD complexity as the one to be considered, i.e. $O(N_U N_B^2 + N_B N_U^2 + N_B^3)$.

It has to be remarked that the aforesaid complexity is for each subcarrier and thus the overall number of operations is multiplied by N_F , where N_F is the number of subcarriers. However, it has to be remarked that the a-priori knowledge that the channel frequency response is significantly correlated in the frequency domain, due to the inherently time-limited structure of the channel impulse response (or equivalently its power delay profile). This implies that an approximate, but satisfactory evaluation can be performed by applying the computations one out 4 or 5 subcarriers and then interpolating the data. The exploitation of the frequency correlation of the data allows the proportional decrease of the overall computational effort by a 4 or 5 factor, but introduces the additional complexity due to interpolation operation. In this case the computation of each matrix element needs about $O(N_U N_B^2 + N_B N_U^2 + N_B^3)$ complex operations, against the 4 complex operations needed by a complex-valued linear interpolation, so the computational saving is worth the associated (small) decrease in estimation accuracy.

3.1.3.3 Pairing metric computation

The conclusions drawn from sections 3.1.1 and 3.1.2 indicate that, although more complex, the best suited metric in a realistic scenario is the best joint SE pairing.

The aggregated spectral efficiency $S_{k,j}$ of the coupling (k, j) can be written as

$$S_{k,j} = S_k(\gamma_{k,j})(1 - P(\gamma_{k,j})) + S_j(\gamma_{j,k})(1 - P(\gamma_{j,k})) \quad (40)$$

where $\gamma_{k,j}$ is the reception SNR of the user k when couples with user j . $S_k(\gamma)$ is the maximum spectral efficiency for the user k with the MCS assigned as a function of the SNR value γ . $P_k(\gamma)$ is the PDU error rate achieved as a consequence of the SNR value γ . To enable the proper selection of the couples to be scheduled in UL, the evaluation of U^2 values of the (39) have to be computed. Each value of (39) involves the sequential computation of a) the SNR $\gamma_{k,j}$. and the SNR $\gamma_{j,k}$ ($\gamma_{k,j}$ and $\gamma_{j,k}$ can be computed in parallel); b) the estimation of the PDU-ER $P(\gamma_{j,k})$ and $P(\gamma_{k,j})$ (in parallel again); c) the computation of the aggregated SE $S_{k,j}$. Stated differently, the computation of $\gamma_{k,j}$. and $\gamma_{j,k}$ cannot be performed in parallel with $P(\gamma_{j,k})$ and $P(\gamma_{k,j})$ and with $S_{k,j}$.

The estimation of $\gamma_{k,j}$. and $\gamma_{j,k}$ is done by employing the following formula

$$\gamma_{k,j} = \frac{P_k |\lambda_k|^2}{P_j |\lambda_j \mathbf{v}_k^H \mathbf{v}_j|^2 + P_n} \quad (41)$$

where P_m is the power transmitted by the m^{th} user, P_n is the received noise power and λ_m and $\mathbf{v}_m$ are the dominant channel eigenvalue and left eigenvalue for the m^{th} user, respectively. In this formulation (32) needs N_B complex multiplications and 6 real multiplications, as λ_k , λ_j , $\mathbf{v}_k$ and $\mathbf{v}_j$ are already computed from the previous processing block (SVD) and P_n , P_k and P_j are already estimated and stored by the normal BS operations.

The overall computational effort for the exhaustive construction of the $\{\gamma_{k,j}\}$ matrix is approximately $O(U^2(N_B+2))$ complex multiplications. The final computation of the metric (39) needs 3 further operations, as the estimation of $S_k(\gamma)$ and $P_k(\gamma)$ can be done by picking their values in lookup tables that are easy to build and store, because they merely represent the performance of the BS in terms of maximal spectral efficiency and PDU-ER for a given SNR.

As we intentionally disregarded subcarrier notation (as each processing block is frequency-wise independent), the complete construction of the metric matrix $\{S_{k,j}\}$ on all the subcarriers involves about $O(N_F U^2 (N_B+3))$ (equivalent) complex operations, where N_F is the number of subcarriers. It is however worth remarking two points. The first deals with the correlation of the frequency response of the channel; as in the case of the previous processing block (as well as for the channel estimation) this implies that we could operate one computation one out 4 or 5 subcarriers and then interpolate the data. The exploitation of the frequency correlation of the data allows the proportional decrease of the overall computational effort by a 4 or 5 factor, but introduces the additional complexity due to interpolation operation. In this case, as seen, the computation of each matrix element needs about (N_B+3) operations, against the 4 operations needed by a real-valued linear interpolation and the computational saving might be not worth the associated (although small) decrease in estimation accuracy. The second point to be highlighted is the strong parallelism of the computations, that can be split either in $O(N_F U^2 (N_B+3))$ operation run sequentially by a single processing unit or by $O(U^2 (N_B+3))$ operations performed by N_F parallel processing units or even by $O(U(N_B+3))$ operation run by UN_F parallel processing units. When the maximal parallelism is exploited, the total computation time (excluding the further benefit coming from the more efficient memory accesses) speeds up at least by a UN_F factor. For instance, for a 1024 carriers and UEs equipped with 2 antennas, the computation time of the coupling metric matrix reduces of 2048 times w.r.t. its sequential processing.

3.1.3.4 Sequential selection of the couples

In the proposed formulation, the selection of the couples is operated by sequentially, in a column wise sense, by finding the maximal value of the joint SE computed in the previous processing block. The operation itself does not involve operations by the FPU (Floating Point Unit), but a max search that is efficiently computed by a comparison unit with 2 digits storage (one for the volatile temporary max value and one for the corresponding index). When operated over the whole matrix $\{S_{k,j}\}$ for each subcarrier, this involves $N_F U^2$ max finding operations in a sequential mode. Once again, also in this

case, the parallelisation implies that the operation can be also performed by U^2 max finding operations by N_F comparison units or also by U max finding operations by $N_F U$ comparison units.

Synopsis

The following points summarise the key elements relevant to the complexity evaluation of the proposed UL user pairing technique.

- Type of optimization.* The SVD decomposition is linear with both the number of UEs and the number of frequency subcarriers, but it is basically between quadratic and cubic in the number of BS or UE antennas (quadratic when the UE have one antenna and cubic when the UEs have the same number of antennas as the BS). The algorithm can be highly optimised by the exploitation of the joint parallelism in both the number of UEs and the number of frequency subcarriers. By exploiting the maximal parallelism offered by the algorithm, the complexity shrinks down to $O(N_U N_B^2 + N_B N_U^2 + N_B^3)$, each operated by UN_F elementary processing units. The coupling metric computation is linear with both the number of frequency subcarriers and the number of BS antennas, but is quadratic with the number of UEs. Also in this case the algorithm is highly optimised by its inherent parallelism on both subcarriers and on the number of UEs and its complexity can be reduced down to $O(U(N_B+3))$ operations, each performed by UN_F parallel processing units. Also the sequential selection of users (quadratic with the number of users and linear with the subcarriers), although not involving floating point operations, but simple comparison and storage, can be parallelised and its execution time can be decreased up to a factor UN_F with respect to its sequential execution.
- Features at receiver side.* The coupling protocol does not involve specific features at the BS (receiver) side, as the focus is on the coupling criteria and the adoption of sequential cancellation, ML (or suboptimal ML) estimators have an impact on the final performance in terms of BER, but have no direct impact on the scheduling and pairing protocol itself. The only requirements posed above the methodology deals with the redesign of the frame structure, to closely accommodate the training phase, the DL of the scheduling and of the UEs beamforming vectors and the subsequent UL phase, in order to better cope with time varying environments and let the time interval between the training phase and the UL phase fall well within the channel coherence time.
- Features at transmitter side.* The pairing methodology poses no specific requirements on the UEs (transmitter) side.

3.2 DOWNLINK MULTI-USER PAIRING

3.2.1 User pairing in OFDM systems

In this section, we consider the DL of a single cell multicarrier system. We focus on the case where a pair of users with diverse mean channel strengths' competes for service from the BS transmitter over two allocated subcarriers. Two possible schemes are considered: exclusive use of subcarriers and simultaneous use of subcarriers. Considering the conventional approach of exclusive use of subcarriers, we let each subcarrier to be used by a maximum of one user. In the simultaneous use of subcarriers (also termed as *reuse* of subcarriers) the two users in the pair simultaneously use the same subcarrier. The user with a stronger channel is in a better position to decode any signal that has been designed for the user with the weakest channel. Hence, the user with the strongest channel performs interference cancellation for the other user's signal before extracting its own desired signal. The weakest user tolerates the interference from the other user. The objective of both allocations is to maximise the WSR of the DL for the given weights of the two users. The instantaneous CSI is assumed to be available at the BS; in order to attain the system's objective of WSR maximisation, the

BS uses this CSI to allocate the available constrained power resource over the two subchannels. We show by using a suitable mathematical model and numerical calculations that the reuse scheme can provide gains in terms of information theoretic rate, as compared to the exclusive subcarrier allocation.

3.2.1.1 Introduction

High data rate services for the cellular mobile systems are becoming increasingly popular. This requires an efficient use of the available spectrum while fairly serving all the system users. Higher data rate services are particularly in demand over the DL of the cellular system due to the asymmetric nature of services, e.g., data download and web-browsing [Goldsmith-97]. In a cellular system, the users that are closer to the BS of the serving cell are in an advantageous situation. Other users, which are located at the boundary of the serving cell, are in a disadvantageous scenario. Over the DL channel, in order to deliver an acceptable SNR at the receiver of the cell-edge user, high signal power needs to be transmitted from the serving BS. This results in very strong interferences to the users closer to the BS of the cell. To avoid this problem, the DL scheduler keeps the transmissions orthogonal in time, frequency or code [Andrews-05].

Let us consider the DL of the OFDM system; usually each subcarrier is allocated to a single user within each cell (termed as OFDMA). Since radio signals decay in strength as they travel a physical distance, the subcarriers can be re-used in any cell that is sufficiently distant from the cell where it is primarily used. This keeps the other cell (inter-cell) interference to a tolerable low value. This scheme performs well in terms of interference minimisation, however each user is restrained to exclusively use specific portion of the spectrum.

The exclusive subcarrier allocation schemes are studied in detail (along with bit and power loading problems) in [Cheong-99, Ying-04]. However, the condition of exclusively using each subcarrier for a single user only is maintained. Bergmans and Cover in [Bergmans-74] introduced the novel idea of cooperative broadcasting where two users can be simultaneously served over the DL channel if their channel gains are significantly diverse: one user having very strong channel and the other having very weak channel. The main idea is that any signal that is transmitted to the user with the weak channel will be more likely to be decoded by the user with the strong channel. Hence, the user with the strong channel can perform interference cancellation by first decoding the interference signal and then re-modulating and subtracting it from the aggregate received signal to recover its desired signal. The user with the weak channel will tolerate the interference caused by the signals that are transmitted to the other user with strong channel (the required power of the transmitted signal over the strong channel is relatively low and causes less interference at the receiver of the weakest channel).

In this work, we compare the performance of a single cell DL system for the two different subcarrier allocation schemes; the first scheme uses exclusive subcarrier allocation; the second scheme lets the two users simultaneously use the same subcarriers by employing the cooperative broadcasting approach. We use the term subcarrier *reusing* or *overloading* to describe this process of allocating the same subcarrier to two users. The two schemes are compared for the information theoretic user rate shares and the sum-rate. Results of this comparison show the advantage of reuse over exclusive allocation.

The rest of the description is organised as follows; the next section describes the system model that has been considered for our study and how the sum-rate and user rate shares are calculated for each of the allocation schemes; this is followed by the presentation of some performance results for the two schemes; then, some conclusions are drawn.

3.2.1.2 System Model

Consider the DL of a single cell, serving K users (indexed by k) with M orthogonal subcarriers (indexed by m). The channel from the BS transmitter to each of the user over all subcarriers is modelled as follows. Let us represent the magnitude squared channel gain from the BS to a user

terminal k , for a subcarrier m , by $h_{k,m}$. For all users and subcarriers, these channel gains are composed of two factors. One factor ($\psi_{k,m}$) models the distance dependent path loss experienced by the signal and the other factor ($g_{k,m}$) models the effect of Rayleigh fading on the received power. We assume a narrow-band, single-tap flat (frequency non-selective), Rayleigh fading channel for each subcarrier and for this work we ignore the log-normal shadowing effects. The (magnitude squared) channel gain can be expressed in terms of these two factors as: $h_{k,m} = \psi_k \times g_{k,m}$ where ψ_k is the distance dependent power loss (same for all subcarriers seen by the user k). Factor $g_{k,m}$ is the magnitude squared value of a Rayleigh fading gain and hence modelled as an exponentially distributed random variable with mean normalised to 1. For ease of explanation the magnitude squared channel gains for each user over each subcarrier can be compactly written as a matrix $\mathbf{H}$ with dimensions $K \times M$ with elements $h_{k,m}$.

To simplify the exposition, we consider the case where the users have been paired to maximise the diversity of their mean channel strength: i.e. very strong channel paired with very weak channel. Each pair has been allocated two subcarriers for their use (assuming $M = K = 2$). The paired users can either use them exclusively (one subcarrier for each of them) or in an overloading manner in the sense that each user uses both subcarriers assigned to the pair. The objective of allocation is to perform a WSR maximisation for a given weight vector $\mathbf{w}$ that defines how much bias each user shall be given in order to share the resources between them. Each user k is given a weight w_k such that $\sum_k w_k = 1$. These two subcarrier-use schemes and the calculations of their corresponding sum-rate and user rate shares for WSR maximisation are described below.

3.2.1.2.1 Exclusive Allocation

For exclusive subcarrier allocation, an allocation vector $\mathbf{a}$ (dimension $M \times 1$) describes how each user is allocated to each subcarrier, where the m^{th} entry $a_m = k$ indicates that the m^{th} subcarrier is allocated to user k . The WSR is given by

$$R_{sum} = \sum_{m=1}^2 w_{a_m} \log_2 \left(1 + \frac{h_{a_m,m} p_m}{\sigma^2} \right), \quad (42)$$

where p_m is the power share allocated to the subcarrier m . The values of the power share for two subcarriers can be expressed as the power vector $\mathbf{p} = [p_1 \ p_2]^T$. The elements of this vector are constrained by $\sum_{m=1}^2 p_m = P_t$. The objective is to maximise the WSR over all possible allocation vectors $\mathbf{a}$ and the power vectors $\mathbf{p}$ and this can be expressed as

$$\begin{aligned} & \max_{\mathbf{a}, \mathbf{p}} R_{sum} \\ & \text{s.t.} \quad \sum_{m=1}^2 p_m = P_t \end{aligned} \quad (43)$$

This allocation is depicted in Figure 43. At any given time, the two subcarriers can be either exclusively shared by the two users or a single user can load both subcarriers depending on which allocation maximises the optimisation objective of the WSR. This maximisation is over the subcarrier allocation as well as the power split between the two allocated subcarriers.

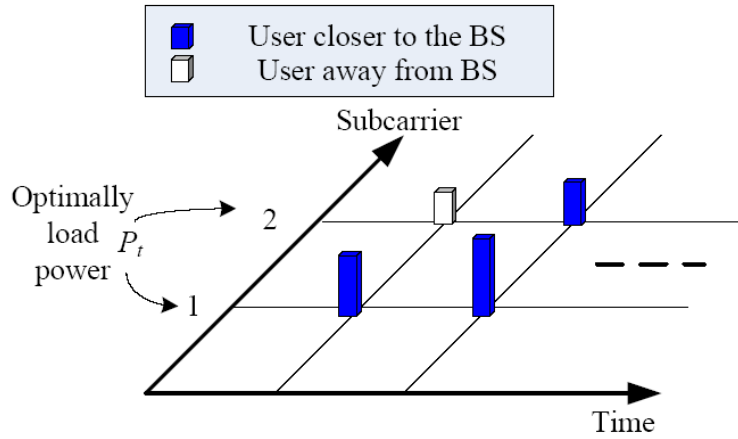


Figure 43: Depiction of orthogonal subcarrier allocation scheme and power optimisation over the pair of subcarriers.

For a given WSR allocation, the actual user rate share can be obtained by using the following equations:

$$R_k = \sum_{m=1}^2 \log_2 \left(1 + \frac{h_{a_m, m} p_m}{\sigma^2} \right), \forall a_m = k, \quad (44)$$

3.2.1.2.2 Subcarrier Reuse Allocation

In subcarrier reuse allocation, both subcarriers are available for use of each user of the pair. For the given channel matrix $\mathbf{H}$, as introduced above, the power allocated on each subcarrier m by each user is given by $p_{k, m}$ and these values form the power allocation matrix $\mathbf{P}$. On each subcarrier, the strong and weak channel indices ($k_{\max, m}$ and $k_{\min, m}$ respectively) are identified by their channel strengths such that

$$\begin{aligned} h_{k_{\max, m}, m} &= \max(h_{1, m}, h_{2, m}) \\ h_{k_{\min, m}, m} &= \min(h_{1, m}, h_{2, m}) \end{aligned} \quad (45)$$

The sum-rate in this scenario is calculated by using

$$R_{sum} = \sum_{m=1}^2 \left(w_{k_{\max, m}} \log_2 \left(1 + \frac{h_{k_{\max, m}, m} p_{k_{\max, m}, m}}{\sigma^2} \right) + w_{k_{\min, m}} \log_2 \left(1 + \frac{h_{k_{\min, m}, m} p_{k_{\min, m}, m}}{h_{k_{\min, m}, m} p_{k_{\max, m}, m} + \sigma^2} \right) \right), \quad (46)$$

where it can be seen that on each subcarrier m , the user with the strongest channel (index $k_{\max, m}$) can perform SIC while the user with the weakest channel ($k_{\min, m}$) tolerates the interference from the signal that is transmitted to the strongest channel user. The objective of WSR maximisation becomes

$$\begin{aligned} \max_{\mathbf{P}} \quad & R_{sum} \\ \text{s.t.} \quad & \sum_{k=1}^2 \sum_{m=1}^2 p_{k, m} = P_t \end{aligned} \quad (47)$$

This scheme is depicted in Figure 44. Both subcarriers are simultaneously shared by both users in the system and the power is split between the users and is further shared between the subcarriers used by these users. The allocation of power is used to achieve the desired objective of WSR maximisation.

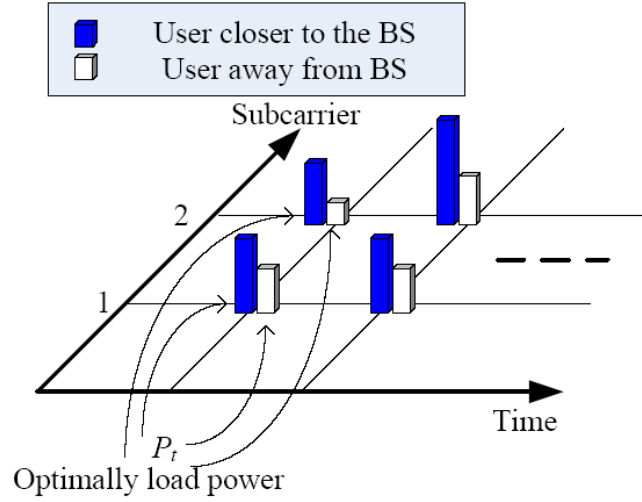


Figure 44: Depiction of reuse subcarrier allocation with power optimisation over the two users and two subcarriers

The rate share of each of the user is then calculated by using

$$R_k = \sum_{m=1}^2 \left(\log_2 \left(1 + \frac{h_{k_{\max,m},m} p_{k_{\max,m},m}}{\sigma^2} \right) + \log_2 \left(1 + \frac{h_{k_{\min,m},m} p_{k_{\min,m},m}}{h_{k_{\min,m},m} p_{k_{\max,m},m} + \sigma^2} \right) \right). \quad (48)$$

$$\forall k_{\max,m} = k, k_{\min,m} = k$$

For each value of m either $k_{\max,m}=k$ or $k_{\min,m}=k$ and the other term is ignored. In the next sub-section, the performance of the two approaches is compared for the obtained sum-rate.

3.2.1.3 Results and discussion

We consider the DL scenario. A given pair of the users with a pair of allocated subcarriers need to be served using either of the above two schemes. Two users have their mean channel strength in the ratio $r = E[h_{1,m}]/E[h_{2,m}] > 1$. We assume this ratio to be 100 for the following results. This implies that the user 1 has the larger mean channel strength and is physically closer to the DL transmitter. The CSI is available at the transmitter and the transmitter optimises the subcarrier allocation and the power share for the two users. Since Rayleigh flat fading is assumed, the magnitude squared channel gains are exponentially distributed [Goldsmith-05].

Each cell has a total transmit power constraint of $P_t = 1W$ per pair of the subcarriers. The noise power is normalised to unity, $\sigma^2 = 1$. Several system snapshots (100 for the results that are presented here) are simulated and their results are averaged to show the mean sum-rate (and user rate share) performance.

In exclusive subcarrier allocation, all possible subcarrier allocations (including the possibility of allocating both subcarriers to one user) are considered for the maximisation of the WSR. The power share between the two subcarriers is also optimised to obtain maximum WSR. In reuse allocation schemes, the strongest channel (for every snapshot) is identified on each subcarrier and the WSR is calculated on both subcarriers using the equations above. The distribution of the available power between the two users and the two subcarriers is optimised to maximise the WSR. The obtained results for the two schemes are presented and discussed below.

In the first results that are presented in Figure 45, we plot the sum-rate that is obtained by performing the WSR maximisation. For the lower weights (left side of the x-axis) of the stronger-channel user

(user 1 in the scenario), the reuse scheme provides a significant increase of the sum-rate. This implies that any attempt to fairly treat the user with the weakest channel (by giving him larger weight) will result in performance loss which is lower in case of reuse scheme. In this fair allocation region the better performance of the reuse scheme makes it a suitable approach for the fair allocation scenario. For a range of weights between 0.17 to 0.32, the orthogonal scheme performs better since the trade-off between the extra rate obtained by the strong-channel user performing interference cancellation is more than offset by the penalty in rate for the weak-channel user that tolerates the interference. This becomes clearer with the other results that will be presented later in the sub-section. Both schemes provide same performance when the weight of the strong-channel user is large (above 0.4 in the result shown). This is due to the reason that all subcarrier and power resources are used by the strong user and the two allocation schemes essentially become the same scheme.

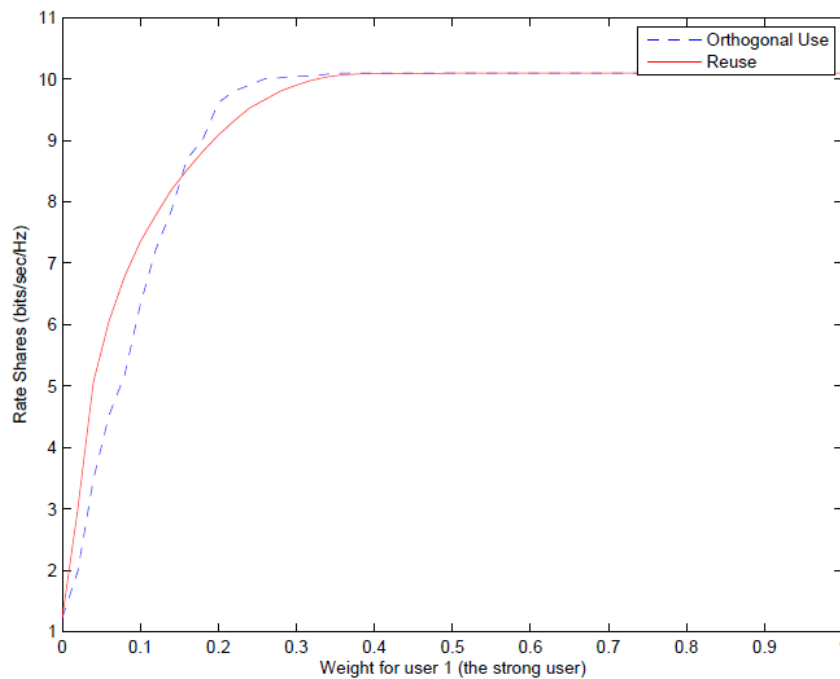


Figure 45: Sum-rate comparison for exclusive allocation (Orthogonal Use: dashed line) and subcarrier reuse allocation (Reuse: solid line) for different weights given to the strong-channel user

To gain more insight in the comparison, we plot the user rate share for the two schemes in Figure 46. As expected, the strongest-channel user gets the larger share of the sum-rate when the weight that is given to the strong-channel user is larger than a certain small value, i.e., around 0.01. It shall be noted that for small values of the strong-channel user weight (a relatively fairer allocation regime), the reuse scheme can provide a higher rate than the exclusive allocation scheme for the strong user, without significantly compromising the performance of the weak-channel user. After a certain value, this increase is not large enough and the performance of the strong-channel user under reuse scheme is not as good as the exclusive allocation scheme. However, in this region a better performance for the weak-channel user is maintained as compared to the exclusive allocation scheme. The overall sum-rate is lower for reuse scheme for this region of weight selection.

The two schemes can achieve a given target for the rate share of a specific user, by allocating specific weights which may be different for the two schemes. An insightful comparison will be to fix the target rate for one user and observe how the rate for the other user compares for the two schemes. In this regard the following result is more insightful.

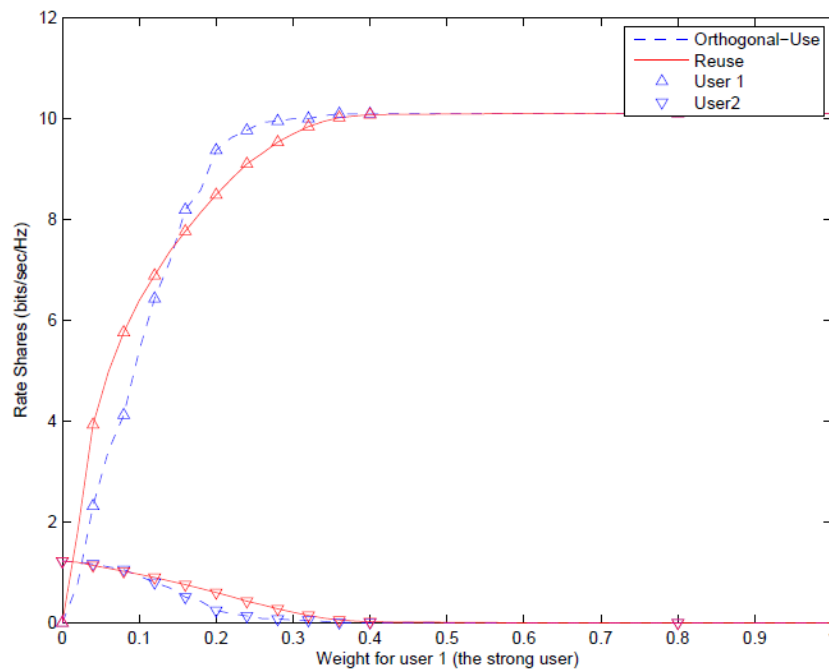


Figure 46: The rate shares (user 1 and user 2) for the two schemes. For exclusive allocation (Orthogonal Use: dashed line) and subcarrier reuse allocation (Reuse: solid line). The rate for strong-channel user (user 1) is indicated by an up-pointing triangle marker. The rate for weak-channel user (user 2) is indicated by a down-pointing triangle marker

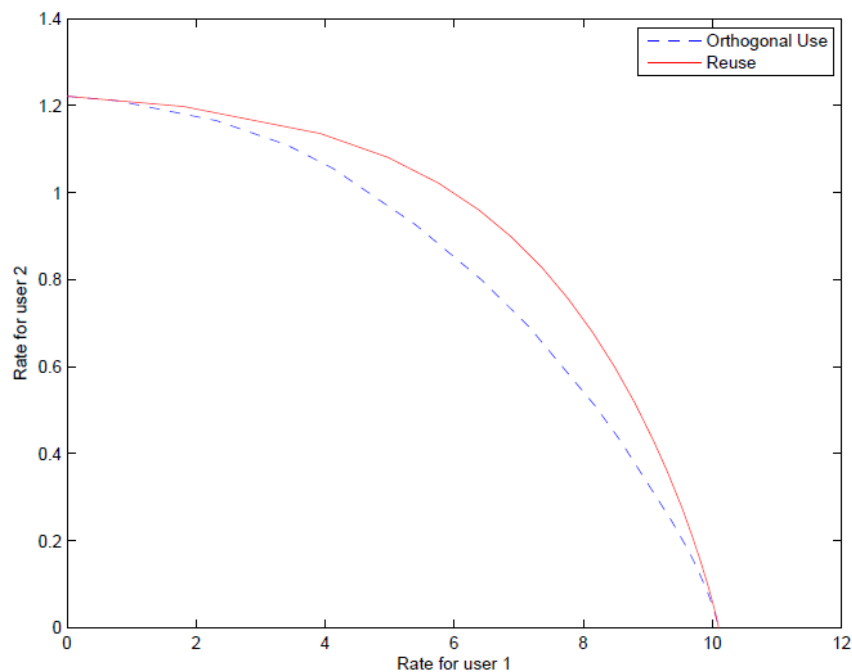


Figure 47: The performance comparison of the two schemes if the target rate for one user is fixed and both schemes can use any weight vector to obtain the targeted rate

In the last result that is presented in Figure 47, we can observe that if the target rate of one user is fixed, then the rate for the other user is larger when applying our reuse scheme instead of the orthogonal allocation scheme, except for the two extreme points (where the weight of either user is

zero implying that only one user is allowed to use all the resources exclusively). The two allocation schemes are essentially the same and give same performance at these two extreme points.

3.2.1.4 Conclusions and outlook

In this section, we have compared the DL WSR of a multicarrier system when two different schemes are used for allocating subcarriers to some users. In one scheme, each subcarrier is exclusively used by only one user at a time. In the other scheme, each subcarrier can be further reused with one additional user sharing the same subcarrier by simultaneously using it along with the other user. In the case that two users are loaded on the same subcarrier, the users with the strongest and weakest channels are identified. For the user with the strongest channel, WSR is evaluated by assuming perfect interference cancellation at the receiver. On the other hand, the WSR is calculated for the user with the weakest channel by assuming that the interference is tolerated. The subcarrier reuse allocation scheme is shown to perform better than the orthogonal scheme, since when the target rate for one user is fixed, then the reuse scheme can provide additional rate for the other user by adjusting the weights of the two users. It shall be noted that the subcarrier reuse allocation scheme requires extra processing at the receiver of the user with the strongest channel (for interference cancellation).

We are currently working on the extension of this work to look at the pair-selection and subcarrier allocation for a large set of users in a single cell environment.

3.2.2 Scheduling criteria for downlink superposed transmissions

The introduction in section 3 has shown that the superposed transmission is superior to the orthogonal transmission for the DL when the served users present a large SNR difference. This section explores the optimal resource allocation under two different scheduling criteria: WSR and PF. To this end, section 3.2.2.1 looks into how to perform optimal resource allocation given a pair of users under the maximisation of those criteria by taking into account the priority of the users (weight factor). It will be shown that given certain priorities, the superposed transmission is not always superior to the orthogonal one in terms of sum-rate. We will set the conditions required, which connect the SNR of the selected pair of users with their priorities, for improving the orthogonal transmission. Afterwards, throughput of the cellular system is evaluated in section 3.2.2.2 for the case where the users are selected under the aforementioned criteria, i.e., the users that present the maximum metric in each transmission time (under *WSR* or *PF*) are selected among all the users present in the cell. Likewise, the resources (transmitted power from the BS) are allocated according to these criteria. Various aspects such as the average throughput, delay and gain over the conventional PF scheduler [PVis-02] with a single-user selection, are investigated.

3.2.2.1 Resource allocation in the downlink under superposed transmission

There are different criteria for selecting the users to be served simultaneously and optimising the resources accordingly in the literature: *WSR*, *Sum-rate* and *PF*.

The *WSR* criterion has been explored in [HVis-03] and [Jaga-07] for MIMO MU configuration. Moreover, [Hott-08] derived the optimal subset of users for maximising the sum capacity by a linear programming algorithm. Moreover, under the *sum-rate* criterion, [Jind-05] has shown that the sum-rate capacity gain of the orthogonal transmission, based on TDMA, over the BC when selecting a single user at each time, is at most

$$\frac{C_{BC}}{C_{TDMA}} \leq \min(M, K), \quad (49)$$

where M stands for the number of antennas at the BS and K is the number of receivers. Therefore, in the single-antenna case, i.e., $M=1$, the superposed transmission does not offer any advantage over the orthogonal transmission. That result is valid at all SNRs when perfect channel knowledge at receivers and transmitters are assumed. For $M>1$, the transmitted signals must be encoded by using Dirty Paper

Coding (DPC), [Cost-81]. Since we assume single antenna devices, the *sum-rate* criterion will not be relevant for our work.

On the other hand, we investigate an additional criterion based on the *PF*, originally introduced by [Kell-98] and applied to wireless systems in [PVis-02]. This criterion ensures some fairness among the users to be selected. Moreover, the scheduling algorithm that was presented in [PVis-02] depends on a parameter that is used to tune the performance of the algorithm; *MU gain* is achieved by selecting always the users with the best instantaneous channel; user fairness is obtained through round robin like scheduler, i.e. distributing all the resources equally to the users.

Our scenario consists of a BS, and two users, MS₁ and MS₂, experiencing different SNRs over the BS-MS link. All terminals have a single antenna element. We classify MS₁ as the *strong* user, having its SNR equal to ρ_1 and MS₂ as the *weak* user, presenting a SNR equal to ρ_2 . Both SNRs are linearly connected by $\rho_1 = \xi \rho_2$ with $\xi > 1$. Moreover, μ_1, μ_2 are the weight factors of MS₁ and MS₂, respectively. They must be taken into account during the resource allocation in order to establish the priority level of each user. For instance, if $\mu_1 \gg \mu_2$, it then means that MS₁ has a higher priority than MS₂, and is therefore more likely to be served. This priority depends on the traffic model and the type of applications considered.

The resource allocation under the *WSR* and the *PF* criteria will be analysed in sections 3.2.2.1.1 and 3.2.2.1.2, respectively. The performance of the superposed transmission will be compared with the orthogonal transmission. Note that both transmission techniques will be implemented by applying TDMA and Frequency Division Multiple Access (FDMA).

3.2.2.1.1 Weighted sum-rate criterion

The resources are optimised in order to maximise the following criterion

$$\begin{aligned} J_{WSR} &= \mu_1 R_1 + \mu_2 R_2 \\ s.t. \quad &(R_1, R_2) \in B_x \end{aligned} \quad (50)$$

where R_1 and R_2 stand for the bitrate of the intended messages to MS₁ (defined as the *strong user*) and MS₂ (defined as the *weak user*), respectively, μ_1, μ_2 are the weights assigned to each user, and finally, B_x denotes the achievable rate region, which depends on the selected transmission strategy and will be introduced in the following subsections.

3.2.2.1.1.1 Gaussian Broadcast Channel

The achievable rate region for the two-user Gaussian BC [Cove-91] is given by

$$B_{BC}(\gamma_1, \xi, \rho_2) = \begin{cases} R_1 \leq \log_2(1 + \gamma_1 \rho_1) & = \log_2(1 + \gamma_1 \xi \rho_2) \\ R_2 \leq \log_2\left(1 + \frac{(1 - \gamma_1) \rho_2}{1 + \gamma_1 \rho_2}\right) & = \log_2(1 + \rho_2) - \log_2(1 + \gamma_1 \rho_2) \end{cases} \quad (51)$$

where ξ is the ratio between the SNR of the *strong* user over the *weak* one and $\gamma_1, 1 - \gamma_1$ stand for the fraction of BS total power that is allocated to each transmitted signal.

The optimal power allocation is obtained by finding the maximum of the utility function J_{WSR} , in (50), with partial derivatives given by

$$\begin{cases} \frac{\partial J_{WSR}}{\partial \gamma_1} = \mu_1 \frac{\xi \rho_2}{1 + \gamma_1 \xi \rho_2} - \mu_2 \frac{\rho_2}{1 + \gamma_1 \rho_2} \\ \frac{\partial^2 J_{WSR}}{\partial^2 \gamma_1} = -\mu_1 \frac{(\xi \rho_2)^2}{(1 + \gamma_1 \xi \rho_2)^2} + \mu_2 \frac{\rho_2^2}{(1 + \gamma_1 \rho_2)^2} \end{cases} \quad (52)$$

Thus, the optimal power allocation for the signal intended to the strong user (γ_1) is expressed as

$$\gamma_1^* = \frac{\mu_1 \xi - \mu_2}{\xi \rho_2 (\mu_2 - \mu_1)} = \frac{\xi - \frac{\mu_2}{\mu_1}}{\xi \rho_2 \left(\frac{\mu_2}{\mu_1} - 1 \right)}. \quad (53)$$

The result in (53) is only valid when

$$\frac{\partial^2 J_{WSR}}{\partial^2 \gamma} < 0 \rightarrow \mu_1 < \mu_2. \quad (54)$$

In other words, (53) is valid only if the *strong* user has a lower priority than the *weak* one. Moreover, if $\gamma_1=0$, then only the *weak* user, MS₂, (ρ_2) is served, and the constraint in (54) is met when

$$\xi < \frac{\mu_2}{\mu_1}. \quad (55)$$

Otherwise, the condition for scheduling the *strong* user MS₁ only is such that $\rho_1 \geq \xi_{strong} \rho_2$, where

$$\xi_{strong} = \begin{cases} \frac{\mu_2}{\mu_1} \frac{1}{1 - \rho_2 \left(\frac{\mu_2}{\mu_1} - 1 \right)} & \text{if } \frac{\mu_1}{\mu_2} > \frac{\rho_2}{1 + \rho_2} \\ \infty & \text{otherwise} \end{cases}. \quad (56)$$

Therefore, the strong user may never be scheduled according to the user configuration. Finally, the optimal power allocation is described by the following piecewise function

$$\gamma_1^* = \begin{cases} 0 & \text{if } \xi \leq \frac{\mu_2}{\mu_1} \\ \frac{\xi - \frac{\mu_2}{\mu_1}}{\xi \rho_2 \left(\frac{\mu_2}{\mu_1} - 1 \right)} & \text{if } \frac{\mu_2}{\mu_1} < \xi < \xi_{strong} \\ 1 & \text{if } \xi \geq \xi_{strong} \end{cases}. \quad (57)$$

Notice that the power allocation depends on three intervals whose limits are defined by the values of μ_1 , μ_2 and ξ . For instance, in the first interval only the *weak* user, MS₂, is served; in the second interval both users are served; in the third interval, only the *strong* user, MS₁, is selected. Our objective is to derive under which circumstances we can remain in the second interval, i.e., serving the two users simultaneously.

To this end, we insert the optimal power allocation into the WSR criterion, having,

$$\begin{aligned} \phi_0(\xi, \rho_1) &= J_{WSR}(\gamma_1^*, \xi, \rho_2) = \\ &= \begin{cases} \mu_2 \log_2(1 + \rho_2) & \text{if } \xi < \frac{\mu_2}{\mu_1} \\ \mu_1 \log_2\left(\frac{\mu_1(\xi - 1)}{\mu_2 - \mu_1}\right) + \mu_2 \log_2\left(\frac{\mu_2 - \mu_1}{\mu_2} \frac{\xi(1 + \rho_2)}{\xi - 1}\right) & \text{if } \frac{\mu_2}{\mu_1} \leq \xi \leq \xi_{strong} \\ \mu_1 \log_2(1 + \xi \rho_2) & \text{if } \xi > \xi_{strong} \end{cases}, \end{aligned} \quad (58)$$

where ξ_{strong} is defined in (56). The dependence of function ϕ_0 with respect to the parameter ξ in the second interval of (58) depends on

$$\frac{\partial \phi_0}{\partial \xi} = \mu_1 \frac{1}{\xi - 1} - \mu_2 \frac{1}{\xi(\xi - 1)} \text{ and } \frac{\partial^2 \phi_0}{\partial^2 \xi} = \mu_1 \frac{-1}{(\xi - 1)^2} + \mu_2 \frac{(2\xi - 1)}{\xi^2(\xi - 1)^2}. \quad (59)$$

Furthermore, we observe a minimum of ϕ_0 at

$$\xi^+ = \frac{\mu_2}{\mu_1}, \quad \left. \frac{\partial^2 \phi_0}{\partial^2 \xi} \right|_{\xi=\xi^+} = \frac{\mu_1}{\mu_2} (\mu_2 - \mu_1) > 0. \quad (60)$$

The above minimum value coincides with the limit of the second interval of equation (57). Figure 48-left depicts the behaviour of the function ϕ_0 , (58), in function of the parameter ξ with $\rho_2=10$ dB and $(\mu_1, \mu_2) = \{(0.3, 0.7), (0.1, 0.9)\}$. We can remark that the WSR is constant for $\xi \leq \mu_2/\mu_1$ because only the weak user is allocated. For larger values of ξ , both users are served. The WSR increases with ξ because the SNR of the strong user also increases. It shall be noted that the condition for serving only the strong user, (58), is not satisfied for the configurations depicted on the left hand side of Figure 48.

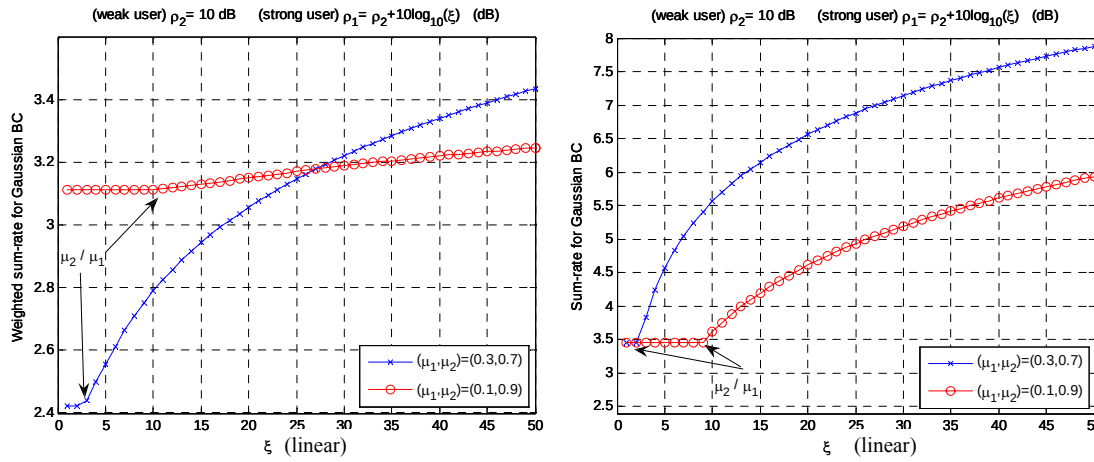


Figure 48: WSR (left) and sum-rate (right) as a function of ratio between SNRs of the two served users (ξ , in linear scale). $\rho_2=10$ dB, $(\mu_1, \mu_2)=\{(0.3,0.7), (0.1,0.9)\}$

On the right hand side of Figure 48, the sum-rate attained by the system under optimal power allocation is also depicted. It can be expressed as a function of the ratio between the SNRs of the served users (ξ) as follows

$$R_1^* + R_2^* = \begin{cases} \log_2(1 + \rho_2) & \text{if } \xi < \frac{\mu_2}{\mu_1} \\ \log_2\left(\frac{\mu_1}{\mu_2} \xi (1 + \rho_2)\right) & \text{if } \frac{\mu_2}{\mu_1} \leq \xi \leq \xi_{strong} \\ \log_2(1 + \xi \rho_2) & \text{if } \xi > \xi_{strong} \end{cases} \quad (61)$$

3.2.2.1.1.2 TDMA

Assuming that we split the resources into two orthogonal time slots with variable length, i.e., one for each transmission, we get the following bitrates

$$B_{TDMA}(R_1, R_2) \in \begin{cases} R_1 \leq \alpha \log_2(1 + \xi \rho_2) \\ R_2 \leq (1 - \alpha) \log_2(1 + \rho_2) \end{cases}, \quad (62)$$

where α is the portion of the time assigned to MS₁ (*strong* user) and $(1-\alpha)$ is the portion of time devoted to MS₂ (*weak* user).

The maximisation of function J_{WSR} in (50) with respect to α , subject to the achievable rate region defined in (62), leads to select one of the users according to μ_1 , μ_2 , ρ_2 and ξ ,

$$J_{WSR}(\alpha_1^*, \xi, \rho_2) = \begin{cases} \mu_2 \log_2(1 + \rho_2) & \text{if } \mu_1 \log_2(1 + \xi \rho_2) < \mu_2 \log_2(1 + \rho_2) \\ \mu_1 \log_2(1 + \xi \rho_2) & \text{otherwise} \end{cases} \quad (63)$$

All the resources are allocated to the *strong* or to the *weak* user depending on the threshold

$$\xi_{TDMA} = \frac{(1 + \rho_2)^{\frac{\mu_2}{\mu_1}} - 1}{\rho_2}, \quad \tilde{\xi}_{TDMA} \approx (\rho_2)^{\frac{\mu_2}{\mu_1} - 1}. \quad (64)$$

After optimising the resources under the WSR, the achieved sum-rate of the TDMA system is given by

$$R_1^* + R_2^* = \begin{cases} \log_2(1 + \rho_2) & \text{if } \xi < \xi_{TDMA} \\ \log_2(1 + \xi \rho_2) & \text{otherwise} \end{cases} \quad (65)$$

The additive sum-rate gain provided by the BC, shown in sub-section 3.2.2.1.1.1, over TDMA is given by the ensuing piecewise function

$$\Delta R_{TDMA}^* = \begin{cases} 0 & \text{if } \xi < \frac{\mu_2}{\mu_1} \\ \log_2\left(\frac{\mu_1}{\mu_2} \xi\right) & \text{if } \frac{\mu_2}{\mu_1} \leq \xi \leq \xi_{TDMA} \\ \log_2\left(\frac{\mu_1}{\mu_2} \xi \left(\frac{1 + \rho_2}{1 + \xi \rho_2}\right)\right) & \text{if } \xi_{TDMA} < \xi \leq \xi_{strong} \\ 0 & \text{if } \xi > \xi_{strong} \end{cases}, \quad (66)$$

where ξ_{strong} is defined for the BC in (56). There are four intervals, where the first and the fourth corresponds when TDMA and BC selects the *weak* and the *strong* user only, respectively. The second interval describes the situation where the BC serves both users, while the TDMA just selects the *weak* one due to the priority described by μ_2 , and finally, the third interval correspond to the case when the BC scheme selects both users, while the TDMA scheme selects the *strong* user. Notice that in this third interval the additive sum-rate gain will be negative, i.e., since the TDMA scheme serves the strong user, it is able to provide a better sum-rate than the BC scheme, which serves both users and the conditions for serving just the strong user are not satisfied (fourth interval).

Equation (66) shows that after optimising the WSR, J_{WSR} in (50), of the TDMA and BC transmission schemes, given weights μ_1, μ_2 , the BC scheme is superior to the TDMA in the second interval $\mu_2/\mu_1 \leq \xi \leq \xi_{TDMA}$. Figure 49 shows the values of the additive sum-rate as a function of ξ when $(\mu_1, \mu_2) = (0.3, 0.7)$ and $\rho_2 = \{10, 3\}$ dB. The maximum gain is attained for $\xi = \xi_{TDMA}$. In addition of being able to serve just a single user with SNR ρ_2 as it is done by the TDMA (depending on the values of μ_1, μ_2, ξ), BC can also serve simultaneously the two users with SNR equal to ρ_2 (*weak user*) and $\rho_2 \times \xi$ (*strong user*). Therefore, equation (64) provides the optimal value of the ratio between SNRs (ξ) as a function of the remaining variables (ρ_2, μ_2, μ_1). Likewise, Figure 50 illustrates the additive sum-rate gain as a function of ξ and the ratio of priorities μ_2/μ_1 . It can be seen that both parameters are important to get the gains that are illustrated in Figure 50, leading to positive and even negative gains, i.e. a configuration where TDMA gets a better sum-rate than the one attained by the BC.

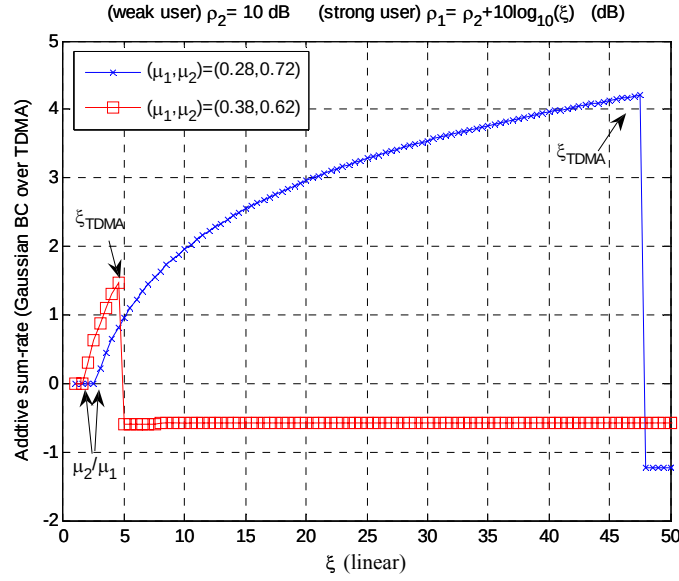


Figure 49: Additive sum-rate gain of BC over TDMA as a function of ratio between SNRs of the served users (ξ)(linear scale). $\rho_2=10$ dB, $(\mu_1, \mu_2)=\{(0.28,0.72),(0.38,0.62)\}$

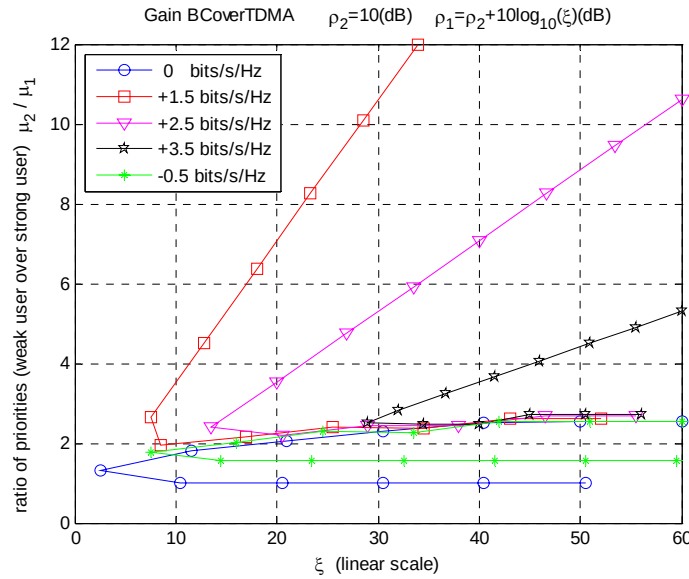


Figure 50: Additive sum-rate gain of BC over TDMA as a function of ratio between SNRs of the users served (ξ)(linear scale) (x-axis) and the ratio between priorities (μ_2/μ_1) (y-axis). $\rho_2=10$ dB,

Let us define the additive sum-rate gain and the percentage-wise sum-rate gain that is obtained by working with the optimal ξ_{TDMA} as

$$\Sigma_{gain} = \Delta R_{TDMA}^*(\xi, \mu_1, \mu_2) \Big|_{\xi_{TDMA}} = \log_2 \left(\frac{\mu_1 (1 + \rho_2)^{\mu_2/\mu_1} - 1}{\rho_2} \right)$$

$$\Psi_{gain} = \frac{\Delta R_{TDMA}^*(\xi, \mu_1, \mu_2) \Big|_{\xi_{TDMA}}}{\log_2(1 + \rho_2) \Big|_{\xi_{TDMA}}} = \frac{\log_2 \left(\frac{\mu_1 (1 + \rho_2)^{\mu_2/\mu_1} - 1}{\rho_2} \right)}{\log_2(1 + \rho_2)} \quad (67)$$

The optimal ratio between SNRs, ξ_{TDMA} , is given in (64) as a function of the weak user (ρ_2) and the user priorities, μ_1, μ_2 . When the weak user has a SNR $\rho_2 \gg 1$, then $\xi_{TDMA} = \tilde{\xi}_{TDMA}$ in (64). However, this optimal value may sometimes be too large, and as a result, it may be difficult to use in real user deployments. For example, if the weak user has $\rho_2 = 10$ dB and the priorities satisfy $\mu_2/\mu_1 = 9$, then the optimal ratio would be $\xi_{TDMA} \approx 80$ dB and the strong user would have a SNR equal to $\rho_1 = 90$ dB. This SNR is very high, and hence, difficult to apply in real systems.

In order to ensure that the SNR of the strong user is realistic, we bound the maximum SNR of the selected strong user as $\rho_1^{\max} = 10 \log_{10}(\nu)$ dB. We will take into account this limitation in our analysis of the additive sum-rate and the percentage-wise sum-rate gains introduced in (67). Consequently, given the *weak* user MS₂ with a SNR equal to ρ_2 and priority denoted by μ_2 , we analyse the different gains that are attained when the strong user presents a priority μ_1 and a SNR that is equal to

$$\rho_1 = \min(\nu, \xi_{TDMA} \rho_2), \quad \xi_{TDMA} \leq \frac{\nu}{\rho_2}, \quad (68)$$

where ν stands for the maximum SNR of the strong user.

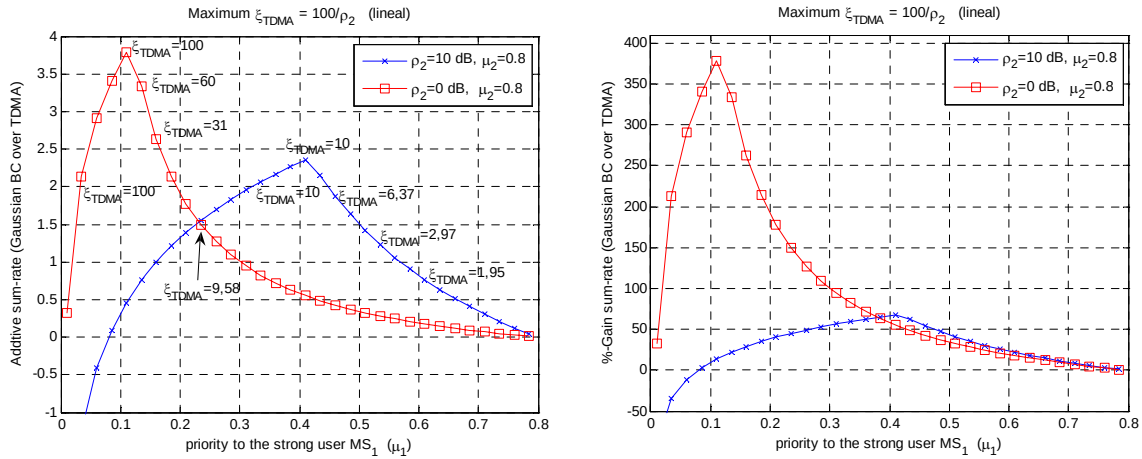


Figure 51: Left) Additive sum-rate and Right) percentage-wise gain of the sum-rate of BC over TDMA (see(67)) as a function of μ_1 when the weak user presents $\rho_2=\{10,0\}$ dB and $\mu_2=0.8$. $\nu=100$ (20 dB).

Figure 51-left depicts the additive sum-rate gain and Figure 51-right shows the percentage-wise sum-rate gain, (67), attained by the BC over the TDMA given a *weak* user (with SNR equal to ρ_2 and priority μ_2) as a function of the *strong* user μ_1 priority. Under such a configuration, the TDMA serves only the *weak* user, while the BC scheme serves both MSs. Results that are presented in Figure 51 demonstrate that the key parameters for attaining the best performance are the priority of the strong user, μ_1 , and the ratio of SNR, ξ_{TDMA} . For instance, assuming a weak user with $\rho_2=10$ dB and $\mu_2=0.8$, the best sum-rate is obtained when the BS finds a strong user with $\mu_1=0.41$ and $\rho_1=10$ (65% gain over the TDMA). Likewise, assuming a weak user with $\rho_2=0$ dB and $\mu_2=0.8$, the strong user should have $\mu_1=0.11$ and $\rho_1=100$ (378% gain over the TDMA which is selecting only the *weak* user). Given a *weak* user with a priority μ_2 , the additive sum-rate gain increases when the priority of the strong user (μ_1) decreases; since, in this case, we are able to find a strong user with the optimal SNR ratio with respect to the weak user (ξ_{TDMA}). Notice that the SNR of the strong user should be different for each value of μ_1 , for instance in Figure 51-left with $\mu_1=0.61$ and $\rho_2=10$ dB, the strong user must have $\rho_1=10 \times 1.95$ (linear scale). Nevertheless, the best strong user has a limitation on its maximum SNR, (68). Hence, we do not work with the optimal SNR ratio for certain conditions, hence, reducing the achievable additive sum-rate gain. The position of the peaks shown in Figure 51 satisfy

$$\mu_1^* = \mu_2 \frac{\log_2(1+\rho_2)}{\log_2(1+\nu)}, \quad \mu_{1,\min} = \frac{\mu_2}{\nu/\rho_2}, \quad (69)$$

where μ_1^* denotes the optimal priority for the *strong* user with SNR equal to ν , given a *weak* user with SNR equal to ρ_2 ; this can be derived from (63). Additionally, $\mu_{1,\min}$ stands for the lowest priority of the strong user that ensure simultaneously scheduling with the weak one. $\mu_{1,\min}$ is obtained from (60) by taking into account the limitation of ξ_{TDMA} in (68).

3.2.2.1.1.3 FDMA

FDMA assumes that the transmission is carried out in two orthogonal frequency slots and that the BS allocates its power among the different frequency bands in order to satisfy an energy (or sum-average power) constraint. In such a case, we have the following bitrates

$$B_{FDMA} = \begin{cases} R_1 \leq \alpha \log_2 \left(1 + \frac{\varphi_1 \xi \rho_2}{\alpha} \right) \\ R_2 \leq (1-\alpha) \log_2 \left(1 + \frac{(1-\varphi_1) \rho_2}{(1-\alpha)} \right) \end{cases}, \quad (70)$$

where α denotes the fraction of bandwidth allocated to MS₁ (*strong* user) and $(1-\alpha)$ is the portion of bandwidth devoted to MS₂ (*weak* user). Likewise, the variable φ_1 and $(1-\varphi_1)$ stand for the fraction of power allocated to MS₁ and MS₂, respectively.

The optimal power allocation for maximising the cost function J_{WSR} , (50), is obtained at

$$\varphi_1^* = \frac{\alpha}{\xi \rho_2} \frac{(1-\alpha)(\mu_1 \xi - \mu_2) + \mu_1 \xi \rho_2}{\mu_1 \alpha + \mu_2 (1-\alpha)} \quad (71)$$

Moreover, the optimal value for the bandwidth allocation with respect to α becomes the solution of

$$\frac{\partial J_{WSR}(\alpha, \xi, \rho_2, \varphi_1^*)}{\partial \alpha} = 0, \quad (72)$$

and its value is found by numerically evaluate

$$\alpha^* = \begin{cases} \mu_1 \log_2 \left(\frac{\mu_1 \mu_2}{\xi} f^2(\alpha) \right) + \frac{1}{\log(2)} \frac{\mu_2 (1 + \xi \rho_2) - \mu_1 (\xi + \xi \rho_2)}{\alpha + (1-\alpha) \xi + \xi \rho_2} - \log_2 \left(\frac{\mu_2}{\xi} f(\alpha) \right) = 0 \\ s.t. \quad f(\alpha) = \frac{\alpha + (1-\alpha) \xi + \xi \rho_2}{\mu_1 \alpha + \mu_2 (1-\alpha)} \end{cases}. \quad (73)$$

We know that only the *weak* user (MS₂) will be served when $\{\alpha^* = 0, \varphi^* = 0\}$. Hence, we get the connection between the parameters ξ and μ_1, μ_2, ρ_2 by solving (73), which will be denoted by the variable S_{weak} . On the other hand, the *strong* user (MS₁) is selected when $\{\alpha^* = 1, \varphi^* = 1\}$, and the values of the different variables that satisfy the previous condition are referenced by S_{strong} . Since it is not possible to get closed-form expressions, we define S_{weak} and S_{strong} as the ratio of SNR, ξ , that solves the following equations

$$\begin{aligned} S_{weak}(\mu_1, \mu_2, \rho_2) &= \{\xi\} \\ s.t. \quad \mu_1 \log_2 \left(\frac{\mu_1}{\mu_2} \right) - (\mu_2 - \mu_1) \log_2(\xi + \xi \rho_2) + \mu_2 \log_2(\xi) + \frac{1}{\log(2)} \frac{\mu_2 (1 + \xi \rho_2) - \mu_1 (\xi + \xi \rho_2)}{\xi + \xi \rho_2} &= 0 \\ S_{strong}(\mu_1, \mu_2, \rho_2) &= \{\xi\} \\ s.t. \quad \mu_2 \log_2 \left(\frac{\mu_1}{\mu_2} \right) - (\mu_2 - \mu_1) \log_2(1 + \xi \rho_2) + \mu_2 \log_2(\xi) + \frac{1}{\log(2)} \frac{\mu_2 (1 + \xi \rho_2) - \mu_1 (\xi + \xi \rho_2)}{1 + \xi \rho_2} &= 0 \end{aligned} \quad (74)$$

The WSR under optimal power allocation and bandwidth assignment is defined by

$$\psi = \begin{cases} \mu_2 \log_2(1 + \rho_2) & \text{if } \xi < S_{weak} \\ \mu_1 \alpha^* \log_2(\mu_1 f(\alpha^*)) + \mu_2 (1 - \alpha^*) \log_2\left(\frac{\mu_2}{\xi} f(\alpha^*)\right) & \text{if } S_{weak} \leq \xi \leq S_{strong} \\ \mu_1 \log_2(1 + \xi \rho_2) & \text{if } \xi > S_{strong} \end{cases} \quad (75)$$

where three different intervals can be distinguished; only the *weak* user, both users or only the *strong* user is served as a function of ξ , depending on S_{weak} and S_{strong} in (74). Additionally, notice that the function f is introduced in (73).

Likewise, the sum-rate that is obtained when the resource allocation has maximised the WSR, which is depicted in equation (50), is equal to

$$R_1 + R_2 = \begin{cases} \log_2(1 + \rho_2) & \text{if } \xi < S_{weak} \\ \alpha^* \log_2(\mu_1) + (1 - \alpha^*) \log_2\left(\frac{\mu_2}{\xi}\right) + \log_2(f(\alpha^*)) & \text{if } S_{weak} \leq \xi \leq S_{strong} \\ \log_2(1 + \xi \rho_2) & \text{if } \xi > S_{strong} \end{cases} \quad (76)$$

Figure 52 depicts the sum-rate that is obtained by the different schemes (FDMA, TDMA and BC) when each scheme maximises the WSR, J_{WSR} in (50). Notice that the maximum gain of the BC over the FDMA is met when this latter scheme serves only the *weak* user MS_2 where $\xi = S_{weak}$ is defined in (74). Figure 52 shows that such point is strictly lower than ξ_{TDMA} (see equation (64)), and hence, the gain attained by the BC over the FDMA is smaller than the gain obtained by the BC over the TDMA. When the FDMA starts serving simultaneously both users (when ξ belongs to the second interval of (76)), the sum-rate difference between BC and FDMA starts to decrease. Therefore, the served users under BC (compared with the FDMA) must satisfy the following ratio between SNRs to get the maximum sum-rate,

$$\xi_{FDMA} = S_{weak} \quad (77)$$

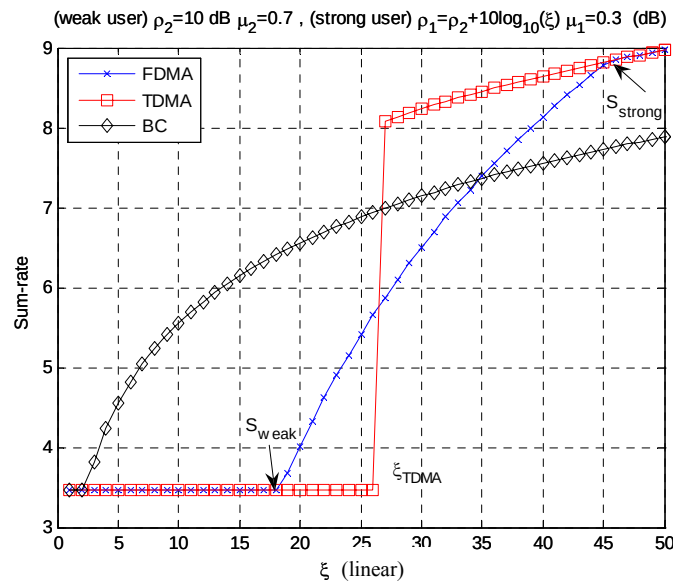


Figure 52: Sum-rate of BC, FDMA and TDMA as a function of ratio between SNRs of the served users (ξ)(linear) $\rho_2=10$ dB , $(\mu_1, \mu_2)=(0.3,0.7)$. ξ_{TDMA} introduced in (64) and S_{weak}, S_{strong} in (74).

Figure 52 has shown that TDMA is able to get a better sum-rate than the BC and FDMA under certain conditions. However, the achievable rate regions of the BC and FDMA are superior to the TDMA. Figure 53 and Figure 54 provide an explanation about this performance behaviour. Figure 53 sketches the achievable rate regions of the different transmission schemes for a configuration where the *weak* user has $\rho_2=10$ dB and the *strong* user has $\rho_1=\rho_2+10\log_{10}(\xi)$ dB, with $\xi=25$ in Figure 53-left and $\xi=40$ in Figure 53-right. These achievable rate regions have been obtained by maximising the WSR, J_{WSR} in (50), for different values of the user priorities (μ_1, μ_2) . In order to compare Figure 52 with Figure 53, we concentrate on the operational point that is described by $\mu_1=0.3, \mu_2=0.7$, as depicted in Figure 53. It can be observed that the sum-rate that is attained by TDMA can be better than the sum-rates that are attained by BC and FDMA, albeit the achievable rate regions of those schemes are higher than the TDMA. However, it must be emphasised that FDMA and BC schemes decide to work at this operational point because it maximises the WSR criterion. In the case that both BC and FDMA schemes decide to work at the same operational point as TDMA, the WSR at such a point will be smaller than the selected one. Figure 54 illustrates this performance.

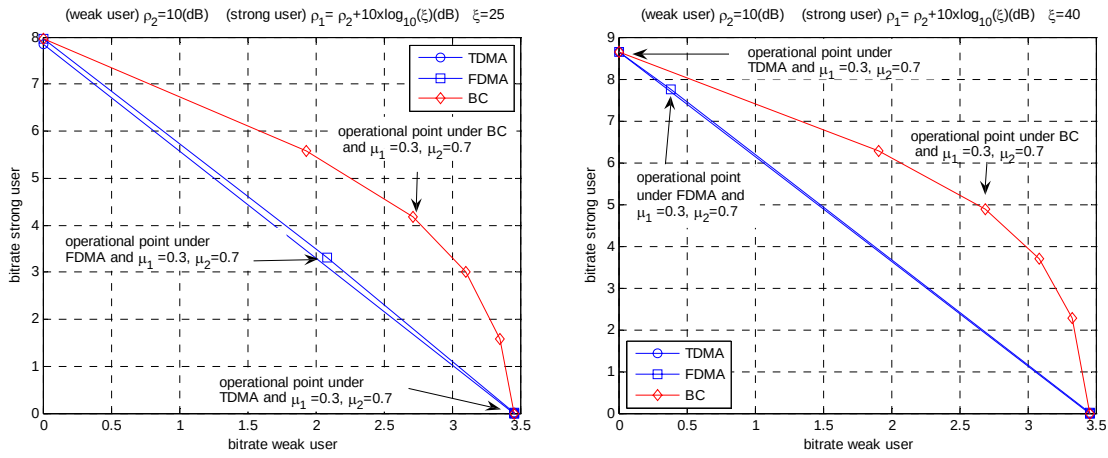


Figure 53: Achievable Rate regions of the TDMA, FDMA and BC for weak user equal to $\rho_2=10$ dB and a strong user with $\xi=25$ (left) and $\xi=40$ (right). The operational points attained due to the priorities $(\mu_1, \mu_2)=(0.3, 0.7)$ are shown for a comparison with Figure 52.

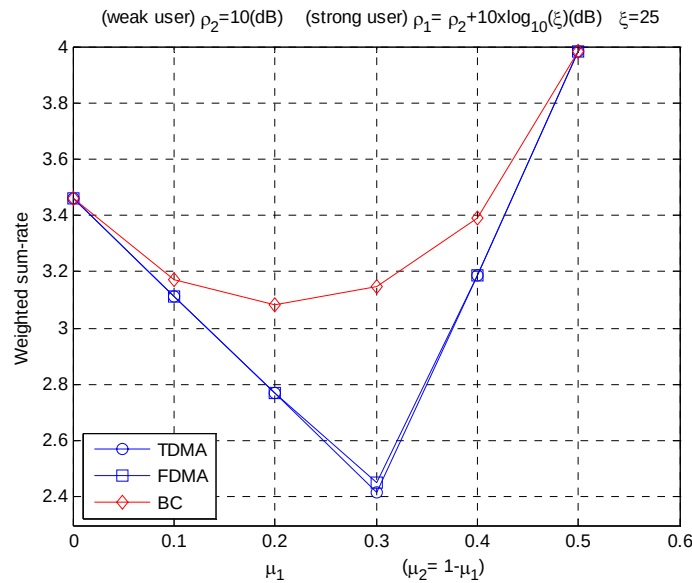


Figure 54: WSR, J_{WSR} in (50), for the TDMA, FDMA, and BC as a function of the priorities of the users. Weak user equal to $\rho_2=10$ dB and a strong user $\rho_1=\rho_2+10\log_{10}(\xi)$ with $\xi=25$.

The additive sum-rate gain of the BC over FDMA becomes

$$\Delta R = \begin{cases} 0 & \text{if } \frac{\mu_2}{\mu_1} > \xi \\ \log_2 \left(\frac{\mu_1}{\mu_2} \xi \right) & \text{if } \frac{\mu_2}{\mu_1} \leq \xi \leq S_{weak} \\ \log_2 \left(\frac{\mu_1}{\mu_2} \xi (1 + \rho_2) \right) - \alpha^* \log_2(\mu_1) - (1 - \alpha^*) \log_2 \left(\frac{\mu_2}{\xi} \right) - \log_2(f(\alpha^*)) & \text{if } S_{weak} \leq \xi \leq S_{strong} \\ 0 & \text{if } S_{strong} < \xi \end{cases} \quad (78)$$

The maximum value of the additive and percentage-wise sum-rate gains are defined by

$$\Sigma_{gain}^{FDMA} = \Delta R^*(\xi, \mu_1, \mu_2) \Big|_{\xi_{FDMA}}, \quad \Psi_{gain}^{FDMA} = \frac{\Delta R^*(\xi, \mu_1, \mu_2) \Big|_{\xi_{FDMA}}}{\log_2(1 + \rho_2)}. \quad (79)$$

Similar to the TDMA case, we assume that the maximum ratio is bounded because the SNR of the strongest user is limited to ν , consequently, the selected strong user must have a SNR equal to

$$\rho_1 = \min(\nu, \xi_{FDMA} \rho_2). \quad (80)$$

Figure 55 depicts the maximum aggregate sum-rate gain and percentage-wise sum-rate, (79), that are obtained by the BC over the FDMA as a function of the user priority of the strong user (μ_1) and for $\mu_1=0.8$ and $\rho_2=\{10,0\}$ dB. Comparing these results with the ones that have been obtained for the TDMA scheme in Figure 51, we observe two important properties:

- The gains in FDMA are lower than in TDMA, as it was already pointed out in Figure 52.
- The maximum gains are obtained for smaller values of user priority μ_1 in FDMA than in TDMA.

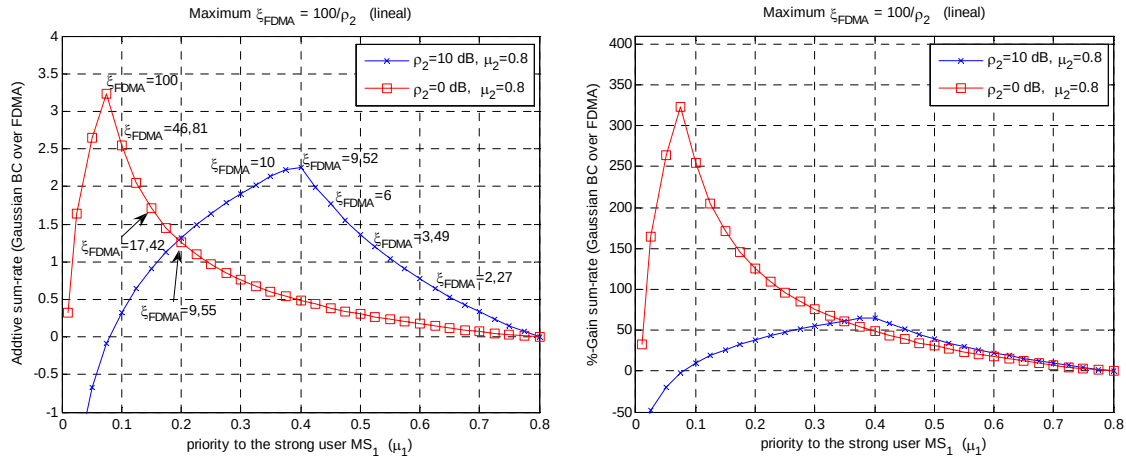


Figure 55: Left) Additive sum-rate and Right) Percentage-wise gain of the sum-rate of BC over FDMA (see (79)) as a function of μ_1 when the weak user presents $\rho_2=\{10,0\}$ dB and $\mu_2=0.8$. $\nu=100$ (20 dB).

3.2.2.1.1.4 Conclusions

In TDMA, all the resources are allocated to just a single, i.e., *strong* or *weak* user as a function of the user priority μ_1 (*strong*), μ_2 (*weak*), when maximisation of the *WSR* is performed,. In contrast, the BC and FDMA schemes are able to serve both users simultaneously. Whenever the *strong* user presents the proper user priority, all the schemes (TDMA, FDMA and BC) allocate all the resources to that user. Nevertheless, when the *weak* user must be served, several differences come up among the

transmission schemes. Whereas the TDMA scheme allocates all the resources to the *weak* user, the BC (and under some circumstances the FDMA) serves both users at the same time if the *strong* user satisfies certain conditions in terms of user priority and SNR.

After applying the resource allocation based on the WSR, the BC is able to get a better sum-rate than TDMA because it also serves the *strong* user, i.e., high bitrate, and at the same time the *weak* user, satisfying its user priority. Nevertheless, the bitrate devoted to the *weak* user is smaller than the one attained under TDMA.

The required condition on the ratio of SNRs in order to schedule two users simultaneously is

$$\frac{\mu_2}{\mu_1} \leq \xi \leq \xi_0 \quad \xi_0 = \begin{cases} \xi_{TDMA} \\ \xi_{FDMA} \end{cases} \quad (81)$$

Moreover, the maximum additive sum-rates that are achieved by the BC over the TDMA scheme, second interval of (66), and over the FDMA scheme, second interval of (78), are

$$\Delta R^* = \begin{cases} 0 & \mu_{strong} \geq \mu_{weak} \\ \log_2 \left(\frac{\mu_{strong}}{\mu_{weak}} \xi_0 \right) & \mu_{strong} < \mu_{weak} \end{cases}, \quad \xi_0 = \begin{cases} \xi_{TDMA} \\ \xi_{FDMA} \end{cases}, \quad (82)$$

where μ_{strong}, μ_{weak} denote the priority of the *strong* and *weak* users and ξ_{TDMA}, ξ_{FDMA} are given in (64) and (77).

3.2.2.1.2 Proportional fairness criterion

The PF criterion was introduced in [Kell-98] and it establishes a desired compromise between the throughput achieved by the system and the *fairness* or opportunities among the different users to have access to the resources of the system.

A scheduling algorithm P is *proportionally fair* if and only if, for any feasible scheduling set S , it satisfies

$$\sum_{i \in U_S} \frac{R_i^{(S)} - R_i^{(P)}}{R_i^{(S)}} \leq 0, \quad (83)$$

where U_S denotes the user set, $R_i^{(S)}$ is the average rate of user i belonging to the set S and $R_i^{(P)}$ stands for the average rate of user i that is considered in the optimal scheduling set P .

[Kell-98] shows that using the following utility function for the resource allocation (selection of users) the proportional fair criterion (83) is satisfied

$$P = \arg \max_S \sum_{i \in U_S} \log R_i^{(S)}. \quad (84)$$

Additionally, [Mo-00] generalised the previous expression and derived a PF criterion that subsumes the *max-min* and *maximum sum-rate* criteria.

The total average rates of selected users and those of users who are not selected are updated according to the following expression,

$$\bar{R}_i = \begin{cases} (1-\tau)\bar{R}_i + \tau R_i & \text{if } i \in U_S \\ (1-\tau)\bar{R}_i & \text{otherwise} \end{cases}, \quad (85)$$

where R_i denotes the instantaneous bitrate, $\bar{R}_i$ stands for the average throughput served to i -th user and τ is a parameter that controls the exponentially decaying window of (85).

When the number of users to be selected at each time is just one, $|U_S|=1$, it must satisfy

$$P_{Qualcomm} = \arg \max_S \frac{R_i}{\bar{R}_i} = \arg \max_S \mu_i R_i. \quad (86)$$

This criterion is employed in Qualcomm's HDR (High Data Rate) system where it is used to adjust the scheduler according to the parameter τ , (85), for getting the *MU gain* (selecting the user with the best instantaneous channel, $\tau \rightarrow 0$) or imposing *fairness* (resources are distributed fairly across the users, $\tau \rightarrow 1$), [PVis-02]. Likewise, the analytical model of the average throughput that is attained by this scheduler is investigated in [Choi-07],[Liu-08].

On the other hand, when there are multiple users to be served, the criterion can be easily derived from [HKim-05], where the proportional fair criterion is derived for a multicarrier system. It turns out to be

$$P = \arg \max_S \sum_{i \in U_S} \log \left(1 + \frac{\tau R_i}{(1-\tau) \bar{R}_i} \right). \quad (87)$$

When two users have to be scheduled $|U_S|=2$, then

$$P = \arg \max_{S, i, j \in U_S} \left(\log \left(1 + \frac{\tau R_i}{(1-\tau) \bar{R}_i} \right) + \log \left(1 + \frac{\tau R_j}{(1-\tau) \bar{R}_j} \right) \right), \quad (88)$$

where S stands for the best user selection, i, j are the users in the user set U_S , R_i, R_j denote the instantaneous rate of each user and they depend on the considered transmission scheme (BC, TDMA or FDMA), $\bar{R}_i, \bar{R}_j$ stand for the average throughput that is served, which is updated according to (85).

The metric derived in (88) imposes that for each pair of users of the system we must optimise their resources according to

$$\begin{aligned} J_{PF} &= \log \left(1 + \frac{\tau R_1}{(1-\tau) \bar{R}_1} \right) + \log \left(1 + \frac{\tau R_2}{(1-\tau) \bar{R}_2} \right) = \\ &= \log \left(1 + \frac{\tau}{(1-\tau)} \mu_1 R_1 \right) + \log \left(1 + \frac{\tau}{(1-\tau)} \mu_2 R_2 \right) \quad , \\ s.t. \quad &(R_1, R_2) \in B_x \end{aligned} \quad (89)$$

where B_x stands for the achievable rate region of the considered transmission scheme, R_1, R_2 are the instantaneous bitrate of the pair of users and $\bar{R}_1, \bar{R}_2$ denote the average served throughput. The scheduling algorithm will select the pair of users with the highest metric (88).

3.2.2.1.2.1 BC versus TDMA

Given two users, we study the benefits of superposed transmission when using the BC under the *proportional fair* metric in (89) (it will be referenced as 2-user PF scheduler) over the TDMA orthogonal transmission, which was introduced in section 3.2.2.1.1.2, and it coincides with the criterion employed by Qualcomm scheduler, (86).

We consider a *weak* user with a SNR equal to ρ_2 and a *strong* user with SNR equal to $\rho_1 = \xi \rho_2$ ($\xi \geq 1$). In contrast to section 3.2.2.1.1.1, the resource allocation must be done as a function of the user's priorities μ_1, μ_2 and the parameter that is used for updating the average served throughput, τ , (85). Figure 56 presents the sum-rate obtained by the two users as a function of the parameter ξ , where the *weak* user has a larger priority than the *strong* user, $\mu_1 < \mu_2$. These results are compared with the sum-

rate performance when only the *strong* or *weak* user is selected. We observe that for certain configurations, the BC scheme allocates all the resources to the *weak* user. Additionally, the minimum value of ξ that is required for having two users simultaneously allocated along with the total sum-rate are dependent on the parameter τ .

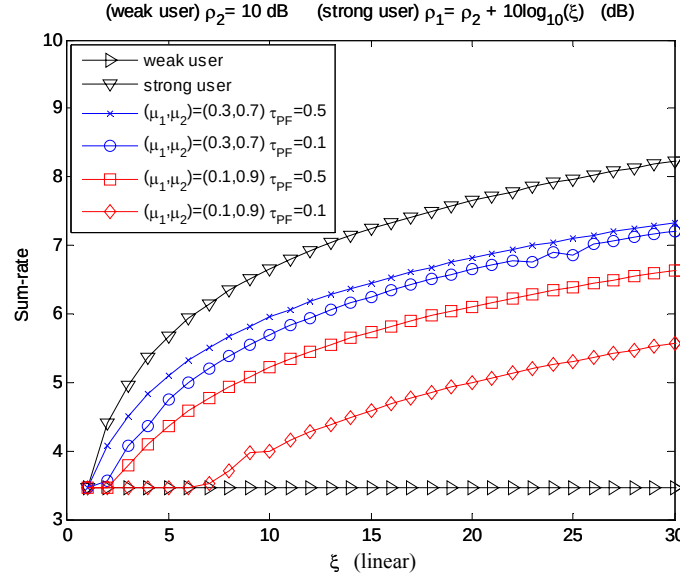


Figure 56: Sum-rate attained by the BC using the PF criterion, (89) as a function of ratio of SNRs between the strong and weak users (ξ), for , $(\mu_1, \mu_2)=\{(0.3,0.7), (0.1,0.9)\}$, $\tau=\{0.5,0.1\}$, (85). $\rho_2=10$ dB.

Taking into account the achievable rate region of the BC that has been presented in (51), the optimal power allocation will be obtained by maximising the proportional fair criterion that is presented in (89)

$$\begin{aligned} \frac{\partial J_{PF}}{\partial \gamma_1} &= \mu_1 \frac{\rho_1}{1 + \gamma_1 \rho_1} - \mu_2 \frac{\rho_2}{1 + \gamma_1 \rho_2} + \mu_1 \mu_2 \left(\frac{\rho_1}{1 + \gamma_1 \rho_1} \log_2 \left(\frac{1 + \rho_2}{1 + \gamma_1 \rho_2} \right) - \frac{\rho_2}{1 + \gamma_1 \rho_2} \log_2 (1 + \gamma_1 \rho_1) \right) = 0 \\ &= \mu_1 \rho_1 (1 + \gamma_1 \rho_2) \left(1 + \mu_2 \log_2 \left(\frac{1 + \rho_2}{1 + \gamma_1 \rho_2} \right) \right) - \mu_2 \rho_2 (1 + \gamma_1 \rho_1) (1 + \mu_1 \log_2 (1 + \gamma_1 \rho_1)) = 0 \end{aligned} \quad (90)$$

From the previous equation, we are able to establish the conditions for satisfyingly serving the *weak* user only, i.e. when $\gamma_1=0$, without considering the *strong* user

$$\rho_1 \leq \frac{\mu_2}{\mu_1} \frac{\rho_2}{1 + \frac{\tau}{1-\tau} \mu_2 \log_2 (1 + \rho_2)} \quad (91)$$

This result leads to define the minimum ratio between SNRs required for serving the two users simultaneously under BC with the PF criterion

$$\xi_{\min} = \frac{\rho_1}{\rho_2} = \frac{\mu_2}{\mu_1} \frac{1}{1 + \frac{\tau}{1-\tau} \mu_2 \log_2 (1 + \rho_2)} \quad (92)$$

When we compare the minimum value of ξ under the PF criterion with the one obtained with the *WSR*, $\xi = \mu_2 / \mu_1$ in (60), we observe that the minimum ξ is smaller with the PF, because the second term of (92) is always lower than one. That means that there are more opportunities to serve two users under this criterion than under the *WSR* criterion.

The condition to be satisfied for allocating all the resources to the *strong* user ($\gamma_1=1$) becomes

$$\rho_2 = \frac{\rho_1}{\frac{\mu_2}{\mu_1}(1+\rho_1)\left(1+\frac{\tau}{1-\tau}\mu_1\log_2(1+\rho_1)\right)-\rho_1}. \quad (93)$$

Hence, under certain circumstances, the *strong* user will not be served alone, and the sum-rate that is obtained by the BC will increase with the SNR of the strong user, see Figure 56. This performance is similar to the one that is obtained by using the *WSR* (see Figure 52). In such a case, the benefit in terms of sum-rate of the BC over the TDMA only is attained when BC serves the two users while the TDMA scheme selects just the *weak* user. Therefore, in order to get a better sum-rate than with the TDMA scheme and to select both users at the same time, the ratio of SNRs should be

$$\xi_{\min} = \frac{\mu_2}{\mu_1} \frac{1}{1+\frac{\tau}{1-\tau}\mu_2\log_2(1+\rho_2)} \leq \xi \leq \xi_{TDMA} = \frac{(1+\rho_2)^{\frac{\mu_2}{\mu_1}}-1}{\rho_2}, \quad (94)$$

where ξ_{TDMA} is defined in (64).

3.2.2.2 Cellular throughput evaluation

After elucidating the benefits of the superposed transmission in the DL for the two-user case in section 3.2.2.1 under different criteria (*WSR* and *proportional fair*), this section investigates the throughput gains that can be obtained in the DL of a cellular scenario. We have seen that the ratio of SNRs for a pair of user should satisfy (81) for the *WSR* and (94) under *PF* criteria.

We assume that the user priorities, or weight factors, which are considered in previous sections, are inversely proportional to the served throughput per user,

$$\mu_i = \frac{1}{\bar{R}_i} \quad (95)$$

where $\bar{R}_i$ is the average served throughput of the i -th user and is measured following equation (85), depending on the parameter τ .

The transmissions are organised in frames. In each frame, the channel coefficients remain constant and the BS decides how to distribute the resources, i.e. how many and which users are selected along with the power allocation. We assume three different *schedulers*,

$$\begin{aligned} \text{(Qualcomm)} \quad [k^*] &= \arg \max_k \mu_k R_k & k \in [1, N_u] \\ \text{(Weighted sum-rate)} [k^*, j^*] &= \arg \max_{k,j,k \neq j} \mu_k R_k + \mu_j R_j & k, j \in [1, N_u], (96) \\ \text{(2-user PF)} \quad [k^*, j^*] &= \arg \max_{k,j,k \neq j} \log\left(1+\frac{\tau}{1-\tau}\mu_k R_k\right) + \log\left(1+\frac{\tau}{1-\tau}\mu_j R_j\right) & k, j \in [1, N_u] \end{aligned}$$

where N_u denotes the number of users in the cell. The Qualcomm scheduler, (86), selects just one user, while the *WSR* and *proportional fair* can serve two users simultaneously with superposed transmissions (BC). It is important to remark that the two latter schedulers must evaluate all the possible combinations of user pairing among the users to be served. In section 3.2.2.1.1.1, we show a closed-form expression for the *WSR* of two users with the optimal power allocation in equation (58), which is reproduced in the following

$$J_{WSR} = \begin{cases} \mu_2 \log_2(1 + \rho_2) & \text{if } \xi < \frac{\mu_2}{\mu_1} \\ \mu_1 \log_2\left(\frac{\mu_1(\xi - 1)}{\mu_2 - \mu_1}\right) + \mu_2 \log_2\left(\frac{\mu_2 - \mu_1}{\mu_2} \frac{\xi(1 + \rho_2)}{\xi - 1}\right) & \text{if } \frac{\mu_2}{\mu_1} \leq \xi \leq \xi_{strong} \\ \mu_1 \log_2(1 + \xi \rho_2) & \text{if } \xi > \xi_{strong} \end{cases} \quad (97)$$

where μ_1, μ_2 are the priorities of the *strong* and *weak* user for a given pair of users, ρ_2 stands for the SNR of the *weak* user, ξ denotes the ratio between the SNR of the two users and finally ξ_{strong} is introduced in (56). Therefore, the WSR for all the combinations can be easily calculated by using (97). On the other hand, no closed-form expression exists for the *proportional fair* criterion (2-user PF scheduler). This implies that for each combination of two users the optimisation of the resources must be carried out. Although we have derived a condition in (94) to be satisfied by the selected users, (94), under the BC, there is still a large number of combinations to be tested. Hence, in order to reduce that number, we will consider a list of L candidates to be scheduled according to the following criterion

$$\Gamma = \{k \in [1, N_u] \mid \mu_k R_k \geq \eta\}, \quad |\Gamma| = L, \quad (98)$$

where μ_k and R_k are the priority and the bitrate of the k -th user when assuming that the BS allocates all the power to this user. The L users with the best metric $\mu_k R_k$ are selected according to the threshold η . In this way, we consider the priority of each user and the current channel conditions, following the same approach as the one tackled in [HVis-03]. Hence, the 2-user PF scheduler introduced in (96) has only to consider the candidate users present in the list Γ

$$\text{(2-user Proportional Fair)} \quad [k^*, j^*] = \arg \max_{k, j, k \neq j} \log\left(1 + \frac{\tau}{1 - \tau} \mu_k R_k\right) + \log\left(1 + \frac{\tau}{1 - \tau} \mu_j R_j\right) \quad k, j \in \Gamma. \quad (99)$$

We assume a circular cell with a radius equal to 200 meters and where the base BS is placed at the centre of the cell. There are N_u users (MSs) uniformly distributed over the cell. The SNR depends on the distance between the BS and each MS, taking into account a propagation exponent equal to 3. The path-loss model has been adjusted for an average SNR equal to 0 dB when the MS is placed at the edge of the cell and a maximum average SNR equal to 30 dB when it is close to the BS (distance less or equal to 10 meters). The channel coefficients have a Rayleigh distribution of zero-mean and unit-variance. The simulation considers 300 different user deployments and for each one the scheduling algorithms are run during 10^4 frames; the channel coefficients are constant during a frame and independent among users.

The aspects analysed are:

- Length of the list Γ (L) for the 2-user PF scheduler
- Comparison of the WSR and PF schedulers (2-user case)
- Dependence on the number of users present in the cell (2-user case)
- WSR and PF schedulers when up to 3 users are served simultaneously

The results obtained by the schedulers working under the BC (WSR and PF) will be compared with the Qualcomm scheduler. The considered metrics for that comparison are

- Throughput: The throughput of each user in each deployment is calculated after the simulation of 10^4 frames. As a consequence, a variable with $300 \times N_u$ throughput values is outputted at the end of the simulation.
- Cell Throughput: It is the sum-throughput delivered by the BS to all users present in a given deployment; a variable with 300 cell throughput values is outputted.

- Delay: Delay has been defined in this work as the average number of frames required by a user for being scheduled. During each simulation of 10^4 frames, one user is scheduled in different time instances. In other word, we define the delay as the average number of frames elapsed between two consecutive selection instants of the same users.

- *Length of the list for the 2-user proportional fair scheduler*

Figure 57 presents the Cumulative Density Function (CDF) of the throughput (Figure 57-left) and CDF of the delay (Figure 57-right) that is experienced by a user under the Qualcomm scheduler (or single-user PF, (96)) and the 2-user PF scheduler, (99). Both algorithms assign the user priority according the inverse of the average served throughput, (95), employing the parameter $\tau=0.05$. The 2-user PF scheduler selects two users from the subset of L -users with highest individual weighted rates, (98). On the other hand, the delay has been measured as the average number of frames required for a user to be selected by the scheduler. The results, which are depicted in Figure 57, show that the 2-user PF scheduler outperforms the Qualcomm algorithm in terms of throughput and delay. The 2-user PF scheduler gets a better throughput, which improves with the number of candidates in the list. For instance, there is a 2% probability (CDF at 98%) of having throughputs higher than 0.5 bits/s/Hz with $L=9$, while with $L=2$, the throughput is higher than 0.4 bits/s/Hz, but it does not reach 0.5 bits/s/Hz. Regarding the delay (Figure 57-right), the benefits of the 2-user PF scheduler are outstanding. The delay reduces approximately by one half. The delay also is improved with the number of candidates in the list. For example by using $L=9$ there is a 10% probability (CDF at 10%) of having a delay smaller than 11 frames, while for $L=3$ or $L=2$ such probability is zero. Nevertheless, there is a high probability of slightly increasing (less than one frame) the delay when $L=9$. The results in terms of throughput and delay suggest that using a list with $L=5$ or $L=9$ candidates seems to be good enough.

Figure 58 depicts the CDF of the throughput and delay, as in the previous figure, but this time for $\tau=0.7$. This parameter controls how the average served throughput is adjusted for each user, (85), and it has influence on the performance of the scheduler [PVis-02]. Notice that the throughput per user that is obtained for $\tau=0.7$ (Figure 58) is smaller than when using $\tau=0.05$ (Figure 57), and the users tend to have a constant delay. For $\tau=0.7$, the length of the candidate list L seems to be of lesser importance than for smaller values of τ .

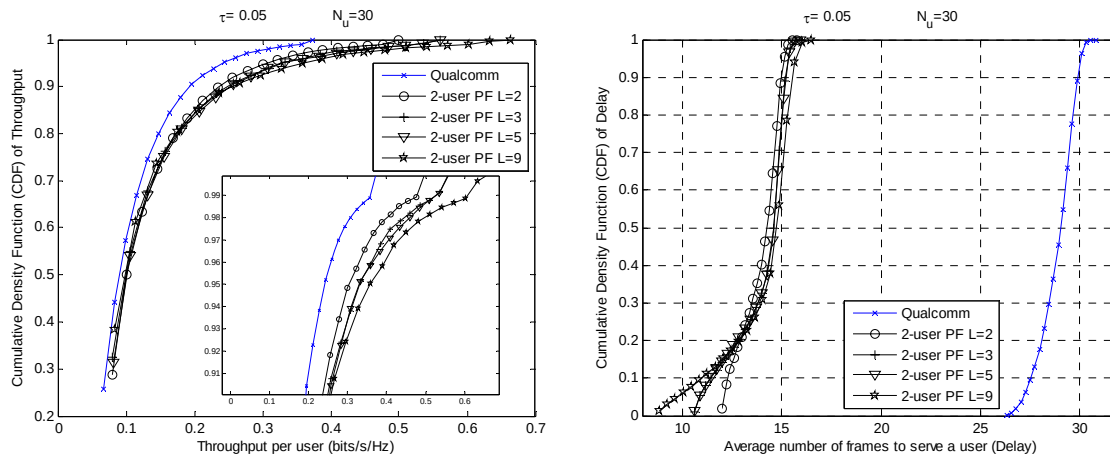


Figure 57: CDF: User throughput (left) and per-user delay (right) for the Qualcomm scheduler (single-user PF) and 2-user PF scheduler with $L=\{2,3,9\}$ candidates. $\tau=0.05$.

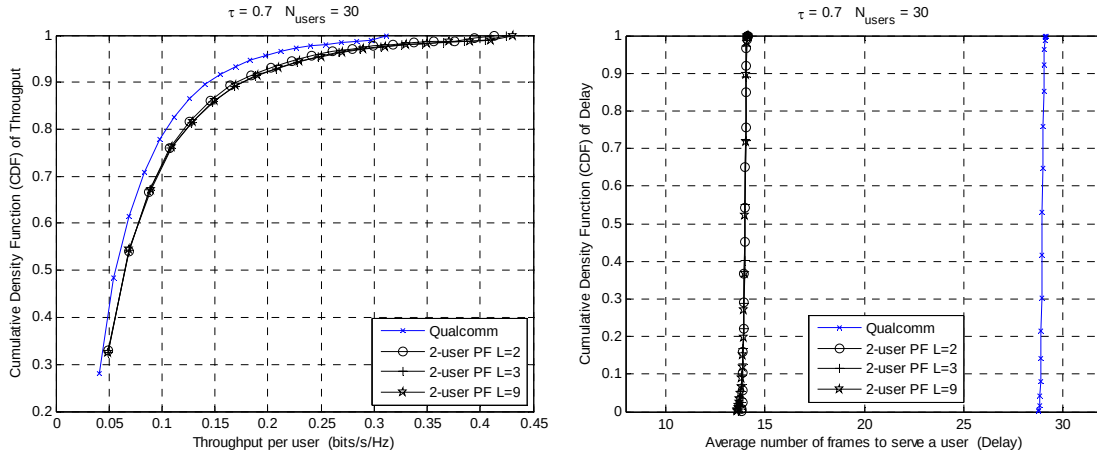


Figure 58: CDF: User throughput (left) and per-user delay (right) for the Qualcomm scheduler (single-user PF) and 2-user PF scheduler with $L=\{2,3,9\}$ candidates. $\tau=0.7$.

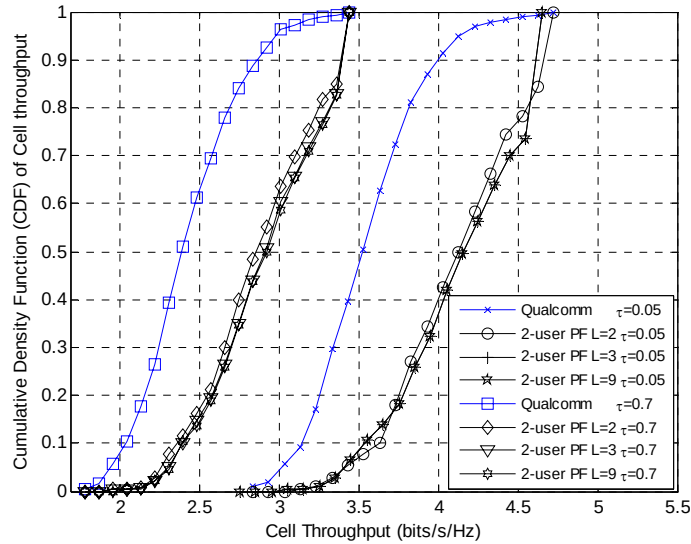


Figure 59: CDF of the Cell Throughput for the Qualcomm scheduler (single-user PF) and 2-user PF scheduler with $L=\{2,3,9\}$ candidates. $\tau=\{0.05, 0.7\}$.

The previous figures have shown that the throughput per user is improved by using the 2-user PF scheduler. Although those gains seem insignificant, they have an important impact on the total cell throughput, i.e. the sum-throughput of the N_u users in the cell. Figure 59 presents the CDF of cell throughput for the Qualcomm and 2-user PF algorithms with $\tau=\{0.05, 0.7\}$. Results show that for a probability of 80% (CDF at 20%), the Qualcomm scheduler gets a cell throughput that is higher than 2.1 bits/s/Hz and 3.3 bits/s/Hz at $\tau=\{0.05, 0.7\}$, respectively. In contrast, the 2-user PF algorithm attains 2.55 bits/s/Hz and 3.75, i.e., gains of around 22% and 14% over the Qualcomm scheduler, respectively.

- *WSR vs. 2-user PF schedulers*

Figure 60 compares the performance of the three schedulers (Qualcomm, WSR and the 2-user PF) as a function of the parameter τ when there are $N_u=30$ users. The results are presented in terms of average throughput per user (Figure 60-left) and average delay per user (Figure 60-right). We observe that the 2-user PF scheduler provides the best average throughput per user, except for $0.002 \leq \tau \leq 0.05$, where the WSR average throughput per user attains a slightly better performance. However, the WSR scheduler provides similar throughput results than the Qualcomm algorithm for $0.05 \leq \tau \leq 1$, but worse results for

$10^{-5} \leq \tau \leq 10^{-4}$. On the other hand, the 2-user PF scheduler obtains the lowest average delay (Figure 60-right), irrespective of the parameter τ . The average delay is around half of the average delay that is obtained with the Qualcomm algorithm. It is important to emphasise that the average delay is not calculated for certain values of τ because the different schedulers are not able to serve all the users in this case. Table 3 shows the average number of users served by each scheduler. The Qualcomm schedulers tend to serve a smaller number of users as τ goes to zero, but the WSR and the 2-user PF are able to increase this number. Nevertheless, the WSR algorithm cannot provide service to all the users for $0.1 < \tau < 0.5$. Therefore the parameter τ controls the delivered throughput (*MU gain*) and the fairness among the users (delay). From Figure 60, we could work with $\tau = 10^{-3}$, because the average delay is almost constant for higher values of τ .

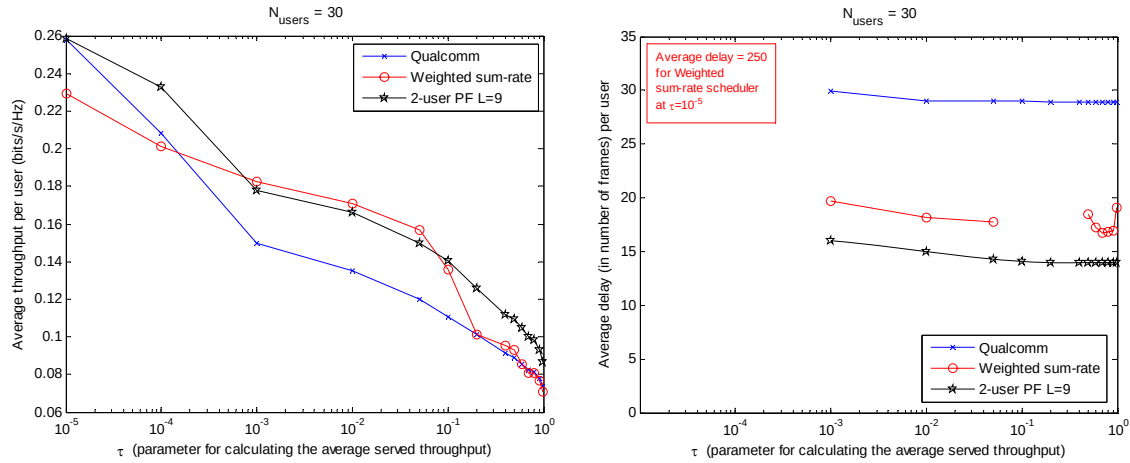


Figure 60: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u = 30$.

Table 3: Average number of served users for the different schedulers (Qualcomm, WSR and 2-user PF)

Average number of served users														
τ	10^{-5}	10^{-4}	10^{-3}	0,05	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	0,99
Qualcomm	13,71	28,38	30	30	30	30	30	30	30	30	30	30	30	30
W sum-rate	30	29,99	30	30	30	29,98	29,88	29,95	30	30	30	30	30	30
2-user PF	29,64	29,99	30	30	30	30	30	30	30	30	30	30	30	30

We have observed in Figure 57 and Figure 58 the CDF of the throughput and delay per user for certain values of τ . We next show the performance in terms of outage throughput and outage delay events defined as

$$\begin{cases} T^{out} = \arg \max_t \Pr(T \geq t) & s.t. \quad \Pr((T \geq t)) \leq (1 - p_{out}) \\ D^{out} = \arg \max_d \Pr(D \geq d) & s.t. \quad \Pr((D \geq d)) \leq (1 - p_{out}) \end{cases} \quad (100)$$

where T , D denote the throughput and delay (in terms of average number of frames), which are random variables. For instance, the CDF of those variables under the Qualcomm and 2-user PF schedulers under $\tau = \{0.05, 0.7\}$ are depicted in Figure 57 and Figure 58. The variable p_{out} stands for the *outage probability*.

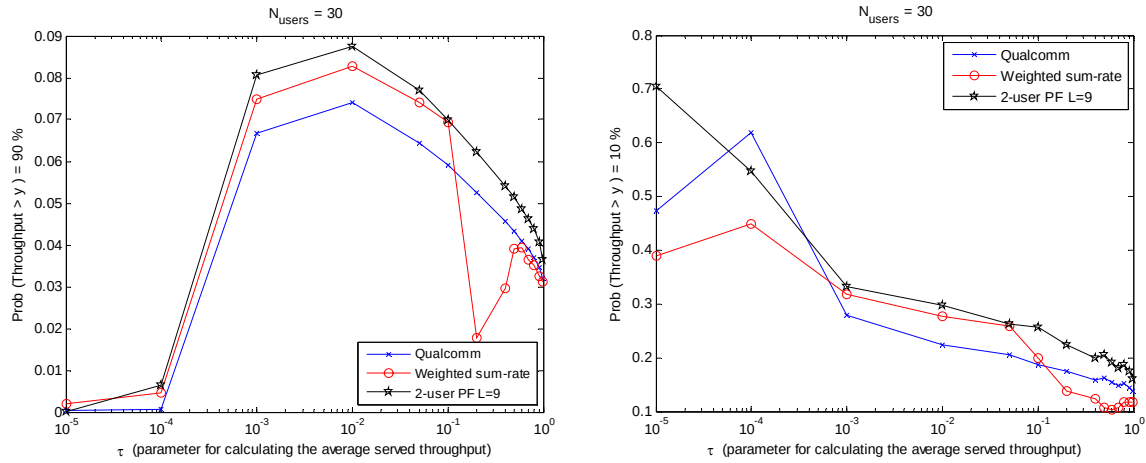


Figure 61: Left) 10%-outage throughput ($p_{out}=0.1$, (100)) and Right) 90%-outage throughput ($p_{out}=0.9$) as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=30$.

Figure 61-left and Figure 61-right present the 10%-outage throughput and 90%-outage throughput, respectively, that are attained by the different schedulers as a function of τ . The results in the first figure indicate that any given user has a 90% probability ($p_{out}=10\%$) of achieving a throughput higher than the one shown in Figure 61-left. The second figure shows that any user has a 10% probability ($p_{out}=90\%$) of getting throughput values better than those depicted in Figure 61-right. In other words, the probability that the throughput is below the values shown in the figure is equal to 90%. These values can be considered as *lower* and *upper bounds* of the throughput in practical systems.

It can be seen that the 10%-outage throughput goes to zero as τ goes to zero, whereas the 90%-outage throughput increases. This performance can be explained by the observations that were presented in [PVis-02], for low values of τ ; the users that are close (with good channel conditions) to the BS are more often selected by the scheduler (large values of the 90%-outage throughput), whereas the other users are rarely served, which implies a reduction of the throughput per user. Likewise, the scheduler can increase the numbers of selected users (i.e. close to median distance to BS) by moderately incrementing τ , and hence, it can increase the statistical minimum throughput of a user. However, after a certain point (the maximum shown in Figure 61-left at $\tau=0.01$), the scheduler serves many users, some of them far from the BS and they get a low throughput. In such a case, the users that are close to the BS (with good channel conditions) have fewer opportunities to be selected and then the 90% outage throughput decreases. The WSR scheduler average throughput performance changes for $\tau \geq 0.2$ (Figure 60), where it tends to be similar to the Qualcomm scheduler.

The 2-user PF scheduler gets the best performance for the 10%-outage throughput and almost the best for the 90%-outage throughput. Hence, the 2-user PF algorithm has the best lower bound, upper bound and average throughput (Figure 60). Notice that the parameter τ is used to select the desired performance.

Figure 62 presents the outage values of the delay per user as a function of τ . Figure 62-left shows the 90% ($p_{out}=0.1$) of the cases where the delay is superior to the values depicted there (some kind of *lower bound*), whereas Figure 62-right considers the 10% ($p_{out}=0.9$) of the cases, i.e. the 90% of the cases the delay per user is upper bounded by Figure 62-right. The 2-user PF scheduler has the lowest upper bound, whereas its lower bound is slightly worse than the one of the WSR scheduler. However, the average delay is better for the 2-user PF scheduler, Figure 60-right.

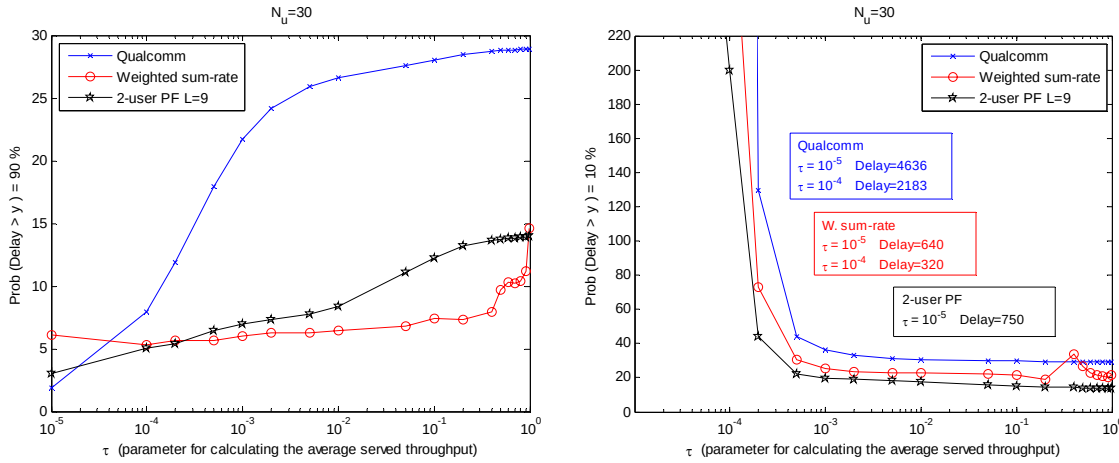


Figure 62: Left) 10%-outage delay ($p_{out}=0.1$, (100)) and Right) 90%-outage delay ($p_{out}=0.9$) as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=30$.

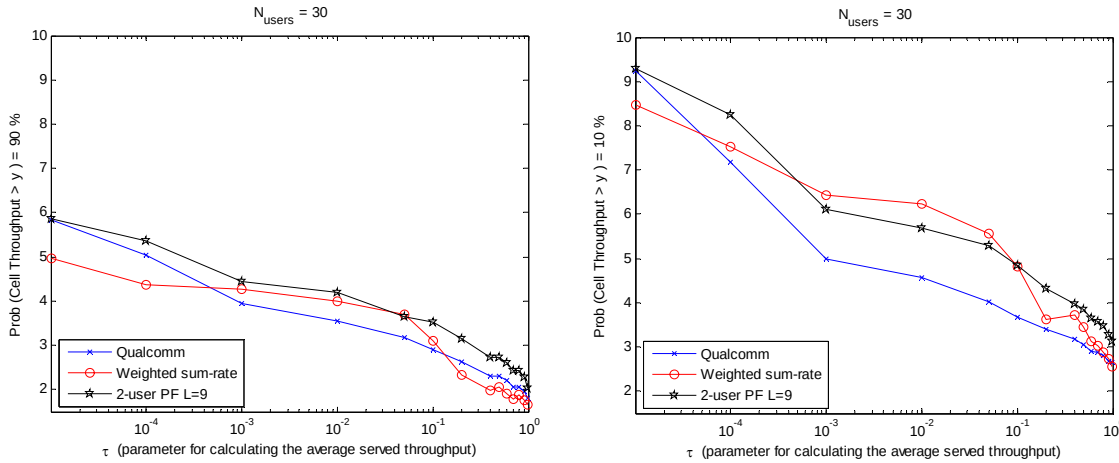


Figure 63: Left) 10%-outage cell throughput ($p_{out}=0.1$, (100)) and Right) 90%-outage cell throughput ($p_{out}=0.9$) as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=30$.

Finally, we plot the outage cell throughput as a function of τ in Figure 63, by using (100). The average cell throughput has not been presented here, but we have observed that under the configuration that has been analysed in this section, it is approximately equal to the average throughput per user multiplied by number of users in the cell, N_u . The performance follows similar trends as the average throughput per user, shown in Figure 60-left. Moreover, Figure 63-left shows that a high cell throughput is achieved for small values of τ , in spite of having a low 10%-outage throughput per user (Figure 61-left). In this case, the users that are close to the BS are more often selected, and hence, they contribute to the cell throughput.

- *Dependence on the number of users in the cell (2-user case)*

The number of users in the cell is increased up to $N_u=90$ in Figure 64, where the average throughput and average delay per users is depicted as a function of the parameter τ in Figure 64-left and Figure 64-right, respectively. The average throughput performance of the different schedulers is similar to the one that has been obtained for $N_u=30$ in Figure 60-left, where the 2-user PF scheduler always outperforms the Qualcomm scheduler; whereas the WSR scheduler can provide better performance than the 2-user PF one for certain values of τ ($0.002 \leq \tau \leq 0.001$), but its performance is equal or even worse than the Qualcomm scheduler for other τ values.

Figure 64-right presents the average delay per user. We can observe that the average delay encompassed by the 2-user PF scheduler is nearly half of the delay due to the Qualcomm scheduler and the average delay that is obtained with the WSR scheduler is worse than the 2-user PF one. Moreover, notice that the average delay is not depicted for $10^{-5} \leq \tau \leq 5 \times 10^{-4}$ because not all schedulers can afford serving all the users of the cell ($N_u=90$) in this case. Additionally, the WSR algorithm cannot serve all the users for $5 \times 10^{-2} \leq \tau \leq 1$, whereas the other schedulers can do it.

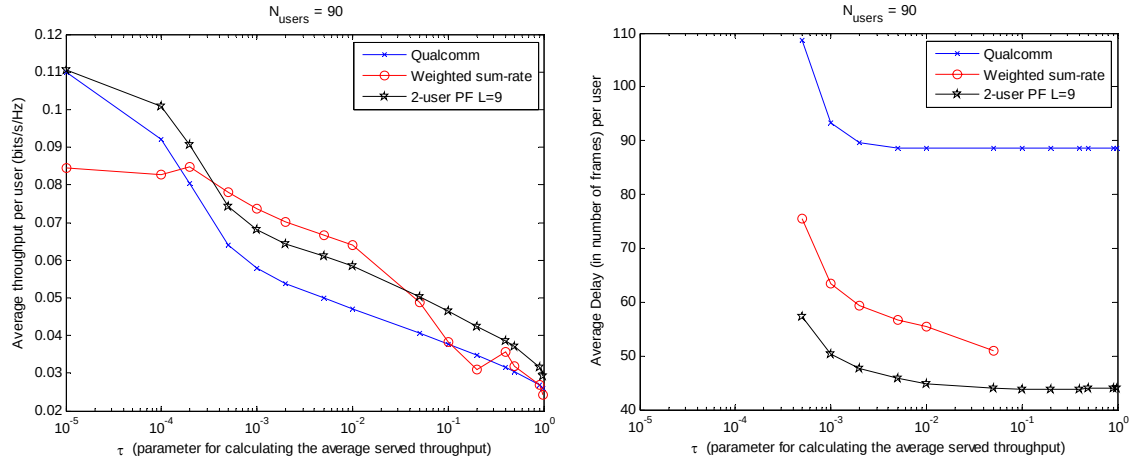


Figure 64: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, WSR and 2-user PF. $N_u=90$.

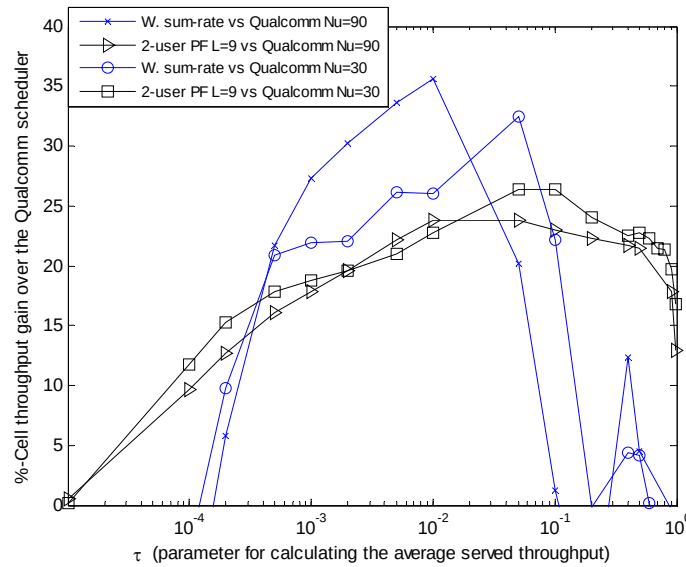


Figure 65. Percentage-wise cell throughput gain of the weighed sum-rate and 2-user PF over the Qualcomm scheduler as a function of the parameter τ , (85). $N_u=\{30,90\}$.

Figure 60-left and Figure 64-left shows that the average throughput that is attained by the 2-user PF scheduler presents a similar shape as the Qualcomm scheduler average throughput, except when $\tau \rightarrow 0$ and $\tau \rightarrow 1$, where both scheduler results coincide. In order to characterise the values of τ where the proposed MU schedulers enhance the Qualcomm algorithm, Figure 65 depicts the %-cell throughput gain that is obtained by the WSR and 2-user PF schedulers over the Qualcomm scheduler as a function of τ . This figure shows that both schedulers have the same %-cell throughput gains over the Qualcomm algorithm irrespective of the number of users, N_u . Moreover, the 2-user PF scheduler always outperforms the Qualcomm algorithm, such that a maximum gain close to 25% is reached. In contrast, the WSR scheduler seems to be superior to the Qualcomm one over the interval $10^{-4} \leq \tau \leq 10^{-1}$, where its maximum gain is around 35%. However, this gain for the WSR implies a larger average

delay than the one obtained by the 2-user PF scheduler (see Figure 60-right and Figure 64-right). Likewise, this gain is due to the fact that the WSR algorithm evaluates all the combinations, whereas the PF scheduler just considers a list of L candidates.

- *WSR and PF scheduler for selecting three users simultaneously*

We have demonstrated the benefits of scheduling two users simultaneously, either by using *WSR* or *2-user PF* schedulers in the previous subsections. We extend here the *WSR* and *PF criteria* schedulers for the 3-user case and analyse their performances. The schedulers are here based on the following criteria

$$\begin{aligned}
 (3\text{-user WSR}) [k^*, j^*, m^*] &= \arg \max_{k, j, m, k \neq j \neq m} \mu_k R_k + \mu_j R_j + \mu_m R_m \quad k, j, m \in \Gamma \\
 (3\text{-user PF}) [k^*, j^*, m^*] &= \arg \max_{k, j, m, k \neq j \neq m} \left\{ \log \left(1 + \frac{\tau}{1-\tau} \mu_k R_k \right) + \right. \\
 &\quad \left. + \log \left(1 + \frac{\tau}{1-\tau} \mu_j R_j \right) + \log \left(1 + \frac{\tau}{1-\tau} \mu_m R_m \right) \right\} \quad k, j, m \in \Gamma
 \end{aligned} \tag{101}$$

where Γ denotes a list of L candidates to be scheduled according to (98) and the bitrates of the different users must satisfy the achievable rate region of the BC for 3 users [Cove-91]. We have considered a list of $L=5$ candidates to be selected for both schedulers, because we have not derived any rule to evaluate all possible combinations for the *WSR* criterion, as it was done for the 2-user case in (97).

Figure 66 and Figure 67 depict the average throughput and delay per user as a function of the parameter τ when there are $N_u=30$ and $N_u=90$ users in the cell, respectively. The Qualcomm scheduler (1 user under PF criterion) is compared with two schedulers trying to serve 3 users simultaneously (*WSR* and *3-user PF*, (101)). It can be observed that the WSR scheduler is always outperformed by the PF scheduler in the 3-user case, which is different from the results that were obtained for the two-user case, e.g., Figure 64. In the 2-user case, the WSR criterion explored all the possible combinations, whereas here, it has the same candidate list as the PF scheduler. In the 3-user case, the performance of the WSR follows similar guidelines as in the two-user case, i.e. it provides the same performance as the PF criterion for certain values of τ , whereas its performance is worse for other values, even worse than the Qualcomm scheduler performance. Regarding the PF criterion, it is able to reduce the average delay per user up to one third of the value that is attained by the Qualcomm scheduler. Likewise, an increment of the average throughput per user for almost all values of τ is also observed.

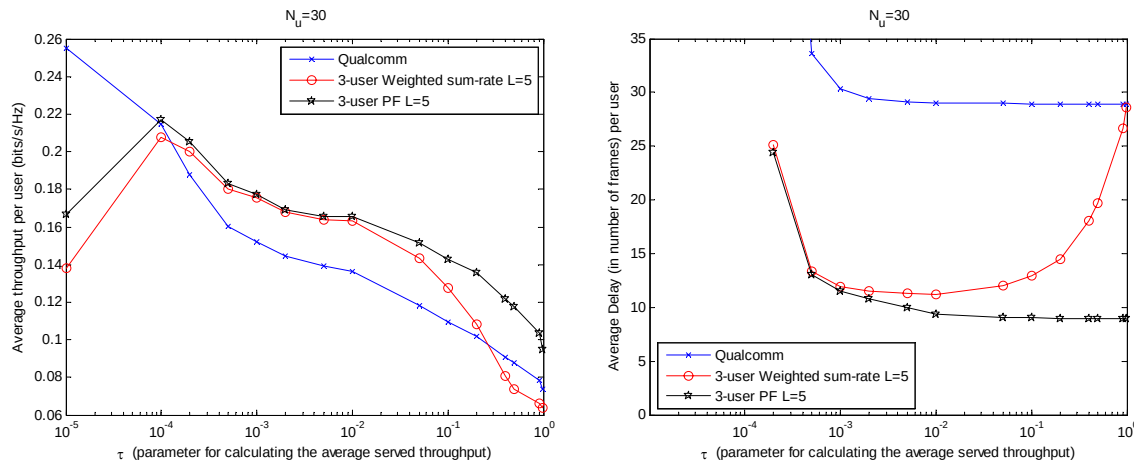


Figure 66: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, 3-user WSR and 3-user PF. $N_u=30$. $L=5$.

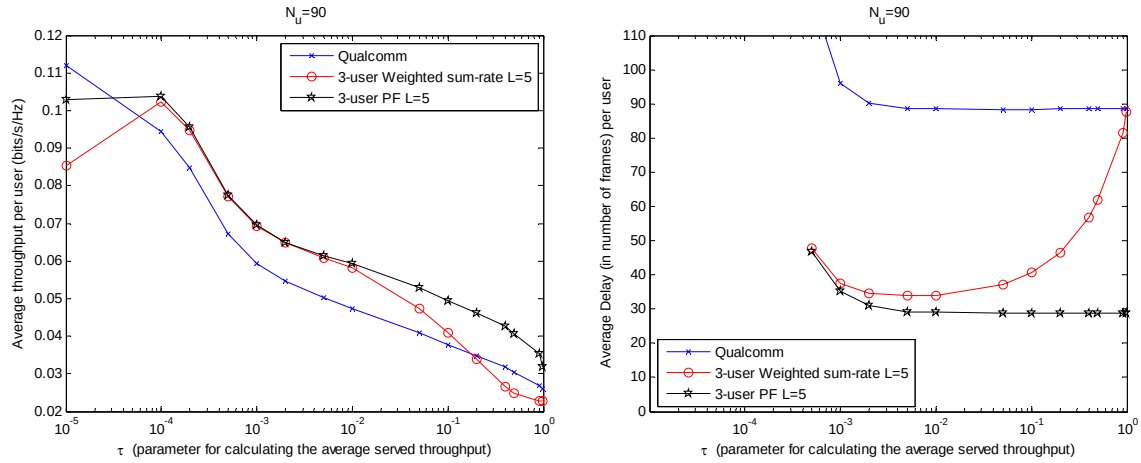


Figure 67: Left) Average throughput per user and Right) Average delay as a function of the parameter τ , (85). Schedulers: Qualcomm, 3-user WSR and 3-user PF. $N_u=90$. $L=5$.

Figure 68 depicts the percentage-wise cell throughput gain that is obtained by the 2-user and 3-user PF schedulers over the Qualcomm algorithm for $N_u=\{30,90\}$. Interestingly, we can identify three zones based on the throughput performance:

- Zone 1 ($10^{-5} \leq \tau \leq 2 \times 10^{-4}$): The 2-user PF is superior to the 3-user PF. We know that the 3-user PF provides a better delay, but at the cost of reducing the average throughput per user.
- Zone 2 ($2 \times 10^{-4} \leq \tau \leq 10^{-2}$): Both algorithms, 2-user and 3-user PF get the same throughput gain.
- Zone 3 ($10^{-2} \leq \tau \leq 1$): The 3-user PF attains the best throughput gain, achieving a 35% gain over the Qualcomm algorithm, while the 2-user PF scheduler only provides a gain of around 22%.

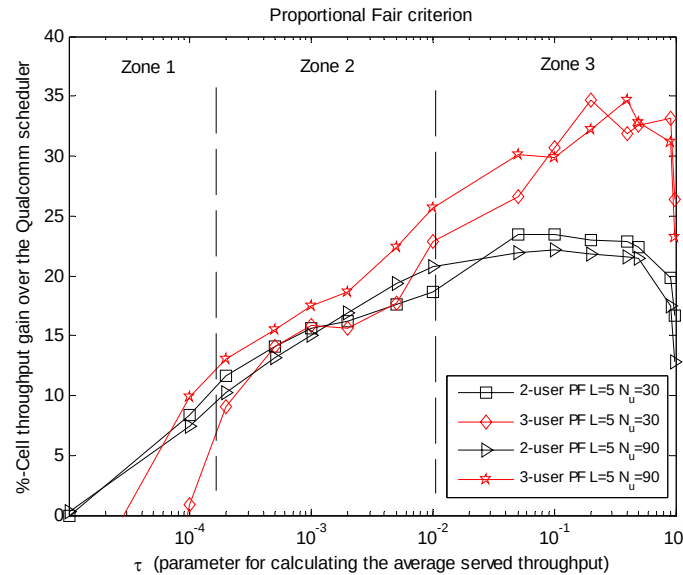


Figure 68: Percentage-wise cell throughput gain of the 2-user PF and 3-user PF over the Qualcomm scheduler as a function of the parameter τ , (85). $N_u=\{30,90\}$. $L=5$.

The three previous zones are analysed for the WSR with 2-user and 3-user selections in Figure 69. It should be recalled that the 2-user WSR algorithm evaluates all the combinations of 2 users, while the 3-user algorithm takes into account just a candidate list of $L=5$. The throughput gain that is observed when using the 2-user algorithm is a consequence of having a worse delay (see Figure 60). The results show that the WSR algorithm only provides interesting results for the zone 2 ($2 \times 10^{-4} \leq \tau \leq 10^{-2}$). The 3-user algorithm reduces the delay by one third, but it provides the same throughput as the 2-user PF

algorithm (and 3-user PF). Henceforth, it seems that the 3-user scheme is interesting for reducing the delay under the WSR criterion, but it does not improve the throughput.

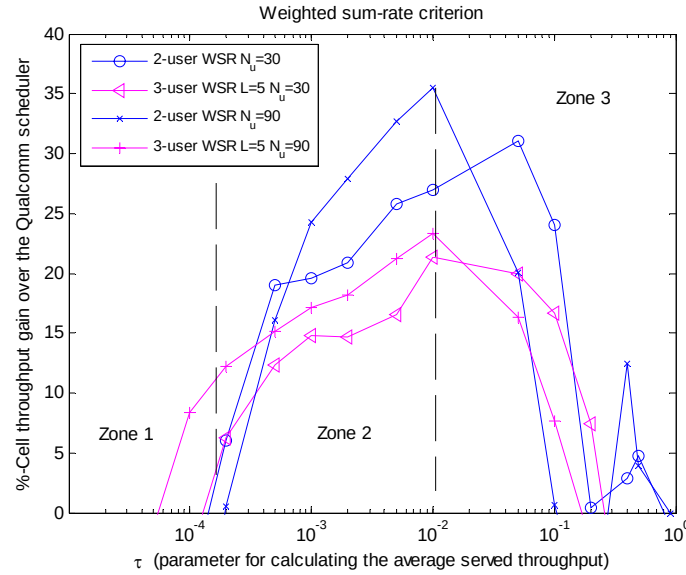


Figure 69: Percentage-wise cell throughput gain of the 2-user and 3-user WSR schedulers over the Qualcomm algorithm as a function of the parameter τ , (85). $N_u = \{30, 90\}$. $L=5$.

3.2.2.3 Conclusions

- As far as the MU PF scheduler (2-user and 3-user) is concerned, a candidate list of length $L=5$ or $L=9$ (populated according to (98)) is sufficient for selecting the best choice of users that are simultaneously scheduled under superposed transmission. Nevertheless, if $\tau > 0.5$, the scheduler performance with a smaller candidate list ($L=3$ or even $L=2$) is almost similar as with a list of $L=9$ candidates. In general, the MU PF scheduler attains the best performance in terms of throughput and delay.
- The WSR schedulers are only suitable for certain values of τ . For the 2-user case, the sum-rate criterion provides a better throughput than the PF criterion because it can be expressed as a closed-form expression, which simplifies the evaluation of all possible combinations. However, the delay per user is worse than when using the MU PF algorithm. In the case that the WSR scheduler has the same candidate list as the MU PF scheduler (i.e. 3-user case), the MU PF always provides a better performance than the WSR. The performance of the WSR scheduler is worse than the Qualcomm algorithm for other values of τ .
- In terms of delay, the 2-user and 3-user PF schedulers reduce the average delay that is attained by the Qualcomm scheduler by one half and one third, respectively.
- Regarding the throughput, we have identified three zones:
 - Zone 1 ($10^{-5} \leq \tau \leq 2 \times 10^{-4}$): It is better to use the 2-user PF scheduler.
 - Zone 2 ($2 \times 10^{-4} \leq \tau \leq 10^{-2}$): The 2-user and 3-user PF algorithms provide the same gains over the Qualcomm scheduler. The WSR scheduler (2-user case) provides a better throughput because it evaluates all the combinations of users to be served.
 - Zone 3 ($10^{-2} \leq \tau \leq 1$): The 3-user PF attains the best throughput gain. Hence, the 3-user PF scheduler offers the best average throughput and delay for this configuration.

- The results have shown that the WSR scheduler is suitable only for Zone 2. In such a case, the maximum throughput is attained by the 2-user WSR algorithm. However, we can use the 3-user WSR algorithm if we are interested in delay reduction, since it provides the same throughput gain as the 2-user scheduler but with a reduced scheduling delay.

4 ADVANCED RELAYING MULTI-USER TECHNIQUES

This section presents a study of some advanced relaying protocols described in the [ROCKET-1D1] document, in a single-cell multi-user OFDMA context. The first two contributions propose innovative resource allocation techniques for single-relay transmission protocols (“one-way relay channel”, “two-way relay channel” and orthogonal amplify and forward, denoted OWRC, TWRC and OAF, respectively), while the last contribution focuses on the performance evaluation of the two-relay assisted “one-way, two-path” cooperative relaying protocol (denoted OWTP-CR), compared with other conventional single-relay or two-relay assisted transmission protocols.

The outline of this section is as follows. The first subsection derives the optimal UL/DL resource allocation for the OWRC and TWRC in both the forwarding [FWRKS-2D2] and protocol I [Agus-08] case. The criterion to maximize is the WSR and the resolution is performed under different power constraints. A mathematical formulation of the optimisation problem to be solved is provided in each case and, when not convex, elementary mathematical transformations are performed to convert it into a convex problem, so that the optimal solution can be derived using conventional solving techniques (e.g. interior-point methods). Simulations results show the achievable rate regions obtained for each one of the studied protocols, under different channel configurations and various power and data traffic constraints. Additional simulations are provided to evaluate the throughput gain provided by the previous relay-assisted protocols in a single-cell, multi-user context, under a PF scheduler.

The second part of this section focuses on the UL and presents five innovative power-allocation methods for cooperative MIMO systems with nonregenerative relays. Each of these allocation strategies is designed to maximize the mutual information (single-user) or the sum-rate/WSR (multi-user) of the cooperative MIMO system and is well-suited to specific levels of CSI knowledge, among those investigated in this study. Simulations are provided to compare the performance, in terms of mutual information or sum-rate, of the new proposed algorithms against other existing power allocation schemes, for different SNR and antenna configurations, in both the single-user and multi-user case.

Finally, the last subsection deals with the study and performance evaluation of various single-relay and two-relay assisted transmission protocols, and in particular, the “one-way, two-path” cooperative relaying protocol (denoted OWTP-CR in the following). Link-level simulations will quantify the degradation induced by inter-relay interference on the OWTP-CR BER performance and demonstrate the efficiency of the proposed interference cancellation mechanism implemented at a single relay. The influence of the relay position will also be analyzed in this context. Simulations conducted in a single-cell multi-user environment will show that, with an adequate relay choice, the OWTP-CR can significantly improve both the average BER performance and the fairness among users.

4.1 MULTI-USER ONE-WAY & TWO-WAY RELAY CHANNEL WITH A COMMON RS

This section takes into account the studies that were carried out in [ROCKET-3D1], where the optimal resource allocation for single user relay-assisted transmission was tackled; we extend here this study for the MU case. The benefits of the superposed transmission in the BC and MAC, which have been shown in section 3.2.2, will be extended to MU relay-assisted transmission with a common RS in section 4.1.1. Afterwards, section 4.1.2 is devoted to investigate the benefits of the MU relay-assisted transmission at system level.

4.1.1 Optimal resource allocation for relay-assisted transmission

The relay-assisted techniques are based on the OWRC and TWRC; analysing a half-duplex decode-and-forward (DF) relay under the *forwarding* and *protocol I* [FWRKS-2D2].

Several strategies were analysed in [Kram-05] for the relay networks, where the Broadcast Relay Channel (BRC) and Multiple Access Relay Channel (MARC) were studied. Furthermore, [Liang-07]

looked into the rate regions of the BRC, and derived the capacity region for some configuration of the terminals. Moreover, the Gaussian degraded BRC was investigated in [Bhas-08], where it was shown that superposition coding is able to achieve the capacity region under the assumption that both receivers are degraded with respect to the relay.

Regarding the MU TWRC, [Weng-07] investigated the TWRC by considering one BS, a single RS and two MSs, where BS and RS are equipped with multiple antennas. However, the authors assumed that each MSs have some knowledge about the message that is transmitted by the other MSs. On the other hand, [Esli-08a],[Esli-08b] looked into the same scenario, and they provided algorithms to design the transmission matrices at the relay that maximises the sum-rate of the system. The protocol that was considered is either two-stage or forwarding.

The objective of our work is to elucidate the benefits of the MU relay-assisted transmission. To this end, we will focus on Gaussian scenarios in order to obtain some criteria for determining when the superposed transmission can outperform the orthogonal transmission in a MU scenario. This section will investigate the achievable rate regions of the relay-assisted transmission schemes.

The considered scenario, which has been introduced in Figure 2, consists of a BS, a half-duplex RS and two MSs. The transmissions must be accommodated in a single frame and the resources must be optimised to consider the DL and UL traffic. The messages transmitted by the BS to the MSs will be encoded at rate R_{bs1} (to MS₁) and R_{bs2} (to MS₂), whereas the messages intended to the BS are transmitted by MSs by employing bitrate R_{ms1} (from MS₁) and R_{ms2} (from MS₂).

Since the relay is half-duplex, the relay-assisted transmission is based on two orthogonal phases where the relay receives data or *listens* in one phase and transmits in the other phase. Moreover, the relay is classified as a function of the type of traffic. We reference as one-way relay when the relay works with DL or UL traffic, whereas the two-way terminology is employed when the relay deals with DL and UL traffic, simultaneously.

The investigated relay protocols are the *forwarding* and *protocol I*, which have both already been introduced in [ROCKET-3D1]. They differ on the way they use the direct link between the BS and MSs. The forwarding protocol assumes that the direct link cannot be used, i.e., when the signal attenuation (or shadowing) of that link is very high. On the other hand, protocol I designs its transmission by taking into account that the destination can make a good use of the source-destination and relay-destination links, i.e., BS-MS and RS-MS in the DL.

The resource allocation must select the values of bitrates, transmitted power and the duration of orthogonal phases that occur during the transmission. The criterion for selecting those values is the maximum achievable rate for that configuration, which is obtained by maximising the following WSR

$$\begin{aligned}
 & \max_{\alpha, P_{ms1}, P_{ms2}, P_{bs}, P_{rs}, R_{ms1}, R_{ms2}, R_{bs1}, R_{bs2}} \mu_1 R_{bs1} + \mu_2 R_{bs2} + \mu_3 R_{ms1} + \mu_4 R_{ms2} \\
 & s.t. \quad (R_{bs1}, R_{bs2}, R_{ms1}, R_{ms2}) \in B(\alpha) \\
 & \quad 0 \leq f(\alpha) \leq 1 \\
 & \quad \Omega(P_{ms1}, P_{ms2}, P_{bs}, P_{rs}, R_{ms1}) \leq P
 \end{aligned} \tag{102}$$

where the vector α contains the duration of the different orthogonal phases that are linearly connected through the function f . The variables P_{ms1} , P_{ms2} , P_{bs} , P_{rs} denote the transmitted power from MS₁, MS₂, BS and RS, respectively. Their values are subject to the function Ω , which gathers different power constraints (i.e. *sum-average* or *individual*). R_{bs1} , R_{bs2} are the data rates of the messages intended to the MS₁, MS₂, whereas R_{ms1} , R_{ms2} are the data rate of the messages transmitted by MS₁ and MS₂. These data rates are constrained by the values of the phase duration and power allocation, which will be described by the function $B(\alpha)$, and it also depends on the selected transmission protocol. The boundary of the achievable rate region is attained by modifying accordingly the weights μ_1 , μ_2 , μ_3 and

μ_4 . As in [ROCKET-3D1], we will assume that the DL and UL traffic are linearly connected through the parameter κ

$$\begin{aligned}\phi(R_{bs1}, R_{ms1}) &= R_{bs1} - \kappa R_{ms1} \leq 0 \\ \phi(R_{bs2}, R_{ms2}) &= R_{bs2} - \kappa R_{ms2} \leq 0\end{aligned}\quad (103)$$

This assumption is based on the observation that the *DL traffic* is in general larger than the *UL traffic*, hence $\kappa > 1$. Likewise, since R_{bs1} (R_{bs2}) and R_{ms1} (R_{ms2}) depend on κ , the optimisation problem can be rewritten as

$$\begin{aligned}\max_{\mathbf{a}, P_{ms1}, P_{ms2}, P_{bs}, P_{rs}, R_{ms1}, R_{ms2}, R_{bs1}, R_{bs2}} \quad & \mu_1 R_{bs1} + \mu_2 R_{bs2} \\ \text{s.t.} \quad & (R_{bs1}, R_{bs2}, R_{ms1}, R_{ms2}) \in B(\mathbf{a}) \\ & 0 \leq f(\mathbf{a}) \leq 1 \\ & \Omega(P_{ms1}, P_{ms2}, P_{bs}, P_{rs}, R_{ms1}) \leq P \\ & R_{bs1} - \kappa R_{ms1} \leq 0 \\ & R_{bs2} - \kappa R_{ms2} \leq 0\end{aligned}\quad (104)$$

In the subsequent sections, we present the signal model and the data rate constraints for the different transmission protocols. The resource allocation problem will be shown to be a *convex* problem [Boyd-04], which means that it exists a single optimal solution that can be for instance obtained by the interior point method. The OWRC and TWRC are analysed in sections 4.1.1.1 and 4.1.1.2, respectively.

4.1.1.1 One-Way Relay Channel

When the relay operates in one-way mode, the frame must be divided in four orthogonal phases in order to accommodate communication in the *DL* and *UL*, see Figure 70. Notice that the *DL* and *UL* communication are both carried out in two phases, because of the half-duplex relay. In each communication way, the relay requires a period of time for listening to the signal from the source/s and another one for transmitting the signals.

Two half-duplex protocols are analysed for the OWRC; the *forwarding* and *protocol I* in [FWRKS-2D2] and [Agus-08], respectively, which will be described in the following sections.

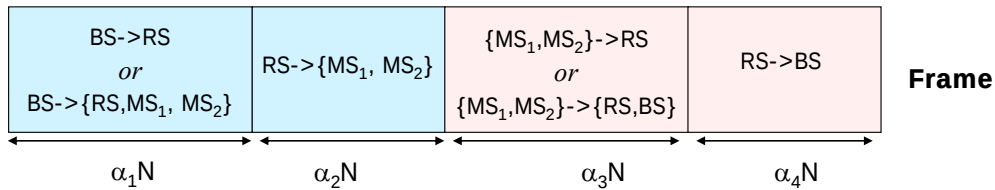


Figure 70. Frame format for the OWRC forwarding and protocol I

We identify four orthogonal phases, which are depicted in Figure 70, and that we reference as

$$\begin{cases} T_1 = [0, \alpha_1 N] \\ T_2 = [\alpha_1 N, (\alpha_1 + \alpha_2) N] \\ T_3 = [(\alpha_1 + \alpha_2) N, (\alpha_1 + \alpha_2 + \alpha_3) N] \\ T_4 = [(\alpha_1 + \alpha_2 + \alpha_3) N, (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) N] \end{cases}\quad (105)$$

4.1.1.1.1 Forwarding

All the terminals are equipped with a single antenna and they transmit Gaussian signals of zero mean and power given by

$$\begin{aligned}
 DL & \begin{cases} x_{bs-rs} \sim N(0, \tilde{\gamma}_1 P), \\ x_{rs-ms1} \sim N(0, \tilde{\phi}_1 P), \quad x_{rs-ms2} \sim N(0, \tilde{\phi}_2 P) \end{cases} \\
 UL & \begin{cases} x_{rs-bs} \sim N(0, \tilde{\psi}_1 P) \\ x_{ms1-rs} \sim N(0, \tilde{\beta}_1 P), \quad x_{ms2-rs} \sim N(0, \tilde{\beta}_2 P) \end{cases},
 \end{aligned} \quad (106)$$

where x_{bs-rs} is the signal transmitted by the BS to the RS, which contains the two messages intended to MS₁ and MS₂, x_{rs-ms1} , x_{rs-ms2} denote the signals transmitted by the relay to the MSs, x_{ms1-rs} , x_{ms2-rs} are the signals transmitted by the MSs to the RS, and finally, x_{rs-bs} is the signal transmitted by the RS to the BS that includes the messages from MSs and the intended ones to the BS.

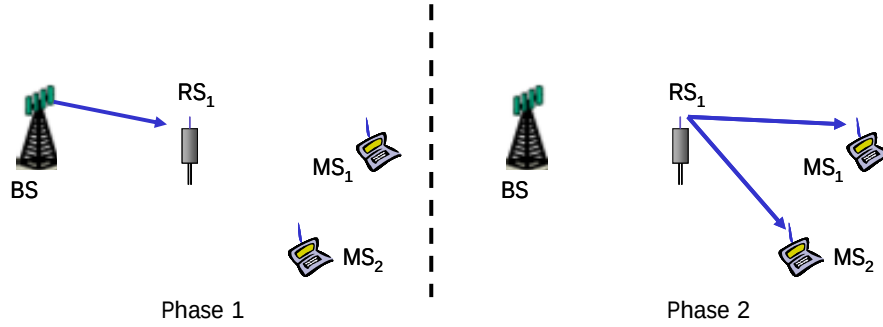


Figure 71. BRC during the first 2 phases of the frame (Figure 70)

During the first two phases of the frame, which is depicted in Figure 70, the DL transmission is carried out. The communication is described by the BRC, which is sketched in Figure 71. In the first phase (phase 1 of Figure 71), the received signal at the RS is defined by

$$y_{rs}(t) = h_{bs-rs} x_{bs-rs}(t) + n_{rs}(t) \quad t \in T_1, \quad (107)$$

where h_{bs-rs} is the channel coefficient of the BS-RS link, x_{bs-rs} is the signal that is transmitted from the BS to the RS, (106), which contains information of the two messages that are intended to both MSs, and T_1 is defined in (105).

The relay decodes the transmission from the BS and finds out the messages to be transmitted to the MSs in the second phase (phase 2 of Figure 71). In such a case, we have a *physically degraded* BC because MSs are equipped with a single antenna and they received the signals from the RS only. It is known [Cove-91] that the maximum achievable rates in that phase of the transmission can be obtained by employing *superposition coding* at the RS. Hence, the received signal at each MS becomes

$$\begin{aligned}
 y_{ms1}(t) &= h_{rs-ms1} (x_{rs-ms1}(t) + x_{rs-ms2}(t)) + n_{ms1}(t) \\
 y_{ms2}(t) &= h_{rs-ms2} (x_{rs-ms1}(t) + x_{rs-ms2}(t)) + n_{ms2}(t) \quad t \in T_2,
 \end{aligned} \quad (108)$$

where h_{rs-ms1} , h_{rs-ms2} stand for the channel coefficients of the RS-MS links, x_{rs-ms1} , x_{rs-ms2} are the Gaussian signals transmitted from the relay, (106), and T_2 is introduced in (105). Notice, that the relay must distribute its available power among the two independent signals.

On the other hand, the UL transmission is done by following the relay-assisted transmission in the last two phases of the frame, which is presented in Figure 72. In such a case, the transmission is modelled by the MARC and the signal model at the RS and the BS are

$$\begin{aligned}
 y_{rs}(t) &= h_{ms1-rs} x_{ms1-rs}(t) + h_{ms2-rs} x_{ms2-rs}(t) + n_{rs}(t) \quad t \in T_3 \\
 y_{bs}(t) &= h_{rs-bs} x_{rs-bs}(t) + n_{bs}(t) \quad t \in T_4,
 \end{aligned} \quad (109)$$

where h_{ms1-rs} , h_{ms2-rs} , h_{rs-bs} are the channel coefficients of the MS₁-RS, MS₂-RS and RS-BS links. It is important to remark that we assume that the channel coefficients of the RS-MS₁ and MS₁-RS can be

different in general. Moreover, the Gaussian signals x_{ms1-rs} , x_{ms2-rs} , x_{rs-bs} , which are defined in (106), denote the transmission from MS₁ and MS₂ to RS and from RS to BS, respectively. The time intervals referenced by T₃ and T₄ are presented in (105).

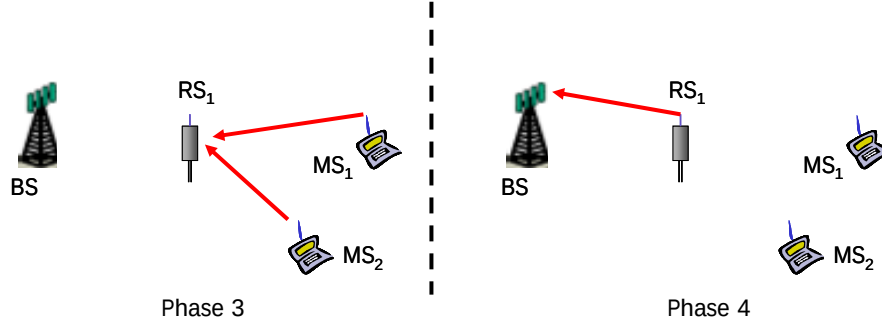


Figure 72. MARC during the last 2 phases of the frame (Figure 70)

The achievable rate region defined by the BRC, (107) and (108), and by assuming that $\rho_{rs-ms1} \geq \rho_{rs-ms2}$ is defined as

$$B_{BRC}^{OWRC, fw} = \begin{cases} R_{bs1} + R_{bs2} \leq \alpha_1 \log_2(1 + \tilde{\gamma}_1 \rho_{bs-rs}) \\ R_{bs1} \leq \alpha_2 \log_2(1 + \tilde{\varphi}_1 \rho_{rs-ms1}) \\ R_{bs2} \leq \alpha_2 (\log_2(1 + (\tilde{\varphi}_1 + \tilde{\varphi}_2) \rho_{rs-ms2}) - \log_2(1 + \tilde{\varphi}_1 \rho_{rs-ms2})) \end{cases}, \quad (110)$$

where R_{bs1} , R_{bs2} are the data rate of the messages that are originated at the BS and are intended to MS₁ and MS₂, respectively, $\tilde{\gamma}_1, \tilde{\varphi}_1, \tilde{\varphi}_2$ are the fraction of powers that are allocated to the signals that are transmitted from the BS to RS, RS-MS₁ and RS-MS₂, α_1, α_2 denote the duration of phases 1 and 2, which are shown in Figure 71, and $\rho_{bs-rs}, \rho_{rs-ms1}, \rho_{rs-ms2}$ stand for the SNR of the BS-RS, RS-MS₁ and RS-MS₂ link, respectively.

Since our scenario is *physically degraded*, therefore under the configuration $\rho_{rs-ms1} \geq \rho_{rs-ms2}$, MS₁ is able to decode the message intended to MS₂, subtract it from the received signal and decode its intended message without interference. However, MS₂ will still see the signal intended to MS₁ as interference [Cove-91].

The data rate inequalities introduced in (110) are not concave over all the variables, as it is required for defining the resource allocation (104) as a convex problem [Boyd-04]. We employ the duality between BC and MACH, [Vish-04], in order to define the resource allocation problem as a convex problem. We will see that after applying the different power constraints in the ensuing subsections A and B, the resource allocation will be defined as a convex problem

$$B_{BRC}^{OWRC, fw} = \begin{cases} R_{bs1} + R_{bs2} \leq \alpha_1 \log_2(1 + \tilde{\gamma}_1 \rho_{bs-rs}) \\ R_{bs1} + R_{bs2} \leq \alpha_2 \log_2(1 + \tilde{\lambda}_1 \rho_{rs-ms1} + \tilde{\lambda}_2 \rho_{rs-ms2}) \\ s.t. \quad \tilde{\lambda}_1 + \tilde{\lambda}_2 = \tilde{\varphi}_1 + \tilde{\varphi}_2 \end{cases}, \quad (111)$$

where new optimisation variables $\tilde{\lambda}_1, \tilde{\lambda}_2$ are introduced. The power allocation at the relay ($\tilde{\varphi}_1, \tilde{\varphi}_2$) depends on the new variables (assuming $\rho_{rs-ms1} \geq \rho_{rs-ms2}$)

$$\tilde{\varphi}_1 = \tilde{\lambda}_1 \frac{1}{1 + \rho_{rs-ms2} \tilde{\lambda}_2}, \quad \tilde{\varphi}_2 = \tilde{\lambda}_2 \left(1 + \frac{\rho_{rs-ms2} \tilde{\lambda}_1}{1 + \rho_{rs-ms2} \tilde{\lambda}_2} \right). \quad (112)$$

During the last two phases of Figure 70 the UL transmission (from MSs to BS) is employed. When the relay participates to the transmission, the equivalent channel becomes the MARC and its signal model

is given in (109). The achievable rate region of that channel can be derived as, see for instance [Agus-08]

$$B_{MARC}^{OWRC, fw} = \begin{cases} R_{ms1} + R_{ms2} \leq \alpha_3 \log_2 (1 + \tilde{\psi}_1 \rho_{rs-bs}) \\ R_{ms1} + R_{ms2} \leq \alpha_3 \log_2 (1 + \tilde{\beta}_1 \rho_{ms1-rs} + \tilde{\beta}_2 \rho_{ms2-rs}) \\ R_{ms1} \leq \alpha_3 \log_2 (1 + \tilde{\beta}_1 \rho_{ms1-rs}) \\ R_{ms2} \leq \alpha_3 \log_2 (1 + \tilde{\beta}_2 \rho_{ms2-rs}) \end{cases}, \quad (113)$$

where R_{ms1} , R_{ms2} and $\tilde{\beta}_1, \tilde{\beta}_2$ are the data rates and the power allocation that are employed by each MS. The variable $\tilde{\psi}_1$ denotes the fraction of power that is allocated for transmitting the signal x_{rs-bs} . Finally, $\rho_{rs-bs}, \rho_{ms1-rs}, \rho_{ms2-rs}$ stand for the SNR of the RS-BS, MS₁-RS and MS₂-RS links, respectively. Notice that the first data rate constraint, which is depicted in (113), establishes the maximum bitrate that is allowed at the BS in the last phase of the transmission, whereas the last three data rate constraints in (113) describe the conventional MACH [Cove-91] that is found in the third phase of the transmission and is shown in Figure 70.

A. Power constraints

The three power constraints, *sum-average*, *individual-average* and *max-power constraints*, can be described with the previous equations using the change of variables depicted in Table 4.

Table 4: Variables employed under the different power constraints

Power constraints	Transmission power variables							
	$\tilde{\gamma}_1$	$\tilde{\varphi}_1$	$\tilde{\varphi}_2$	$\tilde{\lambda}_1$	$\tilde{\lambda}_2$	$\tilde{\beta}_1$	$\tilde{\beta}_2$	$\tilde{\psi}_1$
<i>Sum-average power constraint</i>	$\frac{\gamma_1}{\alpha_1}$	$\frac{\varphi_1}{\alpha_2}$	$\frac{\varphi_2}{\alpha_2}$	$\frac{\lambda_1}{\alpha_2}$	$\frac{\lambda_2}{\alpha_2}$	$\frac{\beta_1}{\alpha_3}$	$\frac{\beta_2}{\alpha_3}$	$\frac{\psi_1}{\alpha_4}$
<i>Individual-average power constraint</i>	$\frac{1}{\alpha_1}$	$\frac{\varphi_1}{\alpha_2}$	$\frac{\varphi_2}{\alpha_2}$	$\frac{\lambda_1}{\alpha_2}$	$\frac{\lambda_2}{\alpha_2}$	$\frac{1}{\alpha_3}$	$\frac{1}{\alpha_3}$	$\frac{\psi_1}{\alpha_4}$
<i>Max-power constraint</i>	1	φ_1	φ_2	λ_1	λ_2	1	1	1

In order to satisfy the different power constraints, the equalities $\Omega_{OWRC, fw}^{sum-ave}, \Omega_{OWRC, fw}^{ind-ave}, \Omega_{OWRC, fw}^{max-pow}$ must be considered in our optimisation problem, (104).

- *Sum-average power constraint* (total power in the four-terminal network)

$$\Omega_{OWRC, fw}^{sum-ave} = \begin{cases} \gamma_1 + \varphi_1 + \varphi_2 + \beta_1 + \beta_2 + \psi_1 = 1 \\ \gamma_1 + \lambda_1 + \lambda_2 + \beta_1 + \beta_2 + \psi_1 = 1 \end{cases}. \quad (114)$$

- *Individual average power constraint* (total power employed by each terminal)

$$\Omega_{OWRC, fw}^{ind-ave} = \begin{cases} \varphi_1 + \varphi_2 + \psi_1 = 1 \\ \lambda_1 + \lambda_2 + \psi_1 = 1 \end{cases}. \quad (115)$$

- *Max-power constraint* (power that is used by each terminal on each phase)

$$\Omega_{OWRC, fw}^{max-pow} = \begin{cases} \varphi_1 + \varphi_2 = 1 \\ \lambda_1 + \lambda_2 = 1 \end{cases}. \quad (116)$$

B. Solving the resource allocation problem

The resource allocation, which is defined in (104), is seen as a convex problem [Boyd-04] for the *sum-average* and *individual power constraint*. In these cases, we can see that the data rate constraints depend on *perspective functions* [Boyd-04], which preserve the convexity property of a given function. For example, given the concave function f , the *perspective* function of f , which is referenced as g_k and is dependent on two variables, is also concave

$$f(x) = \log_2(1 + x\Gamma), \quad g_k(\alpha_k, \beta_k) = \alpha_k f(\beta_k / \alpha_k). \quad (117)$$

Nevertheless, the resource allocation under a *max-power constraint* is not a convex problem. A similar drawback was found in [ROCKET-3D1] for the *superposition coding*. It can be avoided by using the following variables for transforming the resource allocation into a convex problem

$$\begin{cases} v_1 = \alpha_2 \lambda_1 \\ v_2 = \alpha_2 \lambda_2 \end{cases}, \quad \Omega_{OWRC, fw}^{\max-pow} = \{v_1 + v_2 - \alpha_2 = 0\}, \quad (118)$$

where v_1, v_2 are the variables that are connected with λ_1, λ_2 and are obtained after applying the BC-MACH duality in (111). Moreover, α_2 denotes the fraction of time that is devoted for the phase 2 (where the relay transmits), as it is depicted in Figure 70.

Finally, the optimisation problem is given by

$$\begin{aligned} & \max_{\alpha, \eta, R_{ms1}, R_{ms2}, R_{bs1}, R_{bs2}} \mu_1 R_{bs1} + \mu_2 R_{bs2} \\ & s.t. \quad \begin{cases} (R_{bs1}, R_{bs2}, R_{ms1}, R_{ms2}) \in (B_{MARC}^{OWRC, fw} \cup B_{BRC}^{OWRC, fw}) \\ \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1, & R_{bs2} - \kappa R_{ms2} \leq 0 \\ \Omega_{OWRC, fw}, & R_{bs1} - \kappa R_{ms1} \leq 0 \end{cases} \end{aligned} \quad (119)$$

where $B_{BRC}^{OWRC, fw}$ and $B_{MARC}^{OWRC, fw}$ are the data rate constraints that have been introduced in (111) and (113) respectively, vector η denotes the power variables to be optimised and $\Omega_{OWRC, fw}$ stands for the power constraint to be satisfied, establishing a constraint on the power variables. These variables are summarised in Table 5. It is important to remark that the power variables that are assumed in the data rate constraints $B_{BRC}^{OWRC, fw}, B_{MARC}^{OWRC, fw}$ must be modified as a function of the power constraint according to Table 4. Additionally, under the *max-power constraint*, the variable definition that is presented in equation (118) must be considered.

Table 5: Power variables and power as a function of the power constraint

Power constraints	η	$\Omega_{OWRC, fw}$
<i>Sum average power constraint</i>	$\gamma_1, \lambda_1, \lambda_2, \beta_1, \beta_2, \psi_1$	$\Omega_{OWRC, fw}^{sum-ave}, (114)$
<i>Individual average power constraint</i>	$\lambda_1, \lambda_2, \psi_1$	$\Omega_{OWRC, fw}^{ind-ave}, (115)$
<i>Max-power constraint</i>	v_1, v_2	$\Omega_{OWRC, fw}^{\max-pow}, (116), (118)$

4.1.1.1.2 Protocol I

The frame format follows the same structure as the one that is depicted in Figure 70, but in this case, MSs and BS also listens to the medium during the phase 1 and 3, respectively. That means that MSs employ the received signals from BS (in phase 1 of Figure 70) and RS (in phase 2 of Figure 70) to decode the transmit message in the DL. Similarly, the BS takes into account the received signal from MSs (in phase 3 of Figure 70) and RS (in phase 4 of Figure 70) to decode the transmit messages in the UL. Notice that the BRC (in the DL) and MARC (in the UL) will be different from the definitions that are presented in the previous section (*forwarding protocol*).

In references [Kram-05],[Liang-07] and [Bhas-08], the BRC was investigated for a full-duplex relay and achievable rate regions that are attained by block Markov coding were derived. In contrast, our scenario consists of a half-duplex relay. Since all terminals are equipped with a single antenna, our scenario can be considered as a *physically degraded* [Bhas-08] for the different phases where the transmission is carried out, and additionally, *superposition strategy* can be the suitable transmission technique to achieve the largest rates. However, the channel is not always *physically degraded* when we consider the whole transmission (both phases) as the forwarding protocol (section 4.1.1.1.1) and the *superposition coding* are not optimal in these cases. Under those circumstances the DPC technique [Cost-81] has shown its ability to provide significant achievable rates, this technique is indeed used to attain the capacity of the BC [Wein-06].

Given a single transmitter and two destinations, the transmitter designs the first codeword to be transmitted to one of the receivers, while the second codeword is designed based on the previous one. The receiver, which is interested in the first message, will receive interference due to the second codeword, but the other destination will be able to decode its intended codeword without any kind of interference, due to the cunning codeword design at the transmitter side. Notice that the codewords must be encoded in a certain order. In our scenario, DPC will be applied to the signals that are transmitted by the source in the first phase and by the relay during the second phase. The encoding order will be dependant on the SNR that is measured by the destinations in both phases.

Similarly to the previous sections, all the terminals are equipped with a single antenna and the transmit Gaussian signals have zero-mean and powers

$$\begin{aligned}
 DL \left\{ \begin{aligned} x_{bs-ms1} &\sim N(0, \tilde{\gamma}_1 P), & x_{bs-ms2} &\sim N(0, \tilde{\gamma}_2 P) \\ x_{rs-ms1} &\sim N(0, \tilde{\beta}_1 P), & x_{rs-ms2} &\sim N(0, \tilde{\beta}_2 P) \end{aligned} \right. \\
 UL \left\{ \begin{aligned} x_{ms1-bs} &\sim N(0, \tilde{\varphi}_1 P), & x_{ms1-bs} &\sim N(0, \tilde{\varphi}_2 P) \\ x_{rs1-bs} &\sim N(0, \tilde{\nu}_1 P), & x_{rs2-bs} &\sim N(0, \tilde{\nu}_2 P) \end{aligned} \right.
 \end{aligned} \quad (120)$$

where x_{bs-ms1} and x_{bs-ms2} are the signals transmitted by the BS to RS, MS₁ and MS₂, with a fraction of allocated power equal to $\tilde{\gamma}_1, \tilde{\gamma}_2$. The transmit signals from the RS are denoted by x_{rs-ms1} and x_{rs-ms2} , and have a fraction of allocated power of $\tilde{\beta}_1, \tilde{\beta}_2$. The signals x_{ms1-bs} , x_{ms2-bs} are transmitted by the MSs in the UL phase by using the fraction of power $\tilde{\varphi}_1, \tilde{\varphi}_2$, and x_{rs1-bs} , x_{rs2-bs} are the signals that are transmitted by the relay and which contain the messages from MS₁ and MS₂, respectively, with powers proportional to $\tilde{\nu}_1, \tilde{\nu}_2$.

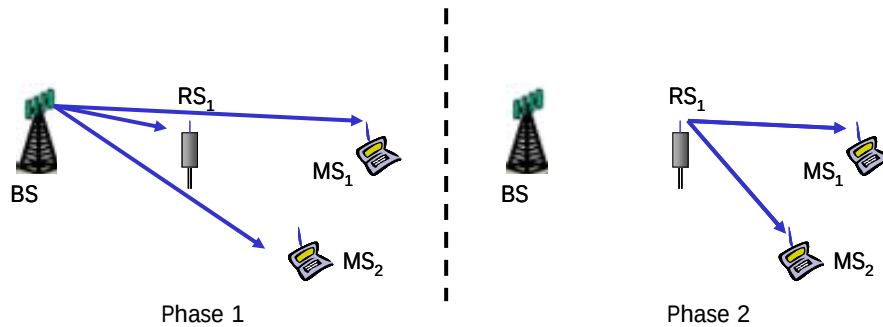


Figure 73. BRC during the first 2 phases of the frame (Figure 70) under protocol I

The DL transmission entails the BRC that is sketched in Figure 73. There are two orthogonal phases where the source transmits to RS and MSs; once the RS has decoded the two messages, it forwards them to the MSs. In phase 1, as it is shown in Figure 73, we consider a degraded scenario with the following constraint on the SNR of the different links $\rho_{bs-rs} \geq \rho_{bs-ms1} \geq \rho_{bs-ms2}$, i.e., the BS-RS link presents a larger SNR than the BS-MS₁ link (and BS-MS₂), and BS-MS₁ link presents a better SNR than BS-MS₂ link. Therefore, the MSs can be classified as *strong* and *weak* users as a function of the BS-MS link SNR during the phase 1. Moreover, in the phase 2 of Figure 73, since only the RS

broadcasts messages to two single-antenna MSs, the resulting scenario is also a degraded one, thus the MSs can also be classified as *strong* and *weak* users with respect to the RS-MS link, and it does not coincide with the classification that is done in phase 1.

The transmitters (BS in phase 1 and RS in phase 2 of Figure 73) employ DPC to transmit the different messages. The bit rate that is allocated to each message will depend on the topology of the scenario in each phase. If the BS-RS link presents a larger SNR than the other links, the RS will be able to help both MSs in receiving the messages from the BS.

According to the transmission schemes that are depicted in Figure 73 and the transmitted signals that are defined in (120), we have the following signal model

$$y_{RS}(t) = h_{bs_rs}(x_{bs_ms1}(t) + x_{bs_ms2}(t)) + n_{rs}(t) \quad t \in T_1 \quad (121)$$

$$y_{ms1}(t) = \begin{cases} h_{bs_ms1}(x_{bs_ms1}(t) + x_{bs_ms2}(t)) + n_{ms1}(t) & t \in T_1 \\ h_{rs_ms1}(x_{rs_ms1}(t) + x_{rs_ms2}(t)) + n_{ms1}(t) & t \in T_2 \end{cases} \quad (122)$$

$$y_{ms2}(t) = \begin{cases} h_{bs_ms2}(x_{bs_ms1}(t) + x_{bs_ms2}(t)) + n_{ms2}(t) & t \in T_1 \\ h_{rs_ms2}(x_{rs_ms1}(t) + x_{rs_ms2}(t)) + n_{ms2}(t) & t \in T_2 \end{cases}$$

At the BS, we will encode the messages as a function of the SNR of the BS-MS₁ and BS-MS₂ links. The data of the MS with the worst BS-MS link quality (*weak user* in phase 1) are encoded first, and afterwards the data of the MS with the best link quality (*strong user* in phase 1) are encoded. Under this configuration, the *strong* user will receive its intended signal without interference, while the *weak* user must deal with the interference that is created by the signal intended to the *strong* user. Since the channel is *physically degraded*, there is only one possible selection for the *strong* and *weak* user. In this phase, the relay, which presents a better link quality than both MSs, is able to decode the both messages; introducing the following rate constraint on the selected rates

$$\left. \begin{aligned} R_{bs1} &\leq \alpha_1 \log_2(1 + \tilde{\gamma}_1 \rho_{bs_rs}) \\ R_{bs2} &\leq \alpha_1 \log_2 \left(1 + \frac{\tilde{\gamma}_2 \rho_{bs_rs}}{1 + \tilde{\gamma}_1 \rho_{bs_rs}} \right) \end{aligned} \right\} \quad R_{bs1} + R_{bs2} \leq \alpha_1 \log_2(1 + (\tilde{\gamma}_1 + \tilde{\gamma}_2) \rho_{bs_rs}) \quad (123)$$

In this phase the MSs cannot decode their intended messages and they must wait until the end of phase 2 for decoding their messages. In the second phase the RS transmits two messages to the MS by using DPC. In this case, the *strong* and *weak* users in the second phase are selected according to the SNR of the RS-MS₁ and RS-MS₂ links. MS will be able to decode the transmit messages from the BS by combining the received signals in each phase.

Assuming that $\rho_{bs_ms1} \geq \rho_{bs_ms2}$ and $\rho_{rs_ms1} \leq \rho_{rs_ms2}$ the achievable rate region is given by

$$\begin{aligned} R_{bs1} &\leq \alpha_1 \log_2(1 + \tilde{\gamma}_1 \rho_{bs_ms1}) + \alpha_2 \log_2 \left(1 + \frac{\tilde{\beta}_1 \rho_{rs_ms1}}{1 + \tilde{\beta}_2 \rho_{rs_ms1}} \right) \\ R_{bs2} &\leq \alpha_1 \log_2 \left(1 + \frac{\tilde{\gamma}_2 \rho_{bs_ms2}}{1 + \tilde{\gamma}_1 \rho_{bs_ms1}} \right) + \alpha_2 \log_2(1 + \tilde{\beta}_2 \rho_{rs_ms2}) \end{aligned} \quad (124)$$

In order to write the rate constraints in a proper way for defining a convex problem, we apply the BC-MACH duality on each phase [Vish-04], the achievable rate region becomes

$$B_{BRC}^{OWRC,pl} = \begin{cases} R_{bs1} + R_{bs2} \leq \alpha \log_2 (1 + \tilde{\kappa}_1 \rho_{bs_ms1} + \tilde{\kappa}_2 \rho_{bs_ms2}) + \alpha_2 \log_2 (1 + \tilde{\psi}_1 \rho_{rs_ms1} + \tilde{\psi}_2 \rho_{rs_ms2}) \\ R_{bs1} \leq \alpha_1 \log_2 (1 + \tilde{\kappa}_1 \rho_{bs_ms1}) + \alpha_2 \log_2 (1 + \tilde{\psi}_1 \rho_{rs_ms1}) \\ R_{bs2} \leq \alpha_1 \log_2 (1 + \tilde{\kappa}_2 \rho_{bs_ms2}) + \alpha_2 \log_2 (1 + \tilde{\psi}_2 \rho_{rs_ms2}) \\ R_{bs1} + R_{bs2} \leq \alpha_1 \log_2 (1 + (\tilde{\kappa}_1 + \tilde{\kappa}_2) \rho_{bs_rs}) \end{cases}$$

, (125)

s.t. $\tilde{\kappa}_1 + \tilde{\kappa}_2 = \tilde{\gamma}_1 + \tilde{\gamma}_2$
 $\tilde{\psi}_1 + \tilde{\psi}_2 = \tilde{\beta}_1 + \tilde{\beta}_2$

where

$$\tilde{\gamma}_1 = \frac{\tilde{\kappa}_1}{1 + \rho_{bs_ms2} \tilde{\kappa}_2}, \tilde{\gamma}_2 = \tilde{\kappa}_2 \left(1 + \frac{\rho_{bs_ms2} \tilde{\kappa}_1}{1 + \rho_{bs_ms2} \tilde{\kappa}_2} \right), \tilde{\beta}_1 = \tilde{\psi}_1 \left(1 + \frac{\rho_{rs_ms1} \tilde{\psi}_2}{1 + \rho_{rs_ms1} \tilde{\psi}_1} \right), \tilde{\beta}_2 = \frac{\tilde{\psi}_2}{1 + \rho_{rs_ms1} \tilde{\psi}_1}. \quad (126)$$

Notice that the new variables that are introduced for defining $B_{BRC}^{OWRC,pl}$ are $\tilde{\kappa}_1, \tilde{\kappa}_2, \tilde{\psi}_1, \tilde{\psi}_2$.

The UL transmission is utilised in the last two phases of the frame, which is depicted in Figure 70, these phases are referenced as phase 3 and phase 4 and they are sketched in Figure 74. In phase 3, both MSs simultaneously transmit independent messages. The signals are received by the RS and BS. Afterwards, the relay transmits two messages to the BS by using *superposition coding*.

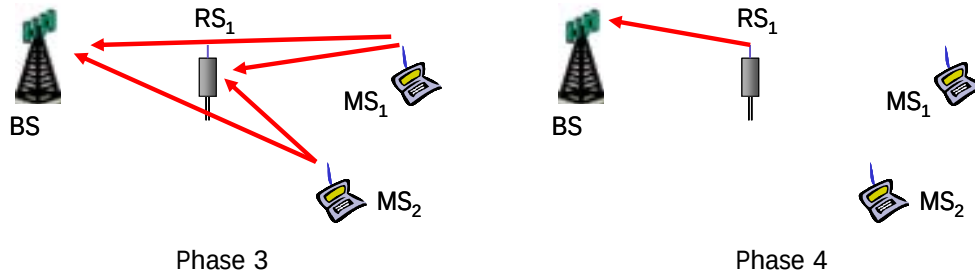


Figure 74. MARC during the last 2 phases of the frame (Figure 70) under protocol I

The received signal at the different nodes is

$$\begin{aligned} y_{RS}(t) &= h_{ms1_rs} x_{ms1_bs}(t) + h_{ms2_rs} x_{ms2_bs}(t) + n_{rs}(t) & t \in T_3 \\ y_{BS}(t) &= \begin{cases} h_{ms1_bs} x_{ms1_bs}(t) + h_{ms2_bs} x_{ms2_bs}(t) + n_{bs}(t) & t \in T_3, \\ h_{rs_bs} (x_{rs1_bs}(t) + x_{rs2_bs}(t)) + n_{bs}(t) & t \in T_4 \end{cases} \end{aligned} \quad (127)$$

where h_{ms1_rs} , h_{ms2_rs} , h_{rs_bs} are the channel coefficients of the MS₁-RS, MS₂-RS and RS-BS links. The signals that are transmitted by the different nodes x_{ms1_bs} , x_{ms2_bs} , x_{rs1_bs} , x_{rs2_bs} are defined in (120). Finally, the time intervals T_3 , T_4 are defined in (105).

The achievable rate region is described by the two MACs that are found at the RS and BS. In this latter case, the BS must consider the signals that are received during the different phases of the transmission, which are depicted in Figure 74. It is given by

$$B_{MARC}^{OWRC,pl} = \begin{cases} R_{ms1} \leq \alpha_3 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-rs}) \\ R_{ms2} \leq \alpha_3 \log_2 (1 + \tilde{\varphi}_2 \rho_{ms2-rs}) \\ R_{ms1} + R_{ms2} \leq \alpha_3 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-rs} + \tilde{\varphi}_2 \rho_{ms2-rs}) \\ R_{ms1} \leq \alpha_3 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-bs}) + \alpha_4 \log_2 (1 + \tilde{v}_1 \rho_{rs-bs}) \\ R_{ms2} \leq \alpha_3 \log_2 (1 + \tilde{\varphi}_2 \rho_{ms2-bs}) + \alpha_4 \log_2 (1 + \tilde{v}_2 \rho_{rs-bs}) \\ R_{ms1} + R_{ms2} \leq \alpha_3 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-bs} + \tilde{\varphi}_2 \rho_{ms2-bs}) + \alpha_4 \log_2 (1 + \tilde{v}_1 \rho_{rs-bs} + \tilde{v}_2 \rho_{rs-bs}) \end{cases} \quad (128)$$

where $\rho_{rs-bs}, \rho_{ms1-rs}, \rho_{ms2-rs}, \rho_{ms1-bs}, \rho_{ms2-bs}$, denote the SNR of the RS-BS, MS₁-RS, MS₂-RS, MS₁-BS and MS₂-BS links, respectively. The first three data rate constraints are motivated by the achievable rate at the RS, whereas the last three data rate constraint are due to the BS, which receives during phase 3 and phase 4 of Figure 70.

A. Power constraints

The three power constraints, *sum-average*, *individual-average* and *max-power constraints*, can be described with the previous equations by using the change of variables that is depicted in Table 6.

Table 6: Variables employed under the different power constraints

Power constraint	Transmission power variables											
	$\tilde{\gamma}_1$	$\tilde{\gamma}_2$	$\tilde{\kappa}_1$	$\tilde{\kappa}_2$	$\tilde{\beta}_1$	$\tilde{\beta}_2$	$\tilde{\psi}_1$	$\tilde{\psi}_2$	$\tilde{\varphi}_1$	$\tilde{\varphi}_2$	$\tilde{v}_1$	$\tilde{v}_2$
<i>Sum-average</i>	$\frac{\gamma_1}{\alpha_1}$	$\frac{\gamma_2}{\alpha_1}$	$\frac{\kappa_1}{\alpha_1}$	$\frac{\kappa_2}{\alpha_1}$	$\frac{\beta_1}{\alpha_2}$	$\frac{\beta_2}{\alpha_2}$	$\frac{\psi_1}{\alpha_2}$	$\frac{\psi_2}{\alpha_2}$	$\frac{\varphi_1}{\alpha_3}$	$\frac{\varphi_2}{\alpha_3}$	$\frac{v_1}{\alpha_4}$	$\frac{v_2}{\alpha_4}$
	$\frac{\gamma_1}{\alpha_1}$	$\frac{\gamma_2}{\alpha_1}$	$\frac{\kappa_1}{\alpha_1}$	$\frac{\kappa_2}{\alpha_1}$	$\frac{\beta_1}{\alpha_2}$	$\frac{\beta_2}{\alpha_2}$	$\frac{\psi_1}{\alpha_2}$	$\frac{\psi_2}{\alpha_2}$	$\frac{\varphi_1}{\alpha_3}$	$\frac{\varphi_2}{\alpha_3}$	$\frac{1}{\alpha_4}$	$\frac{1}{\alpha_4}$
<i>Individual average</i>	$\frac{\gamma_1}{\alpha_1}$	$\frac{\gamma_2}{\alpha_1}$	$\frac{\kappa_1}{\alpha_1}$	$\frac{\kappa_2}{\alpha_1}$	$\frac{\beta_1}{\alpha_2}$	$\frac{\beta_2}{\alpha_2}$	$\frac{\psi_1}{\alpha_2}$	$\frac{\psi_2}{\alpha_2}$	$\frac{\varphi_1}{\alpha_3}$	$\frac{\varphi_2}{\alpha_3}$	$\frac{1}{\alpha_4}$	$\frac{1}{\alpha_4}$
<i>Max-power</i>	γ_1	γ_2	κ_1	κ_2	β_1	β_2	ψ_1	ψ_2	φ_1	φ_2	1	1

In order to satisfy the different power constraints, the equalities $\Omega_{OWRC,pl}^{sum-ave}, \Omega_{OWRC,pl}^{ind-ave}, \Omega_{OWRC,pl}^{max-pow}$ must be considered in our optimisation problem, (104).

- *Sum-average power constraint* (total power in the four-terminal network)

$$\Omega_{OWRC,pl}^{sum-ave} = \begin{cases} \gamma_1 + \gamma_2 + \beta_1 + \beta_2 + \varphi_1 + \varphi_2 + v_1 + v_2 = 1 \\ \kappa_1 + \kappa_2 + \psi_1 + \psi_2 + \varphi_1 + \varphi_2 + v_1 + v_2 = 1 \end{cases} \quad (129)$$

- *Individual average power constraint* (total power employed by each terminal)

$$\Omega_{OWRC,pl}^{ind-ave} = \begin{cases} \gamma_1 + \gamma_2 = 1 \\ \beta_1 + \beta_2 + \varphi_1 + \varphi_2 = 1 \\ \kappa_1 + \kappa_2 = 1 \\ \psi_1 + \psi_2 + \varphi_1 + \varphi_2 = 1 \end{cases} \quad (130)$$

- *Max-power constraint* (power used by each terminal on each phase)

$$\Omega_{OWRC,pl}^{\max-pow} = \begin{cases} \gamma_1 + \gamma_2 = 1 \\ \beta_1 + \beta_2 = 1 \\ \varphi_1 + \varphi_2 = 1 \\ \kappa_1 + \kappa_2 = 1 \\ \psi_1 + \psi_2 = 1 \\ \varphi_1 + \varphi_2 = 1 \end{cases} \quad (131)$$

B. Solving the resource allocation problem

Under protocol I, the resource allocation problem, which is depicted in (104), is a well-defined convex problem for the sum-average power constraint, because the data rate constraint depend on perspective functions, (117).

On the other hand, we must introduce the following variables to transform our problem into a convex problem for the individual average and max-power constraints

- *Individual average power constraint*

$$\begin{cases} \nu_1 = \alpha_1 \kappa_1 \\ \nu_2 = \alpha_1 \kappa_2 \end{cases} \quad \Omega_{OWRC,pl}^{ind-pow} = \{ \nu_1 + \nu_2 - \alpha_1 = 0, \quad (132)$$

where the variables ν_1, ν_2 are connected to the variables that have been obtained by using the BC-MACH duality property at the BS, see (125), and α_1 is the fraction of time that is devoted to phase 1, which is shown in Figure 70.

- *Max-power constraint*

$$\begin{cases} \nu_1 = \alpha_1 \kappa_1 \\ \nu_2 = \alpha_1 \kappa_2 \\ \vartheta_1 = \alpha_2 \psi_1 \\ \vartheta_2 = \alpha_2 \psi_2 \end{cases} \quad \Omega_{OWRC,pl}^{\max-pow} = \begin{cases} \nu_1 + \nu_2 - \alpha_1 = 0 \\ \vartheta_1 + \vartheta_2 - \alpha_2 = 0 \end{cases}, \quad (133)$$

where the variables ν_1, ν_2 and ϑ_1, ϑ_2 are connected to the variables that have been obtained by using the BC-MACH duality property at the BS and RS, respectively, (125).

The optimisation problem is given by

$$\begin{aligned} & \max_{\alpha, \eta, R_{ms1}, R_{ms2}, R_{bs1}, R_{bs2}} \mu_1 R_{bs1} + \mu_2 R_{bs2} \\ & s.t. \quad \begin{cases} (R_{bs1}, R_{bs2}, R_{ms1}, R_{ms2}) \in (B_{MARC}^{OWRC,pl} \cup B_{BRC}^{OWRC,pl}) \\ \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1, & R_{bs2} - \kappa R_{ms2} \leq 0 \\ \Omega_{OWRC,pl}, & R_{bs1} - \kappa R_{ms1} \leq 0 \end{cases}, \end{aligned} \quad (134)$$

where $B_{BRC}^{OWRC,pl}, B_{MARC}^{OWRC,pl}$ are the data rate constraints, which have been introduced in (125) and (128), respectively, vector η denotes the power variables to be optimised and $\Omega_{OWRC,pl}$ stands for the power constraint to be satisfied; it establishes a constraint on the power variables. Those variables are summarised in Table 6. Additionally, the variable definition, which has been presented in equations (132) and (133), must be considered under the *individual-average* and *max-power constraints*.

Table 7: Power variables and power as a function of the power constraint

Power constraints	η	$\Omega_{OWRC,pl}$
Sum average power constraint	$\kappa_1, \kappa_2, \psi_1, \psi_2, \phi_1, \phi_2, \nu_1, \nu_2$	$\Omega_{OWRC,pl}^{sum-ave}, (129)$
Individual average power constraint	$\nu_1, \nu_2, \psi_1, \psi_2, \phi_1, \phi_2$	$\Omega_{OWRC,pl}^{ind-ave}, (130)$
Max-power constraint	$\nu_1, \nu_2, \vartheta_1, \vartheta_2, \phi_1, \phi_2$	$\Omega_{OWRC,pl}^{max-pow}, (131), (133)$

4.1.1.2 Two-way Relay channel

The TWRC was analysed in [ROCKET-3D1] and [Agus-09] for the single user case, i.e. BS, RS and MS, under different power constraints. In this section, our work is devoted to the design of resource allocation procedures in the case of multiple users to be served (BS, RS and multiple MS) by applying the same scenario that is depicted in Figure 2. Although there exists up to three transmission protocols [Kim-08], here only the *forwarding* protocol (or two-step) and *protocol I* (or three-step) will be tackled.

In contrast with the OWRC, the half-duplex relay accommodates *bidirectional* traffic simultaneously (DL and UL). The relay operates in DF mode and its transmission can be based on *superposition coding*, *XOR-coding* and Network Coding Relaying (NCR), as it is shown in [ROCKET-3D1]. Here we only apply the NCR technique, because it gets the best results for the single-user case. Likewise, we anticipate that the RS will utilise two data streams in the MU case, where each one of them, i.e., BS-MS₁ and BS-MS₂, will employ NCR. The RS will distribute its power among these data streams.

4.1.1.2.1 Forwarding

Similarly to the forwarding protocol that has been analysed in the OWRC case, this forwarding protocol assumes that there is no any direct communication between BS and MSs, and the relay must be employed for this purpose. The transmission model is depicted in Figure 75 and it is divided in two orthogonal phases. In the first one, BS and MSs transmit data to the RS, afterwards the RS transmits data to BS and MSs

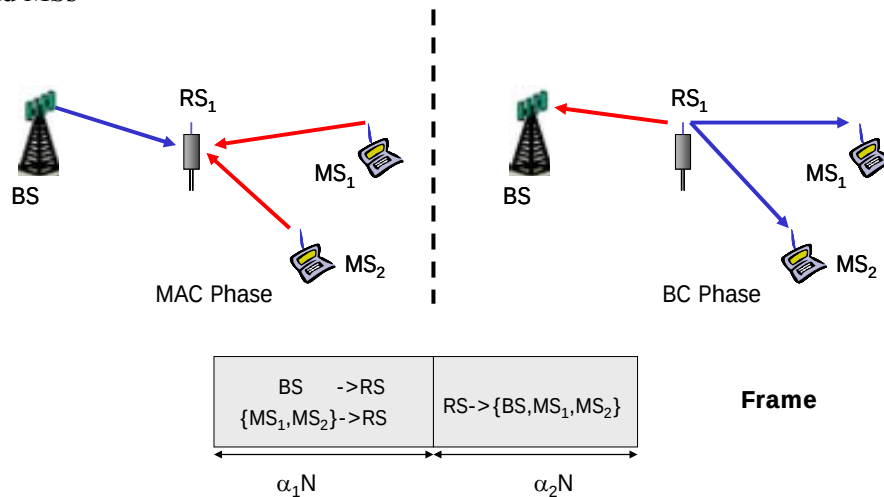


Figure 75. Transmissions in the TWRC under forwarding protocol

The communication is carried out by employing two orthogonal phases that are denoted by T_1 and T_2

$$\begin{cases} T_1 = [0, \alpha_1 N] \\ T_2 = (\alpha_1 N, (\alpha_1 + \alpha_2) N] \end{cases}, \quad (135)$$

where α_1, α_2 denote the fraction of time that is devoted to each phase.

All the terminals have a single antenna and transmit Gaussian signals of zero-mean and power given by

$$\begin{aligned} &DL \{x_{bs-rs1} \sim N(0, \tilde{\gamma}_1 P), \quad x_{bs-rs2} \sim N(0, \tilde{\gamma}_2 P) \\ &DL \& UL \{x_{rs-bsms1} \sim N(0, \tilde{\psi}_1 P), \quad x_{rs-bsms2} \sim N(0, \tilde{\psi}_2 P), \\ &UL \{x_{ms1-rs} \sim N(0, \tilde{\beta}_1 P), \quad x_{ms2-rs} \sim N(0, \tilde{\beta}_2 P) \end{aligned} \quad (136)$$

where x_{bs-rs1}, x_{bs-rs2} are the signals that are transmitted from the BS to the RS by using *superposition coding* and are intended to MS₁ and MS₂, respectively, i.e., DL transmission. The fraction of power that is allocated to each MS is described by $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$. The signals that are transmitted from the MSs to the RS are denoted by x_{ms1-rs} and x_{ms2-rs} ; they have a fraction of power allocation defined by $\tilde{\beta}_1, \tilde{\beta}_2$. Those signals describe the UL transmission. Finally, in the second phase of the communication, which is depicted in Figure 75, the RS transmits the signals $x_{rs-bsms1}$ and $x_{rs-bsms2}$ that are intended to BS-MS₁ and BS-MS₂ and are encoded under NCR, respectively. The fraction of allocated power to these two signals is described by $\tilde{\psi}_1, \tilde{\psi}_2$.

In the first phase of the communication, which is depicted in Figure 75, all the terminals transmit simultaneously to the RS. Hence, the received signal at the relay is defined by

$$y_{RS}(t) = h_{bs-rs}(x_{bs-rs1}(t) + x_{bs-rs2}(t)) + h_{ms1-rs}x_{ms1-rs}(t) + h_{ms2-rs}x_{ms2-rs}(t) + n_{rs}(t), t \in T_1, \quad (137)$$

where $h_{bs-rs}, h_{ms1-rs}, h_{ms2-rs}$ denote the channel coefficients of the BS-RS, MS₁-RS and MS₂-RS links, n_{rs} denote the received noise at the relay, x_{bs-rs1}, x_{bs-rs2} stand for the DL transmission that is intended to MSs and x_{ms1-rs}, x_{ms2-rs} describe the UL transmission that is devoted to the BS.

The relay decodes all the messages that are received in the first phase of the communication and groups the DL-UL messages of BS-MS₁ and BS-MS₂ in two new messages. The encoding technique for each new message is based on the NCR and *random binning* [ROCKET-3D1], [Agus-09]. Afterwards, the relay transmits these messages to all the remaining terminals in the second phase of the communication. The signal received by the different terminals is defined as

$$\begin{aligned} y_{BS}(t) &= h_{rs-bs}(x_{rs-bsms1}(t) + x_{rs-bsms2}(t)) + n_{bs}(t) & t \in T_2 \\ y_{MS_1}(t) &= h_{rs-ms1}(x_{rs-bsms1}(t) + x_{rs-bsms2}(t)) + n_{ms1}(t) & t \in T_2, \\ y_{MS_2}(t) &= h_{rs-ms2}(x_{rs-bsms1}(t) + x_{rs-bsms2}(t)) + n_{ms2}(t) & t \in T_2 \end{aligned} \quad (138)$$

where $h_{rs-bs}, h_{rs-ms1}, h_{rs-ms2}$ are the channel coefficients of the RS-BS, RS-MS₁ and RS-MS₂ links, $x_{rs-bsms1}$ stands for the signal that is intended to the BS and MS₁ and $x_{rs-bsms2}$ denotes the signal that is devoted to the BS and MS₂. It is important to remark that the BS must receive two independent messages from MS₁ and MS₂ (UL).

The achievable rate region that is attained during the first phase of the communication, which is depicted in Figure 75, can be described by a conventional MACH with four independent messages. It is described by the following 15 rate constraints [Cove-91],

$$B_{MAC}^{TWRC, fw} = \begin{cases} R_{bs1} \leq & \alpha_1 \log_2 (1 + \gamma_1 \rho_{bs_rs}) \\ R_{bs2} \leq & \alpha_1 \log_2 (1 + \gamma_2 \rho_{bs_rs}) \\ R_{ms1} \leq & \alpha_1 \log_2 (1 + \tilde{\beta}_1 \rho_{ms1_rs}) \\ R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\beta}_2 \rho_{ms2_rs}) \\ R_{bs1} + R_{bs2} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_1 \rho_{bs_rs} + \tilde{\gamma}_2 \rho_{bs_rs}) \\ R_{bs1} + R_{ms1} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_1 \rho_{bs_rs} + \tilde{\beta}_1 \rho_{ms1_rs}) \\ R_{bs1} + R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_1 \rho_{bs_rs} + \tilde{\beta}_2 \rho_{ms2_rs}) \\ R_{bs2} + R_{ms1} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_2 \rho_{bs_rs} + \tilde{\beta}_1 \rho_{ms1_rs}) \\ R_{bs2} + R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_2 \rho_{bs_rs} + \tilde{\beta}_2 \rho_{ms2_rs}) \\ R_{ms1} + R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\beta}_1 \rho_{ms1_rs} + \tilde{\beta}_2 \rho_{ms2_rs}) \\ R_{bs1} + R_{bs2} + R_{ms1} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_1 \rho_{bs_rs} + \tilde{\gamma}_2 \rho_{bs_rs} + \tilde{\beta}_1 \rho_{ms1_rs}) \\ R_{bs1} + R_{bs2} + R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_1 \rho_{bs_rs} + \tilde{\gamma}_2 \rho_{bs_rs} + \tilde{\beta}_2 \rho_{ms2_rs}) \\ R_{bs1} + R_{ms1} + R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_1 \rho_{bs_rs} + \tilde{\beta}_1 \rho_{ms1_rs} + \tilde{\beta}_2 \rho_{ms2_rs}) \\ R_{bs2} + R_{ms1} + R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_2 \rho_{bs_rs} + \tilde{\beta}_1 \rho_{ms1_rs} + \tilde{\beta}_2 \rho_{ms2_rs}) \\ R_{bs1} + R_{bs2} + R_{ms1} + R_{ms2} \leq & \alpha_1 \log_2 (1 + \tilde{\gamma}_1 \rho_{bs_rs} + \tilde{\gamma}_2 \rho_{bs_rs} + \tilde{\beta}_1 \rho_{ms1_rs} + \tilde{\beta}_2 \rho_{ms2_rs}) \end{cases} \quad (139)$$

where R_{bs1} , R_{bs2} , R_{ms1} and R_{ms2} stand for the data rates of the messages that are transmitted from the BS to MSs and the other way round.

In the second phase of the transmission, the relay transmits two independent messages to BS-MS₁ and BS-MS₂, hence, the signal that is intended to one of the pairs will be seen as interference for the other pairs. In order to minimise the effect of the interference, the relay will apply DPC for encoding those messages. Since our scenario assumes that all the terminals are equipped with a single antenna, we have a *physically degraded* scenario, where the BS-RS link presents the largest SNR and the *strong* and *weak* user categorisation will depend on the SNR of the RS-MS₁ and RS-MS₂ links.

Assuming that we first encode the message that is devoted to MS₂-BS (*weak user*) and then encode the message that is intended to MS₁-BS (*strong user*), we get the following achievable rate region in the broadcast phase of the relay

$$B_{BC}^{TWRC, fw} = \begin{cases} R_{ms1} \leq \alpha_2 \log_2 (1 + \tilde{\psi}_1 \rho_{rs-bs}) \\ R_{ms2} \leq \alpha_2 \log_2 \left(1 + \frac{\tilde{\psi}_2 \rho_{rs-bs}}{1 + \tilde{\psi}_1 \rho_{rs-bs}} \right) \\ R_{bs1} \leq \alpha_2 \log_2 (1 + \tilde{\psi}_1 \rho_{rs-ms1}) \\ R_{bs2} \leq \alpha_2 \log_2 \left(1 + \frac{\tilde{\psi}_2 \rho_{rs-ms2}}{1 + \tilde{\psi}_1 \rho_{rs-ms2}} \right) \end{cases}, \quad (140)$$

where R_{bs1} , R_{bs2} are the rate of the messages that are transmitted from the BS and R_{ms1} , R_{ms2} are the rates of the messages that are transmitted from the MSs. Finally, ρ_{rs-bs} , ρ_{rs-ms1} , ρ_{rs-ms2} stand for the SNR of the RS-BS, RS-MS₁ and RS-MS₂ links. Notice that the RS only transmits two signals with the fraction of power given by $\tilde{\psi}_1$ (for the pair BS-MS₁), $\tilde{\psi}_2$ (for the pair BS-MS₂).

We can further simplify the expression by using

$$B_{BC}^{TWRC, fw} = \begin{cases} R_{ms1} + R_{ms2} \leq \alpha_2 \log_2 \left(1 + (\tilde{\psi}_1 + \tilde{\psi}_2) \rho_{rs-bs} \right) \\ R_{bs1} \leq \alpha_2 \log_2 \left(1 + \tilde{\psi}_1 \rho_{rs-ms1} \right) \\ R_{bs2} \leq \alpha_2 \log_2 \left(1 + \frac{\tilde{\psi}_2 \rho_{rs-ms2}}{1 + \tilde{\psi}_1 \rho_{rs-ms2}} \right) \end{cases}, \quad (141)$$

Moreover, we can use the properties of the BC-MACd duality in order to transform the previous achievable rate region into

$$B_{BC}^{TWRC, fw} = \begin{cases} R_{ms1} + R_{ms2} \leq \alpha_2 \log_2 \left(1 + (\tilde{\lambda}_1 + \tilde{\lambda}_2) \rho_{rs-bs} \right) \\ R_{bs1} + R_{bs2} \leq \alpha_2 \log_2 \left(1 + \tilde{\lambda}_1 \rho_{rs-ms1} + \tilde{\lambda}_2 \rho_{rs-ms2} \right) \\ s.t. \quad \tilde{\lambda}_1 + \tilde{\lambda}_2 = \tilde{\psi}_1 + \tilde{\psi}_2 \end{cases}, \quad (142)$$

where the power allocation at the relay is obtained as (assuming $\rho_{rs-ms1} \geq \rho_{rs-ms2}$)

$$\tilde{\psi}_1 = \frac{\tilde{\lambda}_1}{1 + \rho_{rs-ms2} \tilde{\lambda}_2} \text{ and } \tilde{\psi}_2 = \tilde{\lambda}_2 \left(1 + \frac{\rho_{rs-ms2} \tilde{\lambda}_1}{1 + \rho_{rs-ms2} \tilde{\lambda}_2} \right). \quad (143)$$

A. Power constraints

The three constraints: *sum-average*, *individual average* and *max-power constraint* are satisfied by updating the power variables that are described in the achievable rate regions $B_{MAC}^{TWRC, fw}$, $B_{BC}^{TWRC, fw}$, which are shown in (139) and (142), with the ones that are presented in Table 8.

Table 8: Variables employed under the different power constraints

Power constraints	Transmission power variables							
	$\tilde{\gamma}_1$	$\tilde{\gamma}_2$	$\tilde{\beta}_1$	$\tilde{\beta}_2$	$\tilde{\psi}_1$	$\tilde{\psi}_2$	$\tilde{\lambda}_1$	$\tilde{\lambda}_2$
<i>Sum-average power constraint</i>	$\frac{\gamma_1}{\alpha_1}$	$\frac{\gamma_2}{\alpha_1}$	$\frac{\beta_1}{\alpha_1}$	$\frac{\beta_2}{\alpha_1}$	$\frac{\psi_1}{\alpha_2}$	$\frac{\psi_2}{\alpha_2}$	$\frac{\lambda_1}{\alpha_2}$	$\frac{\lambda_2}{\alpha_2}$
<i>Individual average power constraint</i>	$\frac{\gamma_1}{\alpha_1}$	$\frac{\gamma_2}{\alpha_1}$	$\frac{1}{\alpha_1}$	$\frac{1}{\alpha_1}$	$\frac{\psi_1}{\alpha_2}$	$\frac{\psi_2}{\alpha_2}$	$\frac{\lambda_1}{\alpha_2}$	$\frac{\lambda_2}{\alpha_2}$
<i>Max-power constraint</i>	γ_1	γ_2	1	1	ψ_1	ψ_2	λ_1	λ_2

In order to satisfy the different power constraints, the equalities $\Omega_{TWRC, fw}^{sum-ave}$, $\Omega_{TWRC, fw}^{max-pow}$ must be considered in our optimisation problem, (104).

- *Sum-average power constraint* (total power in the four-terminal network)

$$\Omega_{TWRC, fw}^{sum-ave} = \begin{cases} \gamma_1 + \gamma_2 + \beta_1 + \beta_2 + \psi_1 + \psi_2 = 1 \\ \gamma_1 + \gamma_2 + \beta_1 + \beta_2 + \lambda_1 + \lambda_2 = 1 \end{cases}. \quad (144)$$

- *Individual average power constraint* (total power employed by each terminal)

$$\Omega_{TWRC, fw}^{ind-ave} = \begin{cases} \gamma_1 + \gamma_2 = 1 \\ \psi_1 + \psi_2 = 1 \\ \lambda_1 + \lambda_2 = 1 \end{cases}. \quad (145)$$

- *Max-power constraint* (power used by each terminal on each phase)

$$\Omega_{TWRC, fw}^{\max-pow} = \begin{cases} \gamma_1 + \gamma_2 = 1 \\ \psi_1 + \psi_2 = 1 \\ \lambda_1 + \lambda_2 = 1 \end{cases} \quad (146)$$

B. Solving the resource allocation problem

Under the sum-average power constraint, the resource allocation problem satisfy the required conditions of a convex problem [Boyd-04]. However, we must introduce two variables in order to transform the problem into a convex one for the max-power constraint

$$\begin{cases} \nu_1 = \alpha_2 \lambda_1 \\ \nu_2 = \alpha_2 \lambda_2 \end{cases} \quad \Omega_{TWRC, fw}^{\max-pow} = \{ \nu_1 + \nu_2 - \alpha_2 = 0, \quad (147)$$

where ν_1, ν_2 are the variables that are connected with λ_1, λ_2 and have been obtained after applying the BC-MACH duality in (142), and α_2 denotes the fraction of time devoted for the phase 2 (where the relay transmits), which is depicted in Figure 75. Notice that these variables also are employed for the OWRC forwarding in (118).

The optimal resource allocation is obtained by maximising

$$\begin{aligned} & \max_{\alpha, \eta, R_{ms1}, R_{ms2}, R_{bs1}, R_{bs2}} \mu_1 R_{bs1} + \mu_2 R_{bs2} \\ & s.t. \quad \begin{cases} (R_{bs1}, R_{bs2}, R_{ms1}, R_{ms2}) \in (B_{MAC}^{TWRC, fw} \cup B_{BC}^{TWRC, fw}) \\ \alpha_1 + \alpha_2 = 1, & R_{bs2} - \kappa R_{ms2} \leq 0 \\ \Omega_{TWRC, fw}, & R_{bs1} - \kappa R_{ms1} \leq 0 \end{cases} \end{aligned} \quad (148)$$

where $B_{MAC}^{TWRC, fw}, B_{BC}^{TWRC, fw}$ are the data rate constraints that have been introduced in (139) and (142), respectively, vector η denotes the power variables to be optimised and $\Omega_{TWRC, fw}$ stands for the power constraint to be satisfied, establishing a constraint on the power variables. These variables are summarised in Table 9. Under the *max-power constraint*, the variable definition in equation (147) must be considered.

Table 9: Power variables and power as a function of the power constraint

Power constraints	η	$\Omega_{OWRC, fw}$
Sum average power constraint	$\gamma_1, \gamma_2, \lambda_1, \lambda_2, \beta_1, \beta_2$	$\Omega_{TWRC, fw}^{sum-ave}, (144)$
Individual average power constraint	$\gamma_1, \gamma_2, \lambda_1, \lambda_2$	$\Omega_{TWRC, fw}^{ind-ave}, (145)$
Max-power constraint	$\gamma_1, \gamma_2, \nu_1, \nu_2$	$\Omega_{TWRC, fw}^{\max-pow}, (146), (147)$

4.1.1.2.2 Protocol I

The direct link between BS and the different MSs can be exploited by designing the protocol in three orthogonal phases (three-stage). In Comparison with the forwarding protocol in section 4.1.1.2.1, the terminals can benefit of receiving the intended signals through two links at the cost of an additional orthogonal phase. The transmission protocol is sketched in Figure 76; this figure shows which terminals are active during each phase. In the first phase, the BS broadcast two intended messages to MS₁ and MS₂; they are also received and decoded by the relay. Afterwards, MSs transmit simultaneously their messages to the RS and BS. Finally, after the relay has decoded the four messages (2 in the DL and 2 in the UL), it generates two independent streams that are intended to the pairs BS-MS₁ and BS-MS₂ by using the *random binning* based NCR [ROCKET-3D1] and transmit them to the terminals in the third phase of the communication. Notice that in this configuration, the first phase is devoted to the DL, the second phase to the UL and the third phase contains DL and UL traffic.

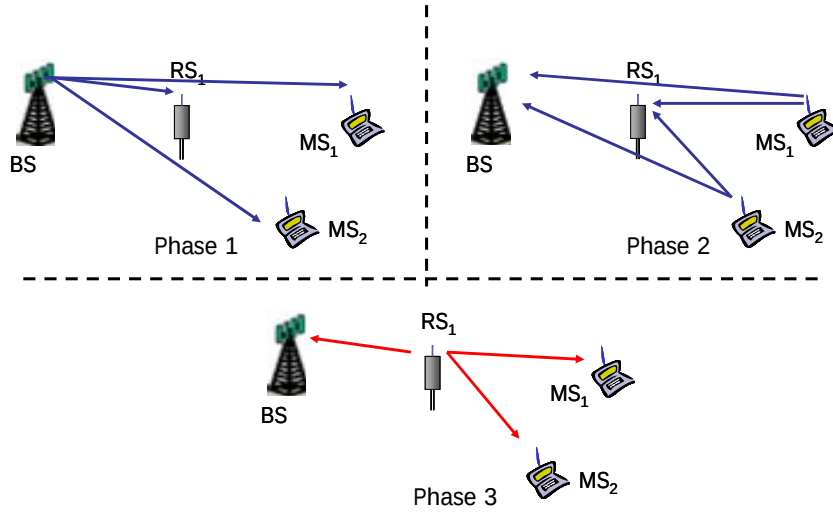


Figure 76. Transmissions in the TWRC under protocol I

The three phases of the communication are defined by the following time intervals T_1 , T_2 and T_3

$$\begin{cases} T_1 = [0, \alpha_1 N] \\ T_2 = (\alpha_1 N, (\alpha_1 + \alpha_2) N] \\ T_3 = ((\alpha_1 + \alpha_2) N, (\alpha_1 + \alpha_2 + \alpha_3) N] \end{cases} . \quad (149)$$

All the terminals transmit Gaussian signals of zero-mean and power given by

$$\begin{aligned} DL \{ x_{bs-ms1} &\sim N(0, \tilde{\gamma}_1 P), \quad x_{bs-ms2} \sim N(0, \tilde{\gamma}_2 P) \\ UL \{ x_{ms1-bs} &\sim N(0, \tilde{\varphi}_1 P), \quad x_{ms2-bs} \sim N(0, \tilde{\varphi}_2 P), \\ DL \& UL \{ x_{rs-bsms1} &\sim N(0, \tilde{\beta}_1 P), \quad x_{rs-bsms2} \sim N(0, \tilde{\beta}_2 P) \end{aligned} \quad (150)$$

where x_{bs-ms1} , x_{bs-ms2} are the signals that are employed by the BS in the first phase and they account for the DL transmission, x_{ms1-bs} , x_{ms2-bs} denote the transmit signals from the MSs in the second phase (UL traffic). Finally, $x_{rs-bsms1}$, $x_{rs-bsms2}$ stand for the transmit signals from the relay in the third phase and they are intended to the pairs BS-MS₁ and BS-MS₂, respectively.

The signal received at the relay is

$$y_{RS}(t) = \begin{cases} h_{bs-rs} (x_{bs-ms1}(t) + x_{bs-ms2}(t)) + n_{rs}(t) & t \in T_1 \\ h_{ms1-rs} x_{ms1-bs}(t) + h_{ms2-rs} x_{ms2-bs}(t) + n_{rs}(t) & t \in T_2, \\ 0 & t \in T_3 \end{cases} \quad (151)$$

where h_{bs-rs} , h_{ms1-rs} , h_{ms2-rs} denote the channel coefficients of the BS-RS, MS₁-RS and MS₂-RS links, x_{bs-ms1} , x_{bs-ms2} are the transmit signals from the BS and x_{ms1-bs} , x_{ms2-bs} are the transmit signals from the MSs, (150). T_1 , T_2 and T_3 are the different time intervals that have been introduced in (149).

On the other hand, the received signals at the BS and MS are

$$\begin{aligned}
 y_{BS}(t) &= \begin{cases} 0 & t \in T_1 \\ h_{ms1-bs}x_{ms1-bs}(t) + h_{ms2-bs}x_{ms2-bs}(t) + n_{rs}(t) & t \in T_2 \\ h_{bs-rs}(x_{rs-bsms1}(t) + x_{rs-bsms2}(t)) + n_{bs}(t) & t \in T_3 \end{cases} \\
 y_{MS_1}(t) &= \begin{cases} h_{bs-ms1}(x_{bs-ms1}(t) + x_{bs-ms2}(t)) + n_{ms1}(t) & t \in T_1 \\ 0 & t \in T_2 \\ h_{rs-ms1}(x_{rs-bsms1}(t) + x_{rs-bsms2}(t)) + n_{ms1}(t) & t \in T_3 \end{cases} , \\
 y_{MS_2}(t) &= \begin{cases} h_{bs-ms2}(x_{bs-ms1}(t) + x_{bs-ms2}(t)) + n_{ms2}(t) & t \in T_1 \\ 0 & t \in T_2 \\ h_{rs-ms2}(x_{rs-bsms1}(t) + x_{rs-bsms2}(t)) + n_{ms2}(t) & t \in T_3 \end{cases}
 \end{aligned} \quad (152)$$

where T_i stands for the i -th phase, (149), h_{ms1-bs} , h_{ms2-bs} , h_{bs-rs} , h_{bs-ms1} , h_{bs-ms2} , h_{rs-bs} account for the channel coefficients of the MS₁-BS, MS₂-BS, BS-RS, BS-MS₁, BS-MS₂ and RS-BS links. Notice that our formulation assumes that in general the MS₁-BS and BS-MS₁ channels are different. The transmit signals from the different terminals are denoted by x_{ms1-bs} , x_{ms2-bs} , x_{bs-ms1} , x_{bs-ms2} , $x_{rs-bsms1}$ and $x_{rs-bsms2}$, which are defined in (150).

The communication from the BS to the MSs can be modelled by a BRC in the phases T_1 and T_3 by following similar guidelines as in section 4.1.1.2. Notice that there is a BC in each phase and all the terminals are equipped with a single antenna. Since we assume that the BS-RS link presents the highest quality, we can identify an *ordered set* of users on each phase (*weak* and *strong* user). The encoding at the BS (in phase 1) and at the RS (in phase 3) will be based on the DPC [Cost-81]. We first encode the signal that is intended to the *weak* user and we afterwards encode the signal that is devoted to the *strong* user. As a consequence, the *strong* user will receive a signal without interference, whereas the *weak* user will see the signal that is intended to the *strong* user as interference. Let us assume that $\rho_{bs-ms1} > \rho_{bs-ms2}$ and $\rho_{rs-ms1} < \rho_{rs-ms2}$, hence, we identify MS₂ as the *weak* user in phase 1 and MS₁ as the *weak* user in phase 3. It means that during the first phase, we first encode the signal x_{bs-ms2} and afterwards x_{bs-ms1} , whereas in the third phase, since we have identified the pair BS-MS₁ as the *weak* user, we first encode $x_{rs-bsms1}$ and afterwards $x_{rs-bsms2}$.

The achievable rate region will be defined with the previous assumption as

$$B_{BRC}^{TWRC,pl} \left\{ \begin{aligned} R_{bs1} + R_{bs2} &\leq \alpha_1 \log_2(1 + (\tilde{\gamma}_1 + \tilde{\gamma}_2)\rho_{bs-rs}) \\ R_{bs1} &\leq \alpha_1 \log_2(1 + \tilde{\gamma}_1\rho_{bs-ms1}) + \alpha_3 \log_2\left(1 + \frac{\tilde{\beta}_1\rho_{rs-ms1}}{1 + \tilde{\beta}_2\rho_{rs-ms1}}\right) \\ R_{bs2} &\leq \alpha_1 \log_2\left(1 + \frac{\tilde{\gamma}_2\rho_{bs-ms2}}{1 + \tilde{\gamma}_1\rho_{bs-ms1}}\right) + \alpha_3 \log_2(1 + \tilde{\beta}_2\rho_{rs-ms1}) \end{aligned} \right. , \quad (153)$$

where ρ_{bs-ms1} , ρ_{rs-ms1} , ρ_{bs-ms2} , ρ_{rs-ms2} denote the SNR of the BS-MS₁, BS-MS₂, RS-MS₁ and RS-MS₂ link, respectively. The first data rate constraint is imposed by the maximum decoded rate at the relay, because it must be able to decode both transmit messages from the BS (x_{bs-ms1} , x_{bs-ms2}). The second data rate constraint is the data rate that is attained at MS₁, where it has been assumed that the MS₁ is the *strong* user in phase 1 (α_1) and the *weak* user in phase 3 (α_3). Hence, there is no interference in phase 1, whereas the transmit signal from the relay to the pair BS-MS₂ ($x_{rs-bsms2}$) must be considered as interference in phase 3. For the data rate of MS₂ described by the third constraint, MS₂ is the *weak* user in phase 1 (so the signal x_{bs-ms1} is seen as interference) and the *strong* user in phase 3.

Likewise, we can apply de BC-MACH duality [Vish-04] on each phase (phase 1 and phase 3) and the achievable rate region becomes

$$B_{BRC}^{TWRC,pl} = \begin{cases} R_{bs1} + R_{bs2} \leq \alpha_1 \log_2 \left(1 + (\tilde{\lambda}_1 + \tilde{\lambda}_2) \rho_{bs-rs} \right) \\ R_{bs1} + R_{bs2} \leq \alpha_1 \log_2 \left(1 + \tilde{\lambda}_1 \rho_{bs-ms1} + \tilde{\lambda}_2 \rho_{bs-ms2} \right) + \alpha_3 \log_2 \left(1 + \tilde{\psi}_1 \rho_{bs-ms1} + \tilde{\psi}_2 \rho_{bs-ms2} \right) \end{cases}$$

$$s.t. \quad \tilde{\lambda}_1 + \tilde{\lambda}_2 = \tilde{\gamma}_1 + \tilde{\gamma}_2$$

$$\tilde{\psi}_1 + \tilde{\psi}_2 = \tilde{\beta}_1 + \tilde{\beta}_2 \quad , \quad (154)$$

where the new variables $\tilde{\lambda}_1, \tilde{\lambda}_2, \tilde{\psi}_1, \tilde{\psi}_2$ have been introduced, which are connected with the fraction of power of the different transmitted signals. In the configuration that is assumed in (153) (i.e. $\rho_{bs-ms1} > \rho_{bs-ms2}$ and $\rho_{rs-ms1} < \rho_{rs-ms2}$), the allocated fraction of power to the signals x_{bs-ms1} , x_{bs-ms2} , x_{rs-ms1} and x_{rs-ms2} is

$$\tilde{\gamma}_1 = \frac{\tilde{\lambda}_1}{1 + \rho_{bs-ms2} \tilde{\lambda}_2}, \tilde{\gamma}_2 = \tilde{\lambda}_2 \left(1 + \frac{\rho_{bs-ms2} \tilde{\lambda}_1}{1 + \rho_{bs-ms2} \tilde{\lambda}_2} \right), \tilde{\beta}_1 = \tilde{\psi}_1 \left(1 + \frac{\rho_{rs-ms1} \tilde{\psi}_2}{1 + \rho_{rs-ms1} \tilde{\psi}_1} \right), \tilde{\beta}_2 = \frac{\tilde{\psi}_2}{1 + \rho_{rs-ms1} \tilde{\psi}_1} \quad (155)$$

On the other hand, the transmission that is originated at MSs and is devoted to the BS can be described by the MARC channel when phases 2 and 3 are considered. According to the signal model in (151) and (152), and the transmitted signals in (150), the achievable rate region becomes

$$B_{MARC}^{TWRC,pl} = \begin{cases} R_{ms1} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-rs}) \\ R_{ms2} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_2 \rho_{ms2-rs}) \\ R_{ms1} + R_{ms2} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-rs} + \tilde{\varphi}_2 \rho_{ms2-rs}) \\ R_{ms1} + R_{ms2} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-bs} + \tilde{\varphi}_2 \rho_{ms2-bs}) + \alpha_3 \log_2 (1 + (\tilde{\beta}_1 + \tilde{\beta}_2) \rho_{rs-ms}) \end{cases} \quad (156)$$

where $\rho_{ms1-rs}, \rho_{ms2-rs}, \rho_{ms1-bs}, \rho_{ms2-bs}$ stand for the SNR in the MS₁-RS, MS₂-RS, MS₁-BS and MS₂-BS links. The first three rate constraints are due to the MACH at the relay, whereas the fourth rate constraint is the maximum sum-rate at the BS when combining the received signals during the second and third phase. Notice that the BS must receive both transmitted messages from the MSs with the help of the relay. Moreover, since we have applied the BC-MAC duality property, when using the variables $\tilde{\psi}_1, \tilde{\psi}_2$ instead of $\tilde{\beta}_1, \tilde{\beta}_2$ for the transmit signals from the relay in the third phase ($x_{rs-bsms1}, x_{rs-bsms2}$), the achievable rate region is modified as

$$B_{MARC}^{TWRC,pl} = \begin{cases} R_{ms1} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-rs}) \\ R_{ms2} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_2 \rho_{ms2-rs}) \\ R_{ms1} + R_{ms2} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-rs} + \tilde{\varphi}_2 \rho_{ms2-rs}) \\ R_{ms1} + R_{ms2} \leq \alpha_2 \log_2 (1 + \tilde{\varphi}_1 \rho_{ms1-bs} + \tilde{\varphi}_2 \rho_{ms2-bs}) + \alpha_3 \log_2 (1 + (\tilde{\psi}_1 + \tilde{\psi}_2) \rho_{rs-ms}) \end{cases} \quad (157)$$

where $\tilde{\psi}_1, \tilde{\psi}_2$ are defined in (155).

A. Power constraints

The power variables that are introduced in the achievable rate regions $B_{BRC}^{TWRC,pl}, B_{MARC}^{TWRC,pl}$, which are shown in (154) and (157), must be updated with the ones shown in Table 10.

Table 10: Variables employed under the different power constraints

Power constraints	Transmission power variables									
	$\tilde{\gamma}_1$	$\tilde{\gamma}_2$	$\tilde{\lambda}_1$	$\tilde{\lambda}_2$	$\tilde{\varphi}_1$	$\tilde{\varphi}_2$	$\tilde{\beta}_1$	$\tilde{\beta}_2$	$\tilde{\psi}_1$	$\tilde{\psi}_2$
Sum-average power constraint	$\underline{\gamma}_1$	$\underline{\gamma}_2$	$\underline{\lambda}_1$	$\underline{\lambda}_2$	$\underline{\varphi}_1$	$\underline{\varphi}_2$	$\underline{\beta}_1$	$\underline{\beta}_2$	$\underline{\psi}_1$	$\underline{\psi}_2$
Individual average power constraint	$\underline{\gamma}_1$	$\underline{\gamma}_2$	$\underline{\lambda}_1$	$\underline{\lambda}_2$	$\underline{\varphi}_1$	$\underline{\varphi}_2$	$\underline{\beta}_1$	$\underline{\beta}_2$	$\underline{\psi}_1$	$\underline{\psi}_2$
Max-power constraint	γ_1	γ_2	λ_1	λ_2	1	1	β_1	β_2	ψ_1	ψ_2

In order to satisfy the different power constraints the equalities $\Omega_{TWRC,pl}^{sum-ave}, \Omega_{TWRC,pl}^{max-pow}$ must be considered in our optimisation problem, (104).

- *Sum-average power constraint* (total power in the four-terminal network)

$$\Omega_{TWRC,pl}^{sum-ave} = \begin{cases} \gamma_1 + \gamma_2 + \beta_1 + \beta_2 + \varphi_1 + \varphi_2 = 1 \\ \lambda_1 + \lambda_2 + \psi_1 + \psi_2 + \varphi_1 + \varphi_2 = 1 \end{cases} \quad (158)$$

- *Individual average power constraint* (total consumed power by each terminal)

$$\Omega_{TWRC,pl}^{ind-ave} = \begin{cases} \gamma_1 + \gamma_2 = 1 \\ \beta_1 + \beta_2 = 1 \\ \psi_1 + \psi_2 = 1 \\ \lambda_1 + \lambda_2 = 1 \end{cases} \quad (159)$$

- *Max-power constraint* (utilised power by each terminal on each phase)

$$\Omega_{TWRC,pl}^{max-pow} = \begin{cases} \gamma_1 + \gamma_2 = 1 \\ \beta_1 + \beta_2 = 1 \\ \psi_1 + \psi_2 = 1 \\ \lambda_1 + \lambda_2 = 1 \end{cases} \quad (160)$$

B. Solving the resource allocation problem

The resource allocation is a convex problem under the *sum-average power constraint*, but we have to employ the following variables for the *max-power constraint*

$$\begin{cases} \nu_1 = \alpha_1 \lambda_1 \\ \nu_2 = \alpha_1 \lambda_2 \\ \vartheta_1 = \alpha_3 \psi_1 \\ \vartheta_2 = \alpha_3 \psi_2 \end{cases} \quad \Omega_{TWRC,pl}^{max-pow} = \begin{cases} \nu_1 + \nu_2 - \alpha_1 = 0 \\ \vartheta_1 + \vartheta_2 - \alpha_3 = 0 \end{cases} \quad (161)$$

where ν_1, ν_2 are the variables connected with λ_1, λ_2 , which have been obtained after applying the BC-MACH duality property in (154), whereas ϑ_1, ϑ_2 are connected with ψ_1, ψ_2 . The fractions of times α_1, α_3 denote the fraction of times that are devoted for the phase 1 and phase 3.

The optimal resource allocation is obtained by maximising

$$\begin{aligned} & \max_{\alpha, \eta, R_{ms1}, R_{ms2}, R_{bs1}, R_{bs2}} \mu_1 R_{bs1} + \mu_2 R_{bs2} \\ & s.t. \quad \begin{cases} (R_{bs1}, R_{bs2}, R_{ms1}, R_{ms2}) \in (B_{MARC}^{TWRC,pl} \cup B_{BRC}^{TWRC,pl}) \\ \alpha_1 + \alpha_2 + \alpha_3 = 1, & R_{bs2} - \kappa R_{ms2} \leq 0 \\ \Omega_{TWRC,pl}, & R_{bs1} - \kappa R_{ms1} \leq 0 \end{cases} \end{aligned} \quad (162)$$

where $B_{BRC}^{TWRC,pl}, B_{MARC}^{TWRC,pl}$ are the data rate constraints, which have been introduced in (154) and (157), respectively, vector η denotes the power variables to be optimised and $\Omega_{TWRC,pl}$ stands for the power constraint to be satisfied, establishing a constraint on the power variables. These variables are summarised in Table 11. Under the *max-power constraint*, the variable definitions that is presented in equation (161) must be considered.

Table 11: Power variables and power as a function of the power constraint

Power constraints	η	$\Omega_{OWRC, fw}$
<i>Sum average power constraint</i>	$\lambda_1, \lambda_2, \psi_1, \psi_2, \phi_1, \phi_2$	$\Omega_{TWRC, pl}^{sum-ave}, (158)$
<i>Individual-average power constraint</i>	$\lambda_1, \lambda_2, \psi_1, \psi_2$	$\Omega_{TWRC, pl}^{ind-ave}, (159)$
<i>Max-power constraint</i>	$\nu_1, \nu_2, \vartheta_1, \vartheta_2$	$\Omega_{TWRC, pl}^{max-pow}, (160) (161)$

4.1.1.3 Results

We define four channel configurations for analysing the achievable rate region that is obtained by the optimal resource allocation, which has been investigated in the previous sections for different transmission techniques. We will reference these channel configurations by A, B, C and D, and the SNR of the different links are defined as

$$\text{Channel Conf} = \begin{cases} \mathbf{A} : \rho_{bs-ms1} = 6 \text{ dB}, \rho_{bs-ms2} = 4 \text{ dB}, \rho_{bs-rs} = 20 \text{ dB}, \rho_{rs-ms1} = 10 \text{ dB}, \rho_{rs-ms2} = 16 \text{ dB} \\ \mathbf{B} : \rho_{bs-ms1} = 6 \text{ dB}, \rho_{bs-ms2} = 4 \text{ dB}, \rho_{bs-rs} = 20 \text{ dB}, \rho_{rs-ms1} = 16 \text{ dB}, \rho_{rs-ms2} = 10 \text{ dB} \\ \mathbf{C} : \rho_{bs-ms1} = 6 \text{ dB}, \rho_{bs-ms2} = 4 \text{ dB}, \rho_{bs-rs} = 20 \text{ dB}, \rho_{rs-ms1} = 12 \text{ dB}, \rho_{rs-ms2} = 12 \text{ dB} \\ \mathbf{D} : \rho_{bs-ms1} = 6 \text{ dB}, \rho_{bs-ms2} = 6 \text{ dB}, \rho_{bs-rs} = 20 \text{ dB}, \rho_{rs-ms1} = 12 \text{ dB}, \rho_{rs-ms2} = 12 \text{ dB} \end{cases}, (163)$$

where reciprocal channels for the reverse links, i.e. MS₁-BS, BS-MS₂, RS-BS, MS₁-RS, and MS₂-RS links, have been assumed. Notice that for the channel *configuration A*, MS₁ is the *strong user* with respect to the BS-MSs link, whereas MS₂ becomes the *strong user* with respect to the RS-MSs link. For *channel configuration B*, MS₁ is the strong user in all links. Additionally, channel *configuration C* and *D* are devoted to investigate the symmetries of the channels. For the *option C*, both MSs present the same SNR with regard to the RS, whereas for *option D*, MSs have the same SNR in the BS-MSs and RS-MSs links.

The transmission techniques analysed will be:

- **TDMA DT.** The *DL* and *UL* transmissions from BS to two MSs are carried out in orthogonal time slots, which durations and transmitted power is properly optimised.
- **BC&MACH DT.** The *DL* transmission is done by using the BC and *UL* with the MACH. There are two orthogonal time slots where the BS transmits to the MSs in the first one, whereas both MSs transmit simultaneously in the second one. Duration of each phase and transmit power is optimised.
- **OWRC protocol I.** This is the protocol analysed in section 4.1.1.1.2.
- **OWRC forwarding.** This is the protocol analysed in section 4.1.1.2.1.
- **TWRC protocol I.** Protocol analysed in section 4.1.1.2.2.
- **TWRC forwarding.** Protocol investigated in section 4.1.1.2.1.
- **Orthogonal TWRC protocol I.** This protocol is based on the orthogonal transmission for two MSs by using the single-user TWRC protocol that we have studied in [ROCKET-3D1] and [Agus-09]. The UL-DL transmission between BS-MS₁ is done in three orthogonal time slots and the transmission between BS-MS₂ is afterwards performed. There are six orthogonal phases in the frame.
- **Orthogonal TWRC forwarding.** In this case, the orthogonal transmission is performed by using the single-user TWRC forwarding [ROCKET-3D1][Agus-09]. The frame consists of four orthogonal phases.

The results will consider different traffic asymmetries between the *UL* and *DL*, as well as the power constraints that have previously been looked into. To this end, the results are divided in two sections;

section 4.1.1.3.1 tackles the dependence of the results on the channel configuration, where the traffic asymmetry is fixed; whereas, section 4.1.1.3.2 takes only into account the channel *configuration A* and it investigates the performance for different values of traffic asymmetry. The power constraints that are analysed in both sections will be the *sum-average* and *max-power constraint*. The results due to the *individual-average power constraint* are depicted in Figure 79 and Figure 81.

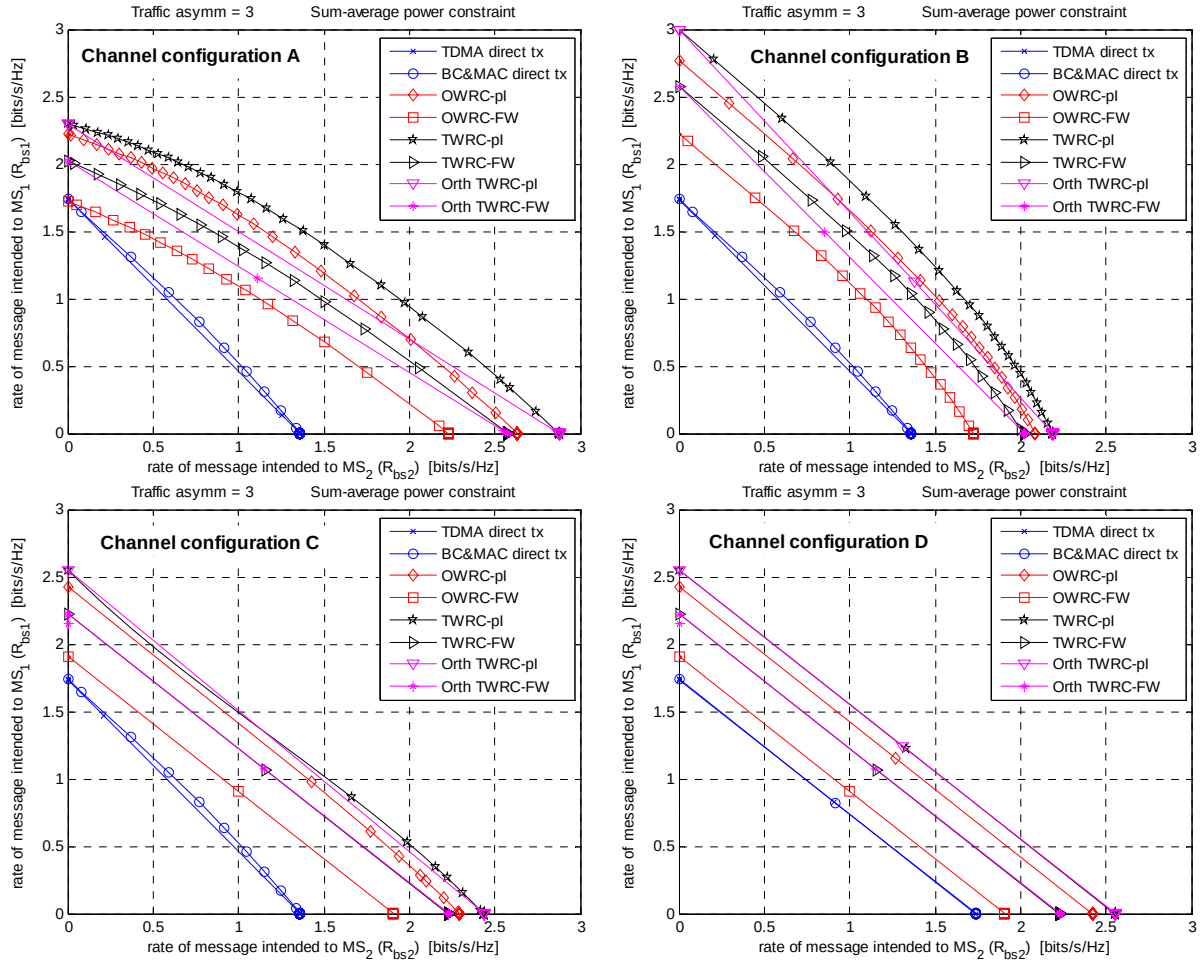


Figure 77. Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Channel configuration A, Top-right) Channel configuration B, Bottom-left) Channel configuration C and Bottom-right) Channel Configuration D, (163). Traffic Asymmetry = 3, Sum-average power constraint.

4.1.1.3.1 Dependence on the channel configuration

Figure 77 presents the achievable rate that is obtained by the intended messages to MS_1 and MS_2 in the *DL*, under the different transmission techniques with a *sum-average power* constraint and a traffic asymmetry equal to 3 (i.e. the data rate of the messages in the *UL* is one third of the data rate that is attained in the *DL*). The different channel configurations are considered in the top-left, top-right, bottom-left and bottom-right of Figure 77. The results indicate that the MU TWRC techniques are better than the MU OWRC techniques for the configuration that is considered here. Moreover, when channel links tend to be symmetric (*channel configuration C and D*, (163)), then the MU TWRC (protocol I and forwarding) and the orthogonal TWRC transmissions tend to deliver the same performance. This behaviour can be explained by the performance of the BC on a symmetric channel configuration [Cove-91], [Tse-05], where the orthogonal transmission attains the best performance. Finally, the OWRC protocol I transmission can outperform the results that are obtained by the orthogonal TWRC for asymmetric channel (*channel configuration A and B*, (163)).

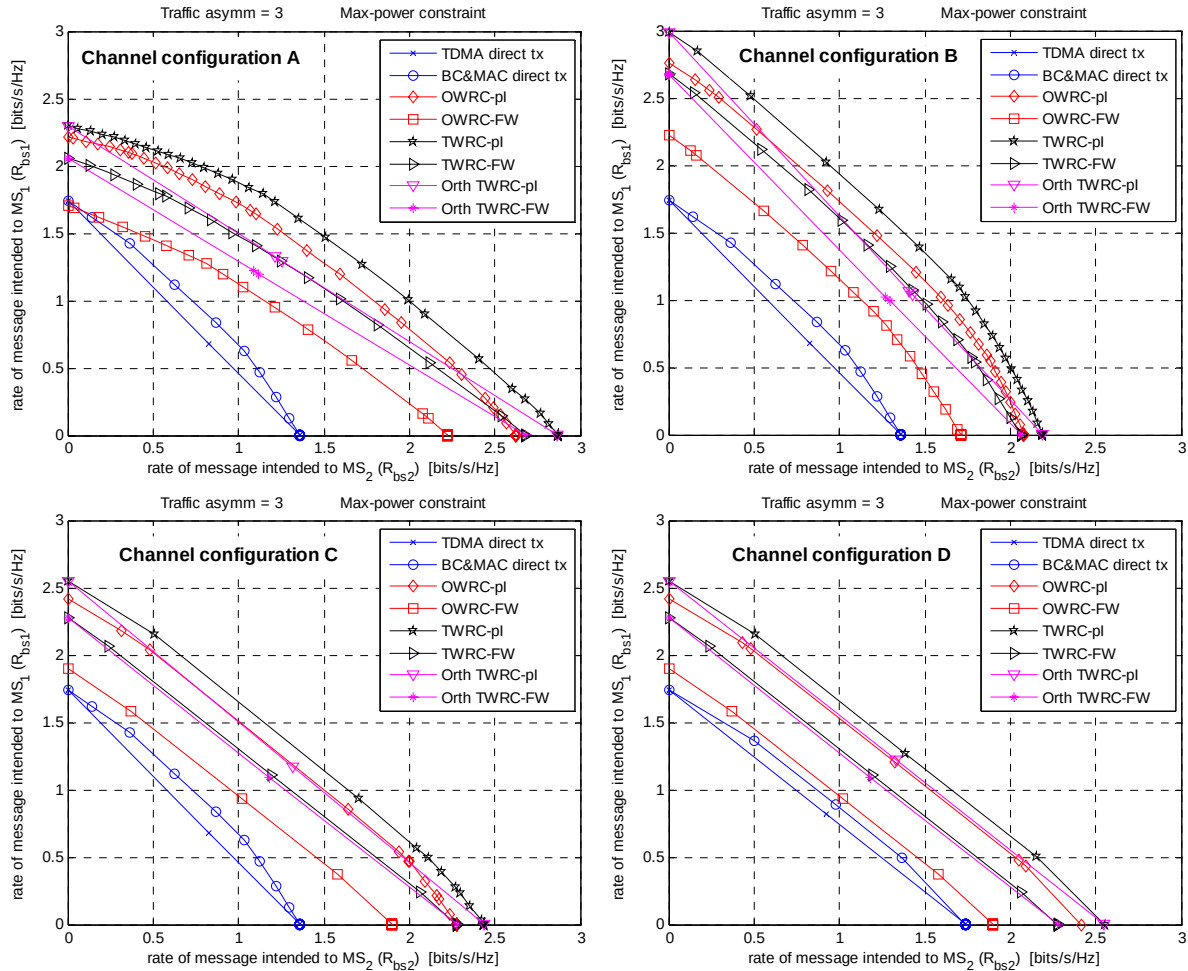


Figure 78. Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Channel configuration A, Top-right) Channel configuration B, Bottom-left) Channel configuration C and Bottom-right) Channel Configuration D, (163). Traffic Asymmetry = 3, Max-power constraint.

A similar performance is obtained when individual average power constraint is considered, as it is shown in Figure 79. In this case, the TWRC protocol I and forwarding achieved fairly different performance.

On the other hand, Figure 78 sketches the same results as the previous figure, but using a *max-power constraint*. In such a case, there are larger differences between the performances that are obtained by the TWRC with the orthogonal TWRC than the ones showed in Figure 77. In this case, when channels are symmetric (*channel configuration C and D*, (163)), the performance does not match. The reason is the different types of considered power constraints. Since the terminals transmit with maximum power, we have the conventional MACH [Cove-91] in the *UL*, and it is known that even in a symmetric scenario it is possible to attain a better performance with *superposed transmissions* than with orthogonal transmissions. In Figure 77, we cannot get the same performance due to the *sum-average power constraint*, the MACH with such a power constraint acts as a BC (duality BC-MACH [Vish-04]), hence, the orthogonal transmission provides the same performance in a symmetric scenario.

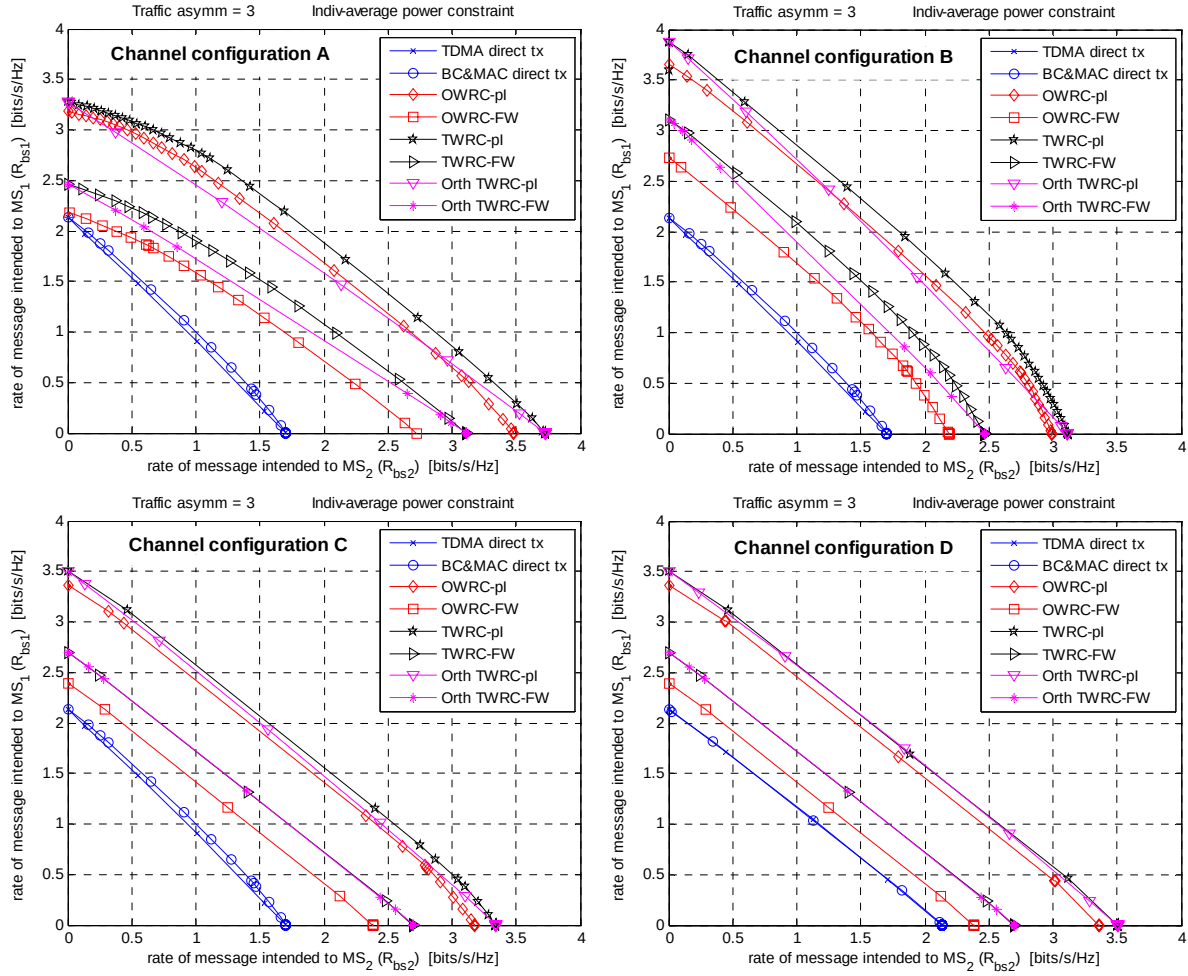


Figure 79. Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Channel configuration A, Top-right) Channel configuration B, Bottom-left) Channel configuration C and Bottom-right) Channel Configuration D, (163). Traffic Asymmetry = 3, Indiv-average power constraint.

4.1.1.3.2 Dependence on the data traffic constraint

Here, we concentrate on the *channel configuration A*, (163), and we analyse the impact of the data traffic constraint on the performance of the different transmission techniques. Figure 80 sketches the achievable rate that is obtained by the intended messages to MS_1 and MS_2 in the *DL* under the different transmission techniques with a *sum-average power* constraint. The traffic asymmetry is equal to 1, 3, 5 and 10 in the top-left, top-right, bottom-left and bottom-right of Figure 80, respectively. We can observe that as traffic asymmetry increases, as the difference between the TWRC and OWRC performances tends to be negligible. Moreover, TWRC forwarding and protocol I attain better performance than OWRC protocol I for low values of traffic asymmetry (equal to 1).

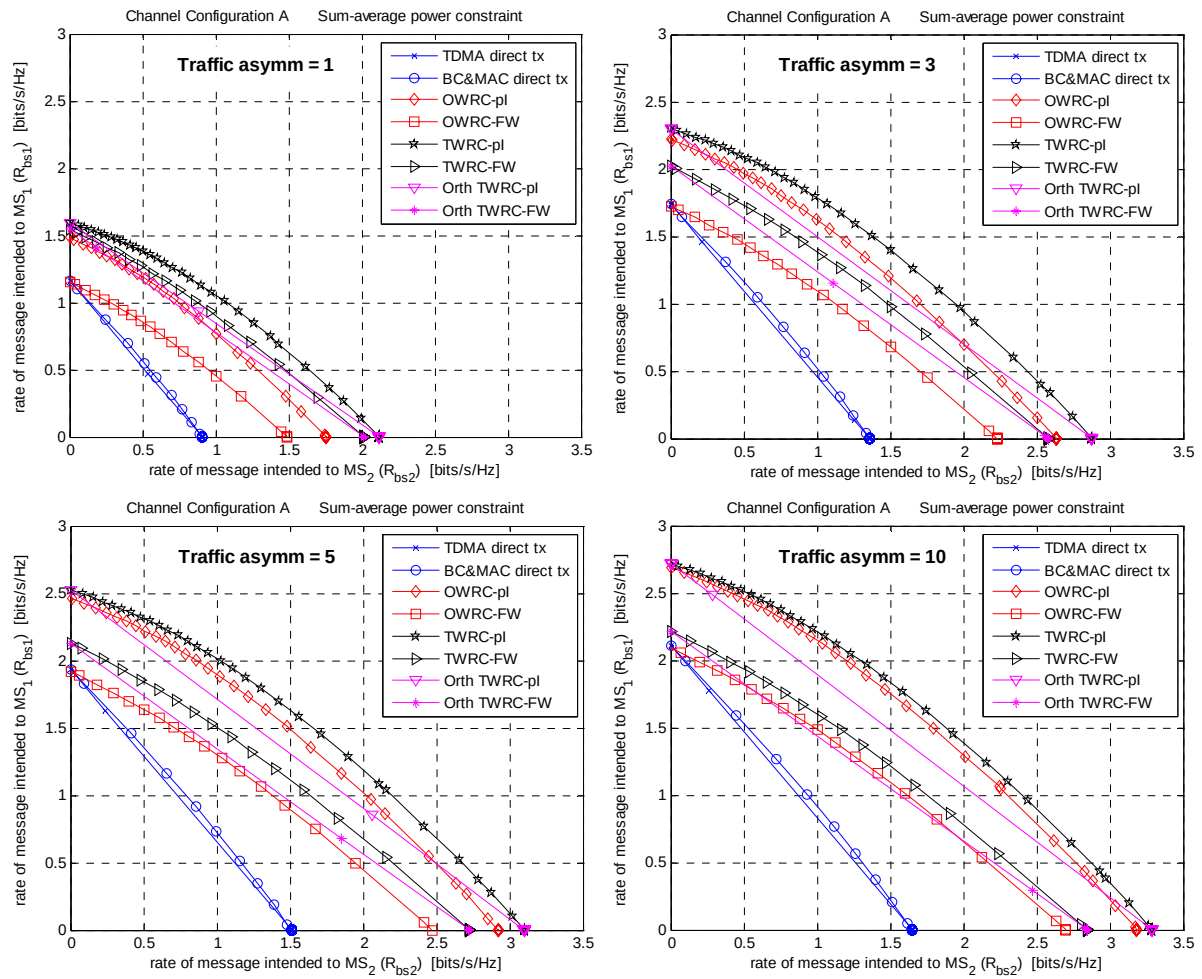


Figure 80. Achievable Rate region attained by the messages intended to MS₁ (y-axis) and MS₂ (x-axis) under the different transmission schemes. Top-left) Traffic Asymmetry=1, Top-right) Traffic Asymmetry=3, Bottom-left) Traffic Asymmetry=5 and Bottom-right) Traffic Asymmetry=10. Channel configuration A, (163), Sum-average power constraint.

A similar performance is obtained when there is an *individual average power constraint*, as it is depicted in Figure 81.

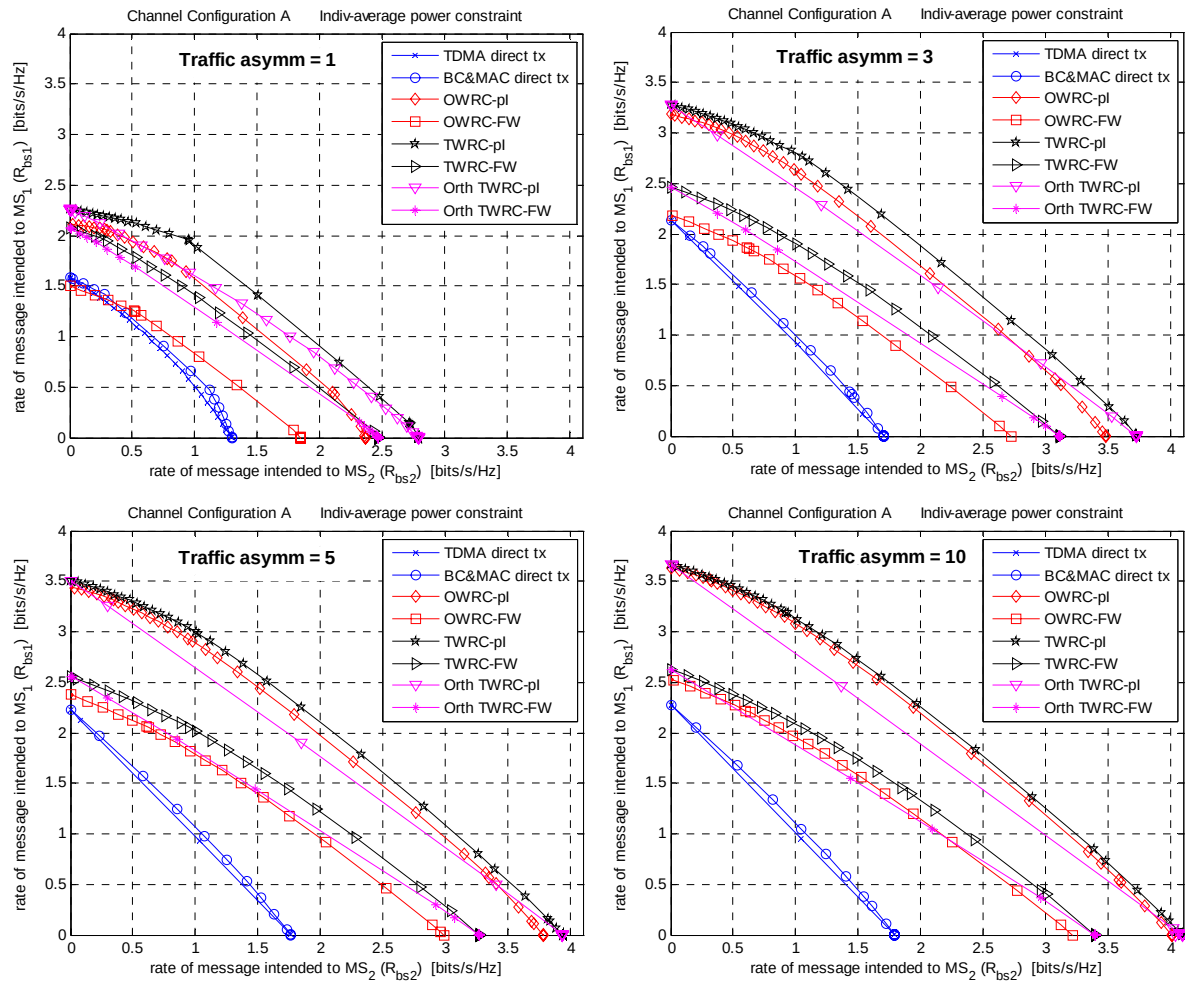


Figure 81. Achievable Rate region attained by the messages intended to MS_1 (y-axis) and MS_2 (x-axis) under the different transmission schemes. Top-left) Traffic Asymmetry=1, Top-right) Traffic Asymmetry=3, Bottom-left) Traffic Asymmetry=5 and Bottom-right) Traffic Asymmetry=10. Channel configuration A, (163), Indv-average power constraint.

Likewise, when we consider a *max-power constraint*, see Figure 82, we can observe some similar results as in Figure 80. However, Figure 82-top-left) depicts that the TWRC forwarding protocol is able to get a better performance than the TWRC protocol I, a property that was already outlined in [ROCKET-3D1] and [Agus-09] for low traffic asymmetry and single-user transmission.

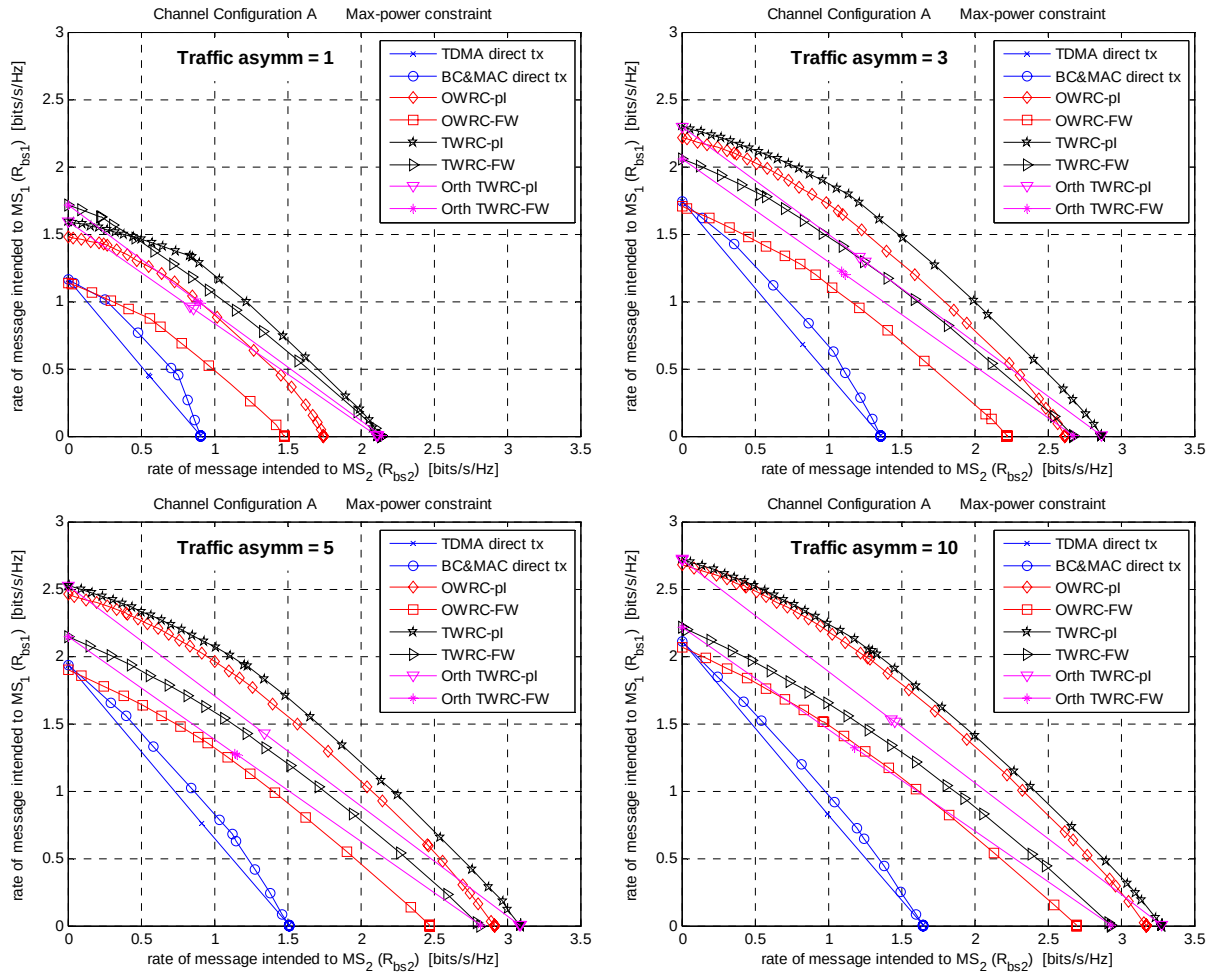


Figure 82. Achievable Rate region attained by the messages intended to MS₁ (y-axis) and MS₂ (x-axis) under the different transmission schemes. Top-left) Traffic Asymmetry=1, Top-right) Traffic Asymmetry=3, Bottom-left) Traffic Asymmetry=5 and Bottom-right) Traffic Asymmetry=10. Channel configuration A, (163), Max-power constraint.

4.1.1.4 Conclusions

- TWRC protocol I is the best protocol under *sum-average* and *individual-average* power constraint.
- TWRC forwarding protocol could outperform protocol I under *max-power constraint* and depending on the SNR of the different channels involved in cooperation.
- TWRC and OWRC transmissions tend to perform similarly for large values of traffic asymmetry.
- The performance that is obtained by the TWRC and orthogonal TWRC becomes generally equal when the channels links are symmetric. However, TWRC provides slightly better performance than orthogonal TWRC under a *max-power constraint*.

4.1.2 Cellular throughput evaluation of multi-user relay-assisted transmissions

Section 4.1 has analysed the optimal resource allocation for the MU relay-assisted transmission with one-way and two-way half-duplex relays. The resource optimization can be formulated as a convex problem and it is solved by interior point methods. This section aims to evaluate the benefits of relay-

assisted techniques investigated in section 4.1 at a system level under a PF scheduler. We would like point out two important aspects considered in this section in order to apply the algorithms developed in section 4.1:

- *Proportional Fair criterion*

The algorithms analyzed in section 4.1 has been derived following the weighted sum-rate (WSR) criterion, see equation (104). The introduction of the PF criterion only changes the objective function to be optimized and the problem remains convex. Taking into account the metric for the PF scheduler derived in (89), the new convex problem will be defined as

$$\begin{aligned}
 & \max_{\alpha, P_{ms1}, P_{ms2}, P_{bs}, P_{rs}, R_{ms1}, R_{ms2}, R_{bs1}, R_{bs2}} \log\left(1 + \frac{\tau}{1-\tau} \mu_1 R_{bs1}\right) + \log\left(1 + \frac{\tau}{1-\tau} \mu_2 R_{bs2}\right) \\
 & s.t. \quad (R_{bs1}, R_{bs2}, R_{ms1}, R_{ms2}) \in B(\alpha) \\
 & \quad 0 \leq f(\alpha) \leq 1 \\
 & \quad \Omega(P_{ms1}, P_{ms2}, P_{bs}, P_{rs}, R_{ms1}) \leq P \\
 & \quad R_{bs1} - \kappa R_{ms1} \leq 0 \\
 & \quad R_{bs2} - \kappa R_{ms2} \leq 0
 \end{aligned} \tag{164}$$

where R_{bs1}, R_{bs2} are the bitrates of the messages transmitted by the BS to MS₁, MS₂ in the DL, while R_{ms1}, R_{ms2} are the bitrates of the messages transmitted by the MS₁ and MS₂ to BS in the UL, B specifies the achievable rate region for the different employed protocols analyzed in sections 4.1.1.1.1, 4.1.1.1.2, 4.1.1.2.1 and 4.1.1.2.2, Ω denotes the transmit power constraint, κ defines the asymmetry in the DL/UL traffic and finally the parameter τ tunes the performance of the PF scheduler.

- *Ergodic capacity bound for multi-antenna terminals*

Although the algorithms presented in section 4.1 assume an AWGN channel with single antenna terminals, we can introduce some minor modifications in order to encompass the OFDM multi-antenna configuration when the power is equally distributed across the carriers. The capacity obtained in a point-to-point scenario with multiple antennas when the transmission is done across many multiple-carriers is close to the ergodic capacity. In this respect, a good lower bound of the ergodic capacity for such configuration is provided in [Calvo-09]

$$E\left\{\log_2 \det\left(\mathbf{I}_{n_r} + \frac{\text{snr}}{n_t} \mathbf{H} \mathbf{H}^H\right)\right\} \geq \sum_{i=1}^{n_r} \log_2\left(1 + \rho_i(f_{\mathbf{H}}) \frac{\text{snr}}{n_t}\right), \tag{165}$$

where n_t, n_r denote the number of receive and transmit antennas, snr is the average signal-to-noise ratio, $\mathbf{H}$ stands for the channel coefficient matrix with dimensions $n_r \times n_t$ and ρ_i is connected with the eigenvalues of the channel and it is defined by [Calvo-09]

$$\rho_i(f_{\mathbf{H}}) \triangleq \exp\left(E\left\{\log\left(\lambda_i(\mathbf{H} \mathbf{H}^H)\right)\right\}\right). \tag{166}$$

Therefore, by introducing equation (165) in the algorithms studied in previous sections we are able to tackle the scenario with multi-antenna terminals assuming that the multi-carrier OFDM transmissions employ a large number of carriers. The ergodic capacity assumption is able to cope with the FUSC and PUSC channelisation modes if the underlying propagation channel is frequency selective.

4.1.2.1 Deployment

For the cellular throughput evaluation we have considered the scenario depicted in Figure 83 which consist of one half of a sector (i.e. 60°). The MSs are served by the BS using the direct or some relay-assisted transmission presented in section 4.1:

- One-Way forwarding protocol.
- One-Way protocol I.
- Two-Way forwarding protocol.
- Two-Way protocol I.

Let us recall that all the resources are optimized in order to accommodate *DL* and *UL* traffic in the same frame.

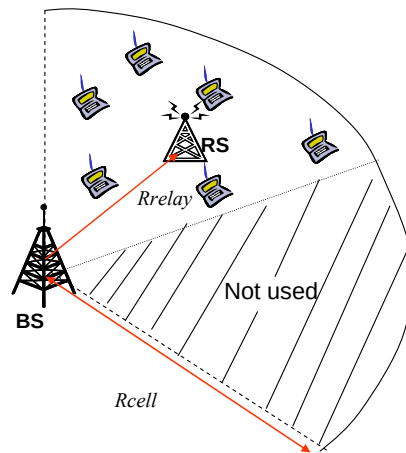


Figure 83. Scenario assumed for the cellular throughput evaluation.

4.1.2.2 Scheduling

We assume that each user is associated a certain priority to be served which is inversely proportional to the served throughput per user

$$\mu_i = \frac{1}{\bar{R}_i}, \quad (167)$$

where $\bar{R}_i$ denotes the served throughput either in the DL or in the UL. We consider a proportional fair (PF) scheduler which calculates the served throughput by

$$\bar{R}_i = \begin{cases} (1-\tau)\bar{R}_i + \tau R_i & \text{if } i \in U_s \\ (1-\tau)\bar{R}_i & \text{otherwise} \end{cases}, \quad (168)$$

where U_s stands for the set of selected users to be served and τ is a parameter to adjust the performance of the scheduler from greedy to round robin performance.

The scheduling algorithm decides in each frame which users have to be simultaneously selected as a function of their priorities and channel state conditions. The scheduler also takes into account the ratio of DL-to-UL traffic requirements. We have assumed that the traffic asymmetry is always constant and modelled by the factor $\kappa \in [1.5, 3]$. Those transmissions employ all the bandwidth, i.e. we are assuming OFDM transmissions.

Because we are grouping users to be served simultaneously, in order to reduce the number of combinations to be evaluated we follow the same guideline that are given in (98). That solution

consists in creating a candidate list of L -users which considers the priority of each user in the DL and UL ($\mu_k^{(DL)}$ $\mu_k^{(UL)}$) and the individual achievable rate attained by a given transmission technique,

$$\Gamma = \left\{ k \in [1, N_u] \mid R_k^{DL} \left(\mu_k^{(DL)} + \frac{\mu_k^{(UL)}}{\kappa} \right) \geq \eta \right\}, \quad |\Gamma| = L \quad (169)$$

where $R_k^{(DL)}$ is the bitrate attained in the DL transmission for the k -th user and it has been designed taking into account that the whole frame is devoted only to that user and it has to encompass the DL and UL transmission. Let me stress the fact that we work under the assumption that $R_k^{(DL)} = \kappa R_k^{(UL)}$.

When the relay is introduced in our scenario, our simulator splits the users in two groups based on the possibility of using relay-assisted transmission or not. Hence, there is one group of users that is going to be served directly by the BS and another group where all the users can employ the relay. That classification is done based on the bitrate attained by the users if the frame were allocated to a single user.

The scheduler must decide which users are allocated to a given frame. Since, there are two groups of users, there are two algorithms working in parallel, one for each group, trying to provide the best user pairing using direct or relay-assisted transmission. The outcome of each algorithm provides one PF metric. The scheduler selects the outcome with the best PF metric, discarding the other option. In a nutshell, our simulator follows the ensuing steps for each user deployment:

1. Assuming that in each frame only one MS can be served, obtain the radio resource allocation (bitrate, duration of phases,...), DL and UL under direct transmission.
2. Assuming that in each frame only one MS can be served, obtain the radio resource allocation (bitrate, duration of phases,...) for the different relay-assisted transmission techniques.
3. Create two groups of users based on the values obtained in step 1 and step 2. One group only employs direct transmission and the other group will be assisted by the RS.

Additionally, at each frame the scheduler decides which users (relay or non-relay) will be allocated:

4. Create two candidate list of L users based on the priority and bitrate, (169), for the non-relay and relay-assisted users.
5. In parallel, obtain the best user pairing for the non-relay and the relay-assisted candidate list.
6. Compare the PF metrics of the non-relay or relay-assisted user pairing, and select that one with the best metric.
7. Once the users have been served, update the average throughput served per user which is connected with priority of each user and return to step 4.

4.1.2.3 Results

Our results consider a MU PF scheduler and aim to elucidate the benefits of the user grouping intended for superposed transmission, which will be denoted by acronym *SP* in the results, over the orthogonal transmission, which will be denoted by *Orth* in the results. In superposed transmission the users are simultaneously allocated in the same resources, while under orthogonal transmission the users employ different resources under time TDD. For example, two users in DL direct transmission may employ the given frame for superposed transmission, while those users use different fractions of that frame under orthogonal transmission. The design of those fractions is decided by the MU PF scheduler.

Another aspect to be analyzed is the impact of the RS in a cellular system based on OFDM transmissions. The transmission schemes analyzed in section 4.1 consider superposed transmission of two terminals in the DL and UL using all the bandwidth (or data carriers) of the system. Thanks to the

goodness of the channel quality introduced by the BS-RS and RS-MS links the communications of those users with bad link qualities in direct transmission can be enhanced. Therefore, in such a case, the users can benefit from the introduction of a RS in the cell and from superposed transmission.

Simulations are carried out in a similar way as it has been described in section 3.2.2.2 for DL transmission. Our simulations consider 100 independent and uniformly distributed user deployments that are evaluated during 1000 frames. The transmissions are done across multiple carriers, approximating the current capacity of the channel by the lower bound of the ergodic capacity presented in (165). Moreover, in order to incorporate the current limitations of the coding schemes, we limit up to 5 bits/s/Hz the maximum capacity attained per a single stream which corresponds to 64QAM and code rate 5/6. Since we are employing a lower bound of the capacity we also consider a *degradation loss factor* [Starr-99] in the SNR of each link devoted to capture the difference between the theoretical bitrate and the actual achievable rate when we use real modulation and coding schemes. In this work this *degradation loss factor* has been fixed to 4.

Because the scenario assumed by the cellular evaluation considers many parameters (multiples users, antenna configuration, RS, channel model, etc) we investigate the performance of the proposed algorithms from different points of view:

- Antenna configuration
- Impact of type of channel model

4.1.2.3.1 Antenna configuration

This section looks into the impact of the antenna configuration in the throughput per sector attained by different transmission techniques. The present section considers the channel model defined in [ROCKET-1D1], and depicted in equations (246) and (255) presented in the appendix, section 8. For the BS-MS and RS-MS links it is assumed the suburban NLOS/LOS Winner II C1, while for the BS-RS link the urban/suburban LOS Winner II B5a is employed. The results have been obtained assuming:

- o 30 users
- o Traffic DL/UL asymmetry $\kappa=3$
- o PF scheduler with $\tau=0.7$
- o Channel model
 - BS-MS, RS-MS: SubUrban NLOS/LOS Winner II C1 (246) in the appendix, section 8.
 - BS-RS: Urban/SubUrban LOS Winner II B5a, see (255) in the appendix, section 8.
- o Antenna configuration (antennas BS $\times$ antennas RS $\times$ antennas MS)
 - $2 \times 2 \times 1$, $2 \times 2 \times 2$, $4 \times 4 \times 2$

Figure 84 depicts the sector throughput in the DL attained by different transmission techniques with the PF scheduler introduced previously. We can observe that the direct transmission (Orth-Direct vs. SP direct) is improved by a throughput gain of 22% thanks of superposed transmission. The one-way forwarding protocol (OW-FW) is able to improve the orthogonal and superposed implementation of direct transmission. The superposed version (SP OW-FW) enhances by 15% the orthogonal implementation of the one-way forwarding. Likewise, the one-way protocol I is able to obtain a gain close to 20% with superposed transmission over the orthogonal implementation of that transmission scheme. The attained relative throughput gains comparing superposed vs. orthogonal transmission for the two-way relay transmission schemes are similar to those showed by one-way relaying. Relay-assisted transmission techniques using protocol I, either with one-way (OW prot I) or two-way relay (TW prot I), get the best results, sketching a total throughput gain over orthogonal direct transmission close to 100%.

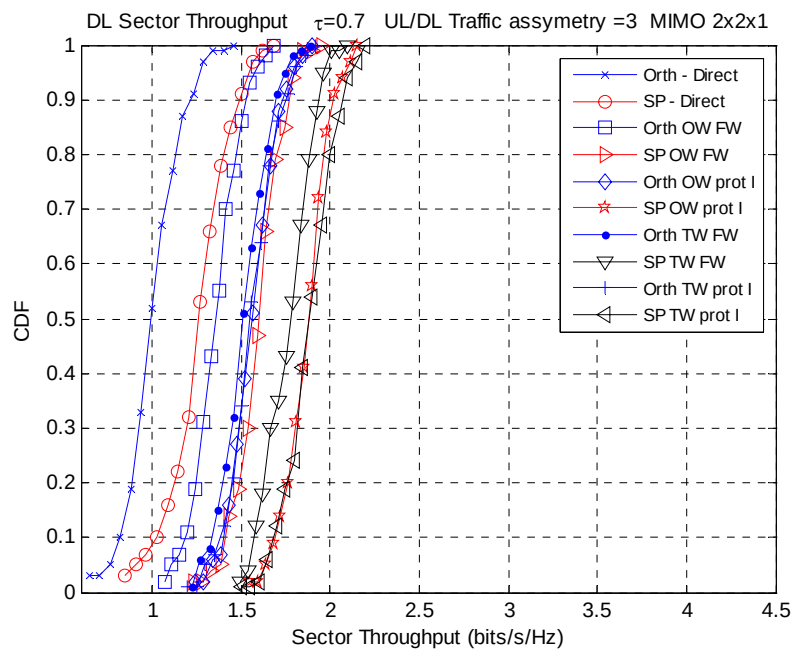


Figure 84. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: 2×2×1.

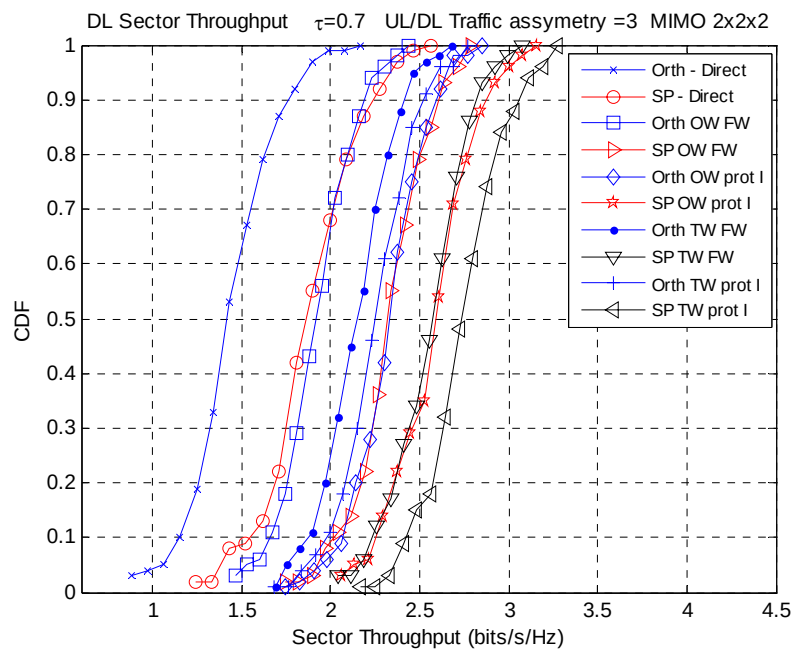


Figure 85. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: 2×2×2.

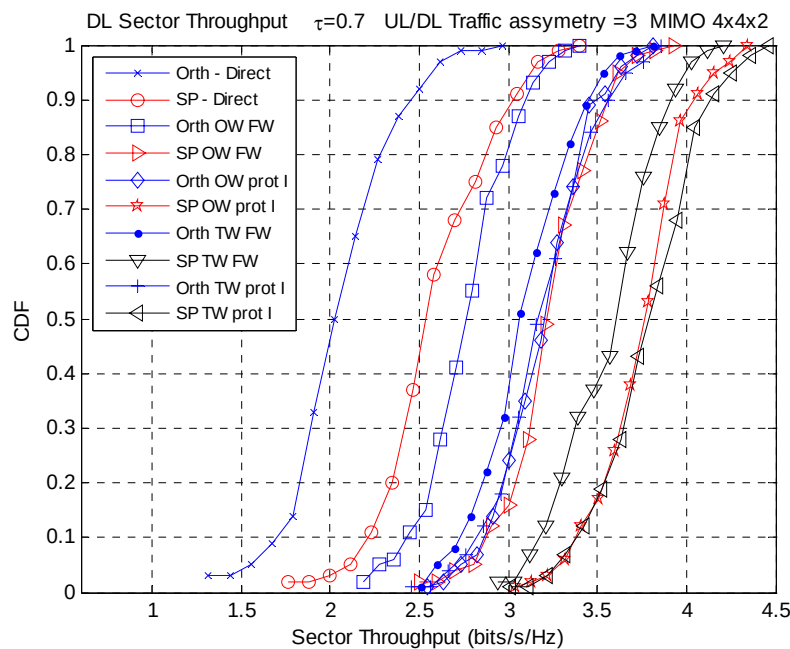


Figure 86. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration 4×4×2.

Figure 85 and Figure 86 present the sector throughput in the DL when the antenna configuration is 2×2×2 and 4×4×2, respectively. The performance obtained under the antenna configuration 2×2×1 which is depicted in Figure 84 is increased by 1.5 times when the antennas at MSs are increased up to 2 (Figure 85), while when we double the antennas at all terminals, configuration 4×4×2, the results are improved by a factor of 2 (Figure 86). The relative throughput gains obtained by superposed transmission compared with the orthogonal one remains the same as the ones shown Figure 84.

4.1.2.3.2 Impact of the type of channel model

This subsection analyses the throughput gains attained by the superposed transmission combined with relays under another channel model different from the one assumed in 4.1.2.3.1. For the BS-MS and RS-MS links the baseline test scenario at 2.5 GHz has been considered, while the channel adopted for the BS-RS link is the 802.16j EVM type D. Both channel models are introduced in the appendix, section 8.

The results have been obtained considering:

- o 30 users
- o Traffic DL/UL asymmetry $\kappa=3$
- o PF scheduler with $\tau=0.7$
- o Channel model
 - BS-MS, RS-MS: Baseline Test Scenario at 2.5 GHz, see (244) in the appendix, section 8.
 - BS-RS: 802.16j EVM type D Baseline Test Scenario, see (254) in the appendix, section 8.
- o Antenna configuration (antennas BS × antennas RS × antennas MS)
 - 2×2×1, 4×4×2

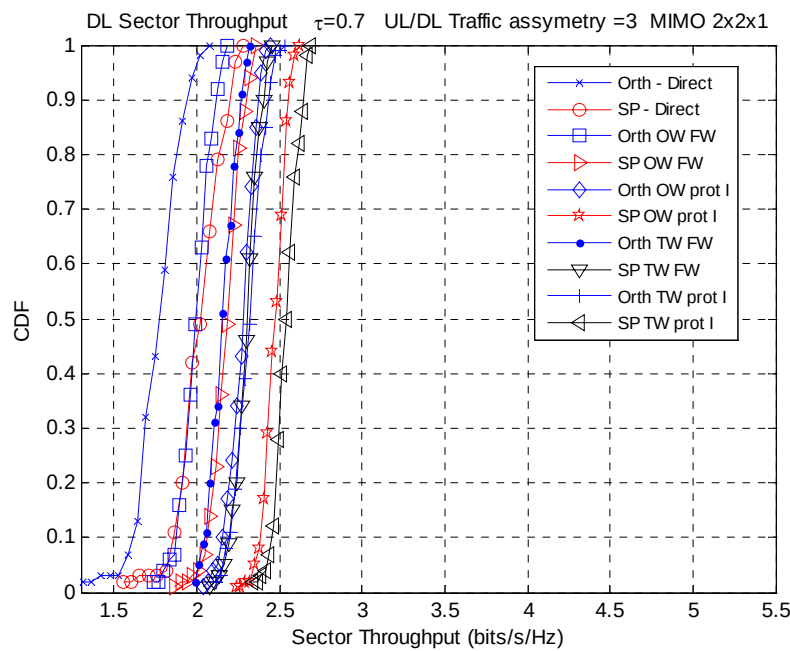


Figure 87. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: 2×2×1.

Figure 87 and Figure 88 sketch the achieved sector throughput in the DL by the different transmission techniques for an antenna configuration 2×2×1 and 4×4×2, respectively, with the channel models described by equations (244) and (254), depicted the appendix (section 8). Its objective is to measure the throughput gain obtained for the new channel configuration and compare with the results obtained under other channel configuration shown in Figure 84 and Figure 86. We can appreciate that with the new channel model the sector throughput values are increased by a factor of 1.6. Nevertheless, the throughput gains due to superposed transmission, for example on direct transmission, are reduced up to 15% (it was 22% in Figure 84 and Figure 86) and the maximum gain due to superposed and relay-assisted transmission becomes close to 51% (100% in Figure 84 and Figure 86).

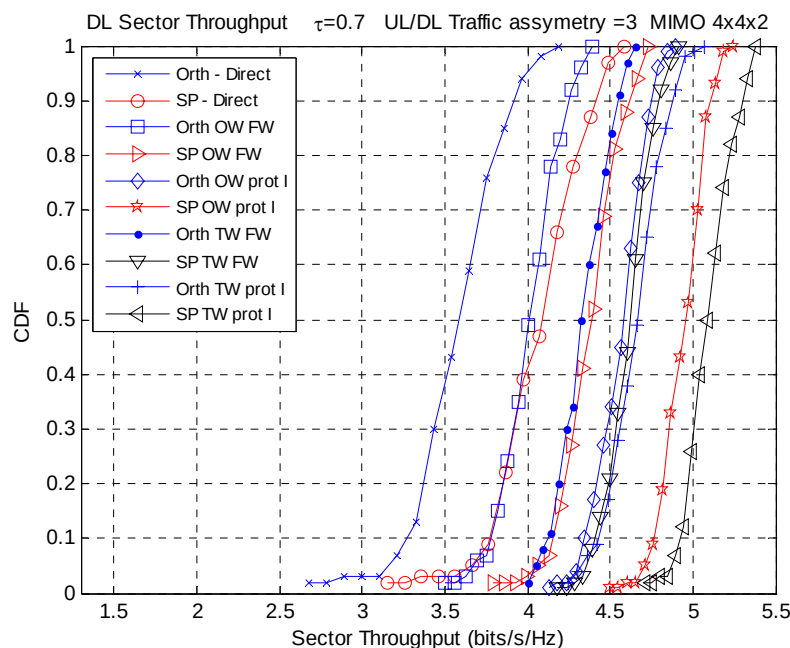


Figure 88. Sector throughput in the DL for different transmission techniques under superposed or orthogonal transmission. Antenna configuration: 4×4×2

4.1.2.4 Conclusions

This section has analyzed the throughput gains provided by the superposed transmission for the different techniques studied in section 4.1 accounting for relay-assisted transmission with one-way and two-way relays and also for direct transmission. All those techniques assume that all the bandwidth is employed for the DL and UL transmissions. The DL and UL traffic is allocated in the same frame under TDD.

We have observed that in a scenario where OFDM-TDMA is the modulation-access scheme for the DL and UL by all users (section 4.1.2.3.2), the best results are obtained by superposed transmission techniques based on one-way and two-way relay with protocol I, getting a gain of 50% over the direct transmission with orthogonal transmission in the baseline test scenario, (244) and (254). However, that throughput gain is close to 100% when the Winner channel models are taken into account, (246) and (255).

On the other hand, the relative percent-wise throughput gain of superposed over orthogonal transmission seems to be almost independent of the antenna configuration (section 4.1.2.3.1). Departing for the configuration $2 \times 2 \times 1$, it has been shown that the throughput values are improved 1.5 times by increasing the number of antennas at the MS to 2. If we double the antennas at all terminals ($4 \times 4 \times 2$), the throughput is enhanced by 2.

The two-way and one way techniques get similar throughput results. Forwarding protocol in the two-way relay is always better than in the one-way relay transmission, but protocol I attains similar results in both cases. However, the energy consumed by the RS depends if one-way or two-way relay coding is employed [ROCKET-3D1]. As a future work, we envision modify our proposed techniques in order to exploit OFDMA transmissions in the UL. With such modifications we expect to enhance the attained throughput in terms of outage, the tails of the CDF of the throughput.

4.2 UPLINK MULTI-USER POWER ALLOCATION FOR AF COOPERATIVE MIMO SYSTEMS

4.2.1 Introduction

In this section, novel power allocation methods for non-regenerative cooperative MIMO scheme will be presented. As a preliminary work, we investigate various schemes operating with different levels of CSI for single user, single and multi antennas, and single and multi carriers, in section 4.2.2. We specifically designed five novel power allocation methods; the first one is designed for the case where both the first and second hop CSI known at the source node; the second one is employed in the case where the first hop CSI is known at the source node and the second hop CSI is known at the RS; the third one is dedicated to the case where the first and second hop CSI are known at the RS; finally, the fourth and fifth algorithms are utilised in the case where the first hop CSI is only known at the RS. These algorithms are designed to maximise the mutual information of the cooperative MIMO system. For instance in [Héliot-09c], we showed that our fourth and fifth algorithms outperform other existing comparable techniques both in terms of mutual information and Bit-Error Rate (BER). In section 4.2.3, we then demonstrate that the UL MU MIMO cooperative system model can be expressed into a similar form as the single-user system model by assuming SIC at the RS. Consequently, the power allocation methods, which have been developed for the single-user case, can be easily applied to the MU case for maximising the sum-rate of the cooperative MU MIMO system. Additionally, we design an algorithm that maximises the WSR in the case where the CSI of the RS-source link is known at the RS.

4.2.2 Single-user power allocation methods for nonregenerative cooperative MIMO communication under various levels of CSI

In cooperative scenario, the relay can be utilised as a smart precoder that allows mutual information improvement through efficient power allocation techniques that are based on the level of CSI available at the RS; instead of being used as a complex encoder/decoder system or as a dumb amplifier, as it is explained in [Fan-07, Muñoz-07]. In our work, the simple non-regenerative approach, i.e., AF, has been chosen since it is well-suited for integrating precoding at the RS, and four different levels of CSI have been investigated with various degrees of feasibility:

1. CSI of both the first and second hop channels are known at the source node;
2. CSI of the first hop channel is known at the source node and CSI of the second hop channel is known at the RS;
3. CSI of both the first and second hop channels are known at the RS;
4. CSI of the first hop channel is only known at the RS.

The first case is hardly feasible in reality but serves as a benchmark. In the second and third scenario, CSI can be obtained by reciprocity, and in the fourth scenario CSI can be easily obtained via pilot symbol. Notice that the fourth scenario is the one with the lowest implementation complexity and is the one that is considered in the traditional AF schemes. To the best of our knowledge, two methods have already been developed for the third case in [Fan-07, Muñoz-07], as well as two techniques for the fourth case in [Fan-07], namely the AF Matched Filter based Relaying (MFR) and AF MMSE Filtering (MMSEF) techniques.

We consider a classic cooperative MIMO communication system that is composed of three nodes, where a source node, which is equipped with n antennas, cooperates with a nonregenerative RS, which is equipped with q antennas, to transmit data to a destination node, which is equipped with r antennas, as it is depicted in Figure 89. Without loss of generality, we consider that the source node is a MS and that the destination node is a BS.

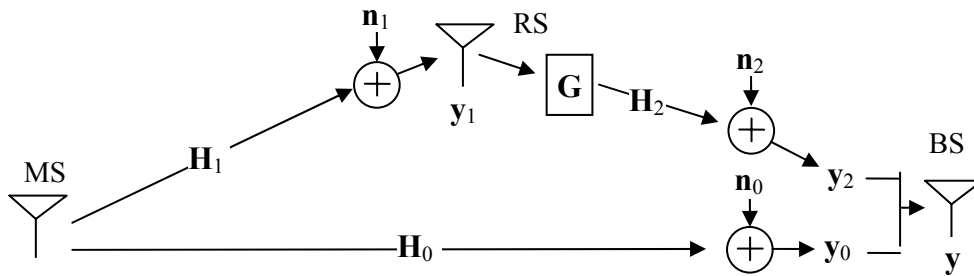


Figure 89: Single-user cooperative MIMO system model

For the simplicity of the introduction, we assume a half duplex relaying scenario with two equal duration phases as in [Fan-07, Muñoz-07], where the MS broadcasts the signal $\mathbf{x}$ to the BS and RS during the first phase, and only the RS transmits to the BS during the second phase. At the end of the first phase, the signal $\mathbf{x}$ is received as $\mathbf{y}_0 = \mathbf{H}_0 \mathbf{x} + \mathbf{n}_0$ and $\mathbf{y}_1 = \mathbf{H}_1 \mathbf{x} + \mathbf{n}_1$ at the BS and RS, respectively, where $\mathbf{H}_0 \in \mathbb{C}^{r \times n}$ and $\mathbf{H}_1 \in \mathbb{C}^{q \times n}$ characterise the MIMO channel of the MS-BS and MS-RS links, respectively. During the second phase, the signal $\mathbf{y}_1$ is modified by using the precoding matrix $\mathbf{G} \in \mathbb{R}^{q \times q}$, then is transmitted towards the BS and is received as $\mathbf{y}_2 = \mathbf{H}_2 \mathbf{G} \mathbf{y}_1 + \mathbf{n}_2$, where $\mathbf{H}_2 \in \mathbb{C}^{r \times q}$ characterises the MIMO channel of the RS-BS link. Moreover, $\mathbf{n}_0 \in \mathbb{C}^{r \times 1}$, $\mathbf{n}_1 \in \mathbb{C}^{q \times 1}$ and $\mathbf{n}_2 \in \mathbb{C}^{r \times 1}$ are vectors of independent zero-mean complex Gaussian noise entries with a variance of σ^2 . The system

model of the single-user cooperative MIMO communication system, which is introduced in Figure 89, can be summarised as follows

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_0 \\ \mathbf{y}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{H}_0 \\ \mathbf{H}_2 \mathbf{G} \mathbf{H}_1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} \mathbf{I}_r & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_2 \mathbf{G} & \mathbf{I}_r \end{bmatrix} \begin{bmatrix} \mathbf{n}_0 \\ \mathbf{n}_1 \\ \mathbf{n}_2 \end{bmatrix}, \quad (170)$$

with $\mathbf{I}_r$ is a $r \times r$ identity matrix. Consequently, the aggregate mutual information of the cooperative MIMO communication system in Figure 89 can be expressed as [Cover-91]

$$I(\mathbf{y}; \mathbf{x}) = \frac{1}{2} \log_2 |\mathbf{I}_{2r} + \mathbf{H} \mathbf{R}_x \mathbf{H}^\dagger \mathbf{R}_n^{-1}|, \quad (171)$$

where

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_0 \\ \mathbf{H}_2 \mathbf{G} \mathbf{H}_1 \end{bmatrix}, \mathbf{R}_n = \begin{bmatrix} \mathbf{R}_{n_0} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_2 \mathbf{G} \mathbf{R}_{n_1} \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{R}_{n_2} \end{bmatrix},$$

$\mathbf{H}^\dagger$ denotes the conjugate transpose of $\mathbf{H}$, $\mathbf{R}_x = E\{\mathbf{x}\mathbf{x}^\dagger\}$ is the transmit signal covariance matrix, $\mathbf{R}_{n_0}$, $\mathbf{R}_{n_1}$ and $\mathbf{R}_{n_2}$ are noise covariance matrices. Notice that the factor 1/2 in (171) accounts for the 2-phases transmission.

Recently in [Muñoz-07], the aggregate mutual information $I(\mathbf{y}; \mathbf{x})$ has been shown to be bounded as $I(\mathbf{y}_0; \mathbf{x}) + I(\mathbf{y}_2; \mathbf{x}) \geq I(\mathbf{y}; \mathbf{x}) \geq I(\mathbf{y}_2; \mathbf{x})$, where $I(\mathbf{y}_0; \mathbf{x})$ is the mutual information of the direct link, i.e., MS-BS link, and $I(\mathbf{y}_2; \mathbf{x})$ is the mutual information of the relay link (2-hop link), i.e., MS-RS-BS link, which are given by

$$\begin{aligned} I(\mathbf{y}_0; \mathbf{x}) &= \frac{1}{2} \log_2 |\mathbf{I}_r + \mathbf{H}_0 \mathbf{R}_x \mathbf{H}_0^\dagger \mathbf{R}_{n_0}^{-1}| \text{ and} \\ I(\mathbf{y}_2; \mathbf{x}) &= \frac{1}{2} \log_2 \left| \frac{\mathbf{R}_{n_2} + \mathbf{H}_2 \mathbf{G} (\mathbf{R}_{n_1} + \mathbf{H}_1 \mathbf{R}_x \mathbf{H}_1^\dagger) \mathbf{G}^\dagger \mathbf{H}_2^\dagger}{\mathbf{R}_{n_2} + \mathbf{H}_2 \mathbf{G} \mathbf{R}_{n_1} \mathbf{G}^\dagger \mathbf{H}_2^\dagger} \right|, \end{aligned} \quad (172)$$

respectively. Hence, the aggregate mutual information $I(\mathbf{y}; \mathbf{x})$ can be increased by maximising $I(\mathbf{y}_2; \mathbf{x})$ in (172), which means by optimising $\mathbf{G}$ at the RS, in the case that $\mathbf{H}_0$ is unknown.

In the next four subsections, we design five power allocation methods for four different levels of CSI and provide algorithm to optimise the precoding matrix $\mathbf{G}$ in any of these cases. In the following, we assume $\mathbf{R}_{n_0} = \mathbf{R}_{n_2} = \sigma^2 \mathbf{I}_r$ and $\mathbf{R}_{n_1} = \sigma^2 \mathbf{I}_q$, with $\sigma = 1$, and define the total transmit power at the MS and RS, as P_1 and P_2 , respectively. We denote as P_S and P_R , the power constraints at the MS and RS, respectively, which can be expressed as follows

$$\begin{aligned} P_S &: E\{\|\mathbf{x}\|_F^2\} \leq P_1, \\ P_R &: E\{\|\mathbf{G}(\mathbf{H}_1 \mathbf{x} + \mathbf{n}_1)\|_F^2\} \leq P_2, \end{aligned} \quad (173)$$

where $E\{\cdot\}$ is the expectation and $\|\cdot\|_F$ is the Frobenius norm.

4.2.2.1 Full CSI at the MS

We first consider the case of Full CSI (FCSI) at the MS, where $\mathbf{H}_1$ and $\mathbf{H}_2$ are both known at the MS. In this case, both matrices can be decomposed via SVD as $\mathbf{H}_1 = \mathbf{U}_1 \sqrt{\hat{\Lambda}_1} \mathbf{V}_1^\dagger$ and $\mathbf{H}_2 = \mathbf{U}_2 \sqrt{\hat{\Lambda}_2} \mathbf{V}_2^\dagger$, respectively, where $\mathbf{U}_1 \in \mathbb{C}^{q \times q}$, $\mathbf{V}_1 \in \mathbb{C}^{n \times n}$, $\mathbf{U}_2 \in \mathbb{C}^{r \times r}$ and $\mathbf{V}_2 \in \mathbb{C}^{q \times q}$ are unitary matrices, and $\hat{\Lambda}_1 \in \mathbb{C}^{q \times n}$ and $\hat{\Lambda}_2 \in \mathbb{C}^{r \times q}$ are rectangular diagonal matrices. Moreover, $\Lambda_1 = \sqrt{\hat{\Lambda}_1} \sqrt{\hat{\Lambda}_1}^\dagger$ and

$\Lambda_2 = \sqrt{\hat{\Lambda}_2} \sqrt{\hat{\Lambda}_2^\dagger}$ are $q \times q$ and $r \times r$ diagonal matrices with nonnegative diagonal elements $\lambda_{1,i}$ and $\lambda_{2,i}$, respectively, which are sorted in descending order as in [Fan-07]. This ordering has been shown to be close to optimal in [Fan-07, Muñoz-07]. Notice also that $\lambda_{1,i} \neq 0$ for $i \in [1, L_1]$ and that $\lambda_{1,i} = 0$ for $i \in [L_1 + 1, q]$, with $L_1 = \min\{q, n\}$, and $\lambda_{2,i} \neq 0$ for $i \in [1, L_2]$ and that $\lambda_{2,i} = 0$ for $i \in [L_2 + 1, r]$, with $L_2 = \min\{r, q\}$. Furthermore, we consider that $\mathbf{G} = \mathbf{V}_2 \tilde{\mathbf{G}} \mathbf{U}_1^\dagger$ with $\tilde{\mathbf{G}} = \text{diag}(\sqrt{p_{2,1}}, \sqrt{p_{2,1}}, \dots, \sqrt{p_{2,q}})$ is a $q \times q$ diagonal matrix, and that $\mathbf{x} = \mathbf{V}_1 \mathbf{D} \mathbf{s}$ where $\mathbf{D} = \text{diag}(\sqrt{p_{1,1}}, \sqrt{p_{1,1}}, \dots, \sqrt{p_{1,n}})$ is a $n \times n$ diagonal matrix, $\mathbf{s} \in \mathbb{C}^n$, $E\{\mathbf{s} \mathbf{s}^\dagger\} = \mathbf{I}$, and we define $L = \min\{L_1, L_2\}$. Consequently, we can re-express (172) after some simplifications as

$$I(\mathbf{y}_2; \mathbf{x}) = \frac{1}{2} \sum_{i=1}^L \log_2(1 + p_{1,i} \eta_{1,i}), \quad (174)$$

$$\text{with } \eta_{1,i} = \frac{\lambda_{1,i} \lambda_{2,i} p_{2,i}}{1 + \lambda_{2,i} p_{2,i}}.$$

In order to obtain the $p_{1,i}$ and $p_{2,i}$ values that maximise (174), we must solve the following optimisation problem

$$\begin{aligned} \max_{p_{1,i}, p_{2,i} \ i \in [1, L]} & \sum \log_2(1 + p_{1,i} \eta_{1,i}) \\ \text{s.t. } & p_{1,i}, p_{2,i} \geq 0, i \in [1, L], P_S, P_R \end{aligned} \quad (175)$$

This problem requires the optimisation of two set of variables at the same time and it cannot be directly solved by using classic concave optimisation [Boyd-04]. However, we solve it in a simple recursive fashion by splitting it into two parts. First, we assume that $p_{2,i}$ is known in $\eta_{1,i}$ (174), and then we optimise the $p_{1,i}$ values by solving the classic water-filling problem

$$\begin{aligned} \max_{p_{1,i} \ i \in [1, L]} & \sum \log_2(1 + p_{1,i} \eta_{1,i}) \\ \text{s.t. } & p_{1,i} \geq 0, i \in [1, L], P_S \end{aligned} \quad (176)$$

for which a straightforward solution is given by $p_{1,i} = [\xi_1^* - 1/\eta_{1,i}]_+$, with ξ_1^* being the water line and $[x]_+ = \max\{0, x\}$. At this point we denote S the set of indices i for which $p_{1,i} \neq 0$. Using the result of (176) in (175), we then optimise the $p_{2,i}$ values by solving the following concave problem

$$\begin{aligned} \max_{p_{2,i} \ i \in S} & \sum \log_2(\xi_1^* \lambda_{1,i}) + \log_2\left(\frac{\lambda_{2,i} p_{2,i}}{1 + \lambda_{2,i} p_{2,i}}\right) \\ \text{s.t. } & p_{2,i} \geq 0, i \in S, P_R \end{aligned} \quad (177)$$

Applying the Lagrange method [Boyd-04] to (176), we obtain after simplifications that

$$p_{2,i} = \left(-1 + \sqrt{1 + \frac{4\lambda_{2,i}\xi_2^*}{1 + p_{1,i}\lambda_{1,i}}} \right) / (2\lambda_{2,i}) \text{ with } \xi_2^* \text{ is such that } f(\xi_2^*) = 0, \text{ where}$$

$$f(\xi_2) = \sum_{i \in S} \left(-1 + \sqrt{1 + \frac{4\lambda_{2,i}\xi_2}{1 + \lambda_{1,i}p_{1,i}}} \right) (1 + \lambda_{1,i}p_{1,i}) / (2\lambda_{2,i}) - P_2. \quad (178)$$

The values of $p_{2,i}$ can then be used in (175) to obtain $p_{1,i}$, and so on and so forth until (174) converge. The algorithm can be summarised as follows

Algorithm 1

```

1: While  $L > 1$  do
2:   Set  $I = 0$ ,  $S = [1, L]$ ,  $p_{2,i} = P_2/L, \forall i \in S$ ;
3:   Calculate  $\eta_{1,i}$ ;
4:   Water-filling:  $p_{1,i} = \left[ \xi_1^* - 1/\eta_{1,i} \right]_+ \Rightarrow L = |\{p_{1,i} \mid p_{1,i} > 0\}|$ , resize  $S$ ;
5:   Solve  $f(\xi_2) = 0$  using the Newton-Raphson method [Wolf]  $\Rightarrow \xi_2^*$ ;
6:   Calculate  $p_{2,i}$  using the value of  $\xi_2^*$ ;
7:   Set  $\hat{I} = I$ ;
8:   Calculate  $I = \sum_{i \in S} \log_2(1 + p_{1,i}\eta_{1,i})/2$ ;
9:   If  $I - \hat{I} < \varepsilon$  then Set  $I_{MAX,L} = \hat{I}$ ,  $L = L - 1$ , go to 1
10:  Else go to 3
11: End While
12:  $I_{MAX,1} = \log_2(1 + P_1\lambda_{1,1})/2 + \log_2(1 + P_2\lambda_{2,1})/2 - \log_2(1 + P_1P_2\lambda_{1,1}\lambda_{2,1})/2$ ;
13:  $I(\mathbf{y}_2; \mathbf{x}) = \max_j (I_{MAX,j})$ ;
14: Obtain  $\mathbf{R}_x = \mathbf{V}_1 \mathbf{D} \mathbf{D}^\dagger \mathbf{V}_1^\dagger$  by using the values of  $p_{1,i}$  that maximises  $I(\mathbf{y}_2; \mathbf{x})$ ;
15: Obtain  $\mathbf{G} = \mathbf{V}_2 \tilde{\mathbf{G}} \mathbf{U}_1^\dagger$  by using the values of  $p_{2,i}$  that maximises  $I(\mathbf{y}_2; \mathbf{x})$ .

```

4.2.2.2 Transmit CSI at the MS and RS

We consider, in this subsection, the case of independent Transmit CSI (TCSI) at the MS and RS, where $\mathbf{H}_1$ is known at the MS and $\mathbf{H}_2$ is known at the RS. As for the previous case both matrices can be decomposed via SVD, i.e., $I(\mathbf{y}_2; \mathbf{x})$ can be simplified as in (174), then the optimisation problem is the same as in (174) and it can also be split into two parts. However, in the first part, we solve the water-filling problem in (176) by using $\lambda_{1,i}$ instead of $\eta_{1,i}$ to obtain the $p_{1,i}$ values, since the $\lambda_{2,i}$ values are not known at the MS. We next solve the optimisation problem in (177) to obtain the $p_{2,i}$ values. Notice that contrarily to *Algorithm 1*, no inner loop is not required in this algorithm since the two optimisation parts are independent. Hence, *Algorithm 2* has a lower implementation complexity than *Algorithm 1*. The algorithm can be summarised as

Algorithm 2

```

1:  $S = [1, L]$ , Flag = 0;
2: While  $L > 1$  do
3:   Water-filling:  $p_{1,i} = \left[ \xi_1^* - 1/\eta_{1,i} \right]_+ \Rightarrow L = |\{p_{1,i} \mid p_{1,i} > 0\}|$ , resize  $S$ ;
4:   If ( $L > 1$ )
5:     If flag == 0 then Set  $p_{2,i} = P_2/L, \forall i \in S$ ; Else
6:       Solve  $f(\xi_2) = 0$  using the Newton-Raphson method [Wolf]  $\Rightarrow \xi_2^*$ ;
7:       Calculate  $p_{2,i}$  using the value of  $\xi_2^*$ ;
8:       Calculate  $I_{MAX,L} = \sum_{i \in S} \log_2(1 + p_{1,i}\eta_{1,i})/2$ ;
9:        $L = L - 1$ ;
10:    End if
11:  End While
12:  $I_{MAX,1} = \log_2(1 + P_1\lambda_{1,1})/2 + \log_2(1 + P_2\lambda_{2,1})/2 - \log_2(1 + P_1P_2\lambda_{1,1}\lambda_{2,1})/2$ ;
13:  $I(\mathbf{y}_2; \mathbf{x}) = \max_j (I_{MAX,j})$ ;
14: Obtain  $\mathbf{R}_x = \mathbf{V}_1 \mathbf{D} \mathbf{D}^\dagger \mathbf{V}_1^\dagger$  by using the values of  $p_{1,i}$  that maximises  $I(\mathbf{y}_2; \mathbf{x})$ .
15: Obtain  $\mathbf{G} = \mathbf{V}_2 \tilde{\mathbf{G}} \mathbf{U}_1^\dagger$  by using the values of  $p_{2,i}$  that maximises  $I(\mathbf{y}_2; \mathbf{x})$ .

```

4.2.2.3 Full CSI at RS

In the case that FCSI is available at the RS, i.e., both $\mathbf{H}_1$ and $\mathbf{H}_2$ are known at the RS, then the optimisation problem simplifies as in (177) with $p_{1,i} = P_1/L, \forall i \in [1, L]$. *Algorithm 2* can then be used without step 3 and with $p_{1,i}$ values set prior to the *while* loop at step 0. We denote this modified version of *Algorithm 2* as *Algorithm 3*.

Algorithm 3

```

0:  $S = [1, L]$ ,  $p_{1,i} = P_1/L, \forall i \in [1, L]$ , Flag = 0;
1: While  $L > 1$  do
2:   If flag == 0 then Set  $p_{2,i} = P_2/L, \forall i \in S$ ; Else
3:     Solve  $f(\xi_2) = 0$  using the Newton-Raphson method [Wolf]  $\Rightarrow \xi_2^*$ ;
4:     Calculate  $p_{2,i}$  using the value of  $\xi_2^*$ ;
5:     Calculate  $I_{MAX,L} = \sum_{i \in S} \log_2(1 + p_{1,i}\eta_{1,i})/2$ ;
6:      $L = L - 1$ ;
7:   End While
8:  $I_{MAX,1} = \log_2(1 + P_1\lambda_{1,1})/2 + \log_2(1 + P_2\lambda_{2,1})/2 - \log_2(1 + P_1P_2\lambda_{1,1}\lambda_{2,1})/2$ ;
9:  $I(\mathbf{y}_2; \mathbf{x}) = \max_j (I_{MAX,j})$ ;
10: Obtain  $\mathbf{G} = \mathbf{V}_2 \tilde{\mathbf{G}} \mathbf{U}_1^\dagger$  by using the values of  $p_{2,i}$  that maximises  $I(\mathbf{y}_2; \mathbf{x})$ .

```

4.2.2.4 Receive CSI only at RS

We considered here that Received CSI (RCSI) only is known at the RS, i.e., only $\mathbf{H}_1$ is known at the RS. This is the most practical case in terms of CSI availability.

4.2.2.4.1 Traditional AF power allocation

In the original AF, the precoding matrix $\mathbf{G}$ design follows an Equal Gain (EG) approach. In other words, $\mathbf{G}$ is designed such that the same amount of power is transmitted on any of the available

eigenmode of the RS-BS link, and such that the total transmit power of the RS node, i.e., P_2 , respects the constraint P_R set in (173). Consequently, in the AF EG approach, $\mathbf{G}$ can be expressed as follows

$$\mathbf{G} = \sqrt{P_2} \mathbf{J} \left(E \left\{ \left\| \mathbf{J}(\mathbf{H}_1 \mathbf{x} + \mathbf{n}_1) \right\|_F^2 \right\} \right)^{-\frac{1}{2}}, \quad (179)$$

with $\mathbf{J} = \mathbf{I}_q$. Lately in [Fan-07], other AF based approaches, which are denoted MFR and MMSEF, have also been proposed. They rely on the same $\mathbf{G}$ matrix formulation as in (179), but where $\mathbf{J} = \mathbf{V}_{1,[1:q]}^\dagger \mathbf{H}_1^\dagger$ and $\mathbf{J} = \mathbf{V}_{1,[1:q]}^\dagger (P_1/n) \mathbf{H}_1^\dagger [\mathbf{I}_q + (P_1/n) \mathbf{H}_1 \mathbf{H}_1^\dagger]^{-1}$, respectively, where $\mathbf{V}_{1,[1:q]}$ are the q first columns of the matrix $\mathbf{V}_1$.

More generally, in the case that $\mathbf{H}_1$ is known at the R node, then $\mathbf{H}_1$ can be decomposed via SVD and thus $I(\mathbf{y}_2; \mathbf{x})$ in (172) can be re-expressed as $I(\mathbf{y}_2; \mathbf{x}) =$

$$\frac{1}{2} \log_2 \left| \mathbf{I}_r + \mathbf{H}_2 \tilde{\mathbf{F}} \tilde{\mathbf{F}}^\dagger \mathbf{H}_2^\dagger \right| - \frac{1}{2} \log_2 \left| \mathbf{I}_r + \mathbf{H}_2 \tilde{\mathbf{G}} \tilde{\mathbf{G}}^\dagger \mathbf{H}_2^\dagger \right|, \quad (180)$$

where $\tilde{\mathbf{G}} = \mathbf{G} \mathbf{U}_1 = \text{diag}(\sqrt{p_{2,1}}, \sqrt{p_{2,2}}, \dots, \sqrt{p_{2,q}})$ is a $q \times q$ diagonal matrix, $\tilde{\mathbf{F}} = \tilde{\mathbf{G}} (\mathbf{I}_q + \mathbf{R}_x \Lambda_1)^{-\frac{1}{2}}$ and $\mathbf{R}_x = (P_1/n) \mathbf{I}_n$.

4.2.2.4.2 Equal eigenmode power allocation method (EEM)

In our Equal EigenMode (EEM) power allocation method, the $\mathbf{G}$ matrix is designed to maximise $I(\mathbf{y}_2; \mathbf{x})$ by assuming that all eigenmodes of $\mathbf{H}_2$ are equivalent. This method is a cunningly modified version of the methods that were proposed in [Fan-07, Muñoz-07] for the case where both MS-RS and RS-BS links' CSI, i.e., FCSI, is known at the RS.

In the case of FCSI at the R node, $\mathbf{H}_1$ and $\mathbf{H}_2$ are known, and $\mathbf{H}_2$ can be decomposed via SVD. We then can re-express (172) after some simplifications as

$$I(\mathbf{y}_2; \mathbf{x}) = \frac{1}{2} \sum_{i=1}^L \log_2 \left(\frac{1 + p_{2,i} \lambda_{2,i} (1 + (P_1/n) \lambda_{1,i})}{1 + p_{2,i} \lambda_{2,i}} \right). \quad (181)$$

The $p_{2,i}$ values that maximise (181) can next be obtained by solving the following concave optimisation problem

$$\begin{aligned} \max_{p_{2,i}} & \frac{1}{2} \sum_{i=1}^L \log_2 \left(\frac{1 + p_{2,i} \lambda_{2,i} (1 + (P_1/n) \lambda_{1,i})}{1 + p_{2,i} \lambda_{2,i}} \right) \\ \text{s.t. } & p_{2,i} \geq 0; P_R : \sum_{i=1}^L p_{2,i} (1 + (P_1/n) \lambda_{1,i}) \leq P_2 \end{aligned} \quad (182)$$

In the case that $\mathbf{H}_2$ is unknown at the RS, this problem cannot be solved since the values of $\lambda_{2,i}$ are unknown. However, in this case, we can reformulate the optimisation problem in (182) by assuming that all the eigenmodes of $\mathbf{H}_2$ are equal to one, i.e., $\lambda_{2,i} = 1$ for all $i \in [1, L_2]$, such that

$$\begin{aligned} \max_{p_{2,i}} & \frac{1}{2} \sum_{i=1}^L \log_2 \left(\frac{1 + p_{2,i} (1 + (P_1/n) \lambda_{1,i})}{1 + p_{2,i}} \right) \\ \text{s.t. } & p_{2,i} \geq 0; P_R : \sum_{i=1}^L p_{2,i} (1 + (P_1/n) \lambda_{1,i}) \leq P_2 \end{aligned} \quad (183)$$

The optimum solution for this problem is obtained by resorting to its Lagrangian dual problem, [Boyd-04], such that

$$p_{2,i} = \left[\frac{-(2 + p_{1,i}\lambda_{1,i}) + \sqrt{4\mu p_{1,i}\lambda_{1,i} + (p_{1,i}\lambda_{1,i})^2}}{2(1 + p_{1,i}\lambda_{1,i})} \right]_+, \quad (184)$$

where $p_{1,i} = (P_1/n)$, $i \in [1, L]$, and μ is the Lagrange multiplier which moves the constraints in (184) into the maximisation cost function P_R of the optimisation problem. The starting point for μ is set to

$$\mu_{\min} = \max_{i \in [1, L]} \left\{ \frac{1 + (P_1/n)\lambda_{1,i}}{(P_1/n)\lambda_{1,i}} \right\}, \quad (185)$$

and μ can be updated by using the Newton-Raphson method [Wolf] until μ^* is obtained. Let the function

$$f(\mu) = \sum_{i=1}^L p_{2,i}(\mu)(1 + (P_1/n)\lambda_{1,i}) - P_2,$$

where $p_{2,i}(\mu)$ is given in (184), then μ^* is a μ value that fulfils the following inequality $f(\mu^*) < \varepsilon$, with $\varepsilon \ll 1$. Our novel power allocation method can be summarised as follows

Algorithm 4 (EEM)
1: $\mu = \mu_{\min}$; 2: Solve $f(\mu) < \varepsilon$ using the Newton-Raphson method [Wolf] $\Rightarrow \mu^*$; 3: Calculate $p_{2,i}$ in (184) using the value of μ^* for $i \in [1, L]$; 4: $\mathbf{G} = \text{diag}(\sqrt{p_{2,1}}, \sqrt{p_{2,2}}, \dots, \sqrt{p_{2,q}}) \mathbf{U}_1^\dagger$, where $p_{2,i} = 0$ for all $i \in [L+1, q]$.

In the case that only RCSI is known at the RS, the *algorithm 4* allows us to obtain swiftly the values of $p_{2,i}$ that maximise (183). However, maximising (183) is not equivalent to maximising (180), and even though this method performs well compared to the EG, MFR and MMSEF methods, as it is shown in section 4.2.2.5, it is not optimal.

4.2.2.4.3 Analytical power allocation method based on the expectation of $\mathbf{H}_2$

We propose here a novel analytical power allocation method to optimise $\mathbf{G}$ based on the expectation of $I(\mathbf{y}_2; \mathbf{x})$ over $\mathbf{H}_2$. Notice that in (180), $I(\mathbf{y}_2; \mathbf{x})$ is expressed as the sum of two terms of the form $\log_2 |\mathbf{I}_r + \mathbf{H}\mathbf{D}\mathbf{H}^\dagger|$, where $\mathbf{D} = \text{diag}(\mathbf{d}_q)$ is a $q \times q$ diagonal matrix with $\mathbf{d}_q = \{\delta_1, \delta_2, \dots, \delta_q\}$.

4.2.2.4.3.1 MIMO case (EXP MIMO)'

The expectation $E \left\{ \log_2 |\mathbf{I}_r + \mathbf{H}\mathbf{D}\mathbf{H}^\dagger| \right\}_{\mathbf{H}}$ of $\log_2 |\mathbf{I}_r + \mathbf{H}\mathbf{D}\mathbf{H}^\dagger|$ is equivalent to the expression of the mutual information of the MIMO semi-correlated Rayleigh fading channel and this expression can be found in [Smith-03]. However, the formulation of this expression is not practical for power allocation purpose. Fortunately, we have recently shown in [Hélot-09a] that the expression $E \left\{ \log_2 |\mathbf{I}_r + \mathbf{H}\mathbf{D}\mathbf{H}^\dagger| \right\}_{\mathbf{H}}$ is asymptotically equivalent to

$$\chi(\mathbf{d}_q, r) = \frac{1}{\ln(2)} \left[-q - r \ln \left(\frac{d_0(\mathbf{d}_q, r)}{r} \right) + \sum_{i=1}^q \ln(d_0(\mathbf{d}_q, r)\delta_i + 1) + \sum_{i=1}^q \frac{1}{d_0(\mathbf{d}_q, r)\delta_i + 1} \right] \quad (186)$$

for any $\mathbf{H} \in \mathbb{C}^{r \times q}$ and large values of r and q , where $d_0(\mathbf{d}_q, r)$ is the only nonnegative root of the polynomial given by

$$P(d, \mathbf{d}_q, r) = (d - r) \prod_{i=1}^q \left(\frac{1}{\delta_i} + d \right) + d \sum_{i=1}^q \prod_{\substack{j=1 \\ j \neq i}}^q \left(\frac{1}{\delta_j} + d \right). \quad (187)$$

Notice that (186) is similar to (19) in [Héliot-09a], for $n=1$, $m=q$, $\alpha=1$, $\beta=r$, $\omega=1$, and $\nu_i = 1/\delta_i$ in [Héliot-09a]. Then, we can obtain the $p_{2,i}$ values that maximise $E\{I(\mathbf{y}_2; \mathbf{x})\}_{\mathbf{H}_2}$ by solving the following problem

$$\begin{aligned} \max_{\mathbf{p}_2} & \frac{1}{2} \left[\chi(\mathbf{p}_2(\mathbf{I}_q + (P_1/n)\Lambda_1), r) - \chi(\mathbf{p}_2, r) \right] \\ \text{s.t. } & p_{2,i} \geq 0; P_R : \sum_{i=1}^{L_1} p_{2,i} (1 + (P_1/n)\lambda_{1,i}) \leq P_2, \end{aligned} \quad (188)$$

where $\mathbf{p}_2 = \{p_{2,1}, p_{2,2}, \dots, p_{2,q}\}$. Let $\mathbf{v}_2 = \{v_{2,1}, v_{2,2}, \dots, v_{2,q}\}$ with $v_{2,i} = p_{2,i}\omega_i$ and $\omega_i = (1 + (P_1/n)\lambda_{1,i})$, then the problem in (188) can be reformulated with (186) as

$$\begin{aligned} \max_{\mathbf{v}_2} f(\mathbf{v}_2) &= \frac{1}{2 \ln(2)} \left[-r \ln \left(\frac{d_0(\mathbf{v}_2, r)}{d_0(\mathbf{p}_2, r)} \right) + \sum_{i=1}^q \ln \left(\frac{d_0(\mathbf{v}_2, r)v_{2,i} + 1}{d_0(\mathbf{p}_2, r)p_{2,i} + 1} \right) + \sum_{i=1}^q \frac{1}{d_0(\mathbf{v}_2, r)v_{2,i} + 1} \right. \\ &\quad \left. - \sum_{i=1}^q \frac{1}{d_0(\mathbf{p}_2, r)p_{2,i} + 1} \right] \\ \text{s.t. } & v_{2,i} \geq 0; P_R : \sum_{i=1}^q v_{2,i} \leq P_2 \end{aligned} \quad (189)$$

In order to solve this problem, we require the expression of $d_0(\mathbf{d}_q, r)$ for any vector $\mathbf{d}_q$ of length q . The closed-form expressions of $d_0(\mathbf{d}_q, r)$ for $q=2$ and $q=3$ have been given in [Héliot-09b], and they are difficult to obtain for $q \geq 3$, therefore, we resort to approximate $d_0(\mathbf{d}_q, r)$. The polynomial in (187) can be re-expressed as a sequence of initial term $P(d, \mathbf{d}_0 = \{\emptyset\}, r) = d - r$ and general term given by

$$P(d, \mathbf{d}_q, r) = \left(d + \frac{1}{\delta_q} \right) P(d, \mathbf{d}_{q-1}, r) + d \prod_{j=1}^{q-1} \left(d + \frac{1}{\delta_j} \right), \quad (190)$$

where the only nonnegative root of $P(d, \mathbf{d}_q, r)$ and $P(d, \mathbf{d}_{q-1}, r)$ are $d_0(\mathbf{d}_q, r)$ and $d_0(\mathbf{d}_{q-1}, r)$, respectively.

An approximation of $d_0(\mathbf{d}_q, r)$

The second term of the polynomial sequence $P(d, \mathbf{d}_q, r)$ is $P(d, \delta_1, r) = d^2 + d \left(1 - r + \frac{1}{\delta_1} \right) - \frac{r}{\delta_1}$ and the only nonnegative root of this polynomial can be straightforwardly given by

$$\begin{aligned} d_0(\delta_1, r) &= \frac{-(1 + \delta_1(1-r)) + \sqrt{(1 + \delta_1(1-r))^2 + 4r\delta_1}}{2\delta_1} \\ &= \frac{2r}{1 + \delta_1(1-r) + \sqrt{(1 + \delta_1(1-r))^2 + 4r\delta_1}}. \end{aligned} \quad (191)$$

Let $P(d, \delta_1, r)$ be re-expressed as $P(d, \delta_1, r) = \left(d + \frac{1}{\delta_1}\right)(d - r) + d$; in the case that $\delta_1 \rightarrow +\infty$, then $P(d, \delta_1, r)$ is such that $\lim_{\delta_1 \rightarrow +\infty} P(d, \delta_1, r) = d[P(d, \mathbf{d}_0, r) + 1] = dP(d, \mathbf{d}_0, r - 1)$, and hence, $\lim_{\delta_1 \rightarrow +\infty} d_0(\mathbf{d}_0, \delta_1, r) = d_0(\mathbf{d}_0, r - 1)$. Similarly, in the case that $\delta_1 \rightarrow 0$, then $P(d, \delta_1, r)$ simplifies as $\lim_{\delta_1 \rightarrow 0} P(d, \mathbf{d}_0, \delta_1, r) = \left(d + \frac{1}{\delta_1}\right)P(d, \mathbf{d}_0, r)$, and therefore, $\lim_{\delta_1 \rightarrow 0} d_0(\mathbf{d}_0, \delta_1, r) = d_0(\mathbf{d}_0, r)$. We can then conclude that

$$r - 1 \leq d_0(\mathbf{d}_0, r - 1) \leq d_0(\delta_1, r) \leq d_0(\mathbf{d}_0, r) \leq r. \quad (192)$$

More generally, it can be shown by recurrence that

$$[r - q]_+ \leq d_0(\mathbf{d}_{q-1}, r - 1) \leq d_0(\mathbf{d}_q, r) \leq d_0(\mathbf{d}_{q-1}, r) \leq r \quad (193)$$

by following the previous reasoning for the rank q of the sequence $P(d, \mathbf{d}_q, r)$. Relying on the results in (193), $d_0(\mathbf{d}_q, r)$ can be well-approximated by the functions of the form $d_0(\mathbf{d}_q, r) \approx d_0(\mathbf{d}_{q-1}, r - 1)(1 - f(\delta_q)) + d_0(\mathbf{d}_{q-1}, r)f(\delta_q)$ where $f(\delta_q): \mathbb{R}_+ \rightarrow [0, 1]$ is a monotonically decreasing function. Consequently, we choose to approximate $d_0(\mathbf{d}_q, r)$ by using $\tilde{d}_0(\mathbf{d}_q, r)$, which is defined as follows

$$\tilde{d}_0(\mathbf{d}_q, r) = \frac{1}{q} \sum_{j=1}^q d_0(\bar{\mathbf{d}}_j, r - 1) \left[(r - 1) - d_0(\delta_j, r - 1) \right] + d_0(\bar{\mathbf{d}}_j, r) \left[d_0(\delta_j, r - 1) - (r - 2) \right], \quad (194)$$

where $\bar{\mathbf{d}}_j = \{\delta_1, \dots, \delta_{j-1}, \delta_{j+1}, \dots, \delta_q\}$.

Gradient ascent approach to solve (189)

The function $f(\mathbf{v}_2)$ in (189) is non-concave, as it can be shown that the Hessian of $f(\mathbf{v}_2)$ is positive for some values of q , r and $\mathbf{v}_2$. Therefore, the usual concave optimisation method for constrained problems cannot be directly applied. The dual function of $f(\mathbf{v}_2)$ does not have a simple expression and resorting to the dual problem may prove to be very complicate. Instead, we develop a sub-optimal efficient algorithm based on the gradient ascent method [Boyd-04], which is simple and quite accurate. Since the eigenvalues $\lambda_{1,i}$ are ordered in descending order, consequently $\omega_1 \geq \omega_2 \geq \dots \geq \omega_q$ and

$v_{2,1} \geq v_{2,2} \geq \dots \geq v_{2,q}$. Moreover, we know that $v_{2,i} \geq 0$ and $\sum_{i=1}^q v_{2,i} \leq P_2$, it then implies that

$$v_{2,i} \in \begin{cases} [P_2/q, P_2] & , i = 1 \\ \left[0, \min \left(\min \left(\frac{P_2}{i}, v_{2,i-1} \right), \max \left(0, P_2 - \sum_{j=1}^{i-1} v_{2,j} \right) \right) \right] & , i \in [2, q] \end{cases} \quad (195)$$

Thus, we can distinguish $q-1$ areas of search for $\mathbf{v}_2$ such that $\mathbf{v}_2 \in [\mathbf{v}_2^k, \mathbf{v}_2^{k+1}]$, where

$$\mathbf{v}_2^k = \begin{cases} v_{2,i} = P_2/k, \text{ for } i \in [1, k] \\ v_{2,i} = 0, \text{ else} \end{cases}, \text{ e.g., } \mathbf{v}_2^3 = \{P_2/3, P_2/3, P_2/3, 0, 0\} \text{ for } q=5. \text{ In our algorithm, we first}$$

evaluate $f(\mathbf{v}_2)$ at each $\mathbf{v}_2^k$ point, then search for the maximum of this point $f(\mathbf{v}_2^*) = \max_{k \in [1, q]} (f(\mathbf{v}_2^k))$

and use this point as a starting point for the gradient search. On the one hand, this initial step requires q evaluation of $f(\mathbf{v}_2)$, which can be time consuming for large values of q ; on the other hand, the gradient search will be shorten by choosing an appropriate starting point. Next, we proceed with a

constrained gradient search around this local maximum in order to find local maxima that are equal or greater than $f(\mathbf{v}_2^*)$. We constraint our search by defining $v_{2,q} = P_2 - \sum_{i=1}^{q-1} v_{2,i}$ and by insuring that each individual constraint on $v_{2,i}$ in (195) is respected. In order to perform the gradient search, we require the expression of the gradient of $f(\mathbf{v}_2)$, which is given by

$$\nabla \mathbf{v}_2 = \left[\frac{\partial f(\mathbf{v}_2)}{\partial v_{2,1}}, \frac{\partial f(\mathbf{v}_2)}{\partial v_{2,2}}, \dots, \frac{\partial f(\mathbf{v}_2)}{\partial v_{2,q-1}}, 0 \right], \quad (196)$$

where

$$\begin{aligned} \frac{\partial f(\mathbf{v}_2)}{\partial v_{2,i}} = & \frac{1}{2 \ln(2)} \left[\frac{\partial \tilde{d}_0(\mathbf{v}_2, r)}{\partial v_{2,i}} \left(-\frac{r}{\tilde{d}_0(\mathbf{v}_2, r)} + \sum_{j=1}^q v_{2,j} \left(\frac{1}{\tilde{d}_0(\mathbf{v}_2, r) v_{2,j} + 1} - \frac{1}{(\tilde{d}_0(\mathbf{v}_2, r) v_{2,j} + 1)^2} \right) \right) \right. \\ & + \tilde{d}_0(\mathbf{v}_2, r) \left(\frac{1}{\tilde{d}_0(\mathbf{v}_2, r) v_{2,i} + 1} - \frac{1}{(\tilde{d}_0(\mathbf{v}_2, r) v_{2,i} + 1)^2} - \frac{1}{\tilde{d}_0(\mathbf{v}_2, r) v_{2,q} + 1} + \frac{1}{(\tilde{d}_0(\mathbf{v}_2, r) v_{2,q} + 1)^2} \right) \\ & - \frac{\partial \tilde{d}_0(\mathbf{p}_2, r)}{\partial v_{2,i}} \left(-\frac{r}{\tilde{d}_0(\mathbf{p}_2, r)} + \sum_{j=1}^q p_{2,j} \left(\frac{1}{\tilde{d}_0(\mathbf{p}_2, r) p_{2,j} + 1} - \frac{1}{(\tilde{d}_0(\mathbf{p}_2, r) p_{2,j} + 1)^2} \right) \right) \\ & \left. - \tilde{d}_0(\mathbf{p}_2, r) \left(\frac{1}{\omega_i} \left(\frac{1}{\tilde{d}_0(\mathbf{p}_2, r) p_{2,i} + 1} - \frac{1}{(\tilde{d}_0(\mathbf{p}_2, r) p_{2,i} + 1)^2} \right) - \frac{1}{\omega_q} \left(\frac{1}{\tilde{d}_0(\mathbf{p}_2, r) p_{2,q} + 1} - \frac{1}{(\tilde{d}_0(\mathbf{p}_2, r) p_{2,q} + 1)^2} \right) \right) \right] \end{aligned}$$

Note that the partial derivatives of $\tilde{d}_0(\mathbf{v}_2, r)$ and $\tilde{d}_0(\mathbf{p}_2, r)$ can be straightforwardly obtained by differentiating (194).

This algorithm is sub-optimal in the sense that it does not guarantee to find the absolute maximum value of $f(\mathbf{v}_2)$, but it provides a sufficient accuracy with a reasonable complexity, as it is shown later in section 4.2.2.5. A parameter ϵ is used in our algorithm to fine-tune the accuracy. Our Algorithm can be summarised as follows

Algorithm 5 (EXP MIMO)

```

1: Set  $\epsilon=1\text{E-}6$ ,  $\beta=0.5$ ,  $\text{it}2=0$ ,  $\text{it}3=0$ ,  $\text{gradient\_search}=0$ ;
2: Evaluate  $f(\mathbf{v}_2)$  at  $\mathbf{v}_2^k$ , for  $k \in [1, q]$ ;
3:  $f(\mathbf{v}_2^n) = \max_{k \in [1, q]} (f(\mathbf{v}_2^k))$ ;
4: Set  $\mathbf{x} = \mathbf{v}_2^n$ ,  $\bar{f} = f(\mathbf{x})$ ;
5: While  $\text{gradient\_search} == 0$  do
6: Evaluate  $\nabla \mathbf{x}$ ;
7: Normalise  $\nabla \mathbf{x}$  according to  $P_2$ ;
8:  $\hat{\mathbf{x}} = \mathbf{x} + \nabla \mathbf{x}$ ;
9: Check that  $\hat{\mathbf{x}}$  fulfils the constraint in (195) and modify  $\nabla \mathbf{x}$  accordingly as  $\bar{\nabla} \mathbf{x}$ ;
10:  $\hat{\mathbf{x}} = \mathbf{x} + \bar{\nabla} \mathbf{x}$ ;
11: Set  $\eta = \bar{\nabla} \mathbf{x} \bar{\nabla} \mathbf{x}^\dagger$ ;
12: If  $(\eta > \epsilon)$  &  $(\text{it}2 < \text{IT}2)$  &  $(\text{it}3 < \text{IT}3)$  do
13:  $t=1$ ,  $\hat{f} = f(\hat{\mathbf{x}})$ ;
14: While  $(\hat{f} < \bar{f}) \& (\text{it}2 < \text{IT}2)$  do
15:  $t = \beta t$ ,  $\hat{\mathbf{x}} = \mathbf{x} + t \bar{\nabla} \mathbf{x}$ ,  $\hat{f} = f(\hat{\mathbf{x}})$ ,  $\text{it}2 = \text{it}2 + 1$ ;
16: End While
17: If  $|\hat{f} - \bar{f}| < 1\text{E-}4$  do
18:  $\text{it}3 = \text{it}3 + 1$ ;
19: End If
20:  $\bar{f} = \hat{f}$ ,  $\mathbf{x} = \hat{\mathbf{x}}$ ,  $\text{it} = \text{it} + 1$ ;
21: Else
22:  $\text{gradient\_search} = 1$ ;
23: End If
24: If  $(\text{it} > > 1)$  do
25:  $\text{gradient\_search} = 1$ ;
26: End If
27: End While

```

Note that it , $\text{it}2$, and $\text{it}3$ are variables use for tuning the complexity, and $\text{IT}1$ and $\text{IT}2$ could be set to 10 and 50, respectively. Notice that the EXP MIMO algorithm complexity increases exponentially because of the way $\tilde{d}_0(\mathbf{d}_q, r)$ is defined, but not because of the structure of the algorithm itself.

4.2.2.4.3.2 MISO case**1st Method (EXP MISO 1)**

In the MISO case, i.e., when $r=1$, The expectation $E\left\{\log_2 \left| \mathbf{I}_r + \mathbf{H} \mathbf{D} \mathbf{H}^\dagger \right| \right\}_{\mathbf{H}}$ is equivalent to

$E\left\{\log_2 \left(1 + \sum_{i=1}^q \delta_i |h_i|^2 \right) \right\}$. Let $Z = \sum_{i=1}^q \delta_i |h_i|^2$ be a random variable, which is a weighted sum of central chi-square random variables, then its probability density function can be expressed as follows

$$p_Z = \sum_{i=1}^q \frac{\delta_i^{q-2} e^{-\frac{Z}{\delta_i}}}{\prod_{\substack{i=1 \\ i \neq j}}^q (\delta_i - \delta_j)}, \quad (197)$$

for any $\delta_i \neq \delta_j$. Consequently,

$$\begin{aligned}
 E \left\{ \log_2 \left(1 + \sum_{i=1}^q \delta_i |h_i|^2 \right) \right\} &= \frac{1}{\ln(2)} \int_0^{+\infty} \ln(1+Z) p_Z dZ \\
 &= \frac{1}{\ln(2)} \sum_{i=1}^q \frac{\delta_i^{q-1}}{\prod_{\substack{i=1 \\ i \neq j}}^q (\delta_i - \delta_j)} e^{-\frac{1}{\delta_i}} E_1 \left(\frac{1}{\delta_i} \right). \quad (198)
 \end{aligned}$$

Then, we can obtain the $p_{2,i}$ values that maximise $E\{I(\mathbf{y}_2; \mathbf{x})\}_{\mathbf{H}_2}$ by using (198) in (180) and by solving the following problem

$$\begin{aligned}
 &\frac{1}{2\ln(2)} \max_{p_{2,i}} \sum_{i=1}^q \frac{p_{2,i} w_i}{\prod_{\substack{i=1 \\ i \neq j}}^q (p_{2,i} w_i - p_{2,j} w_j)} e^{-\frac{1}{p_{2,i} w_i}} E_1 \left(\frac{1}{p_{2,i} w_i} \right) - \sum_{i=1}^q \frac{p_{2,i}^{q-1}}{\prod_{\substack{i=1 \\ i \neq j}}^q (p_{2,i} - p_{2,j})} e^{-\frac{1}{p_{2,i}}} E_1 \left(\frac{1}{p_{2,i}} \right) \\
 &\text{s.t. } p_{2,i} \geq 0; P_R : \sum_{i=1}^q p_{2,i} w_i \leq P_2 \quad (199)
 \end{aligned}$$

This problem turns out to be non-concave, and it cannot be solved for $\delta_i = \delta_j$, thus it is not a very convenient problem to solve. Instead, we design a simpler equivalent problem in the following section.

2nd Method (EXP MISO 2)

In the MIMO case, we have shown how to obtain the $p_{2,i}$ values that maximise $E\{I(\mathbf{y}_2; \mathbf{x})\}_{\mathbf{H}_2}$ in (189) by using the approximation for $d_0(\mathbf{d}_q, r)$ in (194). According to equation (193), $d_0(\mathbf{d}_q, r)$ is such that $0 \leq d_0(\mathbf{d}_q, r) \leq 1$, when $r=1$. In that case, we can approximate $d_0(\mathbf{d}_q, r)$ as in (194), or more simply as

$$\begin{aligned}
 \tilde{d}_0(\mathbf{d}_q, 1) &= \prod_{i=1}^q d_0(\delta_i, 1) \quad \text{since } 0 \leq d_0(\delta_i, 1) \leq 1. \quad \text{In the case of } q=2, \quad \tilde{d}_0(\mathbf{d}_2, 1) = d_0(\delta_2, 1) d_0(\delta_2, 1) \\
 &= 4(1+x_1)^{-1} (1+x_1)^{-2} \quad \text{with } x_i = \sqrt{1+4\delta_i}. \quad \text{It then implies that } \tilde{d}_0(\mathbf{d}_2, 1) \delta_1 + 1 = \frac{x_1 + x_2}{1+x_2} \text{ and} \\
 \tilde{d}_0(\mathbf{d}_2, 1) \delta_2 + 1 &= \frac{x_1 + x_2}{1+x_1}. \quad \text{Inserting the values of } \tilde{d}_0(\mathbf{d}_2, 1), \tilde{d}_0(\mathbf{d}_2, 1) \delta_1 + 1, \text{ and } \tilde{d}_0(\mathbf{d}_2, 1) \delta_2 + 1 \text{ in (189),}
 \end{aligned}$$

the problem simplifies as follows

$$\begin{aligned}
 \max_{\mathbf{v}_2} g(\mathbf{v}_2) &= \frac{1}{2\ln(2)} \left[\ln(h(\mathbf{v}_2)) + (h(\mathbf{v}_2))^{-1} - \ln(h(\mathbf{p}_2)) - (h(\mathbf{p}_2))^{-1} \right] \\
 \text{s.t. } v_{2,i} &\geq 0; P_R : \sum_{i=1}^q v_{2,i} \leq P_2 \quad (200)
 \end{aligned}$$

where $h(\mathbf{x}) = \sum_{i=1}^q \sqrt{1 + \beta x_i}$ for $q=2$. The function $\ln(h(\mathbf{x})) + (h(\mathbf{x}))^{-1}$ is concave by composition, since $\ln(X) + (X)^{-1}$ is concave and non-decreasing for $X \in [1, +\infty)$, i.e., $h(\mathbf{x}) \in [q, +\infty)$, and $\sqrt{x}$ is also concave. Moreover, as the terms ω_i grow, the effect of the function $\ln(h(\mathbf{p}_2)) + (h(\mathbf{p}_2))^{-1}$ in (200) vanishes and the overall function $g(\mathbf{v}_2)$ becomes concave. However, the function $g(\mathbf{v}_2)$ is non-concave for low values of ω_i and we apply the same type of algorithm developed for the MIMO case in *Algorithm 5*, but with $g(\mathbf{v}_2)$ instead of $f(\mathbf{v}_2)$. Notice that our algorithm, which we denote as

Algorithm 6 (EXP MISO 2), is generalised for any q value by setting $\beta = q^2$. The gradient of $g(\mathbf{v}_2)$ can be expressed as in (196) but where the partial derivatives of $g(\mathbf{v}_2)$ are given by

$$\frac{dg(\mathbf{v}_2)}{dv_{2,i}} = \frac{1}{2\ln(2)} \left[\frac{dh(\mathbf{v}_2)}{dv_{2,i}} (h(\mathbf{v}_2))^{-1} (1 - (h(\mathbf{v}_2))^{-1}) - \frac{dh(\mathbf{p}_2)}{dv_{2,i}} (h(\mathbf{p}_2))^{-1} (1 - (h(\mathbf{p}_2))^{-1}) \right], \quad (201)$$

with

$$\begin{aligned} \frac{dh(\mathbf{v}_2)}{dv_{2,i}} &= \frac{\beta}{2} \left[(1 + \beta v_{2,i})^{-\frac{1}{2}} - (1 + \beta v_{2,q})^{-\frac{1}{2}} \right], \\ \frac{dh(\mathbf{p}_2)}{dv_{2,i}} &= \frac{\beta}{2\omega_i} \left[\left(1 + \beta \frac{v_{2,i}}{\omega_i}\right)^{-\frac{1}{2}} - \frac{\omega_i}{\omega_q} \left(1 + \beta \frac{v_{2,q}}{\omega_q}\right)^{-\frac{1}{2}} \right]. \end{aligned}$$

4.2.2.5 Single user performance analysis

4.2.2.5.1 Single Carrier MIMO and MISO results

The various power allocation methods introduced in section 4.2.2, i.e., *Algorithm 1* (Alg1), *Algorithm 2* (Alg2), *Algorithm 3* (Alg3), AF EG, MFR, MMSEF, *Algorithm 5* (EXP MIMO), EXP MISO, and EXP MISO 2 are evaluated here in terms of mutual information performance considering cooperative, i.e., $I(\mathbf{y}; \mathbf{x})$, and non-cooperative, i.e., $I(\mathbf{y}_2; \mathbf{x})$, settings, and are compared against Exhaustive Search (Ex. Search) algorithm and the methods that have been proposed in [Fan-07, Muñoz-07] for the case of FCSI at the RS. In Ex. Search algorithm, all the possible values of $\mathbf{p}_2$ are enumerated for any realisation of the channels. Ex. Search is optimal but very time consuming and it can only be applied for very low values of q . It is here used as a benchmark. Moreover, in our simulation, we consider that SNR_0 is the SNR of the MS-BS link, SNR_1 is the SNR of the MS-RS link, and SNR_2 is the SNR of the RS-BS link.

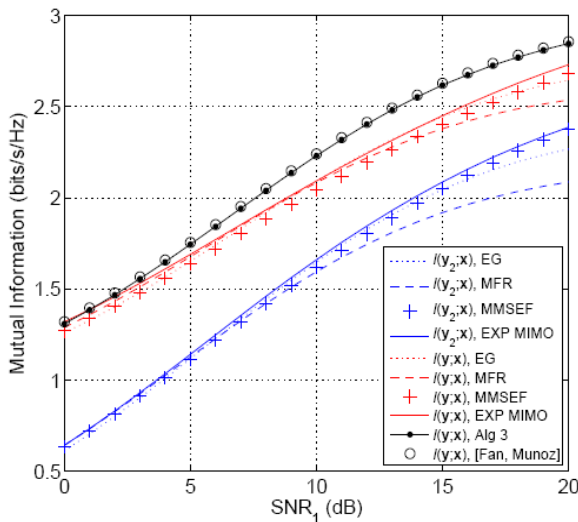


Figure 90 (a)

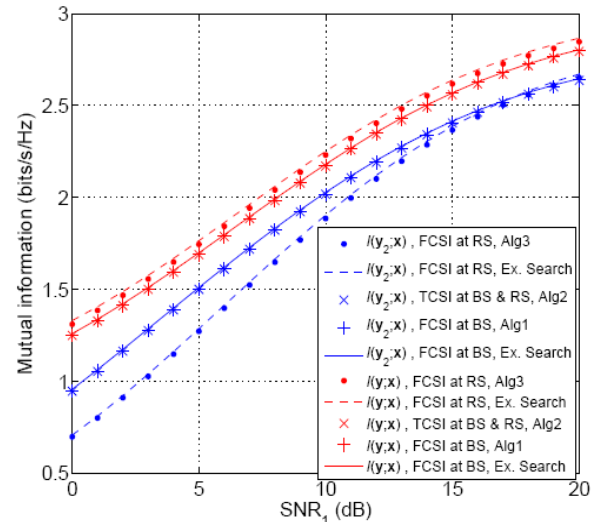


Figure 90 (b)

Figure 90: Mutual information performance of various power allocation algorithms for $n = q = r = 2$, $\text{SNR}_0 = 0$ dB and $\text{SNR}_2 = 10$ dB.

In Figure 90, we compare the cooperative and non-cooperative mutual information performance of various power allocation algorithms for $n = q = r = 2$, $\text{SNR}_0 = 0$ dB and $\text{SNR}_2 = 10$ dB. On the left side of Figure 90, for the case of RCSI at RS studied in section 3.4, our results indicate that our novel

method EXP MIMO outperforms AF EG, MFR, and MMSEF for cooperative and non-cooperative cases. Also, the performance of our novel method is closed to that of Alg3 at low SNR1. Otherwise, the results show that Alg3 performs as well as [Fan-07, Muñoz-07]. Moreover, the results show that Alg1 performance can be achieved with less complexity by using Alg2, since they both perform similarly. As expected Alg1 and Alg2 outperform Alg3 in non-cooperative case since they have been designed to maximise $I(y_2; \mathbf{x})$. However, in the cooperative case, it turns out that Alg3 outperforms both Alg1 and Alg2, but the performance difference decreases as SNR1 (dB) increases. Modifying the power allocation at the BS has an impact on the direct link performance and the results pinpoint that as long as the quality of the relay link is not drastically better than that of the direct link, i.e., $\text{SNR}_1 = \text{SNR}_2 > \text{SNR}_0 + 20$ dB, then it is not efficient to optimise the power allocation solely according to the relay link at the BS in a single-carrier case.

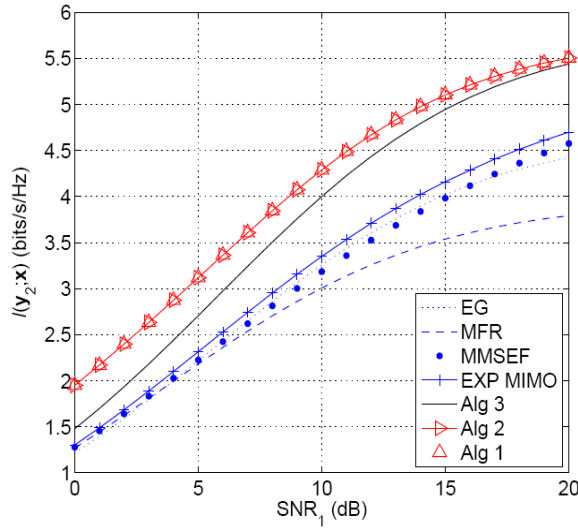


Figure 91 (a)

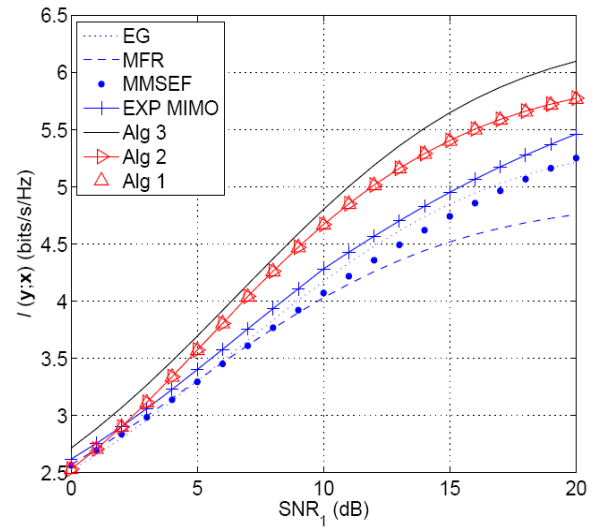


Figure 91 (b)

Figure 91: Non-cooperative (a) and Cooperative (b) Mutual information performance of various power allocation algorithms for $n = q = r = 4$, $\text{SNR}_0 = 0$ dB and $\text{SNR}_2 = 10$ dB.

In Figure 91, we compare the cooperative and non-cooperative mutual information performance of various power allocation algorithms for $n = q = r = 4$, $\text{SNR}_0 = 0$ dB and $\text{SNR}_2 = 10$ dB. For the non-cooperative case, on the left side of Figure 91, the results confirm that EXP MIMO provides the best performance when only RCSI available at RS, Alg2 provides the same performance as Alg1 but with less complexity, and Alg1 and Alg2 outperforms Alg3. Concerning the cooperative case on the right side of Figure 91, it can be noticed that the performance gap between Alg1 and Alg3 is larger than in the $n = q = r = 2$ case. By increasing the number of antennas, the number of eigenmodes increases, and more of these modes are not properly tuned for the direct link when Alg1 or Alg2 is utilised, which further reduces the efficiency of this link in comparison with the $n = q = r = 2$ case.

In Figure 92, Figure 93 and Figure 94, we compare the performances of the EG, MFR, and MMSEF schemes against the performances of our schemes: EEM, EXP MISO 1 and 2, and EXP MIMO, for different SNR settings and number of antennas.

First in Figure 92, we compare these power allocation schemes in the case where the MS-RS link becomes stronger compared to the MS-BS and RS-BS links, i.e., the RS gets closer to the MS. In Figure 92 (a), we show that our scheme EXP MISO 2 can achieve similar performance as EXP MISO 1 but with a lower implementation complexity. The figure also depicts that EXP MISO 1 and 2 outperforms any other schemes and acts as an upper benchmark for MFR at low SNRs, for EG at medium SNRs and for MMSEF at high SNRs. This result is confirmed in Figure 92 (b) and (c) for a larger number of Tx antennas. In the MIMO case, Figure 92 (d), (e) and (f) indicate that both our EXP MIMO and EEM schemes outperforms the usual AF schemes, especially MFR and EG at high SNRS.

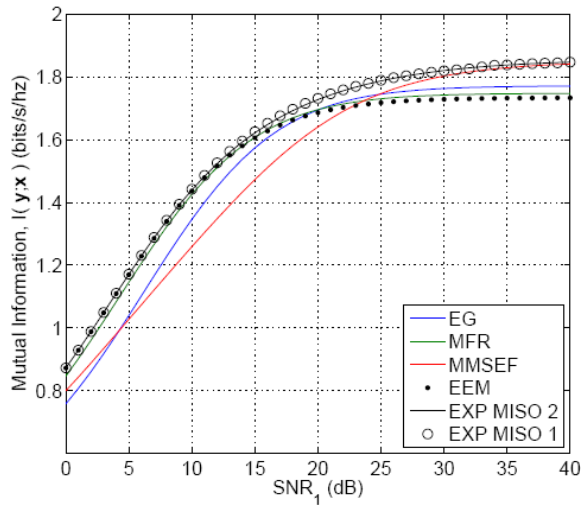


Figure 92 (a)

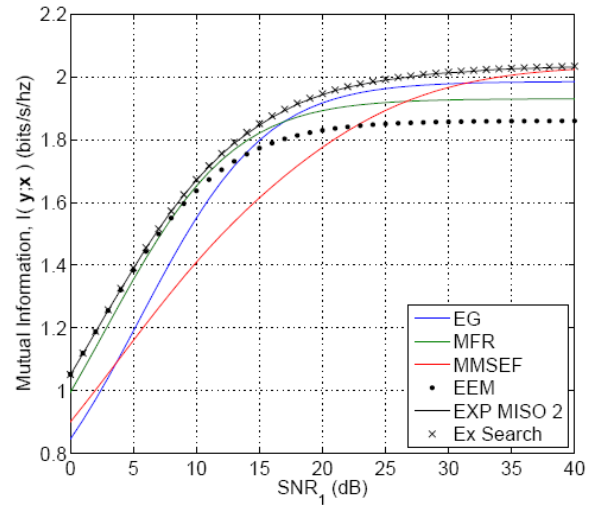


Figure 92 (b)

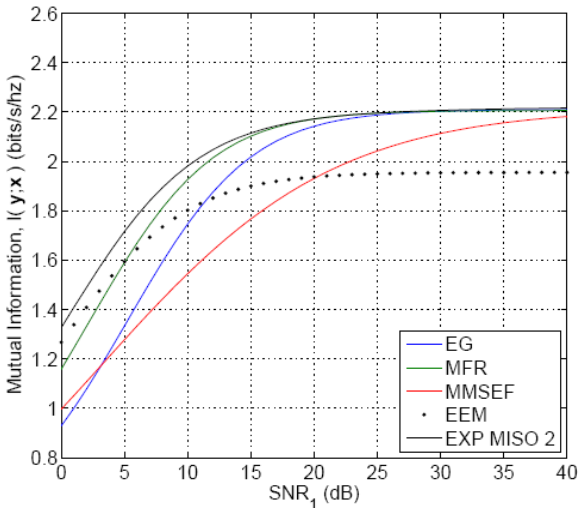


Figure 92 (c)

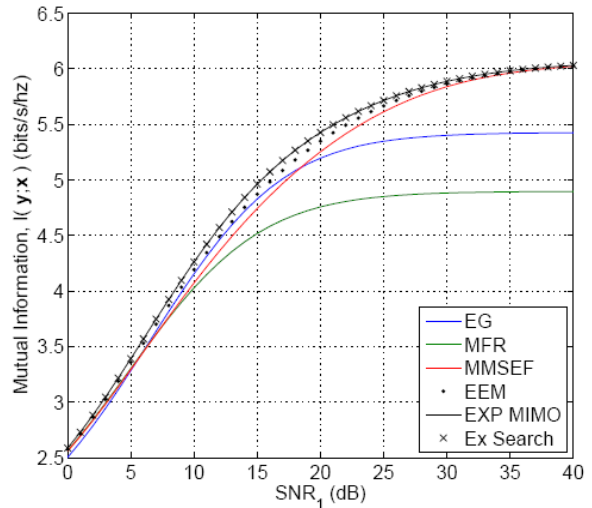


Figure 92 (d)

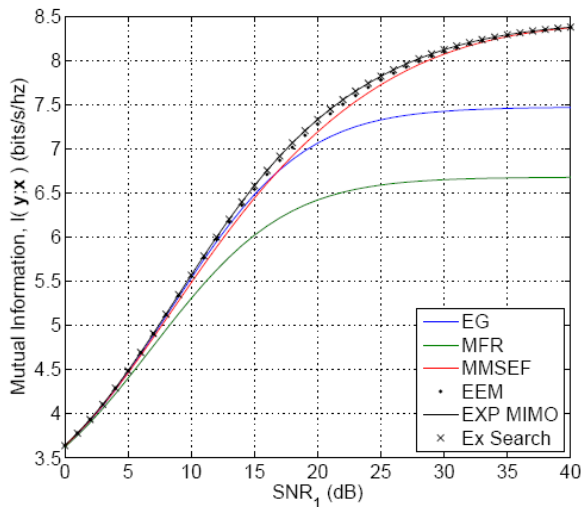


Figure 92 (e)

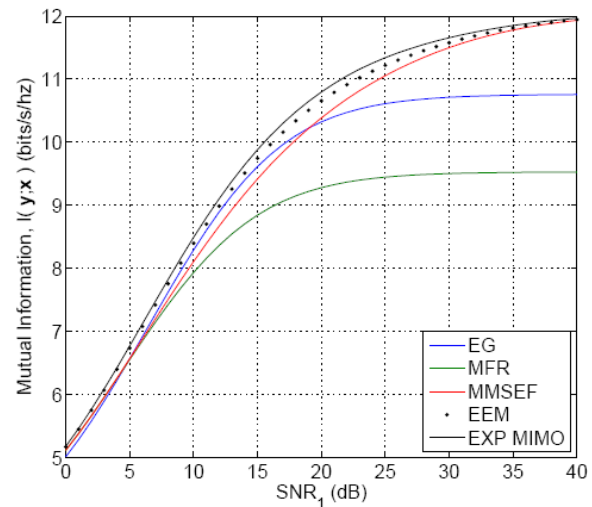


Figure 92 (f)

Figure 92: Cooperative mutual information performance of various power allocation schemes for $SNR_0=0$ dB and $SNR_2=10$ dB, (a) MISO $n=q=2, r=1$, (b) MISO $n=q=4, r=1$, (c) MISO $n=q=64, r=1$, (d) MIMO $n=q=r=4$, (e) MIMO $n=q=4, r=8$, (f) MIMO $n=q=r=8$

The gain between EXP MIMO and the other schemes increases as q increases as it can be seen by comparing Figure 92 (d) and (f). Finally, Figure 92 (b), (d) and (e) shows that our proposed algorithm EXP MISO 2 and EXP MIMO for solving (200) and (189), respectively, are close to optimal since they perform as good as the Ex. Search algorithm but with a lower complexity.

In Figure 93, we assume that the RS-BS link becomes stronger compared to the MS-BS and MS-RS links, i.e., the relay gets closer to the destination. The results again indicate that EXP MISO in Figure 93 (a) and EXP MIMO in Figure 93 (b) and (c) outperform the other schemes. The result also shows that EEM provides at least the same performance as the best of the three original AF scheme.

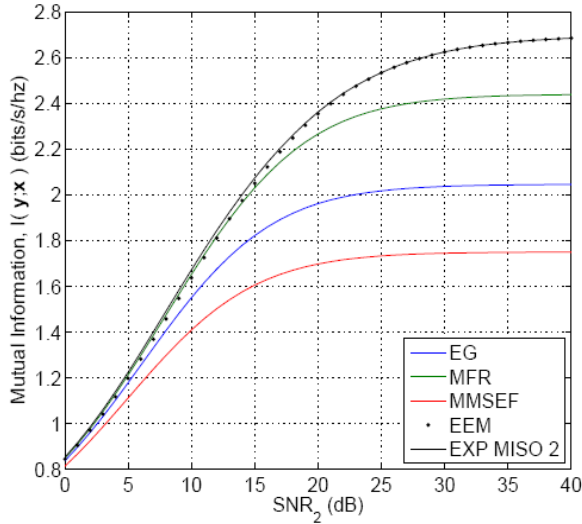


Figure 93 (a)

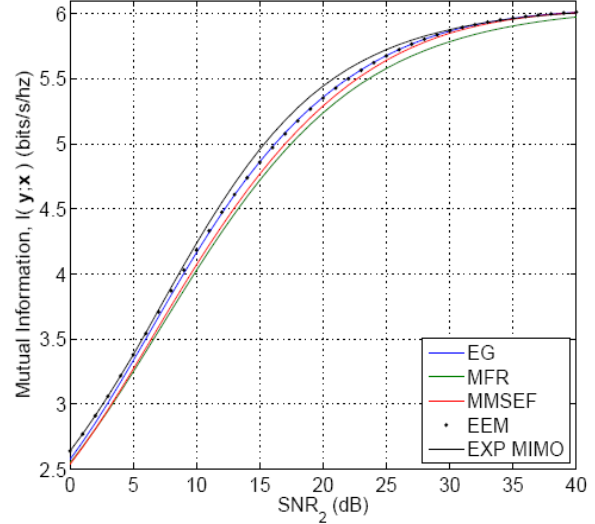


Figure 93 (b)

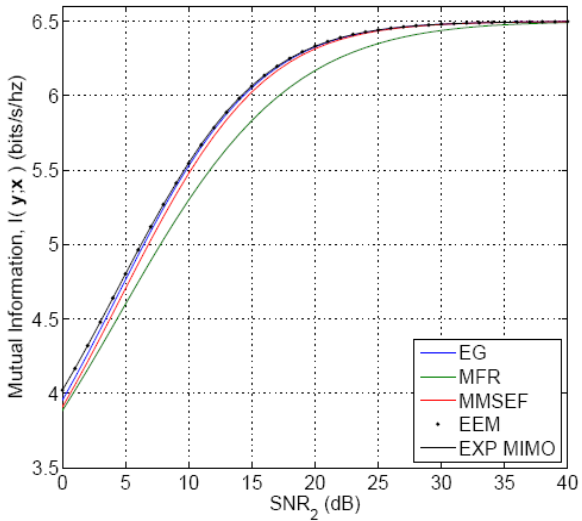


Figure 93 (c)

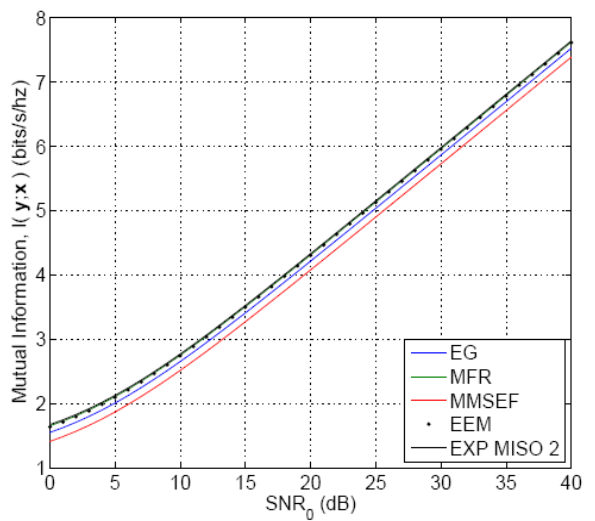


Figure 94 (a)

Figure 93: Cooperative mutual information performance of various power allocation schemes for $\text{SNR}_0=0$ dB and $\text{SNR}_1=10$ dB, (a) MISO $n=q=4$, $r=1$, (b) MIMO $n=q=4$, $r=4$, (c) MIMO $n=q=4$, $r=8$

In Figure 94, we assume that the MS-BS link becomes stronger compared to the MS-RS and RS-BS links, i.e., the direct link becomes predominant compared to the 2-hop link. The graphs (a), (b), (c) show that our scheme performs at least as well as the best of EG, MFR, and MMSE. The performance differences between the schemes are lower in this case; when the direct is stronger than the 2-hop link,

the biggest share of the mutual information is provided by the direct link; therefore, the power allocation at the relay has a lesser impact on the aggregate mutual information.

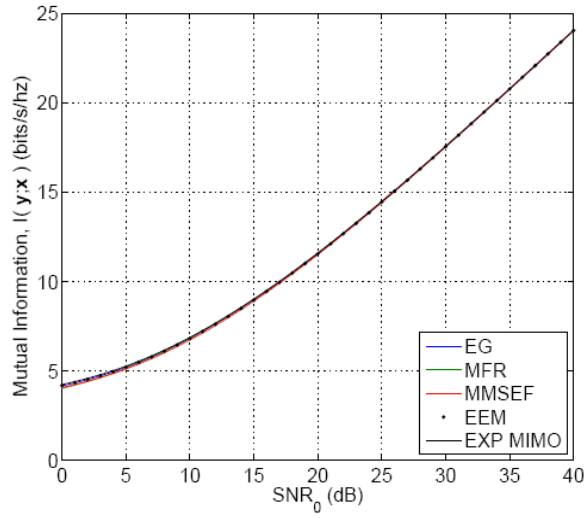


Figure 94 (b)

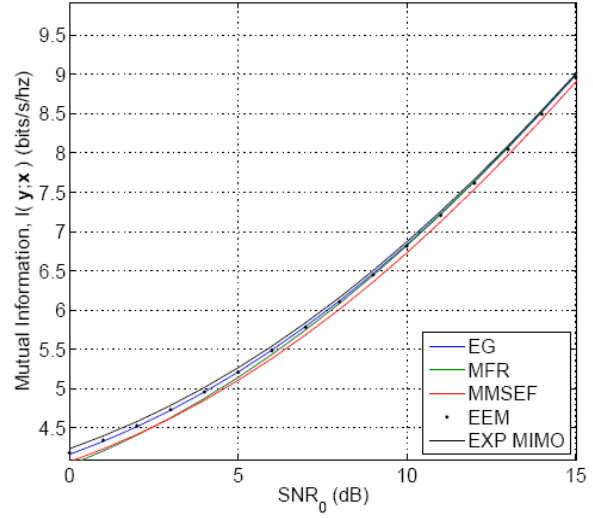


Figure 94 (c)

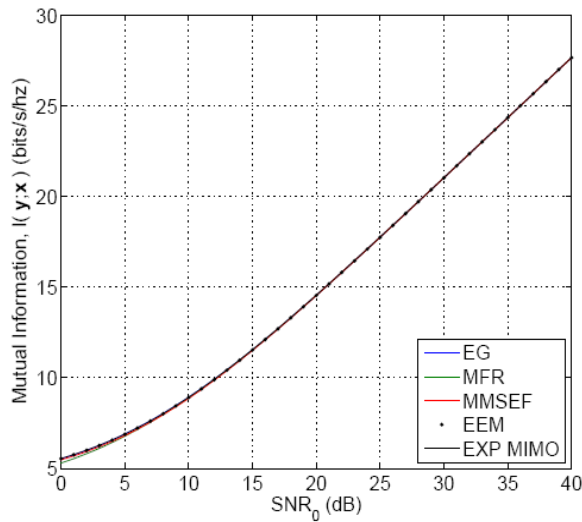


Figure 94 (d)

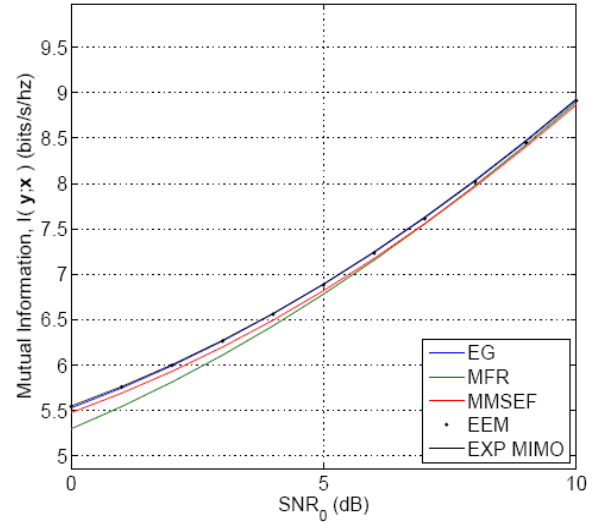


Figure 94 (e)

Figure 94: Cooperative mutual information performance of various power allocation schemes for $\text{SNR}_1=10$ dB and $\text{SNR}_2=10$ dB, (a) MISO $n=q=4, r=1$, (b) MIMO $n=q=4, r=4$, (c) low SNR magnified version of (b), (d) MIMO $n=q=4, r=8$, (e) low SNR magnified version of (d)

4.2.2.5.2 Multi-Carrier SISO and MIMO results

In Figure 95 and Figure 96, we compare the performances of the Alg3, which is equivalent to the schemes in [Fan-07, Muñoz-07], EG, MFR, and MMSEF schemes against the performances of our schemes, i.e., Alg1, Alg2, EEM and EXP MIMO, for different numbers of antenna and subcarrier. Note that Alg1 and Alg2 perform in the same way and, therefore, are both represented by the same curve.

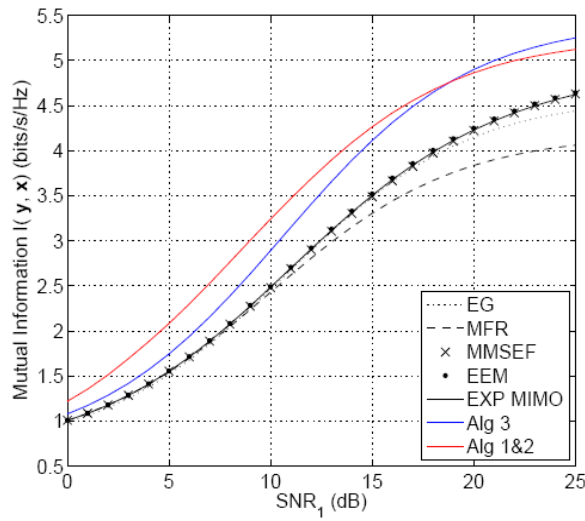


Figure 95 (a)

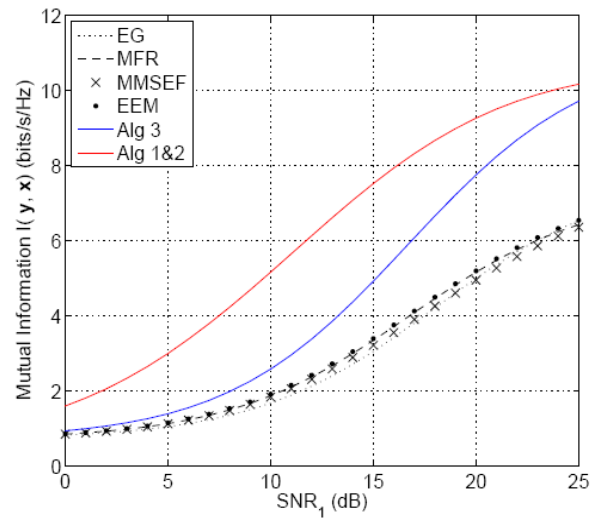


Figure 95 (b)

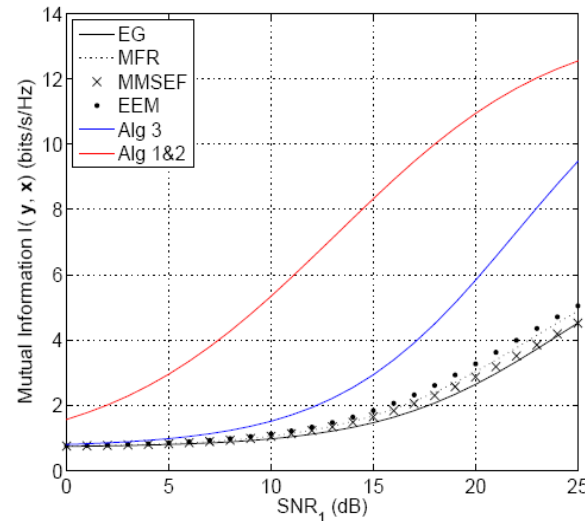


Figure 95 (c)

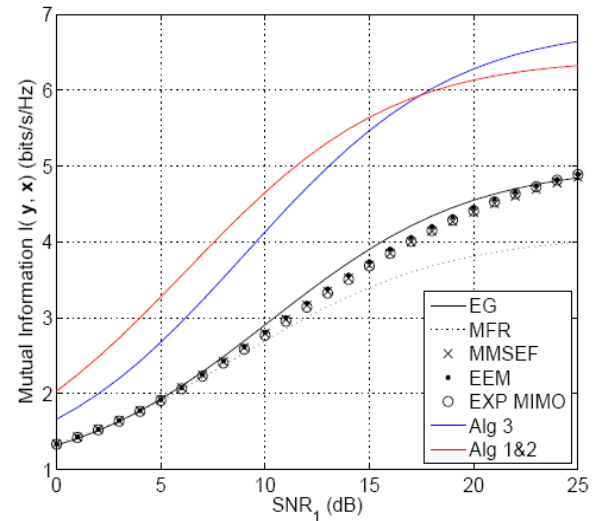


Figure 96 (a)

Figure 95: Cooperative mutual information performance of various power allocation methods for the SISO case with a) 8, b) 64 and c) 256 subcarriers, $SNR_0=0$ dB, and $SNR_2=10$ dB, car scenario, 50km/h.

Figure 95 a) and Figure 96 a) first indicates that our EXP MIMO scheme is not always outperforming the other AF schemes as it is the case for single carrier. It is based on an analytical method that assumes that the channel H_2 is single path Rayleigh fading channel; however, it is not the case in this simulation. Moreover, the EEM method performs well against the EG, MFR, and MMSEF schemes for multi-carrier SISO system. As far as Alg2 and Alg3 are concerned, the results show that a gain in cooperative mutual information can be obtained by using the TCSI at the source, contrarily to the single carrier case in Figure 91 (b). Moreover, this gain increases as the number of utilised subcarriers increases. These results also illustrated that the performance of Alg3 gets closer to these of the RCSI only methods as the number of subcarriers increase, at low SNRs.

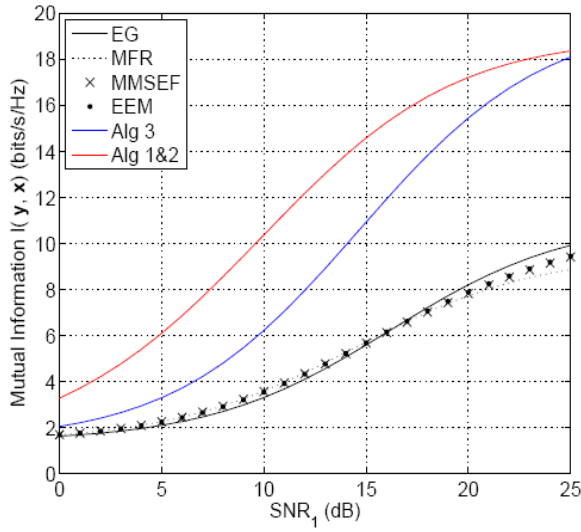


Figure 96 (b)

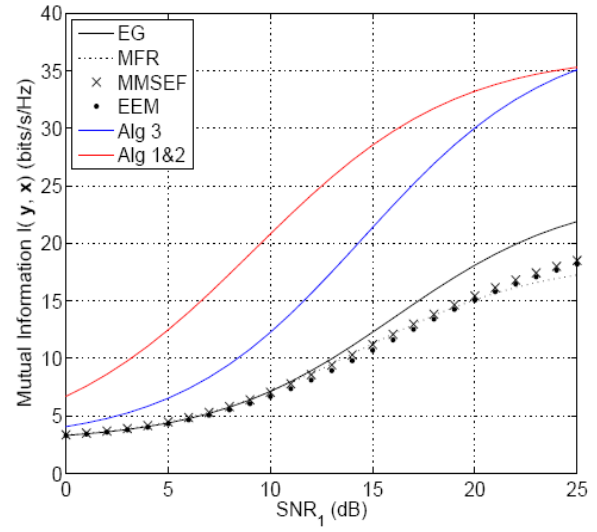


Figure 96 (c)

Figure 96: Cooperative mutual information performance of various power allocation methods for the MIMO case with a) $n=q=r=2$, 4 subcarriers, b) $n=q=r=2$, 32 subcarriers and c) $n=q=r=4$, 32 subcarriers, $\text{SNR}_0=0$ dB, and $\text{SNR}_2=10$ dB, car scenario, 50km/h.

4.2.3 Uplink multi-user power allocation methods for nonregenerative cooperative MIMO communication

In this section, we now consider an UL MU cooperative MIMO system, where K -users terminal equipped with n antennas transmit their data via a nonregenerative R node equipped with q antennas to a D node equipped with r antennas, as illustrated in Figure 97.

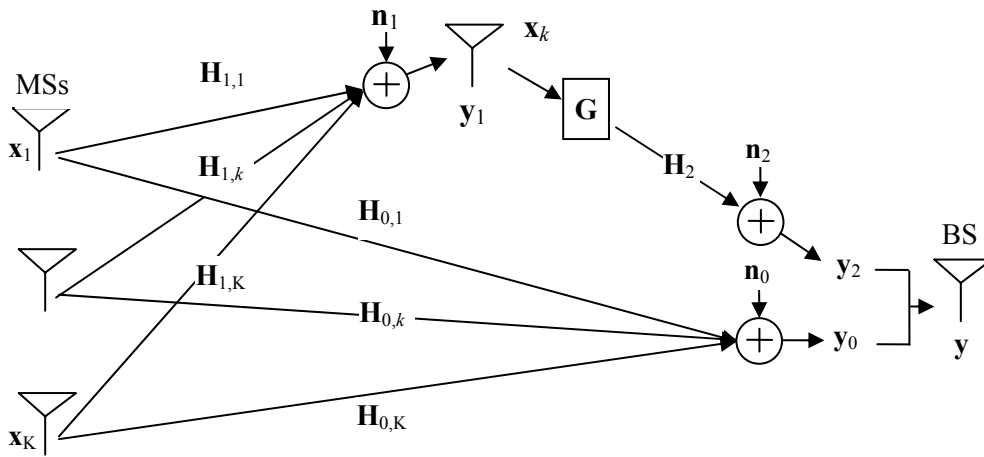


Figure 97: UL MU scenario

As in the single user case, we assume a half duplex relaying scenario with two equal duration phases for the simplicity of the introduction. During the first phase, the signals $\mathbf{x}_k$ sent by each user k is

received as $\mathbf{y}_0 = \sum_{k=1}^K \mathbf{H}_{0,k} \mathbf{x}_k + \mathbf{n}_0$ and $\mathbf{y}_1 = \sum_{k=1}^K \mathbf{H}_{1,k} \mathbf{x}_k + \mathbf{n}_1$ at the D and R, respectively, where

$\mathbf{H}_{0,k} \in \mathbb{C}^{r \times n}$ and $\mathbf{H}_{1,k} \in \mathbb{C}^{q \times n}$ characterise the MIMO channel between the k -th user to D and the k -th user to R links, respectively. During the second phase, the signal $\mathbf{y}_1$ is modified by using the

precoding matrix $\mathbf{G} \in \mathbb{R}^{q \times q}$, then is transmitted towards D and is received as $\mathbf{y}_2 = \mathbf{H}_2 \mathbf{G} \mathbf{y}_1 + \mathbf{n}_2$, where $\mathbf{H}_2 \in \mathbb{C}^{r \times q}$ characterises the MIMO channel between the R-D link. Moreover, $\mathbf{n}_0 \in \mathbb{C}^{r \times 1}$, $\mathbf{n}_1 \in \mathbb{C}^{q \times 1}$ and $\mathbf{n}_2 \in \mathbb{C}^{r \times 1}$ are vectors of independent zero-mean complex Gaussian noise entries with a variance of σ^2 . Inserting the expression of $\mathbf{y}_1$ in $\mathbf{y}_2$, $\mathbf{y}_2$ can be re-expressed as

$$\begin{aligned} \mathbf{y}_2 &= \mathbf{H}_2 \mathbf{G} \left(\sum_{k=1}^K \mathbf{H}_{1,k} \mathbf{x}_k + \mathbf{n}_1 \right) + \mathbf{n}_2 \\ &= \mathbf{H}_2 \mathbf{G} \mathbf{H}_{1,k} \mathbf{x}_k + \mathbf{H}_2 \mathbf{G} \left(\sum_{\substack{i=1 \\ i \neq k}}^K \mathbf{H}_{1,i} \mathbf{x}_i \right) + \mathbf{H}_2 \mathbf{G} \mathbf{n}_1 + \mathbf{n}_2. \end{aligned} \quad (202)$$

Consequently, the mutual information of the k -th user to relay link $I(\mathbf{y}_2; \mathbf{x}_k)$ can be given by

$$I(\mathbf{y}_2; \mathbf{x}_k) = \frac{1}{2} \log_2 \left| \mathbf{I}_r + \frac{\mathbf{H}_2 \mathbf{G} (\mathbf{H}_{1,k} \mathbf{R}_{\mathbf{x},k} \mathbf{H}_{1,k}^\dagger) \mathbf{G}^\dagger \mathbf{H}_2^\dagger}{\mathbf{H}_2 \mathbf{G} \left(\sum_{\substack{i=1 \\ i \neq k}}^K \mathbf{H}_{1,i} \mathbf{R}_{\mathbf{x},i} \mathbf{H}_{1,i}^\dagger \right) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{H}_2 \mathbf{G} \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r} \right|, \quad (203)$$

where $\mathbf{R}_{\mathbf{x},k} = E\{\mathbf{x}_k \mathbf{x}_k^\dagger\}$. Assuming SIC, starting from user K , i.e., the interference from user i to k is negligible for $i > k$, we can then re-formulate (203) as follows

$$I(\mathbf{y}_2; \mathbf{x}_k) = \frac{1}{2} \log_2 \left| \frac{\mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,k} \bar{\mathbf{R}}_{\mathbf{x},k} \bar{\mathbf{H}}_{1,k}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r}{\mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,k-1} \bar{\mathbf{R}}_{\mathbf{x},k-1} \bar{\mathbf{H}}_{1,k-1}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r} \right|, \quad (204)$$

where $\bar{\mathbf{H}}_{1,i} = [\mathbf{H}_{1,1} \ \mathbf{H}_{1,2} \ \dots \ \mathbf{H}_{1,i}]$ and $\bar{\mathbf{R}}_{\mathbf{x},i} = [\mathbf{R}_{\mathbf{x},1} \ \mathbf{R}_{\mathbf{x},2} \ \dots \ \mathbf{R}_{\mathbf{x},i}]$.

4.2.3.1 Weighted sum-rate

According to (204), the expression of the WSR for the relay link can straightforwardly be given as

$$\begin{aligned} R_{\text{wsum},y2} &= \sum_{k=1}^K w_k I(\mathbf{y}_2; \mathbf{x}_k) = \frac{1}{2} \sum_{k=1}^K w_k \log_2 \left| \frac{\mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,k} \bar{\mathbf{R}}_{\mathbf{x},k} \bar{\mathbf{H}}_{1,k}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r}{\mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,k-1} \bar{\mathbf{R}}_{\mathbf{x},k-1} \bar{\mathbf{H}}_{1,k-1}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r} \right|, \\ &= \frac{1}{2} \sum_{k=1}^K \Delta_k \log_2 \left| \mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,k} \bar{\mathbf{R}}_{\mathbf{x},k} \bar{\mathbf{H}}_{1,k}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r \right| - w_1 \log_2 \left| \mathbf{H}_2 \mathbf{G} \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r \right| \end{aligned} \quad (205)$$

where $\Delta_k = w_k - w_{k+1}$, $w_{K+1} = 0$. In order to maximise (205) the users must be sorted according to their priorities such that $w_1 \geq w_2 \geq \dots \geq w_K$, and consequently $\Delta_k \geq 0, \forall k$. The weights are computed according to the channel norm of each user, and the user with the best channel norm, i.e., user K , is decoded first.

For the special case, where all weights are equal to one, the sum-rate of the relay link can be expressed as follows

$$R_{\text{sum},y2} = \sum_{k=1}^K I(\mathbf{y}_2; \mathbf{x}_k) = \frac{1}{2} \log_2 \left| \frac{\mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{\mathbf{x},K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r}{\mathbf{H}_2 \mathbf{G} \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r} \right|. \quad (206)$$

Similarly, it can be easily shown that the sum-rate for the direct link is given by

$$R_{\text{sum},y0} = \sum_{k=1}^K I(\mathbf{y}_0; \mathbf{x}_k) = \frac{1}{2} \log_2 |\mathbf{I}_r + \bar{\mathbf{H}}_{0,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{0,K}^\dagger|, \quad (207)$$

where $\bar{\mathbf{H}}_{\#,i} = [\mathbf{H}_{\#,1} \ \mathbf{H}_{\#,2} \ \dots \ \mathbf{H}_{\#,i}]$. The formulations of the mutual information in the single user case in (172) and of the sum mutual information, or sum-rate, in the MU case in (206) and (207) are similar; consequently, the algorithm developed in section 4.2.2 for maximising the mutual information of the relay link in the single user case can be directly applied here for maximising the sum-rate of several users by considering $\bar{\mathbf{H}}_{0,K}$, $\bar{\mathbf{H}}_{1,K}$ and $\bar{\mathbf{R}}_{x,K}$ instead of $\mathbf{H}_0$, $\mathbf{H}_1$ and $\mathbf{R}_x$, respectively, and $\mathbf{G} = \mathbf{V}_2 \tilde{\mathbf{G}} \mathbf{U}^\dagger$ or $\mathbf{G} = \tilde{\mathbf{G}} \mathbf{U}^\dagger$ in the case that $\mathbf{H}_2$ is known or unknown at the RS, with $\mathbf{U} \in \mathbb{C}^{q \times q}$ is the eigenvector of $(\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q)$, i.e., $(\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) = \mathbf{U} \Lambda \mathbf{U}^\dagger$.

In the WSR case, the problem of maximising the sum-rate under power constraints can be expressed as

$$\begin{aligned} \max_{\mathbf{G}} R_{\text{wsum},y2} &= \frac{1}{2} \sum_{k=1}^K \Delta_k \log_2 |\mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r| - w_1 \log_2 |\mathbf{H}_2 \mathbf{G} \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r| \\ \text{s.t. } \mathbf{G} &\geq 0; \text{tr} \{ \mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \} \leq P_2 \end{aligned} \quad (208)$$

In the case that $\mathbf{H}_2$ is known at the relay, this problem is concave, and can therefore be solved using usual convex optimisation method such as the interior point method or by the gradient method [Boyd-04]. A gradient search can be performed to maximise the cost function with a barrier of (208)

$$L(\mathbf{G}) = R_{\text{wsum},y2} + \frac{1}{t} \left[\log \left(P_2 - \text{tr} \{ \mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \} \right) \right]. \quad (209)$$

The search is conducted over two loops; the inner loop is for a fixed t where the gradient search is performed until the norm of $\nabla L(\mathbf{G})$, i.e., the gradient of $L(\mathbf{G})$, is small enough; the outer loop increases the value of t until it is large enough. The gradient of $L(\mathbf{G})$ is given by

$$\nabla L(\mathbf{G}) = \frac{\partial R_{\text{wsum},y2}}{\partial \mathbf{G}} - \frac{2\mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q)}{t \left[P_2 - \text{tr} \{ \mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \} \right]}, \quad (210)$$

where

$$\begin{aligned} \frac{\partial R_{\text{wsum},y2}}{\partial \mathbf{G}} &= \frac{1}{\ln(2)} \left[\sum_{k=1}^K \Delta_k \mathbf{H}_2^\dagger \left(\mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r \right)^{-1} \mathbf{H}_2 \mathbf{G} (\bar{\mathbf{H}}_{1,K} \bar{\mathbf{R}}_{x,K} \bar{\mathbf{H}}_{1,K}^\dagger + \mathbf{I}_q) \right. \\ &\quad \left. - w_1 \mathbf{H}_2^\dagger (\mathbf{H}_2 \mathbf{G} \mathbf{G}^\dagger \mathbf{H}_2^\dagger + \mathbf{I}_r)^{-1} \mathbf{H}_2 \mathbf{G} \right] \end{aligned}$$

since $\frac{\partial \ln |\mathbf{X} \mathbf{Y} \mathbf{X}^\dagger + \mathbf{I}|}{\partial \mathbf{X}} = 2(\mathbf{X} \mathbf{Y} \mathbf{X}^\dagger + \mathbf{I})^{-1} \mathbf{X} \mathbf{Y}$ if $\mathbf{Y}$ is an Hermitian matrix. This method as low

complexity, but might not always be optimal in the sense that $\mathbf{G}$ will not always have the optimal $\mathbf{G} = \mathbf{V}_2 \tilde{\mathbf{G}} \mathbf{U}^\dagger$ form. In order to keep this formulation, we can utilise the same algorithm structure as in *algorithm 5*, but by evaluating $R_{\text{wsum},y2}$ instead of $f(\mathbf{v}_2)$. Moreover, at step 6: , after calculating

$\frac{\partial R_{\text{wsum},y2}}{\partial \mathbf{G}}$, which is a matrix, we transform this matrix into a vector such

that $\nabla \mathbf{x} = \left[\Re \left\{ \text{diag} \left(\mathbf{V}_2^\dagger \frac{\partial R_{\text{wsum},y2}}{\partial \mathbf{G}} \mathbf{U} \right) \right\} \right]^2 \text{diag}(\Lambda)$. Note that $\tilde{\mathbf{G}} = \text{diag}(\sqrt{\mathbf{x}/\text{diag}(\Lambda)})$.

4.2.3.2 Multi-user performance analysis

Figure 98 depicts the cooperative sum-rate performance of the EG, MFR, and MMSEF schemes against the performances of our two schemes, i.e., EEM and EXP MIMO, for different numbers of user and antenna. The results show that EXP MIMO outperforms other existing schemes, and that EEM provides acceptable performance.

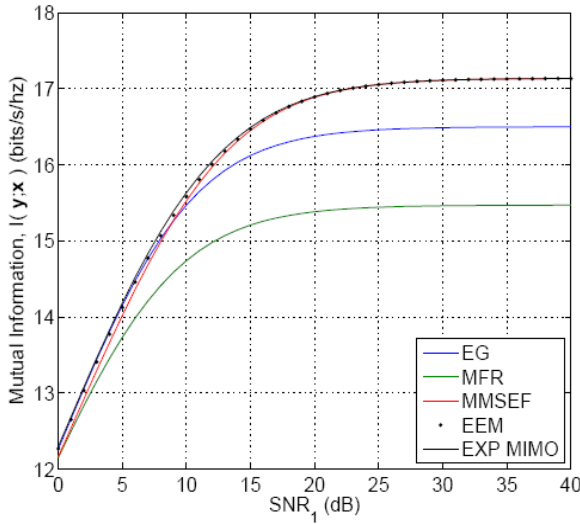


Figure 98 (a)

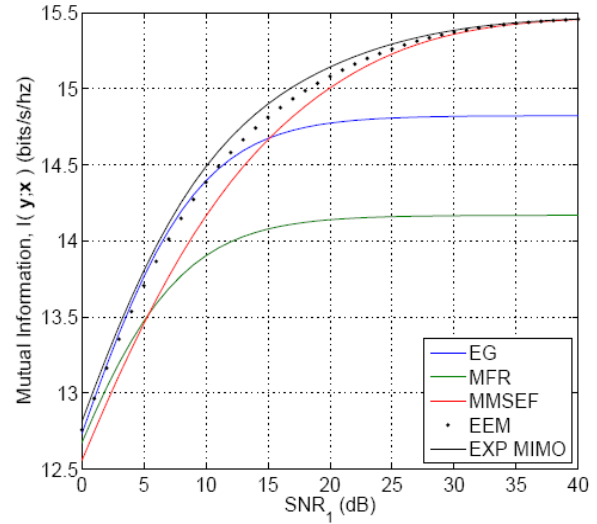


Figure 98 (b)

Figure 98: Cooperative sum-rate performance of various power allocation schemes for $SNR_0=0$ dB and $SNR_2=10$ dB, (a) 4-users MIMO $n=4, q=r=8$, (b) 8-users MIMO $n=1, q=r=8$

In Figure 99 and Figure 100, we compare the cooperative sum-rate performances of the Alg3, which is equivalent to the schemes in [Fan-07, Muñoz-07], EG, MFR, and MMSEF schemes against the performances of our schemes, i.e., Alg1, Alg2, EEM and EXP MIMO, for different numbers of antenna and subcarrier. Note that Alg1 and Alg2 perform in the same way and therefore are both represented by the same curve. We assume that each user has the same number of subcarriers.

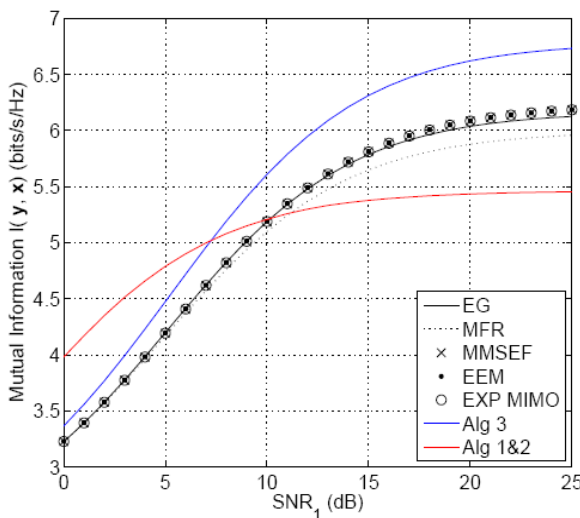


Figure 99 (a)

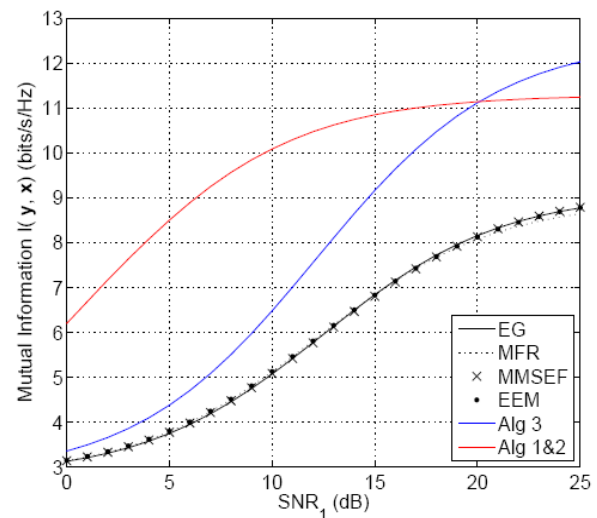


Figure 99 (b)

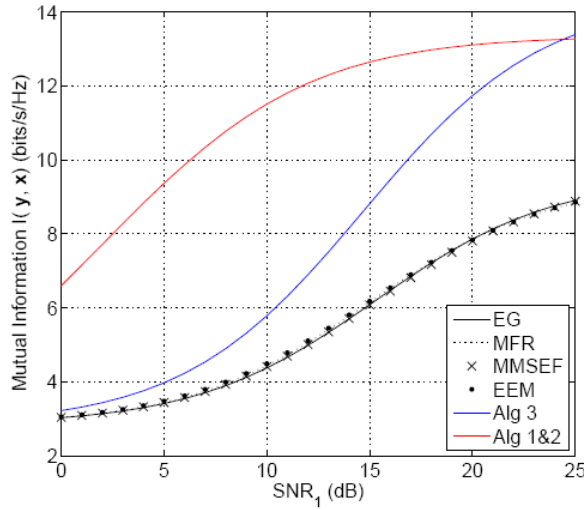


Figure 99 (c)

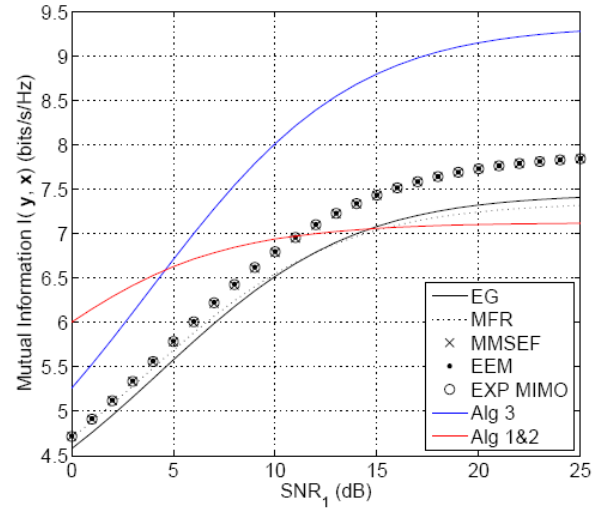


Figure 100 (a)

Figure 99: Cooperative sum-rate performance of various power allocation methods for the SISO case with 4 users and a) 8, b) 64 and c) 128 subcarriers, $SNR_0=0$ dB, and $SNR_2=10$ dB, car scenario, 50km/h.

The results point out that the conclusions that have been made in the single user case still hold in the MU case: Alg2 outperforms Alg3 as the number of subcarrier increases, and the range of SNRs for which this result occurs depends on the number of antenna in the systems, as it is indicated by comparing Figure 100 (b) with Figure 100 (c); EXP MIMO and EEM schemes perform not as well as in the single carrier-case.

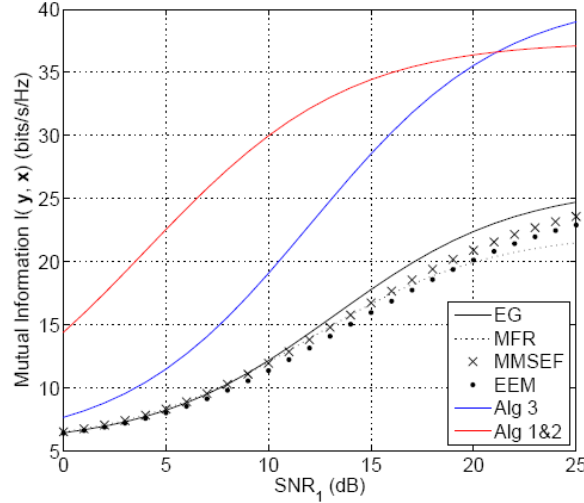


Figure 100 (b)

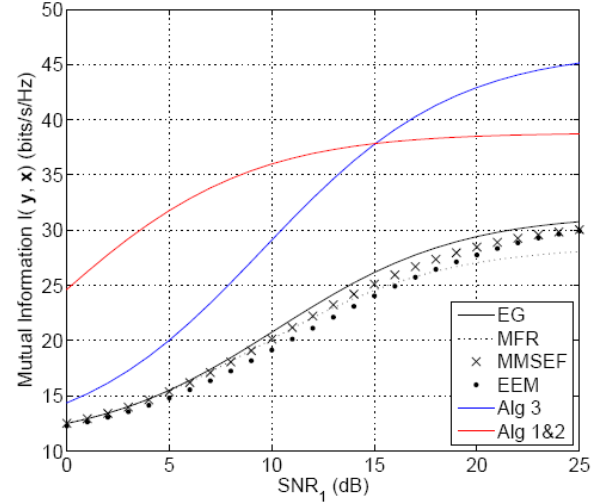


Figure 100 (c)

Figure 100: Cooperative mutual information performance of various power allocation methods for the MIMO case with 4 users and a) $n=q=r=2$, 4 subcarriers, b) $n=q=r=2$, 32 subcarriers and c) $n=q=r=4$, 32 subcarriers, $SNR_0=0$ dB, and $SNR_2=10$ dB, car scenario, 50km/h.

In Figure 101, we compare the cooperative mutual information performances of our weighted-sum rate algorithm with Alg3, i.e., sum-rate, in a MIMO single carrier and multi-carrier settings in Figure 101 (a) and Figure 101 (b), respectively. In the single carrier case, the result shows that our WSR algorithm can improve the fairness between the users by reducing the share of the strongest user and increasing the share of the other users without losing any rate in comparison with the sum-rate allocation. In the multi-carrier case, the selection of the criterion for the rate optimisation does not modify the user rate distribution.

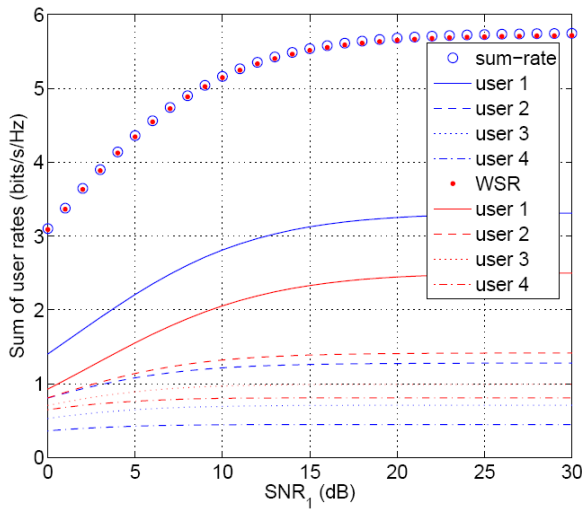


Figure 101 (a)

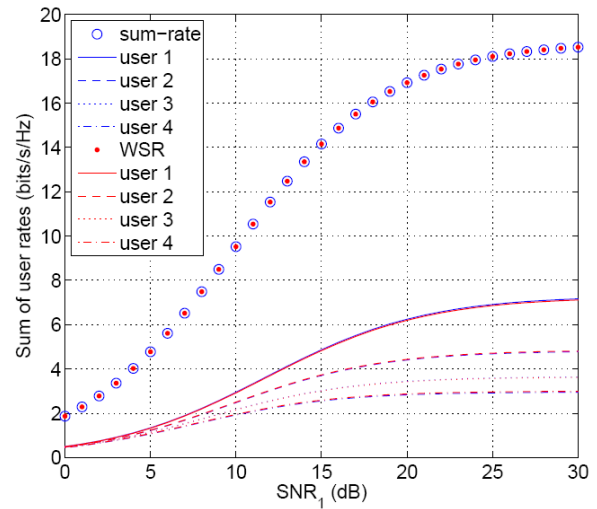


Figure 101 (b)

Figure 101: Cooperative mutual information performance of various power allocation methods for the MIMO case with 4 users and a) 4 users, $n=q=r=4$, 1 subcarriers, Rayleigh fading channel b) $n=q=r=2$, 32 subcarriers, car scenario, 50km/h, $\text{SNR}_0=0$ dB, and $\text{SNR}_2=10$ dB.

4.2.4 Conclusion

In this section, novel power allocation methods for nonregenerative cooperative MU MIMO communication have been proposed. These methods are based on the maximisation of the mutual information of the relay link under various levels of CSI that are available at the RS.

Summary of the single and MU cases:

- Single carrier case:
 - o Our EXP MISO and EXP MIMO schemes outperform any other existing AF schemes for MISO and MIMO settings, respectively, and any link SNR settings, when only RCSI is available at the RS to design the precoding matrix $\mathbf{G}$.
 - o Our EEM scheme provides at least similar performance as the best of the three original AF scheme, but with reduced complexity in comparison with the EXP MISO and EXP MIMO schemes.
 - o Power allocation schemes based on FCSI at the RS, i.e., Alg. 3, [Fan-07] and [Muñoz-07], outperforms our more complex power allocation schemes that requires CSI both at the MS and RS, i.e., Alg. 1&2.
 - o Alg. 2 performs the same as Alg. 1, but with a lower complexity.
 - o Our WSR based algorithm provides same sum-rate performance as Alg. 3 but with an improved fairness.
- Multi-carrier case:
 - o Contrarily to the single carrier case, EXP MIMO scheme does not always outperform any other existing AF schemes, because it has been designed by assuming that $\mathbf{H}_2$ is a single path Rayleigh fading channel.
 - o Contrarily to the single carrier case, Alg. 2 greatly outperforms Alg. 3, especially at low SNRs. Moreover, the performance gap between Alg. 2 and Alg. 3 increases as the number of subcarriers increase.

4.3 MULTI-USER ONE-WAY TWO-PATH RELAY CHANNEL WITH AF

4.3.1 Introduction

The cooperative relaying concept recently appeared in the literature and is not yet deployed in operational cellular networks. The goal of this TCF study is to evaluate the gain that can be expected from the use of advanced cooperative relaying techniques, such as the OWTP-CR protocol, with respect to the DT and other classical single-relay assisted protocols (OWOP-SR and OWOP-CR) in a single-cell MU context and to determine their applicability in WiMAX cellular networks. The expected performance improvement due to cooperation will also be measured by comparing the BER performance of the proposed OWTP-CR scheme with its non-cooperative version, which is denoted OWTP-Simple Relaying (OWTP-SR) in the same scenario.

The first part of this contribution presents the cellular network of interest and the wireless radio channel model. The second part describes the OFDMA radio access, which is designed for ROCKET, and introduces the various possible transmission schemes. The architecture of the OMNeT++ simulation that has been developed in the framework of ROCKET is detailed in section 3. Finally, the last section focuses on the performance evaluation of the OWTP-CR transmission scheme, in comparison with all the other available transmission schemes.

4.3.2 Network

4.3.2.1 Topology

Please refer to section 2 for a description of the network configuration and scenarios that are considered in the TCF ROCKET studies.

4.3.2.2 Nodes hardware characteristics

4.3.2.2.1 Transmitter

In the above-described network and in the DL case, two types of transmitters have to be considered: one for the BS and one for the MS. Both operate at the same carrier frequency $f=2.5$ GHz, use the same 10 MHz transmission bandwidth and the same transmission power of 40 dBm. The overall signal bandwidth of 10 MHz is divided into 1024 elementary OFDMA subcarriers of bandwidth $B_{dB} = 10.94\text{kHz}$ among which there are 184 nulls/DC, 720 data and 120 pilot subcarriers. Assuming a uniform power distribution among subcarriers and a total transmission power of 10 W for the BS and RS, the transmission power A_e^2 on each useful (pilot + data) subcarrier equals $10/840\text{W} = 12\text{mW}$. Both transmitters are equipped with a single antenna that is directional with $G_s = 17$ dBi gain for the BS and is omnidirectional with $G_r = 5$ dBi gain for the relays.

4.3.2.2.2 Receiver

In the DL case, two types of receivers have to be considered: relays and MSs. The thermal noise power at the receiver, N , is given by the following equation

$$N = (K \cdot T)_{dB} + B_{dB} + F_{dB}, \quad (211)$$

where K is the Boltzmann constant, T is the ambient temperature in Kelvin ($K \cdot T = -174 \text{ dBm/Hz}$), B_{dB} is the bandwidth, which is defined in the previous section (in dB) and F_{dB} is the receiver noise factor (in dB). This leads to a receiver thermal noise power level of:

- $-126,6 \text{ dBm}$ (i.e. $2.18 \cdot 10^{-16} \text{ W}$) for terminals ($F_{dB} = 7 \text{ dB}$).
- $-128,6 \text{ dBm}$ (i.e. $1.38 \cdot 10^{-16} \text{ W}$) for RSs ($F_{dB} = 5 \text{ dB}$).

Both receivers use a single omnidirectional antenna with $G_d = 1 \text{ dBi}$ gain for MSs and $G_r = 5 \text{ dBi}$ gain for relays.

4.3.2.3 Wireless radio channel

In this study, we consider the propagation model that is described below (for BS-RS, BS-MS and RS-MS links).

Slow fading:

The signal power is attenuated by a factor called path loss, the expression of which, in dB, is given by the Winner C2 (urban macro-cell) model

$$PL_{dB} = A \log_{10}(d) + B + C \log_{10}\left(\frac{f}{5.0}\right), \text{ with } \begin{cases} A = 26 & B = 39 & C = 20 & \text{for } 10m \leq d \leq 400m \\ A = 40 & B = -1 & C = 6 & \text{for } 400m \leq d \leq 5000m \end{cases}, \quad (212)$$

where c is the speed of light, f the carrier frequency, in GHz and d the distance between the two nodes, in meters (m). The path loss attenuation coefficient that is used in the channel model is defined as

$$PL = 10^{PL_{dB}/20}. \quad (213)$$

Fast fading:

The fast fading term is modelled by a non-correlated and quasi-static Rayleigh process with unit variance, in the form of a complex multiplicative Gaussian process $h = \rho e^{j\theta} = h_I + jh_Q$ in Figure 102.

As shown in Figure 102, each data symbol that is transmitted by the source is thus modified by a channel transfer coefficient (slow and fast fading) and an additive complex Gaussian noise corresponding to the receiver thermal noise; its variance is $\sigma^2 = N/2$, where N is the receiver noise power (see section 4.3.2.2.2).

- *Channel coherence bandwidth:*

Channel transfer coefficients are considered as invariable over a bandwidth corresponding to the channel coherence bandwidth, which is often defined as the inverse of the maximum channel delay spread. Considering a typical maximum delay spread of $2 \mu\text{s}$ in urban macrocell environment, leading to a coherence bandwidth of about 500 kHz, we find that the channel transfer coefficient is the same over blocks of 45 consecutive subcarriers. For the total data transmission bandwidth, only $16 = 720/45$ different values of the channel transfer coefficients have to be considered at each time period (burst/frame).

- *Channel coherence time:*

The channel coherence time is a measure of the duration over which the CIR does not change noticeably. It is related to the carrier frequency and the relative velocity between transmitter and receiver. For a carrier frequency of 2.5 GHz and urban vehicular mobility of 14 m/s, the coherence time is about 8 ms, which is greater than the WiMAX frame duration.

For most of the transmission protocols that are studied in this work, the only requirement is that the channel transfer coefficient shall be constant over a burst time duration. For the OWTP protocol (see section 4.3.3.2.2), we have to assume that it is constant over a whole DL frame duration (3.33 ms), which, given the channel coherence time, is not a stringent restriction.

Gaussian random variables are generated by means of the Box Muller method, which states that, assuming s and t are independent uniform random variables on $[0; 1]$

$$z = \mu + \sigma \sqrt{-2\log(1-s)} \cdot \cos(2\pi t) \quad (214)$$

is a Gaussian random variable with mean μ and variance σ^2 .

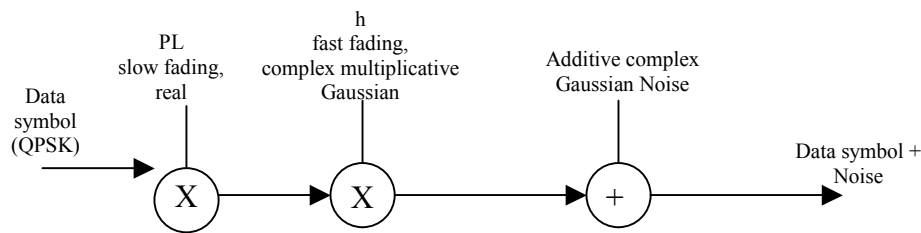


Figure 102: Channel modelling

4.3.3 Waveform design

The first part of this section presents the OFDMA radio resource allocation that is designed for ROCKET on the grounds of the WiMAX 802.16e standard, though not fully compliant with it. The second part describes studied transmission schemes and, in particular, the OWTP protocol that are utilised in ROCKET, whereas the last part provides information about the signalling to be added into the frame beacon, to support the studied transmission schemes.

4.3.3.1 OFDMA resource allocation

For the ROCKET scenarios, we chose: $M=6$ and $N=3$ (see Figure 7, right), leading to a maximum of 5, simultaneous users and 5 bursts of 216 bytes per frame (assuming QPSK modulation). 2 OFDMA symbols are set aside for frame signalling, which roughly corresponds to the amount of resources that is reserved for signalling in the WiMAX frame (preamble, DL_MAP, UL_MAP, Frame Control Header (FCH): see Figure 7). However, it shall be mentioned that our OFDMA simulator uses its own signalling format, which is not compliant with the WiMAX standard. It consists of:

- A beacon message that is used by the BS to inform nodes of time/frequency resource allocations and transmission scheme (see section 4.3.3.2) to be used in each resource. The format of the beacon message is described in section 4.3.3.3.2.
- Some random access messages that are used by terminals to send ‘hello’ messages and are utilised to build neighbourhood tables at MAC-level. In this study, terminals are supposed to be within the range of the BS. Random access messages are also used by terminals to send allocation request messages for data transmission.

In the following, we will call “time resource” the set of $NBMAX_f=5$ bursts synchronous in time and “frequency resource”, the set of $NBMAX_f=5$ bursts using the same frequency sub-channels.

The OFDMA resource allocator was designed to support all the transmission schemes that are described in the next paragraph for the ROCKET MU context. Let us consider a communication requiring the allocation of a set of N bursts.

The allocator will try to find the required time/frequency bursts in one or several consecutive frames, under the following constraints:

1. The first data burst is allocated to the free burst in the time domain which satisfies constraints 2 and 3 below: the allocator starts from the first time resource and searches the frequency domain for the first free burst (see burst numbering in Figure 7). If it does not exist, it repeats this operation for the next time resource until a free burst is found in the frame. If none exists, the communication will be delayed until the next frame.
2. Nodes (and in particular, relays) cannot simultaneously transmit and receive (half duplex time division duplexing relays), even on bursts using orthogonal frequency resources. This means that, before allocating a free burst to a node for data transmission from node A to node B, the allocator will check that node A has not already been allocated with a burst for data reception on the same time resource and also that node B is not simultaneously transmitted on the same time resource.

3. Relays always forward the information by employing the same frequency resource that has been utilised for reception, meaning that the burst chosen for relaying data shall be the first free burst in the time domain after the time burst that is allocated to data transmission from source to relay and on the same frequency resource. If there is no free burst left in the current frame for data relaying, the transmission will be delayed until the next frame.

Note that with such requirements on resource allocation:

- Bursts that are dedicated to a single transmission are not necessarily consecutive, as it will be shown on example 2 below (Figure 104).
- Users' bandwidth allocation requests are formulated by means of dedicated contention slots. A different resource allocation request is required for each user data burst to be sent. User requests with equal priority are processed in the FIFO ("first in first out") order, meaning that the first received request in a priority queue is satisfied first (if possible). The allocator supports different priority levels and always processes all the requests that are associated to a given transmission before processing the next transmission.

The data allocation algorithm implementing the previous constraints is presented in Figure 103. In this figure, $PMAX$ represents the total number of bursts to allocate for a transmission, $NBMAX_t$ is the total number of bursts in the time dimension of a frame (5) and $NBMAX_f$ is the total number of bursts in the frequency dimension of a frame.

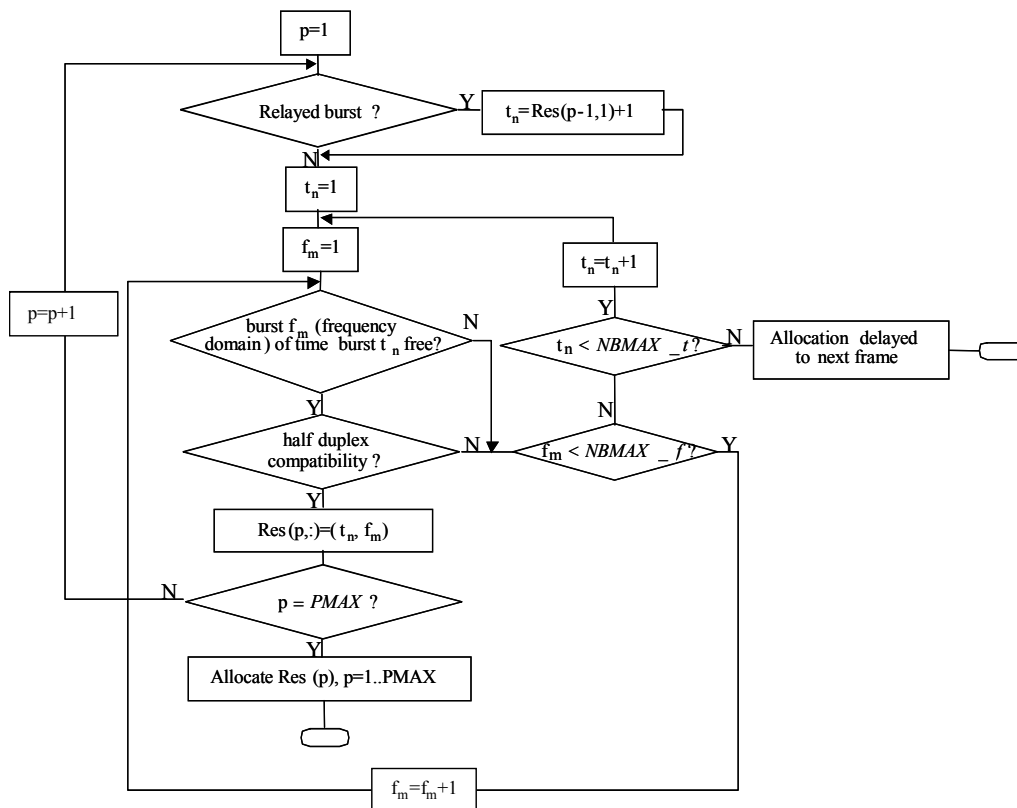


Figure 103: Burst allocation algorithm

Figure 104 illustrates the principles of this algorithm through different examples. In Example 1 of Figure 104, node 2 requests the transmission of seven data bursts to node 3. Bursts are allocated in priority during the first time resource and then during the second one, as it is shown in this figure.

Example 2 of Figure 104 shows the results of resource allocation for a relayed communication from node 1 to node 3 when using node 4 as relay. The first data burst (communication from node 1 to node

4) is allocated to the first free burst of the first time resource (*i.e.* burst 2 using burst numbering conventions of Figure 7). The second data burst (communication from node 4 to node 3) is allocated to the next burst, in the time domain, of the same frequency resource, which is free.

Example 3 of Figure 104 presents a situation in which the burst that is allocated to the relay transmission (from node 3 to node 4) is not contiguous (in the time domain) to the one used for data transmission from source to destination (node 1 to node 3). Indeed, node 3 cannot be allocated the second burst of the second time resource, as it already acts as a receiver during time slot 1 of this same time resource. Due to half duplex constraints, it cannot be simultaneously transmitter and receiver on the same time resource. This explains why the relay retransmission is allocated to the second burst of the third time resource.

In Example 4, the resource allocator receives two requests for relayed data transmission from node 1 to node 3 by the intermediate of node 2, which is used as relay, and a request for direct data transmission from node 2 to node 4. Requests will be processed in their arrival order. Communications from node 1 to node 2 will thus be allocated to two consecutive bursts (in the frequency domain) of the first time resource, whereas forwarding transmission from node 2 to node 3 will use the same bursts on the next time resource. DT from node 2 to node 4 will then be allocated to the first free burst of the first time resource for which node 2 is not a receiver (*i.e.*: burst 3 of time resource 2).

Example 1: Allocate 7 bursts for direct communication 2 → 3

Frame before allocation					Frame after allocation				
free	free	free	free	free	2->3	free	free	free	free
free	free	free	free	free	2->3	free	free	free	free
free	free	free	free	free	2->3	free	free	free	free
free	free	free	free	free	2->3	2->3	free	free	free
free	free	free	free	free	2->3	2->3	free	free	free

Example 2: Allocate bursts for relayed communication 1 → 4 → 3

Frame before allocation					Frame after allocation				
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	1->4	4->3	free	free	free
1->2	2->3	free	free	free	1->2	2->3	free	free	free

Example 3: Allocate bursts for relayed communication 1 → 3 → 4

Frame before allocation					Frame after allocation				
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	1->3	free	3->4	free	free
1->2	2->3	free	free	free	1->2	2->3	free	free	free

Example 4: Allocate bursts for
 relayed communication 1 → 2 → 3
 relayed communication 1 → 2 → 3
 direct communication 2->4

Frame before allocation					Frame after allocation				
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	free	free	free	free	free
free	free	free	free	free	free	2->4	free	free	free
free	free	free	free	free	1->2	2->3	free	free	free
free	free	free	free	free	1->2	2->3	free	free	free

Figure 104: burst allocation: examples

4.3.3.2 Transmission schemes

In this work, we will focus on performance comparison between DT, OWOP-SR and OWOP-CR schemes and the ROCKET OWTP-CR scheme. Below is described the terminology used to characterise and differentiate these transmission schemes.

We call “transmission scheme”, the complete transmission setting that includes the global strategy for the source, relay and destination. It is characterised by three main components and an optional one:

- Modulation and Coding Scheme: defines which modulation and which error correcting code is used to transmit information.
- Protocol: figures the way cooperating nodes (source, relays and destination) coordinate their transmissions. In other words, it refers to the MAC frame format.
- Receiver processing: describes how information received from the nodes (relay and/or source) is processed by the destination.
- Forwarding strategy: only defined if a relay is used in the protocol. It defines how data is processed at the relay node before being forwarded to the destination.

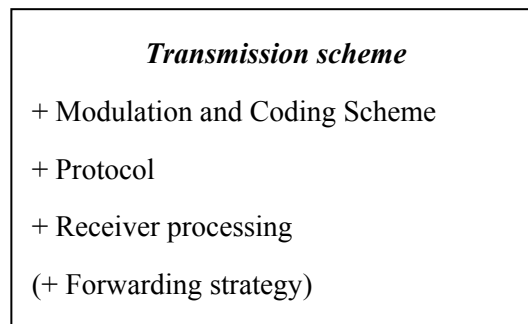
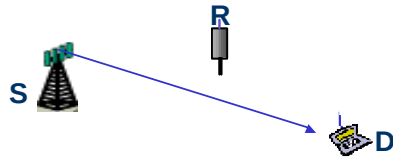


Figure 105: Transmission scheme representation

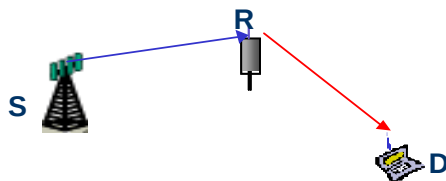
4.3.3.2.1 Modulation and coding schemes

In the following simulations, the only modulation scheme we consider in the simulations is QPSK. Before being mapped to symbols, bits can be encoded by a convolutional code with different possible rates that are obtained from rate-compatible codes, which were defined in [Nabar-04]. The different rates are deduced from mother code of rate $1/n$ ($n = 4$ in our case) by using the puncturing tables that were given in [Nabar-04].

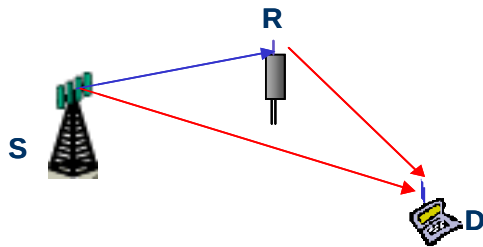
4.3.3.2.2 Protocols



DT protocol: data is sent directly from the source to the destination, without using any relay.

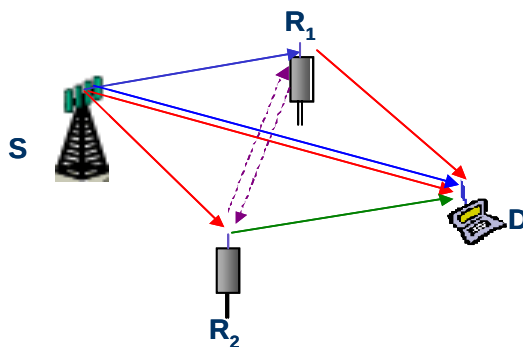


OWOP-SR protocol: data that is sent by the source is relayed by an intermediate (relay) node and is forwarded to the destination. Consequently, the source sends data on a first time slot and the relay amplifies and forwards (using AF forwarding strategy) previously-received data during a second one. The destination only listens for data received during the second time slot.



OWOP-CR: transmission occurs on a two-time-periods basis (*i.e.* four time slots if 2x1 Alamouti distributed space-time coding is used, as it will be shown later). In this study, we focus on non-orthogonal AF cooperative relaying protocols, meaning that the source and relay are allowed to transmit simultaneously towards the destination. The source thus transmits data during both time periods while the relay amplifies and forwards it during the second one. Cooperation can be achieved by letting the destination combine the signal from the two time periods or use only the second one, as it is assumed in our simulations.

OWTP-CR: in this specific ROCKET scenario, transmission is assisted by a pair of alternately-transmitting relays, cooperating as it is described hereafter:



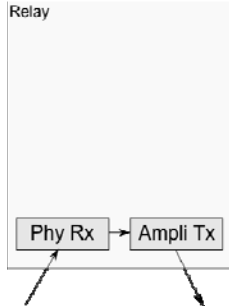
- During a first time-slot, relay 1 (R_1) amplifies and forwards data that are received during the previous time slot and the base station broadcasts new information to both the MS and relay 2 (R_2).
- During the second time slot, relay 2 amplifies and forwards data that are received from the base station during the first time-slot, while the base station broadcasts new information which, will be received by both relay 1 and the MS and forwarded in the next slot.

Using such a process, two different data flows are received by the MS in each time slot: one originating from the BS and the other one from either R_1 or R_2 . Multi-user reception techniques will be used at the receiver for recovering each of these two data flows as

NB: it shall be noted that the original message is forwarded on the same sub-frequency resource as the received one from the BS for relayed schemes.

4.3.3.2.3 Forwarding strategy

As stated in section 4.3.3.2, the forwarding strategy defines how data is processed at the relay node before being forwarded to the destination. In the next simulations, we will focus on the AF strategy, which is sketched in the next figure.



The relay node amplifies the received signal by the relay amplification coefficient (α), which takes into account the receiver noise and the pathloss.

The calculus that is used to determine the amplification factor α at the relay will be detailed in the next sections, for each kind of relaying protocol.

4.3.3.2.4 Receiver processing

As stated in section 4.3.3.2, the receiver processing describes how received information is processed by the destination to recover the transmitted symbols. In this section, we describe the different receiver processing techniques that are implemented for each one of the above-mentioned protocols.

Notation

Let us consider a set of 4 nodes corresponding to a source S (subscript s), a destination D (subscript d) and a maximum of 2 relays (subscripts r_1 and r_2 hereafter) that are potentially assisting the transmission from a source to a destination node.

Let us also use the following notations for the rest of this paragraph :

- N is the total number of data symbols in a time burst (or a time period in the OWOP-CR case).
- $\mathbf{x}_p^r(k) = [x_{p1}^r(k) x_{p2}^r(k) \dots x_{pN}^r(k)]$: received signal by node p ($p = d, r_1$ or r_2) during time burst number k (or time period number k , in the OWOP-CR case).
- $\mathbf{x}_p^t(k) = [x_{p1}^t(k) x_{p2}^t(k) \dots x_{pN}^t(k)]$: transmit signal from node p ($p = s, r_1$ or r_2) during time burst number k (or time period number k , in the OWOP-CR case).
- $\alpha_p(k)$: forwarding amplification coefficient that is used by relay p ($p = r_1$ or r_2) during time burst number k (or time period number k , in the OWOP-CR case).
- $g_{pq}(k) = PL_{pq} \cdot h_{pq}(k)$: channel gain (including slow and fast fading) between nodes p and q ($p, q = s, d, r_1$ or r_2) during time burst number k . It is assumed that the channel gain remains constant during at least one time burst.
- $\mathbf{S}(k) = [s_1(k) s_2(k) \dots s_N(k)]$: k^{th} packet of transmit symbols from the source.
- $\mathbf{n}_p(k)$: Gaussian noise vector at node p ($p = d, r_1$ or r_2) for time burst number k (or time period number k , in the OWOP-CR case).

Direct transmission

In the DT protocol, a new packet of data is received during each time burst (see Table 12).

Table 12: DT protocol

		Time burst 1	Time burst 2	Time burst 3	Time burst 4
Source	Tx	$\mathcal{S}(1)$	$\mathcal{S}(2)$	$\mathcal{S}(3)$	$\mathcal{S}(4)$
Relay	Rx	Not listening	Not listening	Not listening	Not listening
	Tx	Not active	Not active	Not active	Not active
Dest.	Rx	$\mathcal{S}(1)$	$\mathcal{S}(2)$	$\mathcal{S}(3)$	$\mathcal{S}(4)$

Here, we implement a simple receiver processing by using an adaptive filter that is based on the source to destination channel estimate. Consequently, the received signal at the destination during time burst k , $k \geq 1$ is

$$\mathbf{x}_d^r(k) = A_e \cdot G_s \cdot G_d \cdot \mathbf{g}_{sd}(k) \cdot \mathcal{S}(k) + \mathbf{n}_d(k). \quad (215)$$

Symbols forming a burst are independent, and can thus be estimated individually. The estimated symbol $\bar{s}_p(k)$ ($p = 1..N$) is the one, among the QPSK constellation symbols, which satisfies the equation

$$\bar{s}_p(k) = \arg \min_s \left| \mathbf{x}_{dp}^r(k) - A_e \cdot G_s \cdot G_d \cdot \mathbf{g}_{sd}(k) \cdot s \right|^2, \quad (216)$$

where $\mathbf{x}_{dp}^r(k)$ is the p^{th} received data symbol at the destination during time burst k .

One-way one-path simple relaying

In the OWOP-SR, a new data packet is received every two time burst (see Table 13).

Table 13: OWOP-SR protocol

		Time burst 1	Time burst 2	Time burst 3	Time burst 4
Source	Tx	$\mathcal{S}(1)$	Not active	$\mathcal{S}(2)$	Not active
Relay	Rx	$\mathcal{S}(1)$	Not listening	$\mathcal{S}(2)$	Not listening
	Tx	Not active	$\mathcal{S}(1)$	Not active	$\mathcal{S}(2)$
Dest.	Rx	Not listening	$\mathcal{S}(1)$	Not listening	$\mathcal{S}(2)$

As in the DT protocol, a simple receiver processing is employed by using matched filter based channels estimation. Let $\mathbf{x}_{r_i}^r$ be the received signal at the relay (R_1)

$$\mathbf{x}_{r_i}^r(2k-1) = A_e \cdot G_s \cdot G_r \cdot \mathbf{g}_{sr_i}(2k-1) \cdot \mathcal{S}(k) + \mathbf{n}_{r_i}(2k-1), k \geq 1. \quad (217)$$

During even time bursts, the relay retransmits data that are received during the previous time burst

$$\mathbf{x}_{r_i}^t(2k) = \alpha_{r_i}(2k) \cdot G_r \cdot (A_e \cdot G_s \cdot G_r \cdot \mathbf{g}_{sr_i}(2k-1) \cdot \mathcal{S}(k) + \mathbf{n}_{r_i}(2k-1)), k \geq 1. \quad (218)$$

Consequently, the received signal at the destination is

$$\left. \begin{aligned} \mathbf{x}_d^r(2k-1) &= 0 \\ \mathbf{x}_d^r(2k) &= G_d \cdot \mathbf{g}_{rd}(2k) \cdot \mathbf{x}_{r_i}^t(2k) + \mathbf{n}_d(2k) \end{aligned} \right\}, k \geq 1. \quad (219)$$

Estimation of the transmitted symbols is performed by finding the QPSK symbol that minimises the equation

$$\bar{s}_p(k) = \arg \min_s \left| x_{dp}^r(2k) - \left(A_e \cdot G_s \cdot G_r^2 \cdot G_d \cdot g_{sr_i}(2k-1) \cdot \alpha_{r_i}(2k) \cdot g_{rd}(2k) \right) s \right|^2, p=1..N. \quad (220)$$

The amplification coefficient at the relay, $\alpha_{r_i}(2k)$, is computed such that the relay transmission power is kept (before amplification by the relay transmitter antenna gain) constant and equal to A_e^2 .

One-way one-path cooperative relaying

It is well known [Nabar-04][Lane-03][Yang-07] that the use of non-orthogonal cooperation protocols (which allow cooperating nodes to transmit on the same channel), combined with an appropriate distributed coding scheme, provides higher bandwidth efficiency than protocols where collaborative nodes use orthogonal channels (time-slots, frequencies...). Many Distributed Space-Time Codes (DSTC) schemes were proposed, as extensions of traditional MIMO Space-Time Codes (STCs) to the distributed-antenna case, among which the Alamouti DSTC, which is the one considered in this study. Space-Time Block Codes (STBCs) denote a category of STCs that have the particularity to encode the transmit data stream in blocks of symbols. These blocks are distributed among Tx antennas and across time. They are usually represented by a coding matrix C , the rows and columns of which represent transmit time periods and antennas' transmissions over time, as follows

$$C = \begin{matrix} \text{C = time periods} \downarrow & \begin{bmatrix} S(1,1) & S(2,1) & \cdots & S(N_a,1) \\ S(1,2) & S(2,2) & \cdots & S(N_a,2) \\ \vdots & \vdots & \ddots & \vdots \\ S(1,N_t) & S(2,N_t) & \cdots & S(N_a,N_t) \end{bmatrix} \\ & \xrightarrow{\text{Antennas}} \end{matrix}$$

where N_t is the number of time periods over which encoding is performed, N_a is the number of antennas and $s_{i,j}$ denotes the modulated symbol that are transmitted on antenna j during symbol period i . Although several Tx antennas are necessary, this scheme even works with a single Rx antenna, allowing to benefit from receive diversity.

It was previously shown that Alamouti's code [Tarokh-98][Alam-98] is the only complex orthogonal STBC which achieves rate 1 (average number of transmitted symbols per time period), meaning that it can achieve full diversity gain without any sacrifice on data rate (full rate, full diversity). One of the major advantages of Alamouti's scheme is its receiver simplicity, as simple linear combination of the received symbols is enough to recover the original transmitted information.

In a traditional MIMO system based on the Alamouti STC, each row of the 2x2 coding matrix represents data to be sent by each antenna at each time period. Similarly, in a cooperative relaying system employing distributed Alamouti STCs [Lane-03][Anghel-03], each row of the coding matrix refers to the data to be sent by the source towards each node (relay / destination) during two successive time periods (a time-burst).

As presented in Table 14, Alamouti DSTC encoding is defined over a four time-period basis (two time bursts). This four time-period scheme is reproduced for each couple of data symbol packets of size N (half the size of a time burst) to be transmitted, $(S(2k-1), S(2k)), k \geq 1$.

Table 14: Alamouti DSTC cooperative relaying protocol

		Time Burst 1		Time Burst 2	
Time Period		1	2	3	4
Source	Tx	$\mathbf{S}(1)$	$-\mathbf{S}(2)^*$	$\mathbf{S}(2)$	$\mathbf{S}(1)^*$
Relay	Rx	$\mathbf{S}(1)$	$-\mathbf{S}(2)^*$	Not listening	
	Tx	Not active		$\mathbf{S}(1)$	$-\mathbf{S}(2)^*$
Dest.	Rx	Not listening		$\mathbf{S}(1) \mathbf{S}(2)$	$-\mathbf{S}(2)^* \mathbf{S}(1)^*$

When the AF strategy with a standard Alamouti receiver processing is used, the destination does not listen to the signal that is received during the first time period to decode the message. Consequently

$$\mathbf{x}_d^r(1) = \mathbf{x}_d^r(2) = 0. \quad (221)$$

The Alamouti processing assumes that channel gains are identical for the four time periods, so that we can drop the time period index in the channel gains, as well as in the relay amplification coefficients below.

Regarding the second time burst

$$\begin{aligned} \mathbf{x}_d^r(3) &= A_e \cdot G_s \cdot G_d \cdot g_{sd} \cdot \mathbf{S}(2) + \alpha_{r_1} \cdot G_r \cdot G_d \cdot g_{rd} \cdot (A_e \cdot G_s \cdot G_{r_1} \cdot g_{sr_1} \cdot \mathbf{S}(1) + \mathbf{n}_{r_1}(1)) + \mathbf{n}_d(3) \\ \mathbf{x}_d^r(4) &= A_e \cdot G_s \cdot G_d \cdot g_{sd} \cdot \mathbf{S}(1)^* + \alpha_{r_1} \cdot G_r \cdot G_d \cdot g_{rd} \cdot (A_e \cdot G_s \cdot G_{r_1} \cdot g_{sr_1} \cdot (-\mathbf{S}(2)^*) + \mathbf{n}_{r_1}(2)) + \mathbf{n}_d(4) \end{aligned} \quad (222)$$

The amplification coefficient at the relay, α_{r_1} , is computed by using the same formula as in the simple relaying case.

Let us define

$$\begin{cases} h_1 = A_e \cdot G_s \cdot G_r^2 \cdot G_d \cdot \alpha_{r_1} \cdot g_{sr_1} \cdot g_{rd} \\ h_2 = A_e \cdot G_s \cdot G_d \cdot g_{sd} \\ w_1 = A_e \cdot G_r \cdot G_d \cdot \alpha_{r_1} \cdot g_{rd} \cdot \mathbf{n}_{r_1}(1) + \mathbf{n}_d(3) \\ w_2 = A_e \cdot G_r \cdot G_d \cdot \alpha_{r_1} \cdot g_{rd} \cdot \mathbf{n}_{r_1}(2) + \mathbf{n}_d(4) \end{cases} \quad (223)$$

Thus, the above system takes the form

$$\begin{aligned} \mathbf{x}_d^r(3) &= h_1 \cdot \mathbf{S}(1) + h_2 \cdot \mathbf{S}(2) + w_1 \\ \mathbf{x}_d^r(4) &= h_2 \cdot (\mathbf{S}(1))^* + h_1 \cdot (-\mathbf{S}(2)^*) + w_2 \end{aligned} \quad (224)$$

In order to solve this system, we need to conjugate $\mathbf{x}_d^r(4)$. Thus we can write

$$\begin{aligned} \mathbf{x}_d^r(3) &= h_1 \cdot \mathbf{S}(1) + h_2 \cdot \mathbf{S}(2) + w_1 \\ \mathbf{x}_d^r(4)^* &= h_2^* \cdot (\mathbf{S}(1)) - h_1^* \cdot (\mathbf{S}(2)^*) + w_2^* \end{aligned} \quad (225)$$

This linear system is easily solved and the resulting estimated received symbols are

$$\bar{\mathbf{S}}(1) = \frac{h_1^* \cdot \mathbf{x}_d^r(3) + h_2 \cdot \mathbf{x}_d^r(4)^*}{|h_1|^2 + |h_2|^2} \text{ and } \bar{\mathbf{S}}(2) = \frac{h_2^* \cdot \mathbf{x}_d^r(3) - h_1 \cdot \mathbf{x}_d^r(4)^*}{|h_1|^2 + |h_2|^2}, \quad (226)$$

which can be rewritten as

$$\bar{\mathbf{S}}(1) = \frac{(G_r^2 \cdot \alpha_{r_1} \cdot g_{sr_1} \cdot g_{rd})^* \cdot \mathbf{x}_d^r(3) + g_{sd} \cdot \mathbf{x}_d^r(4)^*}{G_r^2 \cdot \alpha_{r_1} \cdot |g_{sr_1} \cdot g_{rd}|^2 + |g_{sd}|^2} \text{ and } \bar{\mathbf{S}}(2) = \frac{g_{sd}^* \cdot \mathbf{x}_d^r(3) - G_r^2 \cdot \alpha_{r_1} \cdot g_{sr_1} \cdot g_{rd} \cdot \mathbf{x}_d^r(4)^*}{G_r^2 \cdot \alpha_{r_1} \cdot |g_{sr_1} \cdot g_{rd}|^2 + |g_{sd}|^2}. \quad (227)$$

The OFDMA resource allocator that has been developed for ROCKET and has been described in section 4.3.3.1 allocates consecutive bursts of transmit data from the source on different frequency resources in priority. The Alamouti-encoded signal is thus distributed on four bursts (two time bursts on two different frequency resources) with such an allocation scheme, as it is depicted in Table 15; $S(1)$ & $S(2)$ and $-S(2)^*$ & $S(1)^*$ are transmitted on frequency resource 1 and 2, respectively.

This does not result in a true Alamouti scheme, as the channel coefficients affecting Alamouti-encoded symbols $S(1)$, $S(2)$, $-S(2)^*$ and $S(1)^*$ are not identical. The performance of such a cooperative relaying protocol, hereafter named OWOP-CR, is not as good as the true OFDMA Alamouti scheme, and could be improved by bringing modifications to the resource allocator, so as to schedule the four transmitted symbols on the same frequency resource. However, such modifications would require a complete redesign of the resource allocator, which will not be done in ROCKET due to lack of time and is left for future work. Section 4.3.5 presents the performance of the OWOP-CR protocol, compared to other transmission protocols described in this section.

As previously, we assume that channel coefficients are supposed to be constant over the 2 time bursts of the transmission, so that we can drop the time burst index in the channel gains and amplification coefficients. Moreover, we use index i for channel coefficients that are related to frequency resource i ($i=1,2$) on each link, as well as for the relay amplification coefficient on each frequency resource.

Table 15: OWOP-CR protocol

			Time Burst 1	Time Burst 2
Frequency resource 1	Source	Tx	$S(1)$	$S(2)$
	Relay	Rx	$S(1)$	Not active
		Tx	Not active	$S(1)$
	Dest.	Rx	Not listening	$S(1) S(2)$
Frequency resource 2	Source	Tx	$-S(2)^*$	$S(1)^*$
	Relay	Rx	$-S(2)^*$	Not active
		Tx	Not active	$-S(2)^*$
	Dest.	Rx	Not listening	$-S(2)^* S(1)^*$

Here we denote $x_d^r(k)$, $k=1,2$ the signal received at the destination on frequency resource k . We obtain

$$x_d^r(1) = A_e \cdot G_s \cdot G_r^2 \cdot G_d \cdot \alpha_{r_1}(1) \cdot g_{sr_1}(1) \cdot g_{r_1d}(1) \cdot S(1) + A_e \cdot G_s \cdot G_d \cdot g_{sd}(1) \cdot S(2) + n_d(1), \quad (228)$$

and

$$x_d^r(2) = A_e \cdot G_s \cdot G_r^2 \cdot G_d \cdot \alpha_{r_1}(2) \cdot g_{sr_1}(2) \cdot g_{r_1d}(2) \cdot (-S(2)^*) + A_e \cdot G_s \cdot G_d \cdot g_{sd}(2) \cdot S(1)^* + n_d(2), \quad (229)$$

where $n_d(1)$ and $n_d(2)$ are the noise terms that are received at the destination on frequency resource 1 and 2, respectively. Taking the complex conjugate of the second equation, this system can be rewritten as

$$\begin{bmatrix} x_d^r(1) \\ x_d^r(2)^* \end{bmatrix} = M \cdot \begin{bmatrix} S(1) \\ S(2) \end{bmatrix} + \begin{bmatrix} n_d(1) \\ n_d(2)^* \end{bmatrix} \text{ with } M = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \text{ and } \begin{cases} h_{11} = A_e \cdot G_s \cdot G_r^2 \cdot G_d \cdot \alpha_{r_1}(1) \cdot g_{sr_1}(1) \cdot g_{r_1d}(1) \\ h_{12} = A_e \cdot G_s \cdot G_d \cdot g_{sd}(1) \\ h_{21} = A_e \cdot G_s \cdot G_d \cdot g_{sd}(2)^* \\ h_{22} = -A_e \cdot G_s \cdot G_r^2 \cdot G_d \cdot \alpha_{r_1}(2) \cdot g_{sr_1}(2)^* \cdot g_{r_1d}(2)^* \end{cases} \quad (230)$$

This linear system is easily solved and the resulting estimated received symbols are

$$\bar{S}(1) = \frac{(G_r^2 \cdot \alpha_{r_1}(2) \cdot g_{sr_1}(2) \cdot g_{r_1d}(2))^* \cdot x_d^r(1) + g_{sd}(1) \cdot x_d^r(2)^*}{C} \text{ and } \bar{S}(2) = \frac{g_{sd}(2)^* \cdot x_d^r(1) - G_r^2 \cdot \alpha_{r_1}(1) \cdot g_{sr_1}(1) \cdot g_{r_1d}(1) \cdot x_d^r(2)^*}{C}, \quad (231)$$

$$\text{with } C = G_s \cdot G_d \cdot (G_r^2 \cdot g_{sr_1}(1) \cdot g_{r_1d}(1) \cdot g_{sr_1}(2)^* \cdot g_{r_1d}(2)^* \cdot \alpha_{r_1}(1) \cdot \alpha_{r_1}(2) + g_{sd}(1) \cdot g_{sd}(2)^*).$$

One-way two-path cooperative relaying

Transmission is based on a period of five time bursts, as it is depicted in Figure 106 and Table 16. Terms between brackets in Table 16 correspond to data symbols that are transported by unwanted inter-relay interference signals.

Table 16: OWTP-CR protocol

		Time burst 1	Time burst 2	Time burst 3	Time burst 4	Time burst 5
Source	Tx	$S(1)$	$S(2)$	$S(3)$	$S(4)$	Not active
Relay 1	Rx	$S(1)$	Not active	$S(3) (S(2)(S(1)))$	Not active	Not active
	Tx	Not active	$S(1)$	Not active	$S(3) (S(2))$	Not active
Relay 2	Rx	Not active	$S(2) (S(1))$	Not active	$S(4) (S(3)(S(2)))$	Not active
	Tx	Not active	Not active	$S(2) (S(1))$	Not active	$S(4) (S(3)(S(2)))$
Dest.	Rx	$S(1)$	$S(2) S(1)$	$S(3) S(2) (S(1))$	$S(4) S(3) (S(2))$	$S(4) (S(3)(S(2)))$

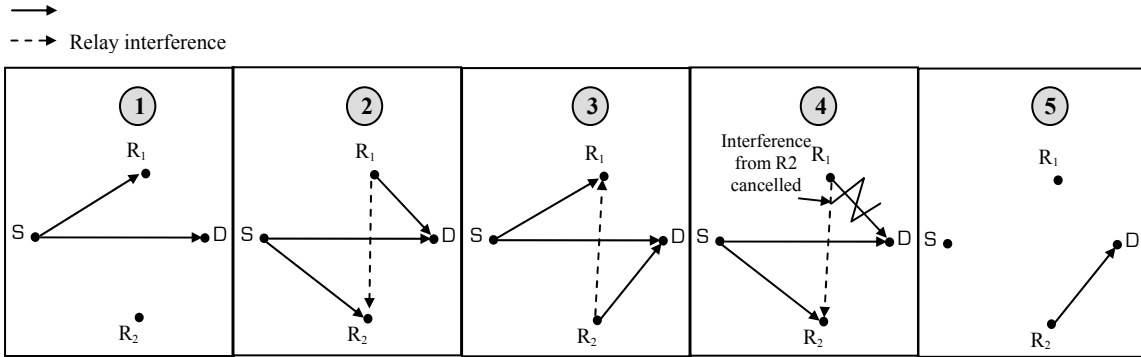


Figure 106: OWTP-CR cooperative relaying protocol

Step 1: (user time burst 1)

During the first time burst, the source broadcasts a packet of symbols $S(1)$, which is received by both relay 1 and destination

$$\begin{aligned} x_{r_1}^r(1) &= A_e \cdot G_s \cdot G_r \cdot g_{sr_1}(1) \cdot S(1) + n_{r_1}(1) \\ x_d^r(1) &= A_e \cdot G_s \cdot G_d \cdot g_{sd}(1) \cdot S(1) + n_d(1) \end{aligned} \quad (232)$$

Step 2: (user time burst 2)

During the second time burst, the source broadcasts $S(2)$. At the same time, relay R_1 forwards data that have been received during the previous burst to the destination. The signal from S is received by both relay R_2 and the destination. This signal from R_1 is received by the destination but also interferes with the received signal by R_2 during the same time period. The signals that are respectively forwarded

by relay R_1 , are received by relay R_2 and are received by the destination are given by the following equations

$$\begin{aligned}
 \mathbf{x}_{r_1}^t(2) &= \alpha_{r_1}(2) \cdot G_r \cdot \mathbf{x}_{r_1}^r(1) \\
 \mathbf{x}_{r_2}^r(2) &= G_r \cdot g_{r_1 r_2}(2) \cdot \mathbf{x}_{r_1}^t(2) + A_e \cdot G_s \cdot G_r \cdot g_{sr_2}(2) \cdot \mathbf{S}(2) + \mathbf{n}_{r_2}(2) \\
 \mathbf{x}_d^r(2) &= G_d \cdot g_{r_1 d}(2) \cdot \mathbf{x}_{r_1}^t(2) + A_e \cdot G_s \cdot G_d \cdot g_{sd}(2) \cdot \mathbf{S}(2) + \mathbf{n}_d(2)
 \end{aligned} \tag{233}$$

Step 3: (user time burst 3)

During the third burst, the source broadcasts $\mathbf{S}(3)$. At the same time, relay R_2 forwards data that have been received during the previous burst to the destination. The signal from S is received by both relay R_1 and the destination. The signal from R_2 is received by the destination but also interferes with the signal received by R_1 during the same time period. The signals that are respectively forwarded by relay R_2 , are received by relay R_1 and are received by the destination are given by the following equations

$$\begin{aligned}
 \mathbf{x}_{r_2}^t(3) &= G_r \cdot \alpha_{r_2}(3) \cdot \mathbf{x}_{r_2}^r(2) \\
 \mathbf{x}_{r_1}^r(3) &= G_r \cdot g_{r_1 r_2}(3) \cdot \mathbf{x}_{r_2}^t(3) + A_e \cdot G_s \cdot G_r \cdot g_{sr_1}(3) \cdot \mathbf{S}(3) + \mathbf{n}_{r_1}(3) \\
 \mathbf{x}_d^r(3) &= G_d \cdot g_{r_2 d}(3) \cdot \mathbf{x}_{r_2}^t(3) + A_e \cdot G_s \cdot G_d \cdot g_{sd}(3) \cdot \mathbf{S}(3) + \mathbf{n}_d(3)
 \end{aligned} \tag{234}$$

Step 4: (user time burst 4)

We notice that $\mathbf{x}_{r_1}^r(3)$ contains the inter-relay interference term $G_r \cdot \alpha_{r_2}(3) \cdot g_{r_1 r_2}(3) \cdot g_{r_1 r_2}(2) \cdot \mathbf{x}_{r_1}^t(2)$, which is assumed to be perfectly known at R_1 and can thus be cancelled. Hence, at time burst 4, the transmit signal from R_1 and the received signal by R_2 and the destination will be (the source transmits $\mathbf{S}(4)$)

$$\begin{aligned}
 \mathbf{x}_{r_1}^t(4) &= A_e \cdot G_s \cdot G_r^4 \cdot \alpha_{r_1}(4) \cdot \alpha_{r_2}(3) \cdot g_{r_1 r_2}(3) \cdot g_{sr_2}(2) \cdot \mathbf{S}(2) + A_e \cdot G_s \cdot G_r^2 \cdot \alpha_{r_1}(4) \cdot g_{sr_1}(3) \cdot \mathbf{S}(3) \\
 &\quad + G_r^3 \cdot \alpha_{r_1}(4) \cdot \alpha_{r_2}(3) \cdot g_{r_1 r_2}(3) \cdot \mathbf{n}_{r_2}(2) + G_r \cdot \alpha_{r_1}(4) \cdot \mathbf{n}_{r_1}(3) \\
 \mathbf{x}_{r_2}^r(4) &= G_r \cdot g_{r_1 r_2}(4) \cdot \mathbf{x}_{r_1}^t(4) + A_e \cdot G_s \cdot G_r \cdot g_{sr_2}(4) \cdot \mathbf{S}(4) + \mathbf{n}_{r_2}(4) \\
 \mathbf{x}_d^r(4) &= G_d \cdot g_{r_1 d}(4) \cdot \mathbf{x}_{r_1}^t(4) + A_e \cdot G_s \cdot G_d \cdot g_{sd}(4) \cdot \mathbf{S}(4) + \mathbf{n}_d(4)
 \end{aligned} \tag{235}$$

Step 5: (user time burst 5)

At time burst 5, relay R_2 forwards data that have been received during the previous time burst

$$\begin{aligned}
 \mathbf{x}_{r_2}^t(5) &= G_r \cdot \alpha_{r_2}(5) \cdot \mathbf{x}_{r_2}^r(4) \\
 \mathbf{x}_d^r(5) &= G_d \cdot g_{r_2 d}(5) \cdot \mathbf{x}_{r_2}^t(5) + \mathbf{n}_d(5)
 \end{aligned} \tag{236}$$

The amplification coefficients at the relays, $\alpha_{r_1}(k)$ and $\alpha_{r_2}(k)$, are computed such that the transmit power of the relays are kept equal to the source transmit power A_e^2 , as it is the case for other relayed protocols.

From the preceding equations, the received signal at the destination during time slot k , $k = 1..5$ can be rewritten as

$$\mathbf{x}_d^r(k) = A(k)\mathbf{S}(k) + B(k)\mathbf{S}(k-1) + C(k)\mathbf{S}(k-2) + D(k)\mathbf{S}(k-3) + \mathbf{v}(k), \tag{237}$$

where A, B, C and D , which are expressed below, are constants over one time burst and $\mathbf{v}$ is a composite noise term (see Table 17).

$$\begin{aligned}
 A(k) &= \begin{cases} 0 & \text{for } k = 5 \\ A_e \cdot G_s \cdot G_d \cdot g_{sd}(k) & \text{for } k = 1..4 \end{cases} \\
 B(k) &= \begin{cases} 0 & \text{for } k = 1 \\ A_e \cdot G_s \cdot G_r^2 \cdot G_d \cdot g_{r_p d}(k) \cdot \alpha_{r_p}(k) \cdot g_{sr_p}(k-1) & \text{for } 1 < k \leq 5 \end{cases}
 \end{aligned} \quad (238)$$

where $p = 1 + (k \% 2)$ is the relay number. Moreover,

$$\begin{aligned}
 C(k) &= \begin{cases} 0 & \text{for } k \leq 2 \\ A_e \cdot G_s \cdot G_r^4 \cdot G_d \cdot g_{r_{p_1} d}(k) \cdot \alpha_{r_{p_1}}(k) \cdot \alpha_{r_{p_2}}(k-1) \cdot g_{r_{p_2}}(k-1) \cdot g_{sr_{p_2}}(k-2) & \text{for } 2 < k \leq 5 \end{cases} \\
 D(k) &= \begin{cases} A_e \cdot G_s \cdot G_r^6 \cdot G_d \cdot g_{r_2 d}(k) \cdot \alpha_{r_2}(k) \cdot \alpha_{r_1}(k-1) \cdot \alpha_{r_2}(k-2) \cdot g_{r_1 r_2}(k-1) \cdot g_{r_1 r_2}(k-2) \cdot g_{sr_2}(k-3) & \text{for } k = 5 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned} \quad (239)$$

where $\begin{cases} p_1 = 1 + (k \% 2) \\ p_2 = 1 + ((k-1) \% 2) \end{cases}$, and “%” denotes the modulo operator.

Table 17: Noise term $\mathbf{v}$

Burst	Noise term $\mathbf{v}$
1	$\mathbf{n}_d(1)$
2	$G_r G_d g_{r_1 d}(2) \cdot \alpha_{r_1}(2) \cdot \mathbf{n}_{r_1}(1) + \mathbf{n}_d(2)$
3	$G_r G_d \cdot g_{r_2 d}(3) \cdot \alpha_{r_2}(3) \cdot \alpha_{r_1}(2) \cdot \mathbf{n}_{r_1}(1) + G_r \cdot G_d \cdot g_{r_2 d}(3) \cdot \alpha_{r_2}(3) \cdot \mathbf{n}_{r_2}(2) + \mathbf{n}_d(3)$
4	$G_r^3 \cdot G_d \cdot g_{r_1 d}(4) \cdot \alpha_{r_1}(4) \cdot \alpha_{r_2}(3) \cdot g_{r_1 r_2}(3) \cdot \mathbf{n}_{r_2}(2) + G_r \cdot G_d \cdot g_{r_1 d}(4) \cdot \alpha_{r_1}(4) \cdot \mathbf{n}_{r_1}(3) + \mathbf{n}_d(4)$
5	$G_r^5 \cdot G_d \cdot g_{r_2 d}(5) \cdot \alpha_{r_2}(5) \cdot \alpha_{r_1}(4) \cdot \alpha_{r_2}(3) \cdot g_{r_1 r_2}(4) \cdot g_{r_1 r_2}(3) \cdot \mathbf{n}_{r_2}(2) + G_r^3 \cdot G_d \cdot g_{r_2 d}(5) \cdot \alpha_{r_2}(5) \cdot \alpha_{r_1}(4) \cdot g_{r_1 d}(4) \cdot \mathbf{n}_{r_1}(3) + G_r \cdot G_d \cdot g_{r_2 d}(5) \cdot \alpha_{r_2}(5) \cdot \mathbf{n}_{r_2}(4) + \mathbf{n}_d(5)$

Assuming that all channels are constant over the duration of a frame (which according to paragraph 4.3.2.3 is a reasonable assumption), the time index following the channel coefficients $g_{sd}, g_{sr_1}, g_{sr_2}, g_{r_1 d}, g_{r_2 d}, g_{r_1 r_2}$ will be dropped hereafter. The received signal at the destination during a frame can thus be rewritten as follows

$$\mathbf{X} = \mathbf{S}\mathbf{H} + \mathbf{N}, \quad (240)$$

where $\mathbf{S}$ is a matrix consisting of normalised transmitted QPSK symbols, which can be viewed as an equivalent STC matrix, $\mathbf{H}$ is the equivalent channel vector and $\mathbf{N}$ is the noise matrix

$$\begin{aligned}
 X &= \begin{bmatrix} x_d^r(1) \\ x_d^r(2) \\ x_d^r(3) \\ x_d^r(4) \\ x_d^r(5) \end{bmatrix}, S = \begin{bmatrix} S(1) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ S(2) & S(1) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ S(3) & 0 & S(2) & 0 & 0 & S(1) & 0 & 0 & 0 \\ S(4) & 0 & 0 & S(3) & 0 & 0 & S(2) & 0 & 0 \\ 0 & 0 & 0 & 0 & S(4) & 0 & 0 & S(3) & S(2) \end{bmatrix}, H = \begin{bmatrix} A(1) \\ B(2) \\ B(3) \\ B(4) \\ B(5) \\ C(3) \\ C(4) \\ C(5) \\ D(5) \end{bmatrix}, \\
 N &= \begin{bmatrix} n_d(1) \\ n_d(2) \\ n_d(3) \\ n_d(4) \\ n_d(1) \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ \alpha_1 & 0 & 0 & 0 \\ \beta & \gamma_1 & 0 & 0 \\ 0 & \delta_1 & \alpha_2 & 0 \\ 0 & \varepsilon & \delta_2 & \gamma_2 \end{bmatrix} \cdot \begin{bmatrix} n_{r1}(1) \\ n_{r2}(2) \\ n_{r1}(3) \\ n_{r2}(4) \end{bmatrix}, \text{ with } \begin{cases} \alpha_1 = G_r \cdot G_d \cdot g_{r1d} \cdot \alpha_{r1}(2) \\ \alpha_2 = G_r \cdot G_d \cdot g_{r1d} \cdot \alpha_{r1}(4) \\ \beta = G_r \cdot G_d \cdot g_{r2d} \cdot \alpha_{r2}(3) \cdot \alpha_{r1}(2) \\ \gamma_1 = G_r \cdot G_d \cdot g_{r2d} \cdot \alpha_{r2}(3) \\ \gamma_2 = G_r \cdot G_d \cdot g_{r2d} \cdot \alpha_{r2}(5) \\ \delta_1 = G_r^3 \cdot G_d \cdot g_{r1d} \cdot g_{r1r2} \cdot \alpha_{r1}(4) \cdot \alpha_{r2}(3) \\ \delta_2 = G_r^3 \cdot G_d \cdot g_{r2d} \cdot g_{r1r2} \cdot \alpha_{r2}(5) \cdot \alpha_{r1}(4) \\ \varepsilon = G_r^5 \cdot G_d \cdot g_{r2d} \cdot g_{r1r2}^2 \cdot \alpha_{r2}(5) \cdot \alpha_{r1}(4) \cdot \alpha_{r2}(3) \end{cases} \quad (241)
 \end{aligned}$$

The joint decoding process consists in finding the quartuple of estimated packets $(\bar{S}(1), \bar{S}(2), \bar{S}(3), \bar{S}(4))$ which minimises the mean-squared error

$$E = \sum_{k=1}^5 E_k = \sum_{k=1}^5 |X_k - (SH)_k|^2, \quad (242)$$

where $|Z|$ denotes the modulus of complex number Z and Y_k and $(XH)_k$ are the k^{th} element of vectors Y and (XH) , respectively. As symbols forming a burst are independent, they can be estimated individually. The decoding process thus aims at finding, for each $p = 1..N$, the quartuple of estimated symbols $(\bar{s}_p(1), \bar{s}_p(2), \bar{s}_p(3), \bar{s}_p(4))$, among the four QPSK constellation symbols, which minimises E_p

$$\begin{aligned}
 E_p &= \left[x_d^r(1) - A(1)_{s_p(1)} \right]^2 + \left[x_d^r(2) - (A(1)_{s_p(2)} + B(2)_{s_p(1)}) \right]^2 + \left[x_d^r(3) - (A(1)_{s_p(3)} + B(3)_{s_p(2)} + C(3)_{s_p(1)}) \right]^2 \\
 &+ \left[x_d^r(4) - (A(1)_{s_p(4)} + B(4)_{s_p(3)} + C(4)_{s_p(2)}) \right]^2 + \left[x_d^r(5) - (B(5)_{s_p(4)} + C(5)_{s_p(3)} + D(5)_{s_p(2)}) \right]^2 \\
 &, p = 1..N.
 \end{aligned} \quad (243)$$

4.3.3.3 Signalling information for cooperative relaying

4.3.3.3.1 Hypotheses

In this section, we focus on the frame beacon definition and its contents requirements for the communication protocols that has been described above (see 4.3.3.2.2). In other words, we assume that nodes that are willing to communicate have already sent a request allocation message. They do not wait for the beacon that will indicate how and on which resources they can communicate. We also assume that the BS has already identified which nodes will communicate in each slot and with which transmission scheme. We only consider point to point transmissions, therefore, IP-level routing aspects are out of the scope of this study. Moreover, all nodes that are involved in a transmission are

required to be in the same cell. This means that they are all synchronised with each other and with the BS. We finally assume that the propagation channel has a constant value during a whole OFDMA DL frame.

The amount of information that is required in the frame beacon for each time burst depends on the selected transmission scheme. As some fields would be useless for some transmission scheme whereas necessary for other ones, we defined specific beacon structures for each transmission scheme. In order for nodes to be able to identify which one of these structures have to be used, the first field of a beacon will always be the transmission scheme identifier denoted **TS_ID**, which defines the type of transmission that is used for a communication (DT, OWOP-SR, OWOP-CR or OWTP-CR).

The beacon shall provide information on communications for all Guaranteed Bursts (GBs) of the frame. We consequently need a GB Identifier **GB_ID**. For the rest of this document, we consider this field as implicit. The beacon is divided into n parts, with n equal to the number of GBs in the frame, the first section corresponding to the first GB and so on.

Note: In the different examples of this document, we do not represent the whole frame beacon but only the information for the GBs that are involved in each example.

4.3.3.3.2 Frame Beacon definition

- DT protocol

In this case, we only need to know the transmitter identifier **Tx_ID** and the receiver identifier **Rx1_ID**. Below is an example in which node 1 sends a message to node 3 in the 5th GB when using DT protocol.

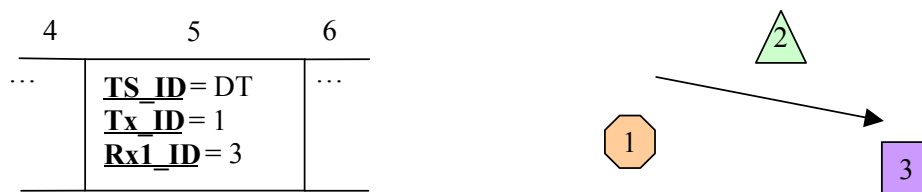


Figure 107: left: DT beacon, right: DT transmission

Node 1 knows that it can send his message in the 5th GB and node 3 knows that it has to listen during this same GB with this beacon.

- One-way one-path simple relaying transmission protocol

In this case, there are one transmitter and two destinations: The intermediate destination (relay) and the final one. Consequently, we define one emitter identifier **Tx_ID** and two receiver identifiers **Rx1_ID** (corresponding to the relay) and **Rx2_ID** (representing the final destination).

As it has been previously stated, this transmission scheme requires two different time bursts: a first time burst to transmit the information from the BS to the relay and a second one to relay the received information to the destination..

Note: the second GB can be in the same frame or in two different frames (it depends if the relay has time to process received data before the end of the frame) but the principle remains the same. The only difference between these two scenarios is that, in the second one, the BS has to keep in memory that the relay has to communicate with the final destination during the next frame. Up to the end of this section, we assume that the second time burst is sent in the same frame as the first one.

For instance, let us consider the situation where node 1 sends a message to node 3 by using OWOP-SR and node 2 as a relay. The 5th GB is allocated to the first part of the transmission and the 8th GB to the second part.

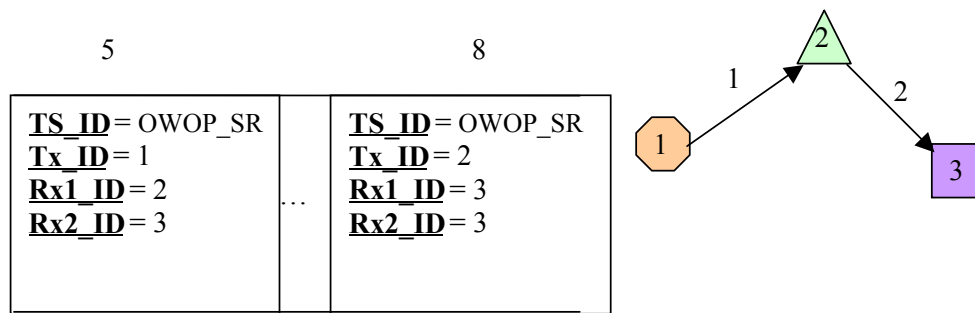


Figure 108: left: OWOP-SR beacon, right: OWOP-SR transmission

When utilising such a beacon information, for the first part of the transmission, node 1 knows that it can transmit on the 5th GB, node 2 knows that it has to listen on this same GB and to forward the received message, as it is not the final destination ($Rx1_ID \neq Rx2_ID$). Finally, node 3 knows that it does not have to listen on the first GB, as $Rx1_ID \neq Rx1_ID$.

For the second part, node 2 knows that it can transmit on the 8th GB, node 3 knows it has to listen on this GB and that it is the final destination, as $Rx1_ID = Rx2_ID$.

- One-way one-path cooperative relaying protocol

In this case, two transmitters send information on the same GB during the second period of the transmission. Indeed, both the relay and the source send information. Consequently, we need a new field, which we call **Tx2_ID**, which indicates a possible second transmitter on a GB. Henceforth, we will call **Tx1_ID** the first transmitter identifier and **Tx2_ID** the second transmitter identifier.

As an example: node 1 sends a message to node 3 by using OWOP-CR protocol and node2 as a relay. The 5th GB is allocated to the first part of the transmission and the 8th GB to the second part.

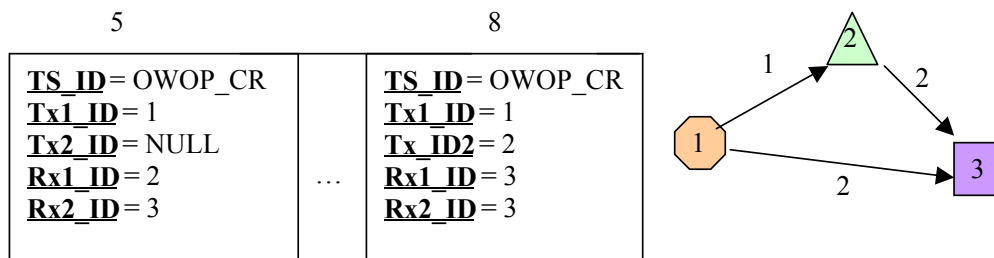


Figure 109: left: OWOP-CR beacon, right: OWOP-CR transmission

When utilising a beacon of information, for the first part of the transmission, node 1 knows that it can transmit on GB 5, node 2 knows that it has to listen on GB 5 and forward the received message, as it is not the final destination ($Rx1_ID \neq Rx2_ID$). Finally, node 3 knows it is the first GB, as $Rx1_ID \neq Rx2_ID$, so he does not need to listen.

For the second part, node 1 and node 2 know they can transmit on the 8th GB, node 3 knows it has to listen on this GB, as he has his identifier in the **Rx1_ID** field and that it is the final destination, as $Rx1_ID = Rx2_ID$.

- One-way two-path cooperative relaying protocol

Each OWTP-CR transmission uses a number $N = 3 + 2P$ of time bursts, as shown on the example of Figure 110.

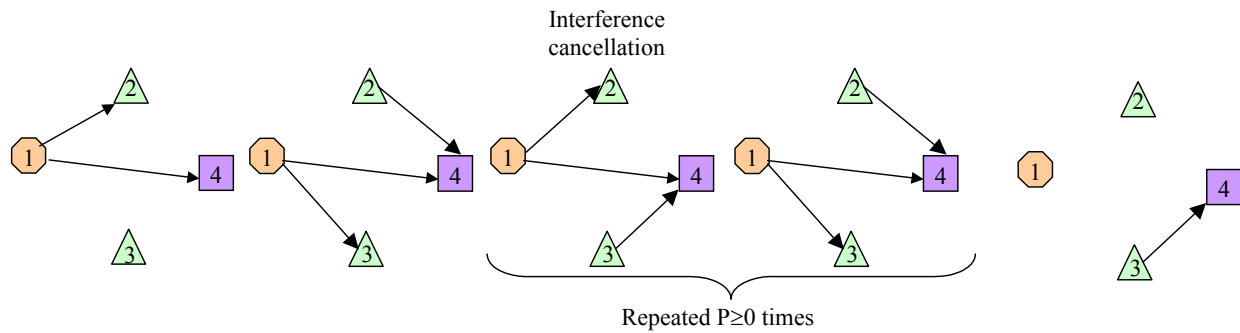


Figure 110: OWTP-CR transmission

As the receiver processing performs joint estimation and demodulation of symbols that have been received during a number M of time bursts, the value of M has to be incorporated in the frame beacon. A new Nb_Bursts field indicating the number of time bursts joint-demodulated by the receiver, has thus been added to the beacon definition, with respect to the previous definition (OWOP-CR). In our simulation settings, we consider DL frames with 5 time bursts per frame (see section 4.3.3.1 for burst dimensioning), so that Nb_Bursts will always be equal to 5.

Moreover, as in a given OWTP-CR transmission, the processing that is performed by network nodes depends on the burst number (for instance, as it is shown in Figure 110, interference cancellation only occurs at transmitted bursts number $N=3+2P$); nodes have to know the burst ID of the current OWTP-CR transmission. This requires the use of a new $Burst_ID$ field in the frame beacon, for this purpose. Below is an example of beacon frame information for the OWTP-CR transmission represented in Figure 110.

<u>TS_ID</u> = OWTP_CR <u>Tx1_ID</u> = 1 <u>Tx2_ID</u> = -1 <u>Rx1_ID</u> = 4 <u>Rx2_ID</u> = 2 <u>Burst_ID</u> =1	<u>TS_ID</u> = OWTP_CR <u>Tx1_ID</u> = 1 <u>Tx2_ID</u> = 2 <u>Rx1_ID</u> = 4 <u>Rx2_ID</u> = 3 <u>Burst_ID</u> =2	<u>TS_ID</u> = OWTP_CR <u>Tx1_ID</u> = 1 <u>Tx2_ID</u> = 3 <u>Rx1_ID</u> = 4 <u>Rx2_ID</u> = 2 <u>Burst_ID</u> =3	<u>TS_ID</u> = OWTP_CR <u>Tx1_ID</u> = 1 <u>Tx2_ID</u> = 2 <u>Rx1_ID</u> = 4 <u>Rx2_ID</u> = 3 <u>Burst_ID</u> =4	...	<u>TS_ID</u> = OWTP_CR <u>Tx1_ID</u> = -1 <u>Tx_ID2</u> = 3 <u>Rx1_ID</u> = 4 <u>Rx2_ID</u> = -1 <u>Burst_ID</u> =N
--	---	---	---	-----	---

Figure 111: OWTP-CR beacon information

4.3.4 OMNeT++ simulation framework for ROCKET

In order to study the performance of the proposed transmission schemes, at various protocol layer levels and in the particular ROCKET MU cellular context, an OMNeT++ simulation was built. This simulation is composed of two different kinds of entities : network nodes and services. Each network node includes an instance of the protocol stack to be simulated, with its own data. Inside the protocol stack, each layer communicates with its homologues by using dedicated services that are provided by lower layers (down to the PHY layer). This explains the distributed aspect of the simulation. The simulation environment also provides other kinds of services that are shared between nodes and will be described hereafter.

4.3.4.1 Simulation architecture

Figure 112 presents a conceptual view of the protocol architecture of the OMNeT++ simulation built for ROCKET. Each component of this protocol stack will be described hereafter.

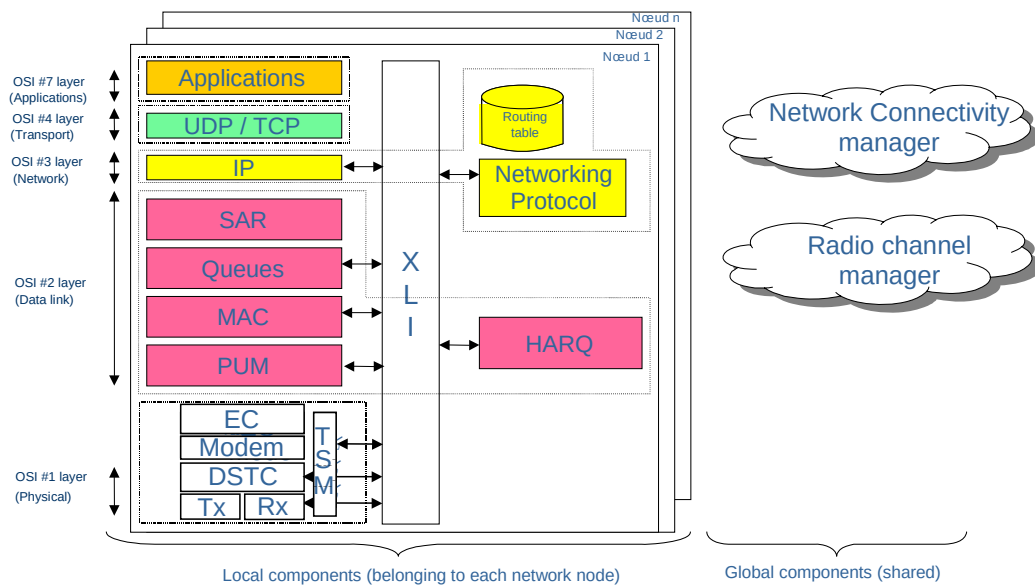


Figure 112: OMNeT++ simulation architecture

4.3.4.2 Simulation environment

The simulation environment incorporates a set of services that are shared by network nodes, including:

- Network connectivity management,
- Mobility management,
- Specific Service Data Unit (SDU)/ PDU data exchange process between layers,
- A dedicated interface for signalling exchange between layers, i.e., a cross-Layer Interface (XLI).

4.3.4.2.1 Network connectivity management

Radio connectivity management relies on the OMNeT++ mobility framework, which provides a set of protocol entities and are shared between all network nodes, to support node mobility and dynamic connection management. The network connectivity manager computes the distance between all the nodes by using mobile terminals location information, and decides whether they are in range of each other based on a maximum threshold distance, i.e., nodes are considered as neighbours if the distance between them is lower than the threshold value. This optimisation considerably improves simulation times in the case of networks with a great number of nodes, as it avoids processing signals coming from out of range nodes.

4.3.4.2.2 Mobility handling

Mobility is handled using the OMNeT++ mobility framework functionalities. In the case of static nodes, the only required information is the location of each network node for a given configuration. For instance:

```
sim.playgroundSizeX = 250
sim.playgroundSizeY = 175

sim.host[0].mobility.x = 50
sim.host[0].mobility.y = 50
```

means that node 0 is located at coordinates (50,50) inside a rectangular space of dimensions 250 by 175. Nodes trajectories are specified by a series of line segments (Bonn mobility model), which are stored in a text file. Each line of this file describes the trajectory of a node. It consists of a set of 3-

uples (t,x,y) of real numbers corresponding to times (t) and their associated node position in a planar (x,y) coordinate system. Let us take the example of a node moving along the perimeter of a square. This motion will be specified by the following text line:

```
0 50 50 100 100 50 200 100 100 300 50 100 400 50 50
```

Node positions are interpolated periodically on each line segment, depending on the value of the *updateInterval* parameter of the mobility framework.

4.3.4.2.3 Data exchange between protocol layers

Each data message is allocated a memory area that is large enough to contain all the information from the protocol layer that generates the message, as well as the headers that are successively added by lower protocol layers, as the message goes down the protocol stack (see Figure 113). On reception of a message from an upper layer, each protocol entity adds its own header before the first data byte of the message and transfers the encapsulated message to the lower layer. When receiving a message from lower layers, a protocol entity reads the header that has been added by its homologous layer at the transmitter side and forwards the message to the upper layer.

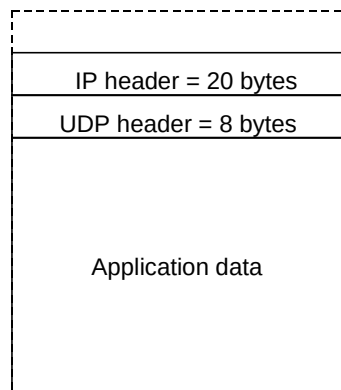


Figure 113: Data message encapsulation

4.3.4.2.4 Cross-layer signalling interface

The XLI is a generic module, which allows the local (*i.e.* node-level) exchange of control information between layers of a protocol stack. Each layer has inputs for control data originating from the XLI layer and outputs for control information that is intended to it. The identity of the destination layer is attached to each control message, in addition to control information itself. For instance, when the IP forwarding layer requires the identity node for the next hop, a NEXT_HOP_REQUEST_MESSAGE request is sent to the routing layer through the XLI layer.

4.3.4.3 Simulation backbone

The ROCKET simulation builds on an existing OMNeT++ framework, is composed of a stack of previously-designed protocol layers, which form the backbone of the simulation. Some of these layers, in particular the PHY and datalink layers, had to be completely redesigned for the project needs. Such a layered architecture has the advantage to allow independent validation of the protocol components that have been developed in the framework of the project. It encloses the following elements:

- A Transmission Control Protocol (TCP)/User Datagram Protocol (UDP) traffic generator,
- An IP layer,
- A centralised routing protocol,
- A Segmentation And Reassembly (SAR) layer,
- A FIFO Queues stage,

- A MAC sub-layer,
- A Packing Unpacking Manager (PUM) sub-layer,
- A modular PHY,
- A flexible channel model.

4.3.4.3.1 Traffic generator

Three different kinds of traffic flows can be modelled:

- TCP-based peer-to-peer data transfer,
- Constant Bit Rate (CBR) peer-to-peer transfer by using UDP transport protocol,
- Peer to peer UDP burst transmission with random transmission times, which is generated according to a Poisson distribution.

The TCP layer is the TCP Reno (RFC 2581) variant, which is derived from the IPSuite implementation that has been created and developed at the University of Karlsruhe in 2001 and precedes the actual INET framework.

4.3.4.3.2 IP Layer

The IP layer is divided into two OMNeT++ modules: IPModule and IPForwarding.

The role of the IPModule is to add an IP header to each SDU that is received from the transport layer and remove it from received IP datagrams. The IPForwarding module is in charge of determining the MAC address of the next hop to the destination. Figure 114 illustrates the next-hop discovery process, which is performed in the following 7 stages:

1. An IP message, which is intended to node ID Ipd, is received by the IP forwarding layer.
2. A request to find the next hop to Ipd receives no answer (no matching entry in the routing table).
3. A request for the next hop is then sent to the centralised networking protocol.
4. The networking protocol computes the new route and updates the routing table.
5. The next hop answer is provided to the IPForwarding module.
6. A request for the next hop to Ipd is sent and the next hop ID is retrieved from the routing table.
7. The IP message is forwarded to the next hop node.

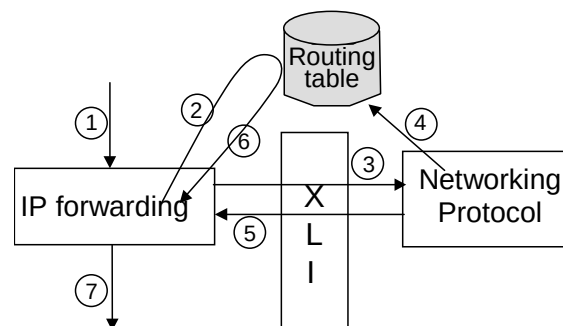


Figure 114: Routing table updating process

4.3.4.3.3 Centralized networking protocol

The centralised networking module computes the next hop to the destination by using a Dijkstra algorithm and global network topology information that is provided by the ChannelControl object. This module updates the routing table on demand, which means that it does not insert any entry in the routing table, unless explicitly stated. This kind of resource-sparing strategy is particularly efficient in the case of very large networks.

4.3.4.3.4 SAR layer

The SAR layer:

- Adds a header to each SAR-SDU that is received from the interface with its neighbour upper layer.
- Divides the resulting PDU into several SAR-PDU and adds a header to each one of these SAR-PDUs. SAR-PDUs have equal sizes, except for the last one.

Figure 115 illustrates this process.

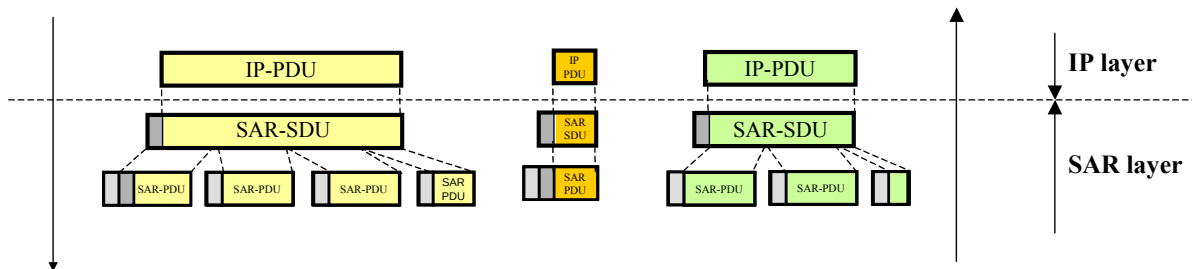


Figure 115: SAR layer operations

The header that is added to each SAR-SDU (4 bytes) has the following format:

- A 16 bits CRC for the whole SAR-SDU (including the header) – 2 bytes
- The SAR-SDU length (including the header) – 2 bytes

The header that is added to each SAR-PDU (4 bytes) has the following format (see Figure 116):

- Bit F : equal to 1 if the SAR-PDU is the last one of a SAR-SDU, 0 otherwise – 1 bit
- SAR-SDU ID – 11 bits
- Required QoS for the SAR-PDU – 6 bits
 - o Priority - 2 bits
 - o Other QoS information – 4 bits
- Reserved bits (set to 0) – 4 bits
- SAR-PDU size, including header – 10 bits

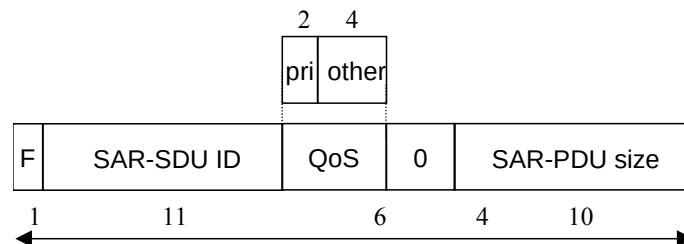


Figure 116: SAR-PDU header

A timer is started on reception of the first SAR-PDU of a SAR-SDU. If the timer expires before the SAR-SDU has been reassembled, all the previously-received SAR-PDU are dropped. SAR-PDU headers do not include sequence number information, as they are supposed to be received in the right order. The 16 bits CRC or the SAR-SDU header protects against possible corruption of one or several SAR-PDUs. The SAR-SDU identification number is bound together with the SAR-PDU sender's MAC address, in order to be able to differentiate SAR-SDUs originating from different neighbour nodes.

4.3.4.3.5 Queues

Each neighbour of a given network node is allocated a set of priority queues (one per priority level : see Figure 117). Each of these queues is of FIFO type and stores variable-length Q-SDUs. Broadcast transmissions do not have any associated queue.

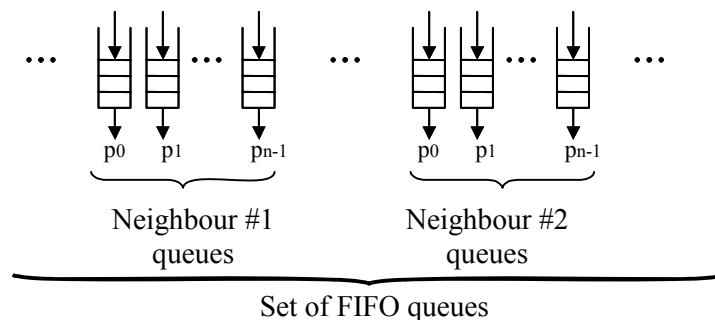


Figure 117: Structure of the FIFO queues stage

- **Adding data to a queue**

Each Q-SDU that is received by the queue layer from the above layer is added to the end of the appropriate queue (if not full), according to its priority and destination ID. If the selected queue is full, the Q-SDU is rejected. Let us keep in mind that a Q-SDU may be a SAR-PDU, and that rejecting a Q-SDU belonging to a set of SAR-PDU all originating from the same SAR-SDU will compromise SDU reassembly at the receiver side. To avoid unneeded data transmissions over the radio channel, a SAR-PDU belonging to a given SAR-SDU will not be added to a priority queue unless there is enough space, in the queue, to contain all the SAR-PDUs originating from the SAR-SDU of interest.

- **Removing data from a queue**

The following algorithm is implemented to remove nbBytes bytes from a set of priority queues:

```

priority = highest
current queue = highest priority queue
While (nbBytes > 0)
  If(( the current queue contains at least one Q-PDU qPdu) and (nbBytes - size(qPdu) ≥ 0))
    Extract a Q-PDU from the current queue
    nbBytes = nbBytes - size(dPdu)
  else
    priority = next lower priority
    if (priority < lowest priority)
      exit while loop
    end if
    current queue = queue with priority priority
  end if
end while

```

Several Q-PDU (=Q-SDU) can thus be extracted from queues at each transmission over the air. In order to allocate resources in the most adequate way, the MAC layer has to search the queues for destination IDs and priorities of messages waiting to be sent, so as to schedule first the neighbours

waiting for data with the highest priority. The required Q-PDU are thus extracted from the corresponding queues and are transmitted through the radio channel.

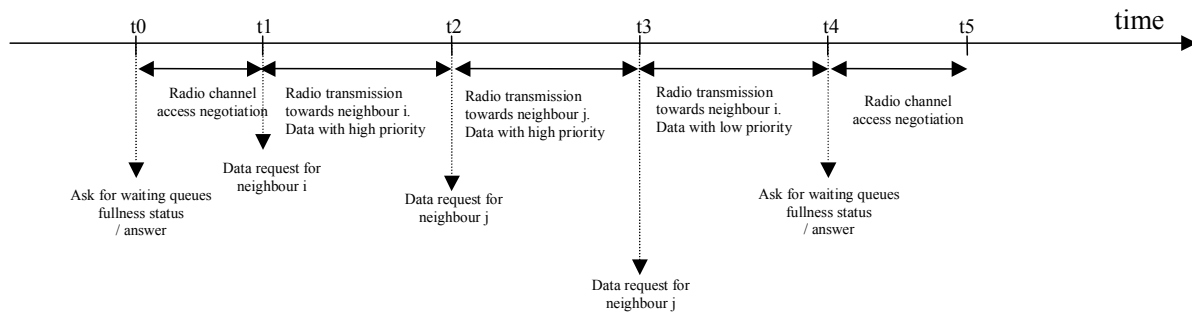


Figure 118: data retrieving from FIFO queues

Figure 118 provides an example illustrating this mechanism.. At time t_0 , the MAC layer requests the queue layer to provide the vacancy status of priority queues for three transmissions, *i.e.* three (neighbour address, priority) information pairs. The response indicates that a first high-priority transmission to neighbour i that is followed by a high-priority transmission to neighbour j and by another transmission to neighbour i with low priority could be performed. The MAC layer thus negotiates channel access for these three transmissions, up to time t_1 . In case of success, the MAC layer immediately retrieves data from all the neighbour i queues and the first high-priority transmission to neighbour i begins. At time t_2 , the previous transmission is over and the MAC layer extracts and transmits to lower layers data from the whole neighbour j queues. At time t_3 , the MAC layer extracts data for neighbour i . It shall be stressed that if, between t_1 and t_3 , data with high priorities are received in the corresponding neighbour i queue, then, these data will be provided in priority to the MAC layer, even if channel access has been negotiated for a lower priority transmission.

4.3.4.3.6 MAC sub-layer

The MAC sub-layer is in charge of:

- Updating the simulation clock at each time burst,
- Building signalling (beacon) messages,
- Fetching data from queues at regular fixed time interval for transmission,
- Processing allocation requests messages and allocating bursts for data transmissions based on their arrival order, priority and resource availability,
- Building data bursts to be sent and transfer it to the PHY.

As it has been previously stated in section 2.2.2, the MAC sub-layer implements both TDMA and OFDMA-based multiple access, with an OFDMA resource allocation strategy that is described in section 4.3.3.1.

4.3.4.3.7 PUM sub-layer

As it has been explained in section 4.3.4.3.5, several Q-PDU can be extracted from queues for each opportunity of transmission over the air. The role of the PUM sub-layer is to aggregate the contents of these Q-PDUs so as to fit them into data bursts (=DL-PDUs). On the receiver side, it extracts the PDUs from received bursts and forwards it to upper layers. The PUM sub-layer is also in charge of creating DL-PDU headers, which enclose all the information that are required to ensure correct decoding of data that are contained in each burst as well as signalling information that is intended to upper layers. DL-PDU headers have variable lengths and a format that is described in Figure 119.

They are composed of a fixed-length part, which is 7 bytes long, and a variable part, with lengths that are comprised between 0 and 119 bytes. The variable part describes the M-PDUs that is included in the DL-PDU (a maximum of 15) and also includes some reserved fields for future needs.

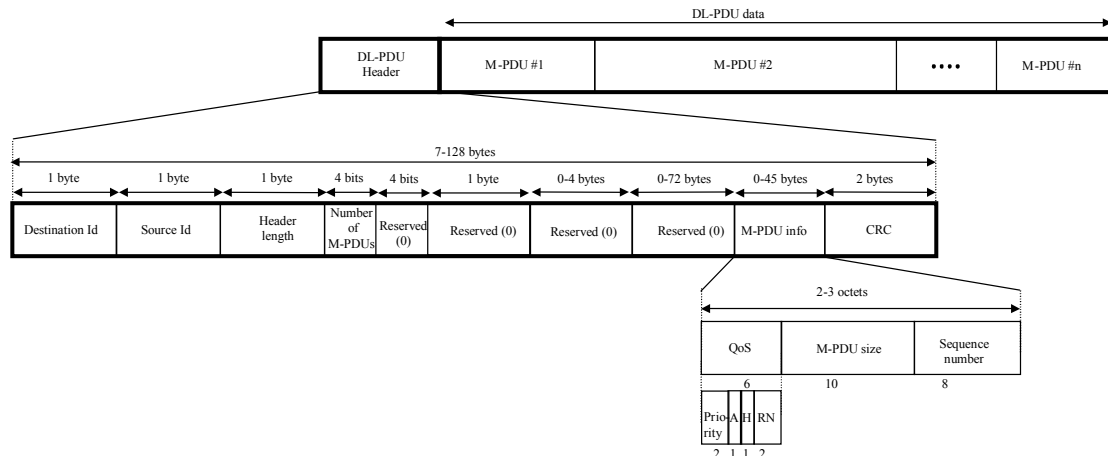


Figure 119: DL-PDU header

4.3.4.3.8 Physical layer

Figure 120 provides a detailed view of the PHY architecture and gives the data format of messages that are exchanged between Physical sub-layers and between the PHY and its upper and lower layers. The PHY implements the following functions: Error Correction Code (ECC), modulation / demodulation (Modem), DSTC, and transmission to (Ampli) / reception (Receiver) from the radio channel. Each of these tasks is performed by a dedicated functional sub-layer, the names of which are listed in the previous sentence. These sub-layers are in charge of the following operations:

- ECC: at the transmitter side, the ECC layer applies convolutional encoding for adding redundancy bits to the message to be transmitted through the radio channel. Available codes are Rate Compatible Punctured Convolutional Codes (RCPCs) with rates 1/4, 1/3, 2/5, 1/2, 2/3, 4/5 et 1, which are obtained by puncturing the output of the rate 1/4 mother code. At the receiver side, bits are decoded using the Viterbi algorithm.
- Modem: at the transmitter side, bits received from upper layers are modulated into complex symbols, depending on the selected modulation scheme (among possible ones: BPSK, QPSK). At the receiver side, the Modem sub-layer demodulates received symbols into bits. In the case of the OWTP protocol, the receiver processing, which is described in section 4.3.3.2.4, is also implemented in the Modem entity.
- DSTC: this sub-layer implements distributed space-time coding and decoding. The only DSTC scheme available in the ROCKET simulation is the Alamouti-like DSTC, which is described in the receiver processing of section (4.3.3.2.4).
- Ampli: before being transmitted, symbols are amplified by a constant amplification factor (depending on the required transmit power) and by the square root of the transmitter antenna gain. Both are tuneable simulation parameters.
- Receiver: the radio receiver checks whether the received signal is the user signal of interest or not. If not so, the received signal is dropped; otherwise, it is amplified by the receiver antenna gain and is forwarded to upper sub-layers (DSTC or Modem, depending on the selected transmission scheme). The radio channel manager (section 4.3.4.3.9) is invoked to get the value of the channel coefficients that are used to recover the received symbols.

Two additional management entities ensure the correct operation of the whole system:

- PHY manager (Manager, in Figure 120): it configures the software components which perform the various functions of the PHY (refer to the above list). In the current version of the simulation, this entity cannot instantiate more than a single object per sub-layer, which limits transmissions to a single modulation and coding scheme. In future versions (out of the scope of the ROCKET project), the PHY manager will be enhanced for allowing the coexistence of user flows with different MCS.
- Transmission Scheme Manager (TSM, in Figure 120): it acts like a switch, its role is to direct messages to the various functional sub-layers of the PHY. It also computes amplification coefficients at relays, in the case of relayed protocols (see section 4.3.3.2.3) and performs matched filter processing.

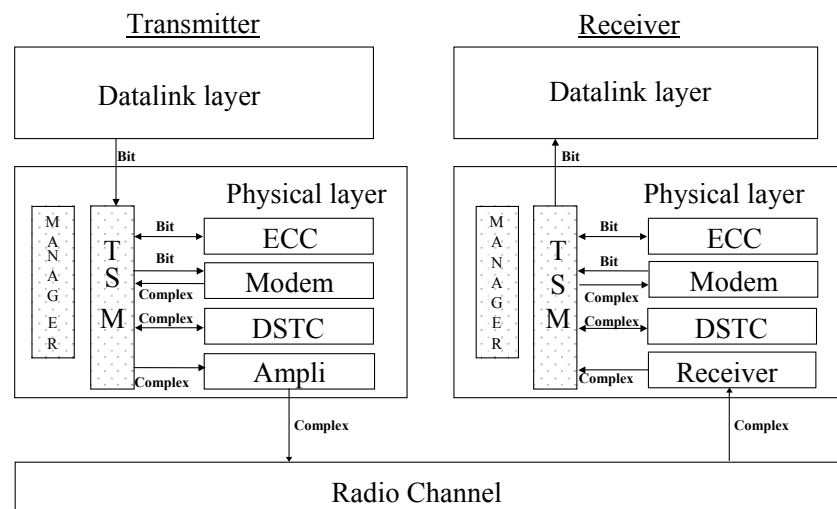


Figure 120: PHY architecture

Sequence diagram of functional sub-layers

Figure 121 and Figure 122 draw the path followed by messages through the PHY, at the transmitter and receiver side, respectively. These diagrams show that MAC signalling, which is not coded in bit format, follows a dedicated path and is not affected by wireless radio channel effects.

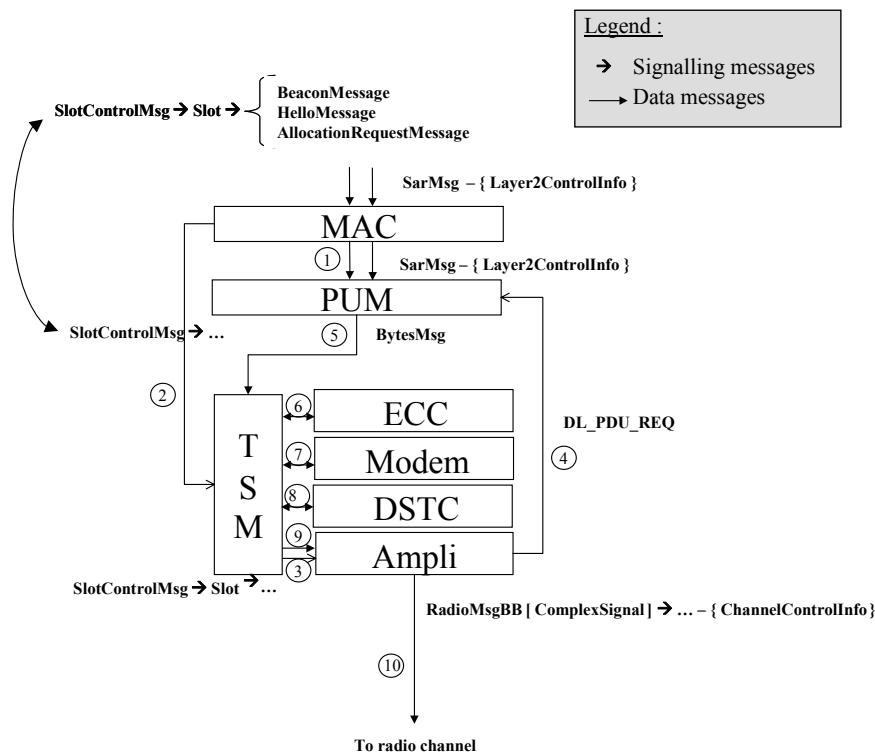


Figure 121: PHY – sequence diagram in transmission

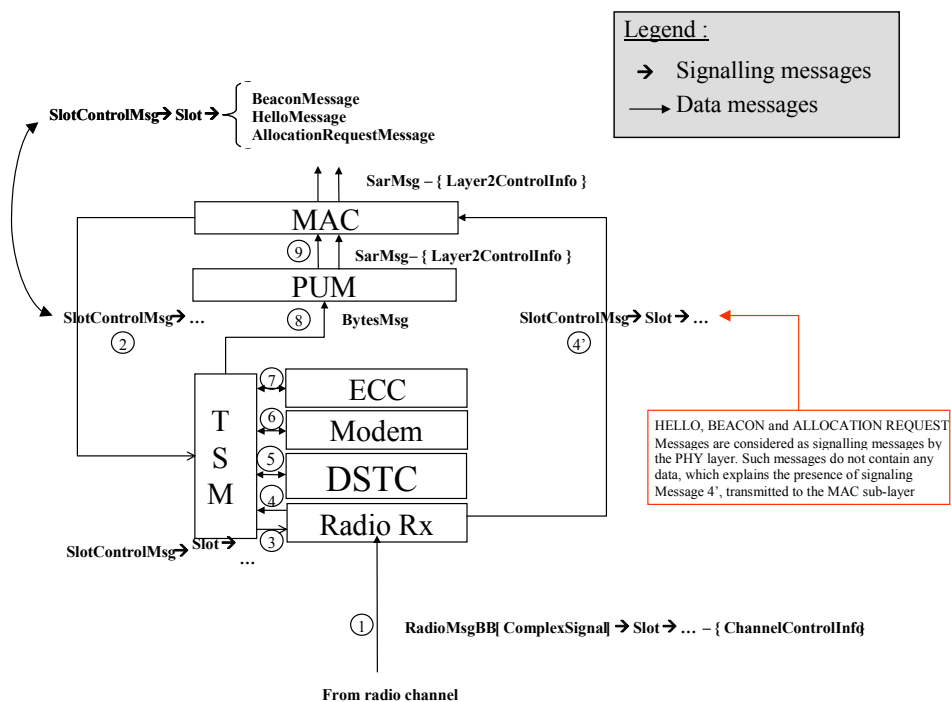


Figure 122: PHY – sequence diagram in reception

4.3.4.3.9 Radio channel manager

The radio channel manager implements a modular and evolutive wireless radio channel model. This model subdivides the effect of the wireless radio channel on transmitted signals into four components:

- Attenuation due to slow fading / Path loss: this attenuation coefficient depends on carrier frequency and distance between transmitter and receiver. The path loss model used in the

ROCKET simulations is the Winner urban macrocell path loss, which is detailed in section 4.3.2.3.

- Attenuation due to slow fading / shadowing: the radio channel model that is used in ROCKET does not include any shadowing component. A log-normal model should be added for more realistic simulations.
- Attenuation due to fast fading: the fast fading model used in the ROCKET OFDMA simulations is a multipath Rayleigh fading model (see section 4.3.2.3). The radio channel manager also implements a basic non frequency-selective Rayleigh model which is mainly used in simulations when using TDMA multiple access.
- Gaussian noise: receiver thermal noise (see section 4.3.2.3)

Each frequency component of the output signal from the Ampli sub-layer is multiplied by the square root of the path loss and then by the fast fading component. Gaussian noise is added to the resulting signal. The principles of the channel model is to propose a generic definition of each one of these four components and to allow the refinement of this definition depending on the simulation radio propagation environment and physical phenomena to model.

4.3.5 Performance evaluation

In this section, we analyse the performances of the specific ROCKET OWTP-CR transmission scheme and compare it to DT and single-relay transmission schemes (OWOP-SR and OWOP-CR). This comparison is realised in both a link and a cellular context.

The link simulation only uses a source, a destination and one or two relays (according to the studied transmission scheme) whereas the cellular simulation is composed of a source, one or two relays (according to the studied transmission scheme) and sixteen destinations placed in a sector of a 2,000m radius WiMAX trisectorial cell with radio channel and stations parameters, as it has been defined in section 4.3.2.

For the sake of readability, we limit the scenarios of link simulations to ones having relays symmetric to the source-destination link. Consequently, for single-relay transmission schemes (OWOP-SR and OWOP-CR) the relay lies on the source-destination axis. For two-relays transmission scheme (OWTP-CR), relays are symmetric to the source-destination axis so that we note $d(sr)$ and $d(rd)$, respectively, as the source-relays and the relays-destination distances. With $d(sr) = d(sr1) = d(sr2)$ and $d(rd) = d(r1d) = d(r2d)$.

In all the following simulations, a QPSK modulation is used and no convolutional encoder is used. The traffic is set to 336 bytes sent to each destination every 0.2s. So that with UDP and IP headers, each message is 364 bytes long and divided in 4 DL-PDU of 110 bytes each, messages are consequently divided in 4 bursts and sent in:

- 4 bursts for DT,
- 8 bursts in OWOP-SR and OWOP-CR transmissions,
- 5 bursts in OWTP-CR transmission.

4.3.5.1 OWTP-CR performance analysis

The first study is realised on a “link simulation” basis. The goal of this study is not to reflect realistic performances but to compare the BER of the OWTP-CR protocol for different transmission scenarios. In order to not create differences between scenarios, we used a reduced set of the ROCKET simulation parameters. As a consequence, the path loss model that is used for this first simulation, is a one slope model path loss, instead of a two-slope one (we use the second part of the WINNER model, which has

been described in section 4.3.2.3, regardless of the distance between two nodes). For the same reason, we also set antenna gains to 0dBi and set the same noise factor at the receiver for all nodes, so that no path loss nor antenna gains nor noise factor affect the results that are presented in this section.

In a cellular context with fixed-position relays, the relay positioning is a key aspect to improve performances. In single-relay transmission schemes, it modifies the source-relay and relay-destination distances (noted $d(sr)$ and $d(sd)$) and consequently the source-relay and relay-destination SNR (noted $snr(sr)$ and $snr(rd)$). In the OWTP-CR protocol (two-relays transmission scheme), which is described in section 4.3.3.2.4, relays interfere together. A judicious relay positioning is consequently very important, as relay positions determine the source-relays, $d(sr1)$ and $d(sr2)$, the relays-destination, $d(r1d)$ and $d(r2d)$, but also the inter-relay distances, $d(r1r2)$, which are critical distances because of interference.

This study is divided into two parts. In the first part, we study the consequence of inter-relay interference on the BER for an OWTP-CR transmission scheme, when $d(sr)$ and $d(sd)$ are kept fixed. Whereas the second part goal is to compare OWTP-CR performances for different values of the source-relays distances.

4.3.5.1.1 Effect of inter-relay interference on BER performance

Simulation configuration

In order to study interferences effect on performances, we set a source at coordinates (0m, 3,000m) and a destination at (2,000m, 3,000m). The two relays remain symmetric to the source-destination link. We compute the BER for different relay positions with a source-relay distance constant (set to 1,000m) as shown in Figure 123. Note that the smaller $d(rd)$ is, the smaller the inter-relay distance is.

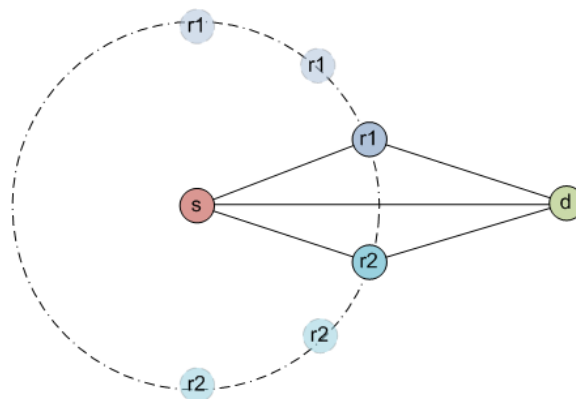


Figure 123: Relay positioning

Results are drawn for:

- a DT,
- an OWTP-SR transmission without interfering signals between the two relays nor DT contribution,
- an OWTP-CR transmission without interfering signals between the two relays,
- an OWTP-CR transmission without interference cancellation processing,
- an OWTP-CR transmission as defined in section 4.3.3.2.4.

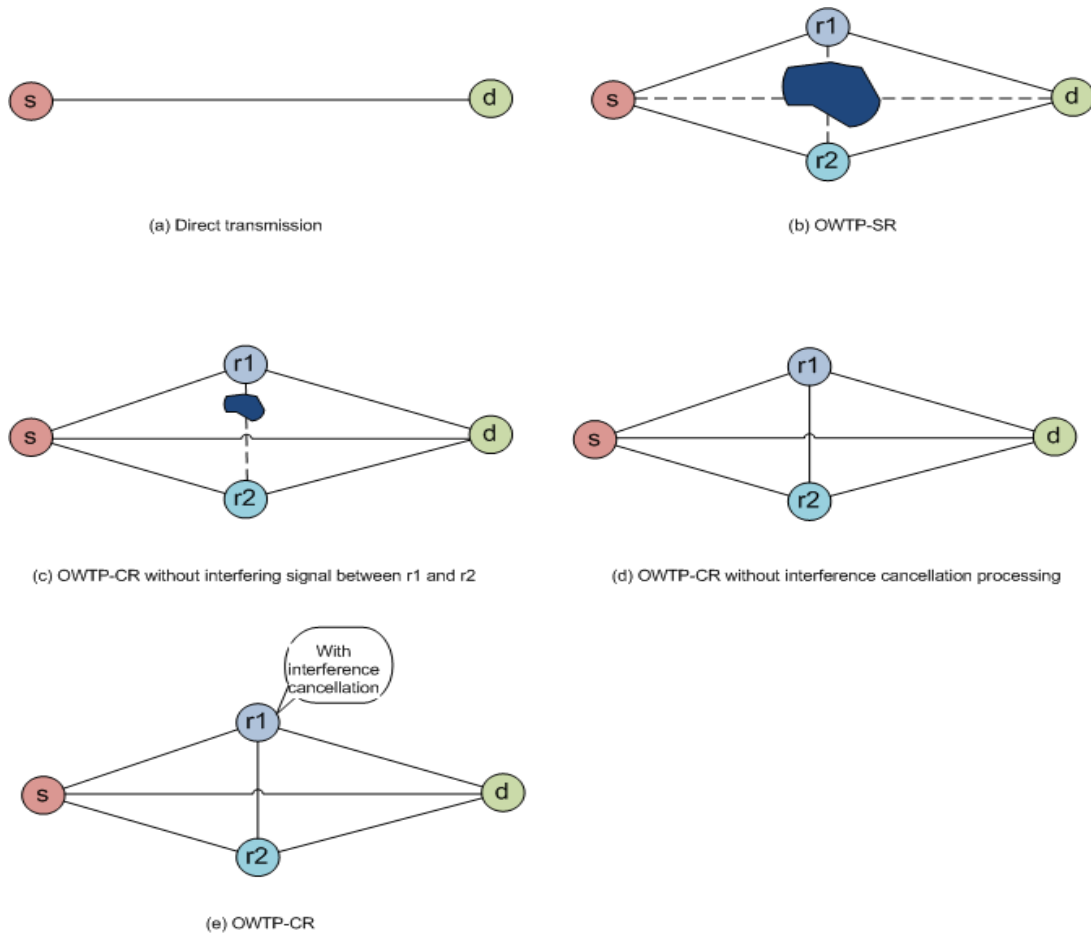


Figure 124: OWTP transmission scenarios

Please, refer to Figure 124 for details on the studied transmission scenarios. The BER is drawn as a function of the relay-destination distance and is presented in Figure 125.

The OWTP-CR transmission with neither interfering signals between the two relays nor DT contribution can be assimilated to a simple relay transmission alternately using two relays. From this point, we call this transmission scheme OWTP-SR. Moreover, the difference between the case without interfering signals (case c) and without interference cancellation (case e) consists in the fact that in case c, relays never receive signals from each other, whereas in case e, both relays receive signals from each other (as in case d) but the first relay cancel interferences. Only the first relay cancels interferences and the interference cancellation is not perfect because of the noise. Case e consequently softens interference in contrast with case d but does not reject all of it as in case c.

First of all, the OWTP-SR (case b) shows, as it was expected, that the smaller $d(rd)$ is, the smaller the BER is. Indeed, this transmission only uses source-relays and relays-destination links. The smaller $d(rd)$ is, the bigger $snr(rd)$ is and consequently, as we set $d(sr)$ (i.e.: $snr(sr)$) constant, the better the BER is.

In the case of no interfering signals between the two relays and by adding the source destination link (case c), it leads to BER improvement. Here is an ideal case in which three nodes (two relays and a source) perfectly cooperate without interfering with each others.

Considering the inter-relay interfering signals but with interference cancellation (case e), we notice that the BER is larger for small $d(rd)$ values than without interfering signals (case c). This is due to the fact that small $d(rd)$ implies small $d(r1r2)$, so that interference is significant for these points. The BER then improves when $d(r1r2)$ (i.e.: $d(rd)$) grows and goes towards the same BER values as in case c

(case without interfering signals). The effect of interference is consequently important when relays are close to each other but quickly decreases when $d(r_1r_2)$ increases. Note that the BER of the OWTP-CR always remains better than the one from the DT in the studied scenario.

On the contrary, in the case where no interference processing is done at the first relay (case d), the effect of interference becomes evident and the BER can become larger than the one of the DT for small inter-relay distances. Moreover, interference effects decrease much more slowly than with interference cancellation.

As a consequence, performances evolve within two different ways according to source-destination and source-relay distances:

- In a first part, the BER decreases when the inter-relay distance increases because of the interference decreasing.
- In a second part, when inter-relay distance becomes so large that interfering signals have no effect on the transmission, the BER increases according to the inter-relay distance increasing.

Simulation results

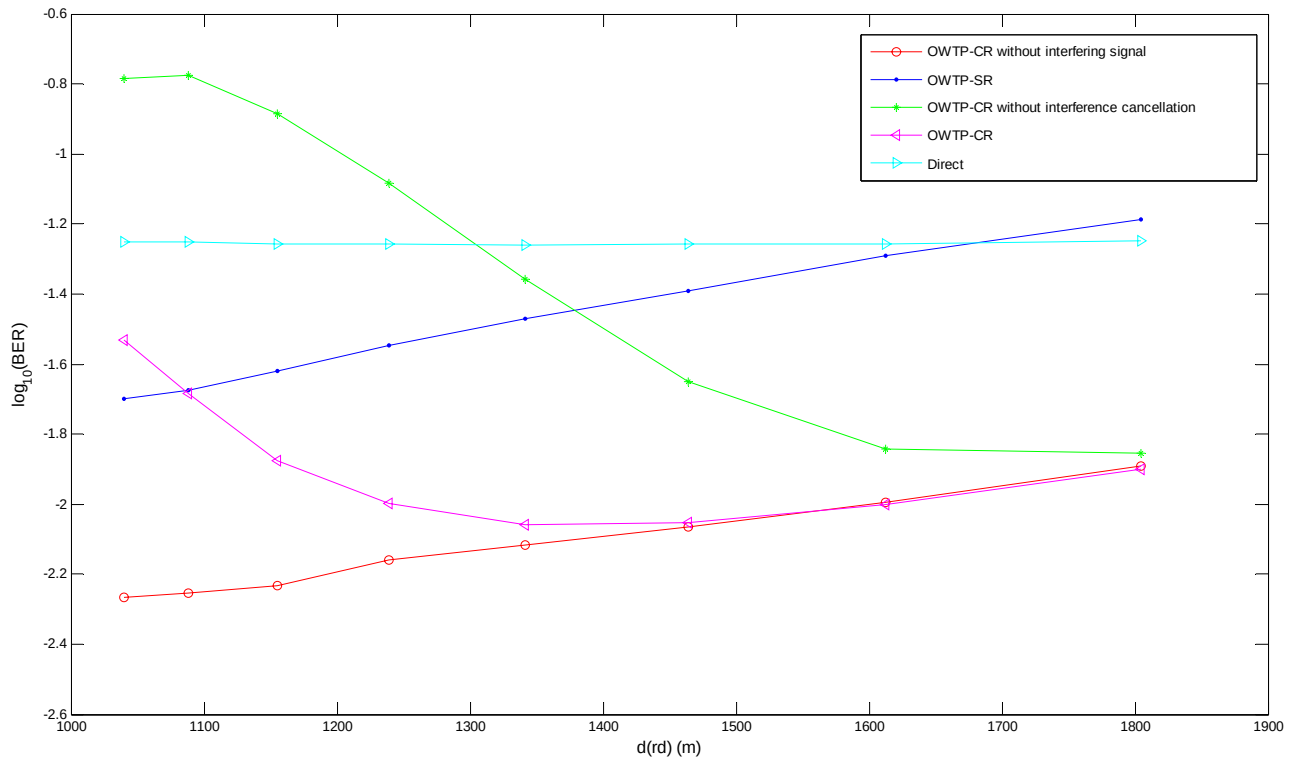


Figure 125: OWTP BER analysis for $d(sr)=1000m$ and $d(sd)=2000m$

4.3.5.1.2 Optimal source-relay and inter-relay distances

Simulation configuration

In this section, we investigate the source-relay distance effect on BER performance. Consequently, we focus on the OWTP-CR transmission scheme, which is described in section 4.3.3.2.4. We use the same scenario as in the first simulation but with different source-relay distances ($d(sr)=\{2,000m, 1,500m, 1,000m, 500m, 250m\}$) such as it is presented in Figure 126. Relays are placed on circles of different radius size ρ with different angles θ so that $r_1 = \rho e^{i\theta}$ and $r_2 = \rho e^{-i\theta}$. BER versus inter-relay distance results are presented in Figure 127.

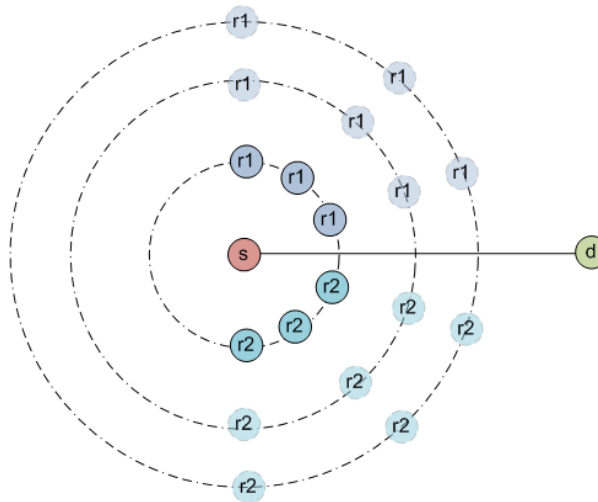


Figure 126: Relay positioning

Simulation results

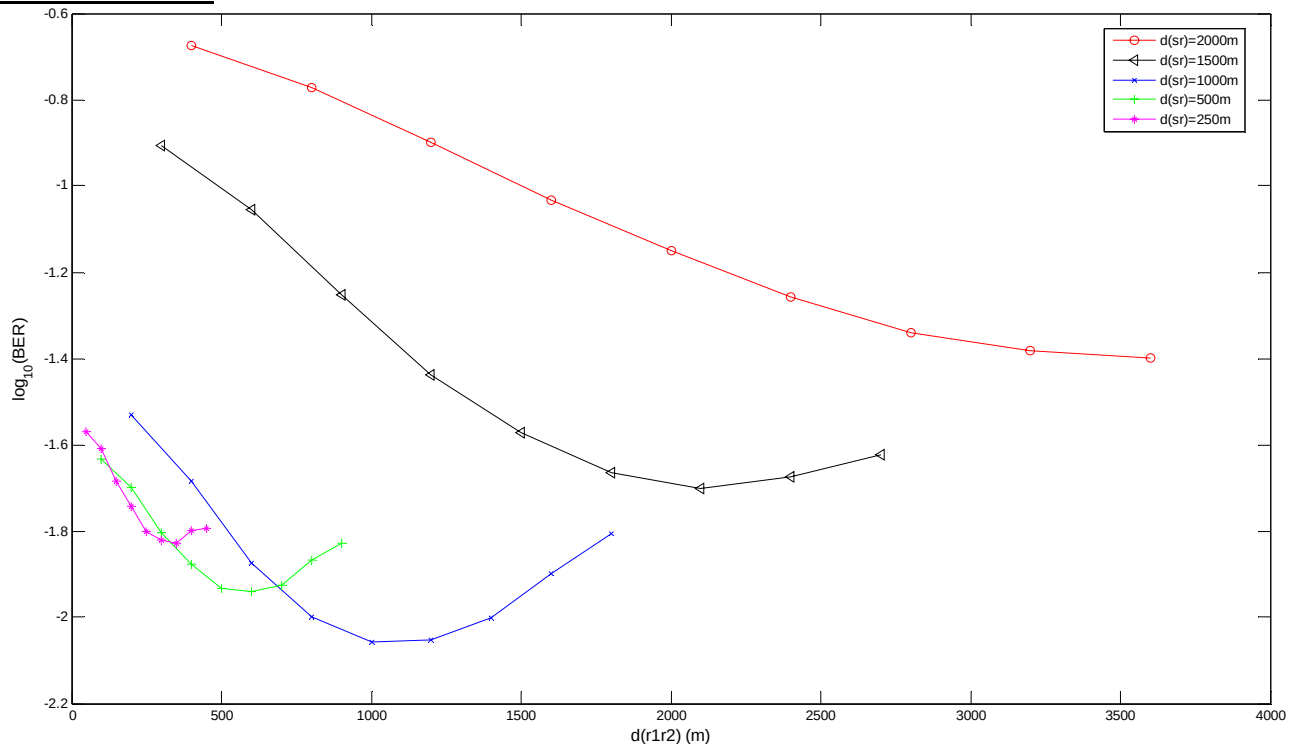


Figure 127: OWTP-CR BER versus the inter-relay distance for different source-relay distances

This figure points out that:

- The same inter-relay distance can be reached from different source-relay distances, but, for a given inter-relay distance, there is only one optimal source-relay distance in terms of BER.
- For small $d(r1r2)$ values, the optimal $d(sr)$ is a small one. Then, when $d(r1r2)$ increases, the optimal $d(sr)$ becomes larger and the BER decreases until reaching an optimum point. From this point, the BER starts increasing but the optimal $d(sr)$ keeps growing within $d(r1r2)$ (see Figure 128).
- Finally, in the given scenario, the optimal points for a source-destination distance of 2,000 meters appear to be around a source-relay and an inter-relay distances equal to 1,000 meters

each. These results, for the studied scenario, in an equilateral triangle, $sr1r2$. Note that, in this simulation, nodes have no antenna gains. The optimal relays position may vary with antenna gains.

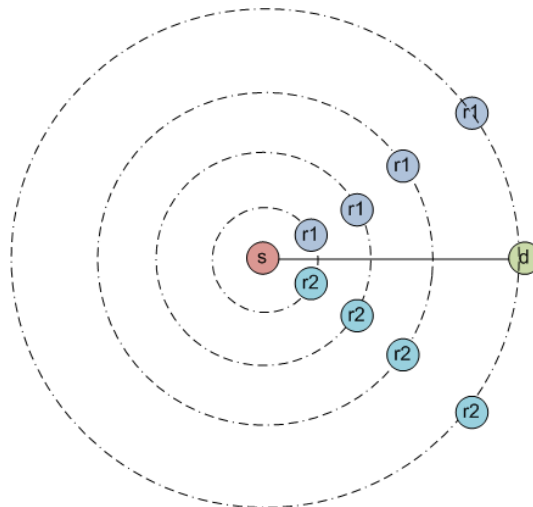


Figure 128: Optimal $d(sr)$ for a given $d(r1r2)$

4.3.5.2 Comparison of OWTP-CR with direct and single-relay transmission schemes in a “link” context

Simulation configuration

In the next simulation, we compare the OWTP-CR transmission scheme with single-relay and DT schemes for a cell-edge user. The cell is supposed to be a WiMAX one with a cell radius of 2,000 meters and with radio channel and stations characteristics, as it is described in section 4.3.2. No restrictions are made on path loss, antenna gains nor noise factor at the receiver as it has been done in previous simulation.

We study the BER of a transmission between a BS at coordinates (0m, 0m) and a static destination (0m, 2,000m). The BER is computed for different relay positions.

For transmission schemes using two RSs, RS positions are symmetrical to (BS-MS) line. Calling I the intersection point between (RS1-RS2) and (BS-MS) lines, I goes from coordinates (0m, 200m) to (0m, 2,400m) with a step of (0m, 200m).

For transmission schemes using a single relay, BS, RS and MS are aligned ($I=R$) and the relay position ranges from (0m, 200m) to (0m, 2,400m) with a step of (0m, 200m) (see Figure 129). In OWTP-CR transmission scheme, different simulations have been realised for different inter-relay distances. Results are presented in Figure 130.

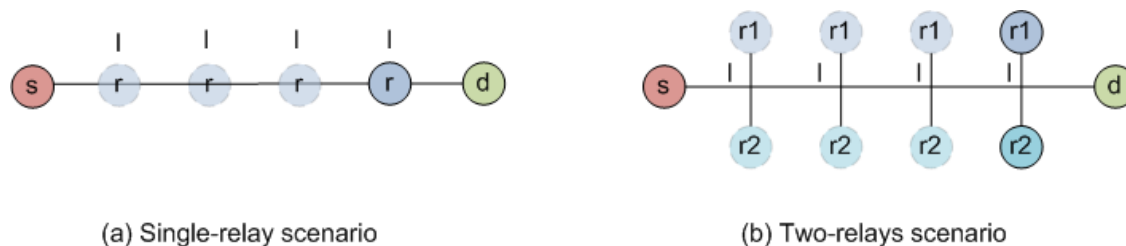


Figure 129: Simulation scenarios

Simulation results

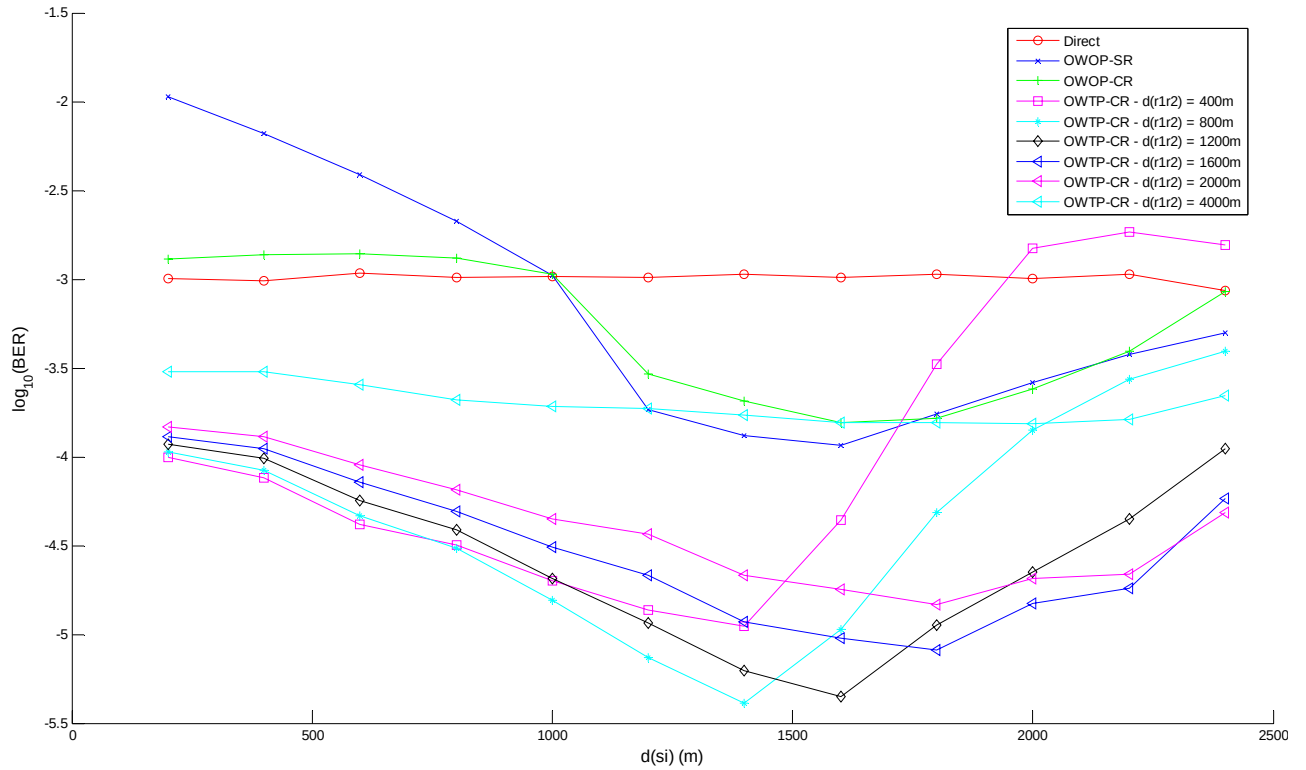


Figure 130: Comparison of different transmission schemes BER versus relay position

As for the OWTP-CR transmission scheme, both OWOP-SR and OWOP-CR have optimum source-relay and relay-destination distances.

All transmission schemes can reach a lower BER than the DT with a good relay positioning. But the one that leads to the best BER appears to be the OWTP-CR one. It can reach a BER of 8.15×10^{-6} whereas OWOP-SR and cooperative OWOP-CR can respectively reach a BER of 1.1×10^{-4} and 1.5×10^{-4} . Defining the relative gain as $G = 100 * \frac{(BER_{direct} - BER)}{BER_{direct}}$, the OWTP-CR transmission scheme can reach a relative gain of 99% whereas OWOP-SR and OWOP-CR only reach a gain of respectively 89% and 85%.

If relays are far enough from each other, the OWTP-CR provides a better BER performance than the DT regardless of the source-relay and relay-destination distances.

Therefore, the OWTP-CR transmission scheme leads most of the time to better performances than single-relay transmission schemes with a better SE. The price to pay for such a performance improvement is higher decoding complexity, increased latency and signalling overhead. Moreover, the channel coherence time shall be long enough so that it can be considered as stationary over a whole frame.

4.3.5.3 Comparison of OWTP-CR with direct and single-relay transmission schemes in a single-cell multi-user context

In this section, we compare the available transmission schemes performances in a single-cell, MU environment.

Simulation configuration

The cell is a WiMAX cell with ROCKET radio channel and stations characteristics that has been defined in section 4.3.2. The cell radius is set to 2,000m, BSs are trisectorial. We focus on the $[-\pi/3; \pi/3]$ sector of the cell. The BS is set at the position (0m; 3,000m). The cell is composed of 16 uniformly distributed mobiles, as it is sketched in Figure 131 below.

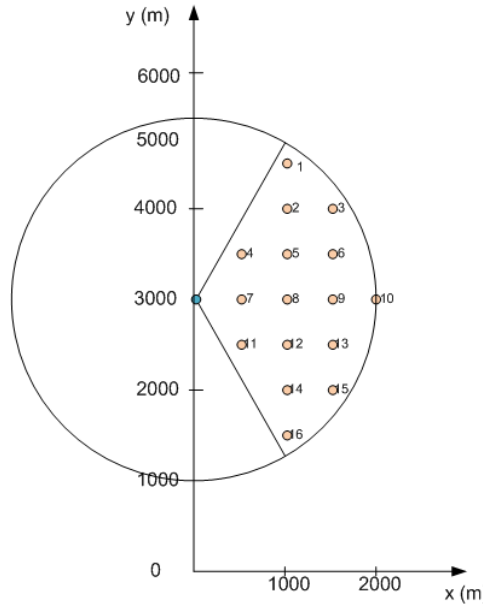


Figure 131: Cell configuration

In order to not overload the BS's queues, all messages are not sent at the same time. Messages for the four first nodes are sent together, then messages for the four next nodes are sent 0.03s after and so on. The total simulation time is set to 10,000 seconds. As a consequence, nearly 176,000 packets are sent to each user during the simulation. This also corresponds to nearly 155,010,000 data bits sent to each user. All messages have the same priority level.

For each simulation, one or two relays (according to the transmission scheme that is used) are inserted in the cell at a static position.

Let N be the number of users in the cell (in our case, $N=16$). For each simulation, we compute:

- the BER per user,
- the average BER in the cell: $\overline{BER} = \sum_{i=1}^N \frac{BER(i)}{N}$,
- the standard deviation of the BER: $SD = \frac{\sum_{i=1}^N (BER - \overline{BER})^2}{N - 1}$,
- the relative gain in percentage per user: $G = 100 * \frac{(BER_{direct} - BER)}{BER_{direct}}$.

As presented in Figure 132, simulations have been realised for:

- a DT,
- an OWOP-SR transmission with a relay placed at coordinates (550m, 3,000m),
- an OWOP-SR transmission with a relay placed at coordinates (1,550m, 3,000m),

- an OWOP-CR transmission with a relay placed at coordinates (1,050m, 3,000m),
- an OWOP-CR transmission with a relay placed at coordinates (1,550m, 3,000m),
- an OWOP-CR transmission with a relay placed at coordinates (1,950m, 3,000m),
- an OWTP-CR transmission with relays placed at coordinates (550m, 2,750m) and (550m, 3,250m),
- an OWTP-CR transmission with relays placed at coordinates (1,050m, 2,550m) and (1,050m, 3,450m),
- an OWTP-CR transmission with relays placed at coordinates (1,050m, 1,750m) and (1,050m, 4,250m),
- an OWTP-CR transmission with relays placed at coordinates (1,550m, 2,250m) and (1,550m, 3,750m),
- an OWTP-CR transmission with relays placed at coordinates (1,950m, 2,550m) and (1,950m, 3,450m).

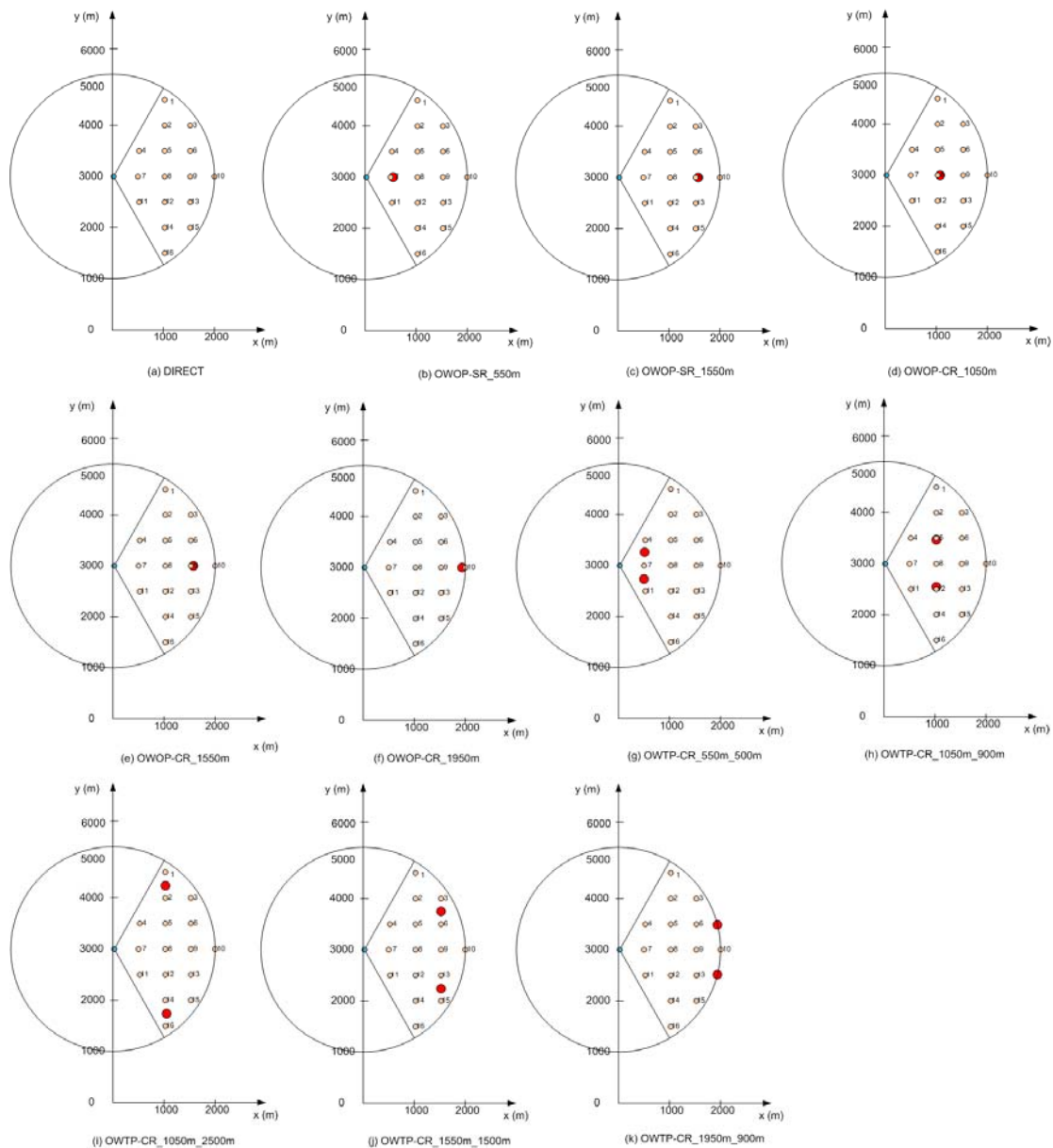


Figure 132: Relay positions in the cell

Up to the end of this report, we name the different scenarios using the transmission scheme and the relay position, as follows: “TRANSMISSION_SCHEME”_“d(si)”m_“d(r1r2)”m.

As an example, scenario 2 is called “OWOP-SR_550m” and scenario 7 is called “OWTP-CR_550m_500m”.

Simulation Results

The next histogram presents the maximum relative gain that can be obtained in the cell by utilising different relay transmission schemes.

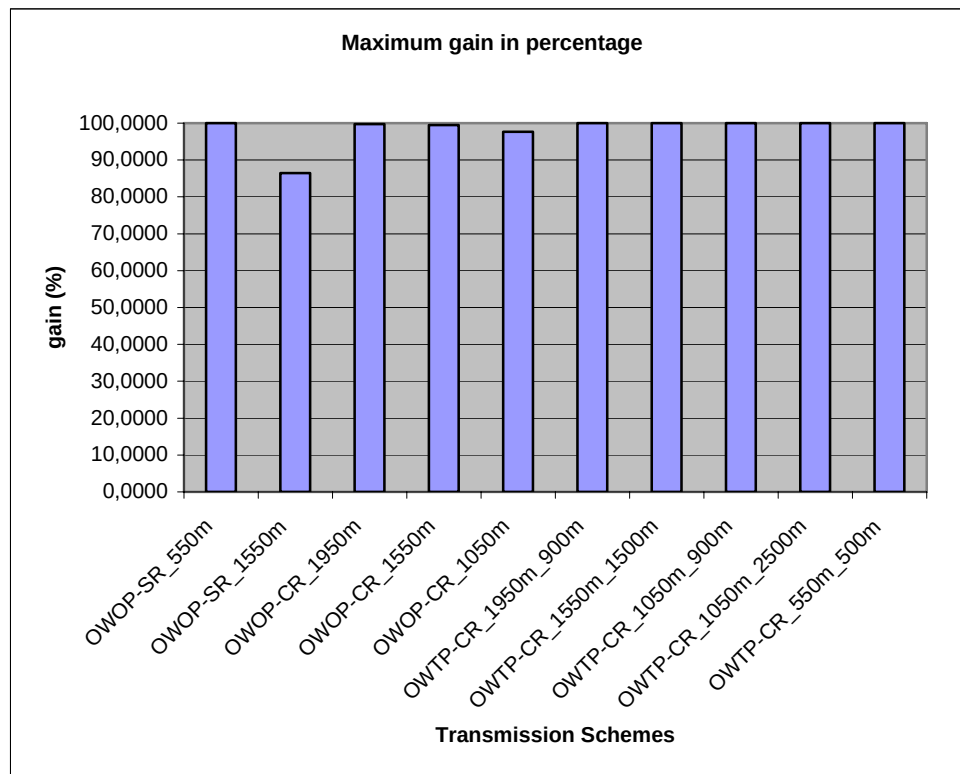


Figure 133: Maximum relative gain in percentage for different scenarios

All transmission schemes, irrespective of the relays position, can provide a BER gain of more than 85% to at least one MS in the cell.

The difference between all these scenarios does not really lie in the gain that can be provided to a specific mobile but on the average and the variability gains in the cell. Consequently, if relaying transmission schemes are used for a specific area in the cell, whereas other areas remain uses the DT scheme, all relaying techniques are nearly equivalent in terms of BER, but not in terms of SE and in implementation complexity. On the contrary, if relaying transmission schemes are used for the whole cell without any choice of relay to use, the choice of the transmission scheme becomes very important according to the mobile position.

Indeed, the next two figures (Figure 134 and Figure 135) present respectively the average BER and the standard deviation of the BER in the cell.

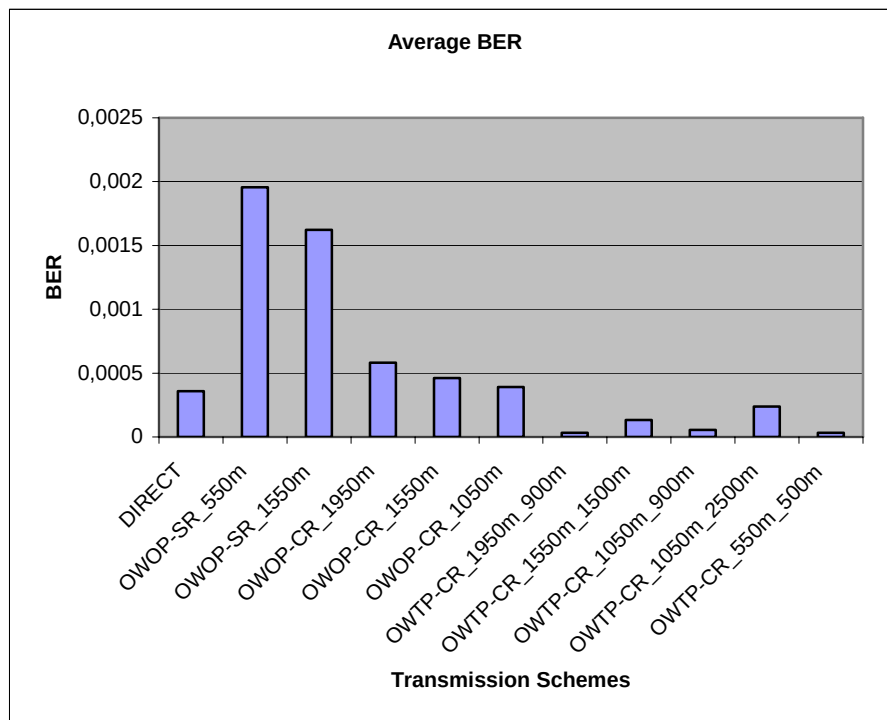


Figure 134: Average BER in the cell for different scenarios

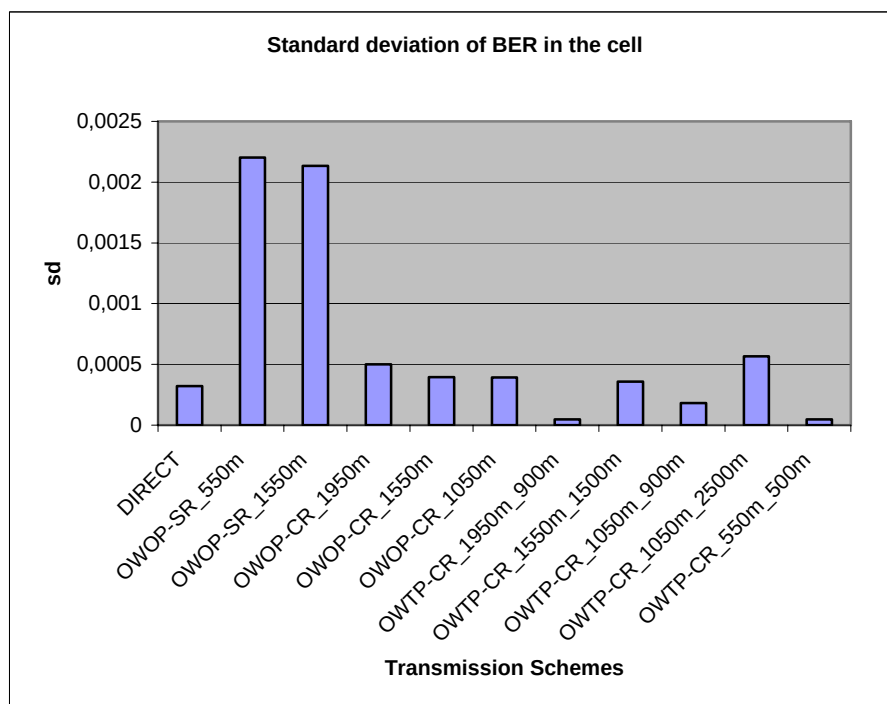


Figure 135: BER standard deviation in the cell for different scenarios

The first histogram shows that only OWTP-CR leads to a better average BER than the DT scheme. In fact, it is also the only transmission scheme that reduces the standard deviation of the BER in the cell (Figure 135).

As a consequence, the transmission scheme that provides the best gain in terms of average BER and fairness in the cell is the OWTP-CR one, with relays placed at the cell border. As an example, the next

figure (Figure 136) presents the BER of each mobile in the cell for some of the transmission schemes presented above.

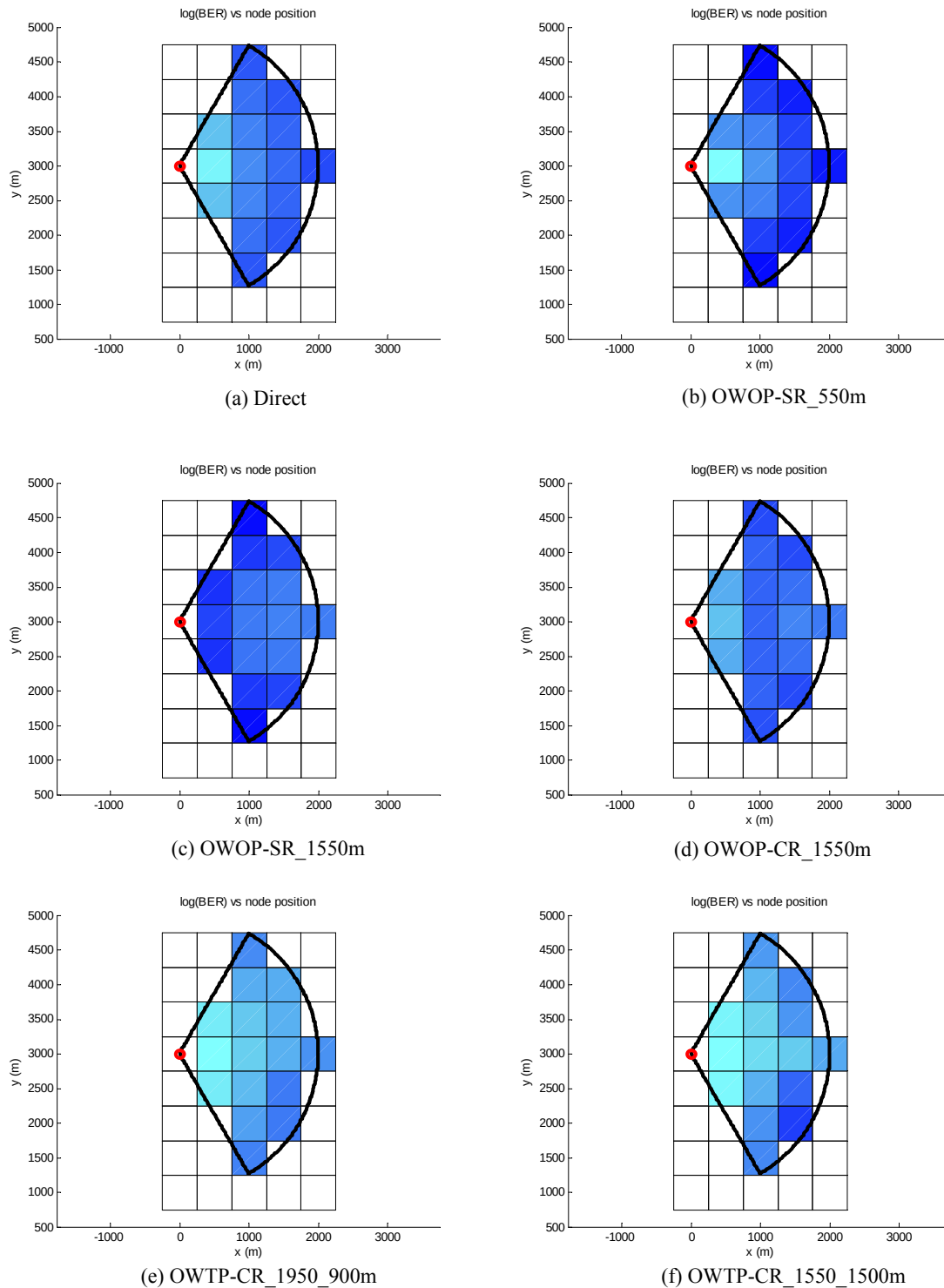


Figure 136: BER per user in the cell

Considering the only transmission schemes that is presented in the above figure, it appears that the OWOP-SR transmission scheme with a relay position at (550m; 3,000m) improves the BER for users near the BS but deteriorates it for cell-edge users. Therefore, the cell becomes less fair than when

using the DT. Using the same transmission scheme with a relay position at (1,550m; 3,000m) inverses the tendency. The BER of cell-edge users near the relay are improved, but BER of users that are far from the relay (those near from the BS included) are deteriorated. The OWOP-SR consequently appears to be a good solution to improve BER in a specific and delimited area.

On the contrary, the OWOP-CR with a relay placed at (1,550m, 3,000m) seems to improve the fairness in the cell but deteriorates the BER of users close to the BS.

The OWTP-CR finally appears as the only transmission scheme, among the studied ones, that can improve both the BER for all users and the fairness. Note that contrary to the other transmission schemes, OWTP-CR does not present symmetry in the BER distribution. This is mainly due to the fact that even if relays are symmetric to the 0 radian axis, only one relay performs interference cancellation. Therefore, there is no symmetry in relay processing. Thus, the nearer from the axis the relays are, the more symmetric is the BER distribution.

4.3.6 Complexity

The simulation results of section 4.3 showed that the OWTP-CR algorithm is the only one, among the protocols studied in this contribution, that can both improve the average BER performance and the fairness among users, provided that the relays locations are adequately chosen. On one hand, such a performance gain makes OWTP-CR an interesting technique for future cellular networks. On the other hand, as already mentioned previously, the price to pay for such performance improvement, against other conventional non-cooperative and single-relay assisted protocols is an increase in processing complexity, transmission latency and signalling traffic load, which may restrict its use the transmission of large data packets with low latency constraints. The specific MAC signalling information required by OWTP-CR and the other studied techniques will be specified in section 5. The signalling complexity will therefore not be addressed in this section. Here, we provide a list of the new specific signal processing functions required by OWTP-CR at the BS, MS and RS sides and give some information about the complexity of the corresponding algorithms (in the downlink case). Keeping in mind that the BS, RS and MS processing resources be very different, we present a separate complexity evaluation for each of these three different types of nodes.

4.3.6.1 New features required at the BS (transmitter)

The OWTP-CR doesn't require any specific features at the BS transmitter side.

However, the OFDMA resource allocator of the BS will have to be modified so as to be able to perform the following operations:

- Group data bursts by P , where P is the number of data bursts jointly-decoded by the receiver (in the case considered in section 4.3.3.2.4, $P=4$).
- Find $P+1$ free consecutive bursts in the same frequency resource of a frame, among a total of $n(\text{time}) \times m(\text{frequency bursts per frame})$.

The complexity of this process doesn't depend on the total number of users but on the number of transmissions to be scheduled in a frame.

4.3.6.2 New features required at the RS

The OWTP-CR doesn't require any specific processing to be implemented at the RS transmitter side.

Apart from channel estimation, which is a functionality that the RS receiver is supposed to have, the following new functions will have to be incorporated in the RS receiver:

- *Amplification coefficient computation*

The relay measures the power of the signal received during each burst and computes the amplification coefficient so as to keep the transmit power constant. This processing shall be performed for each user that is served by the RS. The complexity of this task is linear in the number of users to serve by the relay. Note that, with the frame structure described on Figure 7, a RS can only assist up to five simultaneous users at the same time. This processing may thus only be performed up to five times per time burst by a given relay.

- *Interference cancellation*

The signal transmitted by the relay during time burst 2 (which is supposed to have been stored by the RS) is subtracted from the overall signal received by the RS at time burst 4 (see section 4.3.3.2.4). Note that RSs have to be able to know the burst ID in order to decide whether to perform interference cancellation or not. According to details given in section 5.2.5.2, this information is supposed to be available in the frame beacon. The interference cancellation processing shall be performed for each user that is served by the relay. The complexity of this task is linear in the number of users to be served by the relay.

As different users are scheduled on orthogonal subchannels, there is no need for any multi-user detector at the RS receiver. A single-user receiver is all that is required.

4.3.6.3 *New features required at the MS (receiver)*

Apart from channel estimation, which is a functionality that the MS is supposed to have for normal receiver operation, OWTP-CR requires a multi-burst detector to be embedded in the MS, in order to perform the joint-detection of the data bursts received during a whole frame.

The sub-optimal detector proposed for use in the MS is presented in section 4.3.3.2.4. It was designed so as to minimize the computational cost, though providing tractable solutions, in order to meet the low power consumption requirements of mobile phones.

The complexity of the proposed detection process depends on the order S of the modulation ($S=2$ for QPSK), the number of jointly-decoded data bursts, P , and the number of symbols in a burst, N . To recover P bursts, equation (235) has to be computed $N.2^{S.P}$, which, using the values provided in section 4.3 ($N=864$, $P=4$, $S=4$), amounts to a total of 221184 sum computations involving complex numbers. Improved BER performance could be reached with increased computational complexity using an MMSE [Han-03][Tran-04][Tran-05] detector or other multi-user detection techniques with the same complexity $O((N.P)^3)$, such as MMSE-SIC or MMSE-ML which were proven to outperform MMSE detector in terms of BER [Le-05].

As different users are scheduled on orthogonal subchannels, there is no need for any multi-user detector nor interference canceller at the MS receiver to eliminate unintended signals. A single-user receiver is all that is required.

4.3.7 **Conclusions and outlook**

In this section, we presented the architecture of the OMNeT++ simulation implementing the OWTP-CR technique studied by TCF in the framework of ROCKET. The performance of this two-relays assisted cooperative transmission protocol was compared with the DT and other classical single-relay assisted cooperative (OWOP-CR) and non-cooperative (OWOP-SR and OWTP-SR) relaying protocols in the single-cell, MU WiMAX context. Link-level simulations for a single source-destination transmission revealed that OWTP protocols performances are limited by inter-relay interference, but that, with an adequate relay choice, OWTP-CR outperforms DT, OWOP-SR and OWOP-CR protocols in terms of BER, and it improves the SE with respect to single-relay protocols. Single-cell MU simulations, which has been performed for the scenario that has been proposed by TCF in section 2, showed that the OWTP-CR protocol both improves user fairness and the quality of

service of cell-edge users with well-chosen relay locations, though at the cost of increased latency, decoding complexity and signalling overhead.

Future work will consist in studying an adaptive protocol that automatically selects the transmission scheme leading to the best BER (or frame error rate), among all possible ones, according to the SNR (or Signal-to Interference plus Noise Ratio (SINR), in a multi-cell environment) configuration of all involved links (BS-MS, BS-RS and RS-MS) for taking advantage of cooperative relaying only when needed. Another interesting issue would be to study the performance of the proposed transmission schemes in a multi-cell environment, where the relay position determination is all the more difficult as adjacent cells interfere with each other.

5 MAC PROTOCOL SPECIFICATION

This section will provide the requirements for MAC layer of 802.16m to support our proposed techniques. Some of them may be common to all the techniques proposed in sections 3 and 4, and some of them will be specific. To this end, the present section is organised as follows. First, subsection 5.1 elucidates if those assumptions are tackled in the standard 802.16m. Afterwards, subsection 5.2 describes some practicalities to be considered by the MAC control layer when the techniques will be implemented in real systems.

5.1 MAC FRAME CONTROL

The proposed techniques in sections 3 and 4 investigate the multiuser transmission for non-relay and relay-assisted case based on the following concepts:

- User grouping
- One-Way relay-assisted transmission
- Two-Way relay-assisted transmission
- Two-Path relay-assisted transmission

This section aims to explore if those concepts can be supported in 802.16m or the MAC frame control should be modified:

- *Simultaneous transmission in the DL*

The user grouping in the DL is considered in the standard. Section 11.8.1 of [IEEE802.16m-09] describes that multiple users can be scheduled in one resource unit (RU). There are multiple FEC (Forward Error Correction) blocks at the input of MIMO encoder (horizontal MIMO encoding at transmit side).

The scheduler block depicted in Figure 65 of [IEEE802.16m-09] is responsible of deciding which users should be allocated to the same RU. Among the defined modes for the DL MIMO in Table 8 of [IEEE802.16m-09], those defined as *MIMO mode 3* and *MIMO mode 4*, are the suitable ones for encompassing the superposed transmission in the DL.

- *Simultaneous transmission in the UL*

The user grouping in the UL is also addressed in the standard. Section 11.12.1 of [IEEE802.16m-09] introduces that user grouping in the UL using the same RU is possible. The UL *MIMO modes 3* and *4* (Table 12 of [IEEE802.16m-09]) allow the superposed transmission in the UL.

- *Two-way relay transmission in the current standardised frame format*

Figure 137 depicts the current relay frame format considered in [IEEE 802.16m-09]. We can observe that the one-way forwarding and protocol I can be implemented, i.e. advanced BS (ABS) communicates with advanced RS (ARS) and advanced MS (AMS). For example the first phase of the communication, from BS to RS and MS (for protocol I), is described by zone *a2* in Figure 137, whereas the transmission from RS to MS is done in zones *b1* or *c1* in Figure 137.

On the other hand, the two-way relay protocols are not yet fully supported. It is defined a *Network Coding Transmit* and *Network Coding Receive Zones*, see Figure 137. In the Transmit Zone, the Odd Hop advanced RS (ARS) can transmit network coded transmissions to ABS and Even Hop ARS, see zones *a3*, *b4* and *c4*. Transmissions to AMS in that zone are under study. In the receive zone, the ABS or Even Hop ARS can receive network coded transmissions from the Odd Hop ARS. Moreover, zones *a2*, *b3* and *c3*, tackles the simultaneous transmission of ABS and Even Hop ARS to Odd Hop ARS.

In order to support two-way relay-assisted transmission which has been investigated in this work, the AMS should be able to receive and transmit in the different Network Coding Zones.

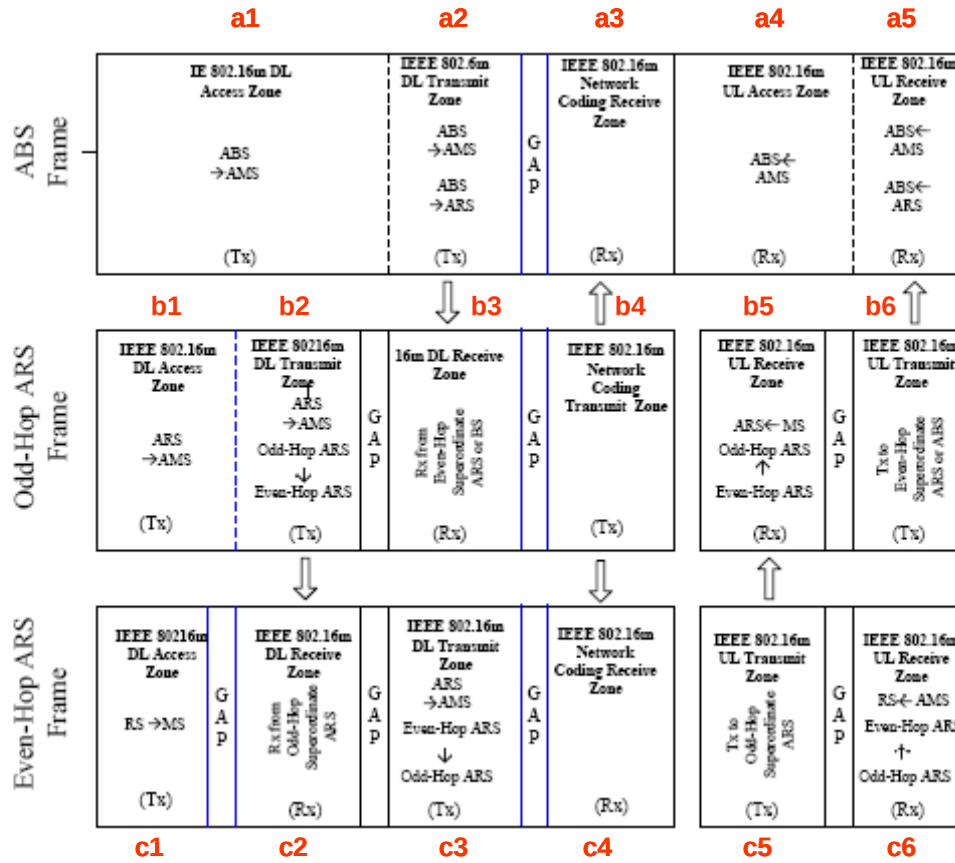


Figure 137: Relay frame format considered in [IEEE802.16m-09]

- *Two-path relay-assisted transmission*

Provided that the signalling modifications specified in section 5.2.5 are made, the current 802.16m standard should be able to support the OWTP-CR protocol studied in section 4.3. Indeed, the only major requirement is that the BS should be able to broadcast a message to one of the two relays and the destination, while the other relay forwards previously-received information. According to the ARS and ABS frame formats given in Figure 137, the transmission steps of the protocol of Figure 106 should be performed as follows:

Step 1:

- The base station broadcasts to the MS and RS1 in zone *a2* of the ABS frame.
- RS1 receives in zone *b3* of the Odd-Hop ARS frame.

Steps 2 and 4:

- The base station broadcasts to the MS and RS2 in zone *a2* of the ABS frame.
- RS2 receives in zone *c2* of the Even-Hop ARS frame while RS1 forwards to the MS in zone *b2* of the Odd-Hop ARS frame.

Step 3:

- The base station broadcasts to the MS and RS1 in zone *a2* of the ABS frame.

- RS1 receives in zone $b3$ of the Odd-Hop ARS frame while RS2 forwards to the destination in zone $c3$ of the Even-Hop ARS frame.

Step 5:

- RS2 forwards to the destination in zone $c3$ of the Even-Hop ARS frame.

5.2 RECOMMENDATIONS FOR MAC PROTOCOL SPECIFICATION

The implementation of the proposed techniques needs of an adequate definition of the MAC protocol for supporting such technique. To this end, this subsection provides additional information to be considered for supporting the proposed algorithms.

5.2.1 Multi-user pairing in the UL

The implementation of the cooperation scheme with user pairing that has been outlined in sections 3.1.1 and 3.1.2 has to face challenges of the fairness at system level as well as the timeliness of the UL BF that has to be performed within a time interval shorter than the channel coherence time from the channel estimation at the BS. We propose to face the former by segmentation of the SNR regions (so that the maximal rate scheduling is approximately a fair solution), but the latter, in the frame of the IEEE 802.16m, has to cope with several requirements that can be summarised in the following operational steps:

- The BS has to be aware of the SNR distribution in order to be able to operate a proper segmentation;
- All the subscribers within a segment have to send preambles (or subset of pilots) to let the BS estimate all the channels of all the users in all the subcarriers (as the method does not solely pairs users, but also operates the best band selection);
- The BS, after the relevant processing and metrics computation, pairs users and selects the best subcarriers allocation for each user;
- The BS transmits the BF vector and the time/subcarriers scheduling that all the active users will have to employ for the next UL;
- The subscribers, based on the data received from the BS, start the UL.

In order to maintain the phase coherence of the processing, i.e., the basic assumption to have nearly-orthogonal pairing and thus the MU gain related to the Maximal Ratio Combiner (MRC), the major constraint is that all the operations from 1) to 5) must be accomplished well within the channel coherence time.

As it stands, the frame format that has been designed in the current standard 802.16e-2005 and is foreseen for the 802.16m can only comply with this requirement for static or pedestrian speeds. Indeed, referring to Figure 14, the channel coherence time for speeds such as 2-3 km/h (2 GHz carrier) is approximately 0.1 s, and is sufficient to accommodate 20 frames, 5 ms each. This provides sufficient room for the phases from 1 to 5 and no specific indication emerges toward the MAC management and/or modifications of the frame format. It can be verified by considering that, without any modification, the relevant preambles for the channel estimation (step 2) need 3 frames, by assuming 100 users and no DL phase and 40 OFDM symbols per frame (the other available symbols are dedicated to the control plane information such as FCH, UL_MAP and DL_MAP). Another frame can be employed for the subsequent DL phase in which the next UL timing and band allocation is communicated to the subscribers. Given this, the next 16 frames are available for the UL scheduling, with no strict need for additional channel estimations.

The above does not hold true for higher speeds such as 50 km/h (typical urban vehicular speeds). In this case, as it has been anticipated in section 3.1.2, the coherence time extends over nearly 60 OFDM symbols and the steps 1) – 5) cannot be accomplished with the current frame format that is employed by the IEEE 802-16e/m. Therefore, an impact has to be expected on the compliance between the frame format that is designed by the current standard 802.16e-2005 and 802.16m and the UL-DL rate that is needed to let the CSI be exploited by the BS in time for its use (we just recall that the CSI has to be estimated, transmitted and employed well within the channel coherence time, otherwise the targeted diversity gain would be lost). The time sequence of the feasible operations for the CSI feedback is reported in figure Figure 138 for the currently adopted frame format. The first estimation of the UL channels can be performed by the BS in the UL transmission phase. As not all the MSs can be accommodated into a single frame in the UL phase, the whole training phase (phase 2 of the list) must extend over more frames (e.g. 3 frames, assuming 40 OFDM symbols for the UL). In a subsequent frame, the DL (broadcast) OFDM symbols will be sent (step 4) to let the MSs adopt the proper bandwidth/timings, which has been computed by the BS (in step 3). The UL data will then be sent (over one or more frames) accordingly to the received BF and time/frequency scheduling.

Leaving unaffected the frame format and MAC management that are adopted by the aforesaid standards, the overall end-to-end analysis of figure Figure 138 indicates that the time interval T_{use} between the estimation of all the MSs channels by the BS and their exploitation by the MSs (assuming 100 users) cannot be less than 20 ms (i.e. 4 frames, 5 ms each). This would correspond to operations upper limited by roughly 12 km/h, in the most optimistic case. This demonstrates that a change in the frame format and MAC management is necessary to cope with typical (higher) urban speeds.

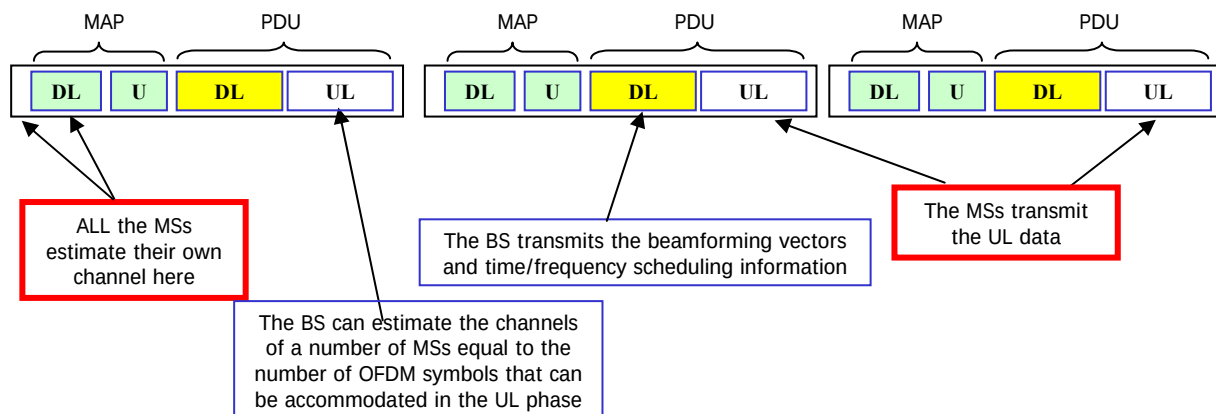


Figure 138: Operations in the currently adopted frame format.

The situation, which is outlined above and comes from the frame structure that is reported in Figure 138, can be improved in some extent by introducing a radical modification of the frame format in order to reduce the value of the interval between the beginning of the channel estimation and the end of the UL phase, so to let it cope at least with typical urban vehicular speed. A possible indication might come from the frame format that is depicted in Figure 139, along with the adoption of the interlaced training sequences that has been introduced in section 3.1.2 (the timing and interlacing sequences are specified in the first UL_MAP of Figure 139). In this case, the BS can estimate the CSI of 100 MSs in about 20 OFDM symbols (see section 3.1.2) and thus a similar set of resources has to be allocated just after the first UL_MAP. After the reception of all the UL preambles, the BS has the information for computing the scheduling, pairing and subcarrier allocation of all the users, along with the eigenvectors to be employed for the UL BF. These information are broadcasted by the BS just after the UL training sequence (maybe with a time lag due to computation time latency) and with or without an additional preamble for the DL channel estimation (depending on the mobility conditions). The remaining resources of the frame are thus entirely dedicated to the UL phase, in which all the selected pairs transmit with their assigned eigenvectors on the same (assigned) time/frequency resources. As the standard-compliant frame duration is not enough to accommodate the whole UL phase, the new

frame should comprise at least 66 OFDM symbols: 2 for the frame preambles (just before the FCH), 1 for the 1st UL map, 3 for the 2nd UL_MAP, 20 for the training phase and 40 for the PDU UL phase.

The whole cycle that is described by the steps from 1) to 5) can be performed for 100 users within the time interval of 60 OFDM symbols by adopting the above scheme, from the 1st channel estimation to the last transmission of a UL-PDU. This corresponds to a time interval (assuming $\sim 100 \mu\text{s}$ per symbol) of 6 ms, and thus all the speeds involving a coherence time below 6 ms (corresponding to about 50 km/h) can be properly managed by the adopted scheme.

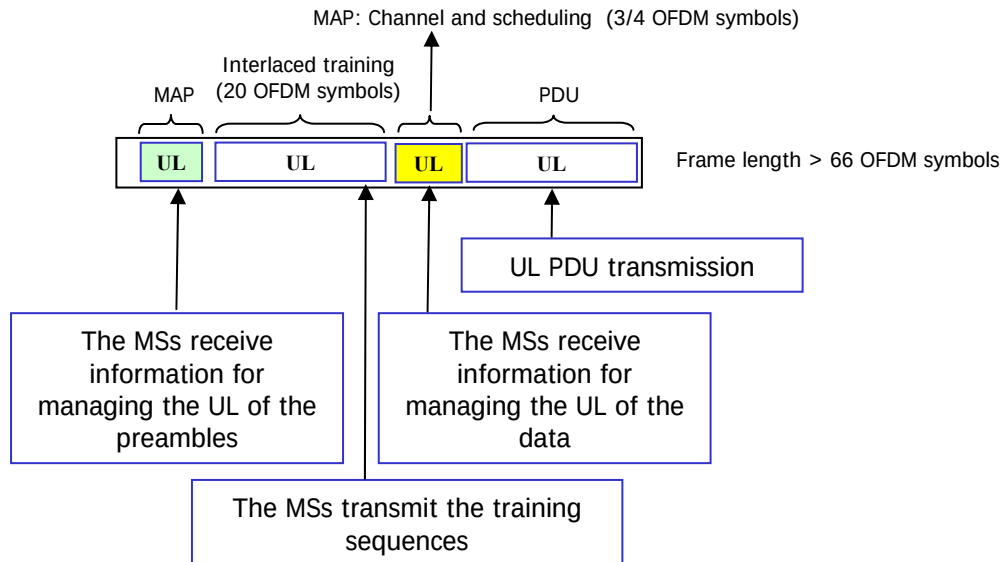


Figure 139: Pictorial view of a possible frame format for reducing the time between the CSI estimation by the BS and its use by the MSs.

The proposed subcarrier overloading scheme in section 3.1.2 will require modifications of the MAC layer protocol in order to allow the simultaneous use of a physical resource block by two users.

- At the BS, modifications will be needed in the existing scheduling and channel allocation services of the MAC, in order to support the subcarrier overloading.
- Appropriate messages will need to be passed over to the PHY layer to encode messages by using superposition coding on the shared subcarriers.
- The user terminal that is required to perform interference cancellation shall be informed of this and shall be provided with the details for the modulation and coding of the sharing user in order to decode and cancel the interference signal.
- Moreover, the proposed scheme requires FCSI at the transmitter end (BS). This information can be obtained by using the channel reciprocity; by assuming that the user terminals are using the subcarriers for the UL transmission and the BS can estimate the channel as the receiver of this UL channel. This assumption requires the use of TDD approach that has an implication on the MAC layer. If FDD is assumed, the CSI will have to be fed-back to the BSs and this will require inclusion of appropriate *channel state feedback* messages to the MAC control messages.

5.2.2 RRM for superposed transmission for DL non-relay techniques

Requirements for the design of RRM algorithms tackling with superposed transmission in multi-user non-relay techniques (section 3.2.2) are:

- The average SNRs of the different links (source-destination, source-relay and relay-destination) is required for scheduling. Therefore, proper pilot tones have to be considered. It is assumed that the transmissions are done using PUSC/FUSC channelisation and the power is equally distributed across carriers. Since we work with the average SNR, it is not required an update of that value at each frame.
- The SNR refresh rate will depend on the velocity of the terminal and the channel statistics, that is, evolution of shadowing, pathloss, etc.
- The MAC protocol [IEEE802.16m-09] allows the user pairing in the UL and DL.
- The scheduling process is centralized at the BS. It controls the average throughput served per user, user priority and it designs the resource allocation (bitrate, power allocation). All MSs have to send their average SNR measured in the BS-MS link.
- Two independent messages with an optimized power allocation are transmitted under superposed transmission in the DL
- For a multicarrier scenario channel state information for all users on each subcarrier (or a group of adjacent subcarriers) will be needed at the transmitter end. The users will be paired at each subcarrier based on their channel realisations.

5.2.3 RRM for superposed transmission for relay-assisted techniques

Requirements for the design of RRM algorithms tackling with superposed transmission in multi-user relay-assisted techniques (section 4.1) are:

- The average SNRs of the different links (source-destination, source-relay and relay-destination) is required for scheduling. Therefore, proper pilot tones have to be considered. It is assumed that the transmissions are done using PUSC/FUSC channelisation and the power is equally distributed across carriers. Since we work with the average SNR, it is not required refresh that value at each frame.
- The SNR refresh rate depends on the velocity of the terminal and the channel statistics (i.e. the evolution of shadowing, pathloss) which may be different for every link,. For instance, if the BS-RS link is in LOS, the average SNR will be quasi constant and its value can be updated in a very low rate. On the other hand, the average SNR of the BS-MS and RS-MS links must be updated according to the mobility of the MSs.
- The relay must be able to deal with network coding when the two-way relay transmissions are assumed.
- The MAC protocol [IEEE802.16m-09] allows implementing one-way protocols, but the two-way protocols are currently under study. Hence, the implementation of the two-way relay techniques implies a modification in the standard.
- The scheduling process is all centralized at the BS. It controls the average throughput served per user, user priority and it designs the resource allocation (bitrate, power allocation, duration of phases of the relay protocols).
- For the multi-users relay techniques, dirty paper coding (DPC) must be considered at the BS and RS.

5.2.4 Uplink multi-user power allocation for non-regenerative cooperative MIMO

The additional parameters to include in the MAC definition to support the power allocation algorithms that are presented in section 4.2 are:

- Channel gains.
- Amplification coefficients at RS.

5.2.5 Multi-user one-way two-path relay channel with AF

The MU relay-assisted transmission techniques that have been investigated in Section 4.1 consider the simultaneous transmission (in UL or DL). It seems that the simultaneous transmissions under the OWRC can be easily adapted to the 802.16m standard. However, the TWRC requires a new definition of the frame format. Currently, a couple of frame format are possible candidates to be standardised for the TWRC. To this end, we will analyse how to adapt the different protocols that are analysed in this work to the current standard by providing the required changes to support these transmission schemes with simultaneous transmissions.

The additional parameters to include in the MAC definition to support the previously-described cooperative OWOP and OWTP transmission protocols of section 4.3.3.2.2 are:

- Channel gains.
- Amplification coefficients at relays.
- Specific fields added to the frame beacon definition.

5.2.5.1 Channel gains/Amplification coefficients at relays

The receiver processing that are associated to the OWOP protocol when using AF forwarding strategy, which are described in section 4.3.3.2.2, requires :

- o The value of the BS-MS channel gain, including slow fading, fast fading and antenna gains (BS, MS) for the two used frequency resources, on the second time burst.
- o The value of the global (BS-RS-MS) channel gain, including slow fading, fast fading, amplification coefficients at relays and antenna gains, for the two used frequency resources, on the second time burst.

These channel coefficients can be directly measured at the destination by using the dedicated pilot subcarriers. The 10 MHz bandwidth WiMAX mode defines subchannels of 24 data and 4 pilot subcarriers, which gives a total of 24 pilot subcarriers per burst when considering bursts of 6 subchannels, as it has been previously stated in section 2.2. The OWOP protocol thus does not need any particular additional signalling to be added to the WIMAX frame.

The OWTP protocol receiver processing (section 4.3.3.2.2) requires:

- o The value of the BS-MS channel gain, including slow fading, fast fading and antenna gains (BS, MS) for the current frame.
- o The value of the global (BS-R1S-MS) channel gain, including slow fading, fast fading, amplification coefficients at relays and antenna gains, for the current frame.
- o The value of the global (BS-R2S-MS) channel gain, including slow fading, fast fading, amplification coefficients at relays and antenna gains, for the current frame.
- o The value of the RS1-RS2 link channel gain, including slow fading, fast fading, and antenna gains, for the current frame.

All these coefficients, except for the RS1-RS2 channel gain, can be directly measured at the destination by using the frame dedicated pilot subcarriers. The only coefficient that requires additional signalling is the RS1-RS2 channel gain, that has to be transmitted from RSs (RS1 or RS2) to the mobile terminal, either directly (in data bursts) or by the intermediate of the BS, in the frame header (under the previous assumptions, channel coefficients are supposed to remain constant during a whole frame).

Considering an average coherence bandwidth of about 500 KHz (45 subcarriers), there are $N_p = 16$ different complex values of channel coefficients per frame (one per coherence bandwidth) for each link. As an OWTP transmission only uses a single frequency resource (a group of 144 subcarriers) among the 5 available this particular case, the maximum number of different complex values to be transmitted for the RS1-RS2 link is 4. Transmitting the whole RS1-RS2 CSI in each frame would thus induce a signalling overhead of 4 complex values (for the considered example), which is too big to fit in a burst header, but it can be incorporated in the frame header, provided quantization is used to further reduce the size of channel estimates. Such a process would however introduce some latency in the transmission process, as relays have to forward the measured channel coefficients to the BS in the previous frame. Here we have to assume that the measured CSI still remains valid in the next frame.

5.2.5.2 *New beacon fields*

As stated in section 4.3.3.3.2, new fields are required to be added in the frame beacon to support the previously-described cooperative and non-cooperative relayed modes (SRT, OWOP, OWTP). The following information has to be incorporated into the beacon, for each burst of the frame:

- Transmission scheme identifier: add new values for the SRT, OWOP and OWTP modes.
- Intermediate and final receiver IDs: in relayed modes, nodes have to know whether they are the final destination of the received message or not, in order to decide how to process the received data (amplify/forward or demodulate, in our case). This justifies the need to incorporate the ID of intermediate and final destinations of each data burst in the frame beacon.
- First and second transmitter ID: in the cooperative modes (OWOP and OWTP), up to 2 nodes are allowed to transmit on the same burst (the source and a relay). The ID of both nodes has to be given in the frame beacon, for each burst.
- Burst ID of the current transmission.
- Number of received bursts before processing the data at the receiver.

6 COMPLEXITY AND ADAPTABILITY OF THE PROPOSED ALGORITHMS

This section evaluates the complexity of the algorithms investigated in section 3 and 4. Although the term *complexity* is an abstract concept that encloses many aspects, here we describe the complexity introduced by our algorithms from three points of view: type of the employed optimization and new features required at receiver and transmitter side:

- *Type of optimization:* We detail if the algorithm is linear or non-linear with the number of users. If there is an analytical solution or the optimization can be formulated as a convex problem.
- *Features at receiver side:* We describe if the algorithms need multiple or single user decoder, if an ML (or approximate ML like sphere decoder), MMSE or a zero-forcing decoder should be adopted, or if the algorithm assumes an interference canceller at the receiver.
- *Features at transmitter side:* The transmission techniques required at the transmitter side are presented here. Aspects like if a dirty paper coding (DPC), designing the covariance transmit matrix or a beamformer will be considered.

Moreover, we also discuss the adaptability of our algorithms with regards to the use of the extra bandwidth, which can be made available through the techniques that have been studied in [ROCKET-2D1] and [ROCKET-2D2], and to their applicability to LTE-A.

- *Additional bandwidth:* Additional bandwidth requirements for the proposed techniques; any issues that may raise and modifications that should be made to the proposed algorithms for accommodating the extra bandwidth
- *Applicability to LTE:* Modifications that should be made to the proposed schemes, which are compliant with 802.16x, to become compatible with LTE and LTE-A

Table 18: Complexity of the proposed techniques

Techniques	Complexity of the optimisation procedures	Complexity at the receiver side	Complexity at the transmitter side	Additional bandwidth	Applicability to LTE
User Pairing in UL (section 3.1.2)	• The knowledge of the principal left eigenvector of the channel is required at the UE side • The complexity associated with the SVD computation linearly scales with the number of UEs and the number of subcarriers, but is quadratic or (nearly) cubic with the number of UE or BS antennas (quadratic when UEs are single-antenna) • The computation of all the SVD relevant to all the UEs can be strongly parallelised, as elementary modules can be run in parallel for each UE and each subcarrier, leading to a maximal exploitable parallelisation factor of UN_F, where U is the number of UEs and N_F is the number of data subcarriers. • The computation of the coupling metric is linear with both the number of frequency subcarriers and the number of BS antennas, but is quadratic with the	• The adoption of sequential cancellation, ML (or suboptimal ML) estimators have an impact on the final performance in terms of BER, but have no direct impact on the scheduling and pairing protocol itself.	• The pairing methodology poses no specific requirements on the UEs (transmitter) side	• The proposed methodology does not require additional bandwidth. • In case of the adoption of additional bandwidth, the method intrinsically works on a subcarrier base; this implies that the optimisation goal is insured; however it must be considered that the associated complexity linearly increases by increasing the employed bandwidth.	• The method does not rely on any specific element belonging to the 802.16x standard, the only basic assumption is that OFDMA is employed (however there are workarounds for this); this implies that it is applicable to LTE-A as well. The proposed frame format modifications are introduced to better cope with mobility and can be suggested in the same fashion for LTE-A as well.

	number of UEs. Also in this case the algorithm is highly optimised by its inherent parallelism on both subcarriers and on the number of UEs and its complexity can be reduced down to $O(U(N_B+3))$ operations, each performed by UN_F parallel processing units.				
User Pairing in DL of OFDM system (section 3.2.1)	- Channel state information is required at the BS transmitter for all users served by the BS. - Closed form solution exists for WSR maximisation for a pair of users. - User pairing need to be done in a heuristic manner to ensure that users with diverse channels are served together.	The receiver needs to perform SIC for a pair of users only at the receiver of the user having strong channel. The user with weak channel will use the conventional receiver.	- Selection of optimum pairs has complexity that grows exponentially with total number of users. However appropriate pairs can be heuristically selected by pairing users with diverse channel states. - Power split between two users of the pair can be optimised to meet certain fairness criteria. - Relative weights of the users need to be calculated to meet fairness requirements.	- There will be some extra parameters to be exchanged through the control plane thus increasing the signalling overhead. However it is believed that the increased overhead is well justified by the obtained performance and boost in system capacity. - The provided concept has flexibility to benefit from the extra spectrum obtained from another system.	The proposed method is generic for an OFDMA system and is applicable to a system like LTE-A and 802.16 with suitable modifications on provision and exchange of extra parameters through control plane. Also the scheduling frame size shall be decided based on mobility and this aspect is not studied in the provided contribution.
User Pairing in DL under different scheduling criteria (section 3.2.2)	- Under the weighted sum-rate (WSR) criterion there is a closed-form solution for the power allocation of two users. - For the proportional fair (PF) criterion the power	This technique requires a multi-user decoder. This decoder can be based on the SIC.	- The simultaneous transmission to two users is currently considered in the 802.16m. - The BS transmits two streams with different power allocation.	The transmission strategy is based on the user pairing along with superposed transmission in systems based on OFDM. Hence, additional bandwidth does not pose any important drawback.	The analyzed algorithms are applicable to LTE and LTE-A, but the frame format should be modified and new information elements be defined.

	allocation is based on equation (90) that has to be evaluated numerically. For the 2-user pairing all the combinations have to be evaluated. In order to reduce the number of evaluations, a list of L candidates is created based on their priority and bitrate. With this approximation the complexity introduced by the search of the best user pairing is independent of the number of users to be scheduled.		Under a multi-carrier transmission, the transmitted power is equally distributed across carriers.	However, if the channel quality seen by each user in the additional bandwidth differs too much from the channel quality in the current bandwidth then two independent scheduling algorithms should be developed in each bandwidth. How the power is distributed among the two types of bandwidth should be investigated.	
User Pairing for relay-assisted transmission (section 4.1)	- The resource allocation (bitrate, power, duration of phases) is obtained a solution of a convex problem that can be solved by interior point methods. - DL and UL traffic is allocated in the same frame. - The 2-user pairing is explored over a list of L candidates that has been created as a function of the priority and bitrate showed by the MSs. With this approximation the complexity introduced by the search of the best user pairing is independent of the number of users to be scheduled.	This technique requires a multi-user decoder. This decoder can be based on the successive interferer cancellation	- The RS can deal with one-way and two-way protocols. - The two-way relay protocols analyzed in this work are not supported by the current 802.16m. - The transmission done at BS and RS must employ DPC in some cases. - Under a multi-carrier transmission, the transmitted power is equally distributed across carriers.	- This technique designs the resource allocation (duration of phases for relay-assisted transmission) and selects a pair of users to be served simultaneously. In case the channel quality seen by each user in the additional bandwidth differ too much from the channel quality in the current bandwidth then some cases should be investigated: a) Employ the additional bandwidth only for the transmission of the relay (see section 3.3 of [ROCKET-2D1]). b) Design an independent scheduler using the additional bandwidth using	- The type of investigated transmissions was the one-way and two-way relay-assisted transmission under OFDM. It has been assumed that DL and UL transmissions are done using all the available data carriers. These are the drawbacks foreseen in LTE and LTE-A: a) The UL transmission in LTE is single-based carrier. However, LTE-A is considering the possibility of allowing multicarrier transmissions in the UL. In this latter case, the investigated techniques are applicable.

				either the relay-assisted transmission or just direct transmission. How the power is distributed among the two types of bandwidth should be investigated.	b) The two-way relay transmission are not considered in LTE or LTE-A. The frame format should be modified in order to allow this type of transmission.
Power allocation methods for non-regenerative MU MIMO cooperative scheme (section 4.2)	- Algorithms are based either on the Newton-Raphson method or gradient ascent method. The former method has a very low computational complexity; the latter method has a complexity according to the number of user - The Exp_MIMO algorithm is the most complex of the proposed algorithms, because it is formulated in terms of a sequence.	This technique requires a SIC multi-user decoder.	- In some of the proposed algorithms, CSI is required at the transmitter. - SVD is needed at the RS and for some proposed algorithms at the BS as well.	- The proposed algorithms do not require any additional bandwidth. - Power allocation is performed over all the subcarriers, extra bandwidth should be taken in the allocation at the RS. It will induce extra complexity.	MU MIMO power allocation techniques have been developed for the UL based on a single carrier assumption and then adapted to multi-carrier system. Thus, our techniques can easily be applied in LTE, in which single-carrier based transmission is utilised in the UL, or even in LTE-A if multi-carrier UL transmission would be employed.
Multi-user one-way, two-path relay channel with AF (section 4.3)	- Adaptations of the OFDMA resource allocator specific to the OWTP-CR protocol: group data bursts by P and allocate $(P+1)$ consecutive bursts to each transmission of P data bursts in the frame. - The complexity introduced by these new functions does not depend on the number	New features have to be added in the RS and MS receivers: RS: amplification coefficient computation and interference cancellation algorithms (complexity: linear in the number of users to be served by the RS).	The OWTP-RC does not require any particular feature at the transmitter side.	Each DL OWTP transmission uses a single frequency resource (i.e a group of contiguous subchannels): relays forward the signal on the same frequency resource as it was received. Different users are scheduled on orthogonal frequency resources.	The proposed technique was designed for generic OFDMA and simulations performed in the framework of the WIMAX 802.16e OFDMA DL. It is therefore applicable to LTE, which also uses OFDMA in the DL, but with different frame format and parameter settings.

	of users but on the number of transmissions occurring in a frame.	• MS: performs the joint detection of $P(P=4)$ assumed in section 4.3) data bursts. Reduced-complexity sub-optimal detector (with respect to conventional sub-optimal detectors such as the MMSE) proposed for power-efficiency reasons. • Single-user detection at the RS and MS.		• Additional bandwidth could be used to schedule additional users, in the same way as in the primary bandwidth.	
--	--	--	--	---	--

7 CONCLUSIONS

This deliverable has presented advanced MU non-cooperative and cooperative techniques for single cell communication systems. The main findings of this deliverable are summarised in the following

In section 3.1.1 the problem of the UL user pairing in conjunction with the best band selection has been faced. The adopted pairing strategy is based on a two steps (joint or disjoint) approach: the band is subdivided in 100 frequency chunks of 14 adjacent subcarriers each, so that in each chunk the channel transfer function is nearly constant. Then the selection is operated by of the most orthogonal pairs based on ZF capacity metrics and the band allocation is assigned by adopting an average capacity maximisation criteria. Five different schedulers are proposed to select the best pairing and frequency chunk per each user. It is assumed that all the MS are at equal distance from the BS and thus experience the same pathloss (the case of inhomogeneous distribution of pathloss among the MSs is faced in section 3.1.2). The proposed pairing algorithms are to be seen as suboptimal in the sense that the coupling process is a tradeoff w.r.t. the full matrix dual user MMSE approach, achieving a search complexity of order n^2 instead of $n!$ (n being the number of MSs). The performance have been assessed by adopting detection schemes based on ZF, MMSE, Joint MMSE, SIC, also proposing reduced-complexity schemes for the orthogonality metric computation. The results indicate that the proposed pairing/scheduling and band selection technique provides the restoring in MIMO of the MU diversity gain when considering user pairing on the top of best band allocation, along with the indications that pairing is always better than single user selection and that best band allocation gives huge improvements in SIMO configurations.

In section 3.1.2, we focussed on the sub-optimal adaptation of the of the user pairing in UL based on coherent joint BF at the UEs and at the BS (introduced in section 3.1.1.2.2) in the presence of mobility and realistic inhomogeneity in the UEs SNR distribution in the scenario with the pathloss models introduced in section 2. It has been shown that the presence of mobility poses severe limitations to the effectiveness of the methodology. In addition, also the inhomogeneous distribution of the SNR in a realistic scenario poses additional challenges to the proper working of the scheduling/pairing protocol outlined in section 3.1. The presence of mobility steered toward a re-design of the frame format (specified in section 5) and a sub-optimal aggregate methodology for the MU channel estimation at the BS. The strong inhomogeneity in realistic scenarios has been faced by introducing a different pairing and scheduling algorithm with respect to the one proposed in section 3.1.1.2.2. It has been demonstrated that, in the absence of mobility, the proposed scheduling/pairing protocol is capable of (nearly) insuring the maximal spectral efficiency offered by the system, meanwhile maintaining an acceptable level of fairness among the active users. However the presence of mobility indicates that the achievable aggregate spectral efficiency gains are jeopardised for speeds distribution above 10 km/h (for 2 GHz carrier), thus limiting the effectiveness of the technique to a pool of UEs (selected among all the active UEs) fulfilling this requirement.

In section 3.2.1 we compare the DL WSR of a multicarrier system when two different schemes are used for allocating subcarriers to some users. In one scheme, each subcarrier is exclusively used by only one user at a time. In the other scheme, each subcarrier can be further reused with one additional user sharing the same subcarrier by simultaneously using it along with the other user. In the case that two users are loaded on the same subcarrier, the users with the strongest and weakest channels are identified. For the user with the strongest channel, WSR is evaluated by assuming perfect interference cancellation at the receiver. On the other hand, the WSR is calculated for the user with the weakest channel by assuming that the interference is tolerated. The subcarrier reuse allocation scheme is shown to perform better than the orthogonal scheme, since when the target rate for one user is fixed, then the reuse scheme can provide additional rate for the other user by adjusting the weights of the two users. It shall be noted that the subcarrier reuse allocation scheme requires extra processing at the receiver of the user with the strongest channel (for interference cancellation).

In section 3.2.2 different scheduling criteria have been investigated for the superposed transmission in the DL: WSR and PF. Those scheduling algorithms depends on the parameter τ which controls the performance of the algorithm, imposing a greedy performance when τ is close to zero and a round robin performance when τ is close to one. It has been observed that the WSR scheduler only is able to improve the performance of the Qualcomm scheduler for certain values of τ , while the MU PF scheduler is always superior to the Qualcomm scheduler, in terms of throughput and delay. In fact, the throughput gain over the Qualcomm scheduler depends on the number of users that are simultaneously scheduled (2 or 3) and the parameter τ . For values of τ close to 1, the throughput gains are better with a higher number of scheduled users.

Section 4.1 has investigated the optimal resource allocation for the MU relay-assisted transmission dealing with one-way and two-way relays, where the scenario is defined by one BS, one RS and two MSs. Different transmit power constraints have been considered. It has been observed that under the sum-average and individual average power constraint the TWRC protocol I gets the best performance. But under a max-power constraint, the TWRC forwarding is able to enhance the TWRC protocol I. The benefits of MU relay techniques are suitable for asymmetric scenarios, where the two served MSs present different SNRs. Since we know from section 3.2.2 that the average served throughput (or priority to be served for each user) also has an important role for the user selection, a cellular throughput evaluation has been done in section 4.1.2. There, the scheduler is based on the PF criterion, and the effect of the antenna configuration and channel model are investigated. The techniques looked into in section 4.1 are based on OFDM accommodating UL and DL transmissions under TDD.. Results presented in section 4.1.2 elucidate the cellular throughput gains of the different techniques under a multi-user PF scheduler. The benefits of using superposed transmission in direct transmission provide throughput gains in the range of [11%-30%] depending on the channel model assumed. When the RS is introduced, the throughput gains over direct transmission without supersposed transmission attain 50% or 100% depending on the channel model for both the OWRC and TWRC protocol I techniques.

Section 4.2 has presented novel power allocation methods for non-regenerative cooperative scheme. Various schemes operating with different levels of CSI for single user, single and multi antennas, and single and multi carriers have been investigated. New MU MIMO power allocation methods relying on receive and transmit knowledge for the UL setting of single or multi-carriers AF systems have been proposed under two optimisation criteria, namely, the sum-rate and the WSR. In the single carrier case, we have seen that if only receive CSI is available at the RS then our novel algorithms outperforms the existing ones. Results have also indicated that scheme that uses both transmit CSI at the BS and RS performs worse than these that only rely on FCSI at the RS in the single carrier case. However, in the multi-carrier case, the former schemes provide a huge performance improvement, more than twice as much as sum-rate, especially at low SNRs. Moreover, our WSR based algorithm provides same sum-rate performance as the sum-rate based algorithm but with an improved fairness in single carrier setting.

The studies tackled in section 4.3 have focused on the performance evaluation of the OWTP-CR protocol, compared with the DT and other conventional single-relay assisted protocols, in a single-cell, MU OFDMA context. Simulations performed in an OMNeT++ framework showed that with an adequate relay choice, the OWTP-CR protocol outperforms all the studied protocols regarding the BER performance and improves the quality of service of cell-edge users and the fairness.

8 APPENDIX: CHANNEL MODEL AND SYSTEM CONFIGURATION

The system parameters considered for the simulation have been defined in [ROCKET-1D1] and they are summarized in Table 19.

Table 19: System parameters

Requirement	Value	Requirement	Value
Site-to-site distance	1.5 Km	RS height	32 m
Carrier Frequency	2.5 GHz	RS Tx Power	30 dBm
BW	10 MHz	Number of tx antennas	2,4
Site-RS distance	60% of cell radius	Number of rx antennas	2,4
		Antenna spacing	4*lambda
BS height	32 m	Antenna type	omni
BS Tx Power per sector	40 dBm	Antenna Gain (boresight)	5 dBi
Antenna Gain (boresight)	17 dBi	Noise Figure	5 dB
Number of sectors	3	Cable loss	0 dB
Front-to-back power ratio	20 dB		
3-dB beamwidth	S=3, Phi=70°	Penetration Loss BS-RS	0
Number of tx antennas	2,4	Penetration Loss RS-RS	0
Number of rx antennas	2,4	Penetration Loss BS-MS	10
BS antenna spacing	4*lambda	Penetration Loss RS-MS	10
Noise Figure	5 dB		
Cable loss	2 dB		
MS height	1.5 m	Error Vector Magnitud (EVM)	30 dB
MS Tx Power	23 dBm		
Antenna Gain (boresight)	0 dBi		
Antenna type	omni		
Number of tx antennas	1,2		
Number of rx antennas	1,2		
MS antenna spacing	lambda/2		
Noise Figure	7 dB		
Cable loss	0 dB		

The considered path loss channel models are:

- BS-MS and RS-MS links:

o Baseline Test Scenario at 2.5 GHz [IEE802.16m-EMD-09]

$$PL(d) = 130.19 + 37.6 \log_{10}(d) \quad [dB] \quad (244)$$

Standard deviation of shadow fading: $\sigma = 8 \text{ dB}$

o Sub-Urban LOS/NLOS Winner II C1 Model [ROCKET-1D1]

Probability of LOS as a function of distance d ,

$$P_{LOS} = \min\left(\frac{18}{d}, 1\right) \left(1 - \exp\left(-\frac{d}{63}\right)\right) + \exp\left(-\frac{d}{63}\right) \quad (245)$$

Path loss for the LOS case:

$$PL(d, f) = \begin{cases} 41.2 + 23.8 \log_{10}(d) + 20 \log_{10}\left(\frac{f}{5 \times 10^9}\right) & 20 \leq d \leq d_{BP} \\ 40 \log_{10}(d) + 11.65 - 16.2 \log_{10}(h_{BS} h_{MS}) + 3.8 \log_{10}\left(\frac{f}{5 \times 10^9}\right) & d > d_{BP} \end{cases} \quad (246)$$

where d_{BP} is the breakpoint distance defined by

$$d_{BP} = \frac{4 h_{MS} h_{BS}}{3 \times 10^8} f$$

Standard deviation of shadow fading in the LOS case:

$$\sigma_{LOS}(d) = \begin{cases} 4 \text{ dB} & 20 \leq d \leq d_{BP} \\ 6 \text{ dB} & d > d_{BP} \end{cases} \quad (247)$$

Path loss for the NLOS case:

$$PL(d, f) = (44.9 - 6.55 \log_{10}(h_{BS})) \log_{10}(d) + 23.46 + 5.83 \log_{10}(h_{BS}) + 20 \log_{10}\left(\frac{f}{2 \times 10^9}\right) \quad (248)$$

Standard deviation of shadow fading in the NLOS case:

$$\sigma_{NLOS}(d) = \sigma_{\mu} \left(1 - \exp\left(-\frac{PL(d) - PL_{free}(d)}{4}\right) \right) + 1.5 \quad [\text{dB}] \quad (249)$$

where σ_{μ} is equal to 8.2 dB and PL_{free} is given by,

$$PL_{free}(d, f) = 20 \log_{10}\left(\frac{4\pi d}{3 \times 10^8} f\right) \quad [\text{dB}]$$

o COST Hata model for urban areas

From 1.5 GHz to 2.0 GHz, MS height from 1 to 10m, BS height from 30m to 200m, link distances from 1 to 20 km.

$$L = 46.3 + 33.9 \log f - 13.82 \log h_B - a(h_R) + (44.9 - 6.55 \log h_B) \log d + C, \quad (250)$$

$$a(h_R) = (1.1 \log f - 0.7) h_R - 1.56 \log f - 0.8$$

$$\text{with } C = \begin{cases} 0 \text{ dB} & \text{for medium-size cities and suburban areas} \\ 3 \text{ dB} & \text{for metropolitans areas} \end{cases},$$

And where L is the mean pathloss (dB), h_B is the BS height (m), h_R is the UE height (m), d is the link distance (km), f is the frequency (MHz).

o Hata model for open areas:

$$L_O = L_U - 4.78 (\log f)^2 + 18.33 \log f - 40.94, \quad (251)$$

where L_O is the open area mean pathloss (dB), L_U is the urban area mean pathloss (dB), f is the frequency (MHz).

o Hata model for suburban areas:

$$L_S = L_U - 2 \left(\log \frac{f}{28} \right)^2 - 5.4, \quad (252)$$

where L_O is the open area mean pathloss (dB), L_U is the urban area mean pathloss (dB), f is the frequency (MHz).

- **Classic two slopes model**

$$PL = \begin{cases} 20\log_{10}\left(\frac{4\pi f}{c}\right) + 20\log_{10}(d) & \text{if } 0 < d \leq d_{BP} \\ 20\log_{10}\left(\frac{4\pi f}{c}\right) + 20\log_{10}(d_{BP}) + 10\alpha \log_{10}\left(\frac{d}{d_{BP}}\right) & \text{if } d \geq d_{BP} \end{cases}, \quad (253)$$

where d , d_{BP} and α are the distance, the break-point distance and the pathloss exponent, respectively. For both the IEEE 802.16n and 16m $d_{BP} = 10\text{m}$, whereas $\alpha=3.5$ and $\alpha=2.5$ for carrier frequencies $f=5.2\text{ GHz}$ and $f=2.4\text{ GHz}$, respectively.

Figure 140 gives a synoptic view of the effect of the adopted pathloss models: two-slopes, Hata Urban, Hata Suburban and Hata Open, assuming a BS Tx power of 30 dBm and the other relevant parameters specified in the figure caption.

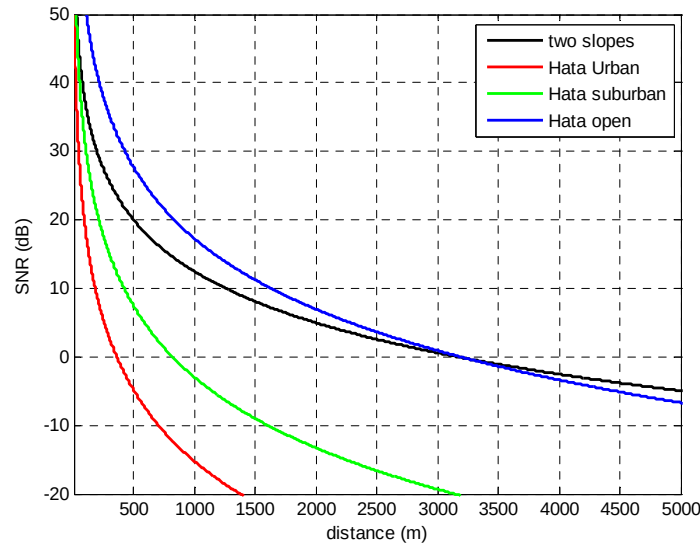


Figure 140: mean SNR as a function of the distance from the BS for the shadowing models Hata urban suburban and open and for the simplified two-slope model. BS height 30 m, UE height 2 m, 2 GHz, Tx power 30 dBm, no antenna gain.

- BS-RS link:

- o **802.16j EVM type D Baseline Test Scenario** [section 14.3.1.1.2 of IEEE802.16m-EMD-09]

$$PL(d) = \begin{cases} 20\log_{10}\left(\frac{4\pi d}{\lambda}\right) & \text{if } d \leq d_{BP} \\ 20\log_{10}\left(\frac{4\pi d_{BP}}{\lambda}\right) + 10\gamma \log_{10}\left(\frac{d}{d_0}\right) + 6\log_{10}\left(\frac{f}{2 \times 10^9}\right) + \Delta PL_h & \text{if } d_{BP} > d \end{cases} \quad (254)$$

where

$$d_0 = 100m, \quad d_{BP} = d_0 10^{-\frac{6 \log_{10} \left(\frac{f}{2 \times 10^9} \right) + \Delta PL_h}{10\gamma}}$$

$$\Delta PL_h = \begin{cases} -10 \log_{10} \left(\frac{h_{RS}}{3} \right) & \text{if } h_{RS} \leq 3 \\ -20 \log_{10} \left(\frac{h_{RS}}{3} \right) & \text{if } h_{RS} > 3 \end{cases}, \quad \gamma = a - b h_{BS} + \frac{c}{h_{BS}}, \quad a = 3.6, b = 0.005, c = 20$$

o Urban LOS Path Loss Winner II B5a Model

$$PL(d) = 36.5 + 20 \log_{10} \left(\frac{f}{2.5 \times 10^9} \right) + 23.5 \log_{10}(d) \quad [dB] \quad (255)$$

Standard deviation of shadow fading: $\sigma = 3.4 \text{ dB}$.